Updated Renard Diamond Project Mine Plan and Mineral Reserve Estimate
Québec, Canada

NI 43-101 Technical Report

March 30, 2016

Qualified Persons
Mr. Patrick Godin, Eng., Stornoway Diamond Corporation
Mr. Robin Hopkins, P.Geol. (NT/NU), Stornoway Diamond Corporation
Mr. Paul Bedell, P.Eng., Golder Associates Ltd

Effective Date
December 31, 2015

CAUTIONARY STATEMENT REGARDING FORWARD-LOOKING STATEMENTS

This document contains forward-looking information and forward-looking statements (collectively referred to herein as “forward-looking information” or “forward-looking statements”). All, statements, other than statements of historical fact regarding Stornoway or the Renard Diamond Project, are forward-looking statements. Any statements that express or involve discussions with respect to predictions, expectations, beliefs, plans, projections, objectives, assumptions or future events or performance (often, but not always, using words or phrases such as “expects”, “anticipates”, “plans”, “projects”, “estimates”, “assumes”, “intends”, “strategy”, “goals”, “objectives”, “schedule” or variations thereof or stating that certain actions, events or results “may”, “could”, “would”, “might” or “will” be taken, occur or be achieved, or the negative of any of these terms and similar expressions) are not statements of historical fact and may be forward-looking statements. This report contains forward-looking statements including with respect to, without limitation: (i) the amount of Mineral Reserves, Mineral Resources and exploration targets; (ii) the amount of future production over any period; (iii) assumptions relating to recovered grade, average ore recovery, internal dilution, mining dilution and other mining parameters; (iv) assumptions relating to gross revenues, operating cash flow and other revenue metrics; (v) mine expansion potential and expected mine life; (vi) the expected time frames for the completion of construction, start of mining and commercial production at the Renard Diamond Project and the financial obligations or costs incurred by Stornoway in connection with such mine development; (vii) the expected time frames for the completion of the open pit and underground mine at the Renard Diamond Project; (viii) future exploration plans; (ix) future market prices for rough diamonds; and (x) future market prices for liquefied natural gaz and diesel.

All forward-looking statements in this report are made based upon certain assumptions and estimates made as of the date such statements are made and are subject to risk factors and uncertainties, many of which may not be controlled or predicted. Material assumptions regarding forward-looking statements include without limitation: (i) required capital investment and estimated workforce requirements; (ii) estimates of net present value; (iii) receipt of regulatory approvals on acceptable terms within commonly experienced time frames; (iv) anticipated timelines for completion of construction, commencement of mine production and development of an open pit and underground mine at the Renard Diamond Project, which heavily depend, among other things, on adequate availability and performance of skilled labour, engineering and construction personnel, performance of mining and construction equipment and timely delivery of components; (v) anticipated geological formations; (vi) market prices for rough diamonds and the potential impact on the Renard Diamond Project; (vii) the satisfaction or waiver of all conditions under each of the Forward Sale of Diamonds, the Senior Secured Loan, the COF and the Equipment Facility to allow Stornoway to draw on the funding available under those financing elements for the completion of the development and construction of the Renard Diamond Project; (viii) Stornoway’s interpretation of the geological drill data collected and its potential impact on stated Mineral Resources and mine life; and (ix) future exploration plans and objectives.

By their very nature, forward-looking statements involve inherent risks and uncertainties, both general and specific, and risks exist that estimates, forecasts, projections and other forward-looking statements will not be achieved or that assumptions do not reflect future experience. We caution readers not to place undue reliance on these forward- looking statements as a number of important risk factors could cause the actual outcomes to differ materially from the beliefs, plans, objectives, expectations, anticipations, estimates, assumptions and intentions expressed in such forward-looking statements. These risk factors may be generally stated as the risk that the assumptions and estimates expressed above do not occur, including the assumption in many forward-looking statements that other forward-looking statements will be correct, but specifically include, without limitation: (i) risks relating to variations in the grade, kimberlite lithologies and country rock content within the material identified as Mineral Resources from that predicted; (ii) variations in rates of recovery and breakage; (iii) the uncertainty as to whether further exploration of exploration targets will result in the targets being delineated as Mineral Resources; (iv) developments in world diamond markets; (v) slower increases in diamond valuations than assumed; (vi) risks relating to fluctuations in the Canadian dollar and other currencies relative to the US dollar; (vii) increases in the costs of proposed capital and operating expenditures; (viii) increases in financing costs or adverse changes to the terms of available financing, if any; (ix) tax rates or royalties being greater than assumed; (x) uncertainty of results of exploration in areas of potential expansion of resources; (xi) changes in development or mining plans due to changes in other factors or exploration results; (xii) risks relating to the receipt of regulatory approvals;

(xiii) the effects of competition in the markets in which Stornoway operates; (xiv) operational and infrastructure risks; (xv) execution risk relating to the development of an operating mine at the Renard Diamond Project; (xvi) failure to satisfy the conditions to the effectiveness, funding or availability, as the case may require, of each of the Forward Sale of Diamonds, the Senior Secured Loan, the COF and the Equipment Facility.

Many of these uncertainties and contingencies can affect Stornoway's actual results and could cause actual results to differ materially from those expressed or implied in any forward-looking statements made by, on or behalf of, Stornoway. All of the forward-looking statements made in this report are qualified by these cautionary statements. Stornoway and the Qualified Persons who authored this report undertake no obligation to update publicly or otherwise revise any forward-looking statements whether as a result of new information or future events or otherwise, except as may be required by law.

CERTIFICATE OF QUALIFIED PERSON

Robin Hopkins

I, Robin Hopkins, P.Geol. (NT/NU) of North Vancouver, British Columbia, Canada, do hereby certify that:

1. I am Vice President, Exploration, for Stornoway Diamond Corporation (118-980 West 1st Street, North Vancouver, British Columbia, V7P 3N4, Canada).

2. I am a co-author of the report prepared by Stornoway Diamond Corporation titled "Updated Renard Diamond Project Mine Plan and Mineral Reserve Estimate, Québec, Canada, NI 43-101 Technical Report" (the Technical Report), dated March 30, 2016, with an effective date of December 31, 2015.

3. I graduated from the University of Waterloo, Waterloo, Ontario in 1986 (B.A.Sc. Honours Earth Sciences) and have practiced my profession continuously since graduation.

4. I am a member in good standing of the Northwest Territories and Nunavut Association of Professional Engineers and Geoscientists (Licence #1027).

5. I have read the definition of "qualified person" set out in National Instrument 43-101 – *Standards of Disclosure for Mineral Projects* (NI 43-101) and certify that, by reason of my education, past relevant work experience and affiliation with a professional association, I fulfil the requirements of a "qualified person" for the purposes of NI 43-101.

6. I most recently visited the Renard property on October 22, 2014.

7. I am responsible for Sections 7, 8, 9, 10, 11, 12, 13 and 14 of the Technical Report.

8. I am not independent of Stornoway Diamond Corporation due to my position as an Officer of the Corporation, as defined in Section 1.5 of NI 43-101.

9. I have been involved with exploration activities at the Renard Diamond Project, including drilling, bulk sampling and micro/macro diamond analyses, since January of 2007.

10. I have read NI 43-101 and Form 43-101F1, and the sections of the Technical Report for which I am responsible have been prepared in compliance with that instrument.

11. As of the effective date of this Technical Report, to the best of my knowledge, information and belief, the sections of the Technical Report for which I am responsible contain all the scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Signed and dated this 30th day of March, 2016 at North Vancouver.

(Signed and sealed) "Robin Hopkins", P.Geol. (NT/NU)

Robin Hopkins, P.Geol. (NT/NU)
Vice President, Exploration
Stornoway Diamond Corporation

CERTIFICATE OF QUALIFIED PERSON
Patrick Godin

I, Patrick Godin, Eng., do hereby certify that:

1. I am employed by Stornoway Diamond Corporation as the Chief Operating Officer, which office is situated at 1111 St-Charles Street West, Suite 400, West Tower, Longueuil, Québec J4K 5G4.
2. I am a co-author of the report prepared by Stornoway Diamond Corporation titled "Updated Renard Diamond Project Mine Plan and Mineral Reserve Estimate, Québec, Canada, NI 43-101 Technical Report" (the Technical Report), dated March 30, 2016, with an effective date of December 31, 2015.
3. I graduated from Université Laval (Québec) with a Bachelor of Applied Sciences – Mining Engineering in 1991.
4. I am a Professional Engineer, member of the Professional Engineers of Québec (#106565).
5. I have practiced my profession for 25 years. I have mining experience in both open pit and underground operations, including five years specific to the diamond industry where I have been involved in the design, construction and commissioning aspects of diamond mines. I have worked for Stornoway Diamond Corporation for five years.
6. I have read the definition of "qualified person" set out in National Instrument 43-101 – Standards of Disclosure for Mineral Projects (NI 43-101) and certify that, by reason of my education, past relevant work experience and affiliation with a professional association, I fulfil the requirements of a "qualified person" for the purposes of NI 43-101.
7. I have not previously authored a technical report on the Renard Project. I have been involved with the operation for the past five years in my role as Chief Operating Officer.
8. I am responsible for Sections 1, 2, 3, 4, 5, 6, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 (at the exception of 18.4.1 and 18.4.2), 25, 26 and 27 of the Technical Report
9. I most recently visited the Renard property on March 22, 2016.
10. I am not independent of Stornoway Diamond Corporation due to my position as an Officer of the Corporation, as defined in Section 1.5 of NI 43-101.
11. I have read NI 43-101, and the sections of the Technical Report for which I am responsible have been prepared in compliance with that Instrument.
12. As of the effective date of the Technical Report, to the best of my knowledge, information and belief, the sections of the technical report for which I am responsible contain all scientific and technical information that is required to be disclosed to make those sections of the Technical Report not misleading.

Signed and dated on this 30th day of March, 2016, in Montreal.

(Signed and sealed) "*Patrick Godin*", Eng.

Patrick Godin, Eng.
Chief Operating Officer
Stornoway Diamond Corporation

CERTIFICATE OF QUALIFIED PERSON

Paul Micheau Bedell

I, Paul M. Bedell, P.Eng., do hereby certify that:

1. I am a Principal and Senior Geotechnical Engineer employed at Golder Associates Ltd., located at 2nd Floor 2920 Virtual Way, Vancouver, British Columbia, Canada.
2. I am a co-author of the report prepared by Stornoway Diamond Corporation entitled "Updated Renard Diamond Project Mine Plan and Mineral Reserve Estimate, Québec, Canada, NI 43-101 Technical Report" (the Technical Report), dated March 30, 2016, with an effective date of December 31, 2015.
3. I graduated with a Bachelor of Engineering Science degree in Civil Engineering in 1994, and a Master's degree in Engineering Science in 1997, both from The University of Western Ontario in London, Ontario.
4. I am a registered Professional Engineer in British Columbia (member #28511), Ontario (member #100026653), Saskatchewan (member #11859), and the Northwest Territories and Nunavut (member #L1746).
5. I have worked as an engineer since my graduation from The University of Western Ontario. For the past 20 years I have been employed with Golder Associates Ltd. During this period I have fulfilled the role of engineer on mining projects directing and completing geotechnical engineering for mine waste management and soils. I currently hold the position of Principal, Senior Geotechnical Engineer.
6. I have read the definition of "qualified person" set out in National Instrument 43-101 – Standards of Disclosure for Mineral Projects (NI 43-101) and certify that, by reason of my education, past relevant work experience and affiliation with a professional association, I fulfil the requirements of a "qualified person" for the purposes of NI 43-101.
7. I am responsible for Sections 18.4.1 and 18.4.2 of the Technical Report.
8. I visited the Renard Diamond Project property from September 29 to October 1, 2010, inclusive.
9. I am independent of Stornoway Diamond Corporation applying the test set out in Section 1.5 of NI 43-101.
10. I have had no prior involvement with the property that is subject of the Technical Report.
11. As of the effective date of this Technical Report, to the best of my knowledge, information, and belief, the sections for which I am responsible in this technical report contain all the scientific and technical information that is required to be disclosed to make this technical report not misleading.

Signed and dated on this 30th day of March, 2016.

(Signed and sealed) "*Paul M. Bedell*", M.E.Sc., P.Eng.

Paul M. Bedell, M.E.Sc., P.Eng.

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1. SUMMARY

1.1 Introduction

The Renard Diamond Project is located in north-central Québec. Stornoway Diamond Corporation, a company listed on the TSX, holds 100% interest in the Renard Diamond Project through its wholly-owned subsidiary, Stornoway Diamonds (Canada) Inc./ Les Diamants Stornoway (Canada) inc. (SDCI). In this report, Stornoway may be used to refer either to Stornoway Diamond Corporation or to Stornoway Diamonds (Canada) Inc.

On July 8, 2014, Stornoway announced that it had entered into a series of transactions intended to provide a comprehensive funding package for the construction of the Renard Diamond Project. In total, gross proceeds of $946 million (assuming a conversion rate of 1 USD = 1.10 CAD) has been funded or committed for funding through the combination of senior and subordinated debt facilities, equity issuances, a purchase and sale agreement (or streaming agreement) for the forward sale of diamonds, and an equipment facility (collectively referred to as the Financing Transactions). The construction of the Renard Diamond Project commenced on July 10, 2014, immediately following the closing of the Financing Transactions.

This technical report is prepared by Stornoway Diamond Corporation (Stornoway) and is intended to be a Form 43-101F1 Technical Report for the purpose of National Instrument 43-101 to provide an updated mine plan as well as provide background and supporting information on the Mineral Reserves estimate at the Renard Diamond Project, as of December 31, 2015.

This report is an update of the previous technical reports on the Renard Diamond Project filed by Stornoway, which include: (i) the technical report filed on January 3, 2012, dated December 29, 2011, presenting a feasibility study for the Renard Diamond Project, (ii) the technical report dated February 28, 2013, which updated the previous feasibility study, and (iii) the technical report updating the Mineral Resource estimate of the Renard Diamond Project dated January 11, 2016.

1.2 Property Location and Site Description

The Project site is located in the James Bay region, central Québec, Canada, approximately 70 km north of the Otish Mountains and some 360 km north-northeast from the mining town of Chibougameau. The site is located at longitude 72°11' West and latitude 52°49' North. The principal access point for employees, contractors and visitors to the Renard mine is via the Clarence and Abel Swallow Airport (a 1,497-m-long gravel landing strip with associated instrumentation, maintenance facilities and passenger terminal operated by Stornoway), with goods and services delivered to site by road on the all-weather Renard Mine Road/Route 167 Extension. The closest settlement is Témiscamie, on the shore of Lac Albanel. The regional centres of Chibougamau and Mistissini act as staging posts for the trans-shipment of materials and personnel. Direct flights to the mine site may also be chartered from southern cities such as Montreal. Power is currently provided to the construction site via diesel generators. A liquefied natural gas storage facility (LNG) and power plant were built on the Renard Project site to provide for future power requirements of the mine operations in line with the LNG Feasibility Study on the viability of an LNG fuelled power plant for the Renard Diamond Project, the results of which were announced by Stornoway in October 2013.

1.3 Property Ownership

The Renard Diamond Project, referred to as the Renard Project in this document, is situated on the Foxtrot Property.

The Foxtrot Property currently covers a total area of 33,825.66 ha with SDCI as the registered holder of a 100% interest. The property comprises 650 claims (33,629.95ha) in four blocks (one large contiguous landholding of 630 claims plus three smaller blocks), mining lease BM 1021 (143.71 ha) and surface lease number 1303 10 000 (199.85 ha). At the effective date of this report, all claims are reported by Stornoway to be in good standing.

SDCI currently holds 100% property interest in the Foxtrot Property, subject to a direct royalty on future diamond production of 2% in favour of DIAQUEM.

1.4 Geology and Mineralization

The Renard Diamond Project is located within the eastern portion of the Superior Craton. The Craton is considered to be an amalgamation of small continental fragments of Mesoarchean age and Neoarchean oceanic plates, with a complex history of aggregation.

There are five known episodes of kimberlitic volcanism in Québec; from south to north, the kimberlite fields are Témiscamingue, Desmaraisville, Otish, Wemindji and Torngat. The Renard Cluster is considered to be part of the Otish kimberlitic volcanic event.

To date, nine (9) kimberlite pipes have been identified over a 2-km^2 area in the Renard Cluster (Renard 1 to Renard 10, with Renard 5 and Renard 6 being one kimberlite body known as Renard 65). The kimberlites are typically spaced between 50 m and 500 m from each other. Whole-rock trace element compositions suggest a closer affinity to Group I kimberlite (olivine-rich, monticellite, serpentine, calcite kimberlites), with derivation from a garnet-bearing mantle. The Lynx and Hibou kimberlite dyke systems are situated to the west of the Renard pipes.

In this report, reference is made to the Renard Core Area, which is defined as a 37-ha area that contains the Renard 2, Renard 3, Renard 4, Renard 65 and Renard 9 kimberlite pipes as well as the Lynx and Hibou dyke systems. Kimberlites are discussed individually in Section 7 of this report.

Geophysical data and drill information from delineation and bulk sampling programs indicate that the Renard pipes are irregular and elliptical in plan view. Surface areas of the kimberlite portion of the pipes range from 0.1 ha to 2.0 ha, and they are situated within a larger halo of fractured country rock.

The pipes comprise root zone to diatreme facies rocks characterized by complex internal geology, with the dominant phase composed of massive volcaniclastic kimberlite classified as tuffisitic kimberlitic breccia (TKB). The TKB contains 15% to 90% by volume, fresh to moderately altered granitoid, country rock clasts in a matrix that is generally dominated by serpentinized olivine macrocrysts, carbonates and serpentine. A minor amount of coherent “hypabyssal” kimberlite (HK) with a small proportion of highly altered and digested country rock xenoliths is also present. These are considered to be later-stage intrusions that occur throughout the pipe-like bodies, on the periphery, or as standalone dyke systems. This material is characterized by abundant, disseminated calcite, olivine macrocrysts and less than 15% by volume crustal xenoliths. Extensive sampling programs conducted between 2001 and 2015 have demonstrated that the pipes and dykes are diamondiferous.

1.5 Mineral Processing

The process plant design capacity is 2,160,000 tpa (dry solids basis). The plant overall utilization is estimated to be 78%, equivalent to an operating time of 6,833 hours per year or 315 tph and 6,000 tpd. Stornoway expects to increase the plant throughput to 323 tph, already within initial plant equipment capacity, and increase the plant availability to 83.5% by optimizing plant maintenance sequences. Once the plant achieves sustainable nameplate throughput, further optimization work will be conducted focusing on liberation, diamond breakage and increasing the overall plant utilization. Taking the above into consideration by July 2018, Stornoway expects the plant will operate at a throughput of 7,000tpd at an overall plant utilization of 83,5%.

The plant is designed to liberate, concentrate and recover diamonds in the size range of 45 mm to 1 mm. Diamond liberation is accomplished through a three-stage crushing process using a primary jaw crusher, a secondary cone crusher and a tertiary high-pressure grinding roll (HPGR). The crushed ROM material is washed in a rotary scrubber after the primary crusher stage, and the ore is then sized in preparation for the concentration processes and further liberation. The concentration process occurs in two different stages: 1) the Large Diamond Recovery (LDR) process, and 2) the Dense Medium Separation (DMS). Both produce a concentrate that will be treated in a secure diamond recovery facility using diamond differentiation techniques based on magnetic, X-ray, laser Raman and ultra-violet technologies, with hand-sorting as the final de-falsing step to produce a nominally 98% diamond by weight product at final sorting. Both stages will produce a non-diamond product that will be further sized, and the coarser size fraction (+6mm) will be recirculated through the HPGR to release locked diamonds. Recovered diamonds will be further cleaned through the diamond cleaning process and prepared for valuation in a purpose-built facility within the recovery plant.

Plant rejects are dewatered and trucked to the processed kimberlite containment facility for disposal.

Process plant design and equipment selections were based on bulk sample plant operational data, test work, experience from other operations and proven technology to minimize diamond breakage.

1.6 Mineral Resources

Following the 2014 deep directional drill program at the Renard 2 kimberlite, and extensive related sampling activities both at Renard 2 and other project kimberlites, an updated Mineral Resource Estimate was completed for Renard 2, Renard 3 and Renard 4 in accordance with the CIM Mineral Resource and Mineral Reserve definitions referred to in NI 43-101, Standards of Disclosure for Mineral Projects. The Mineral Resource Estimate was reported on September 24, 2015. That work, as well as a Target for Further Exploration for the Renard 1, Renard 7 and Renard 10 kimberlite pipes and the Hibou kimberlite dyke system, was more fully documented in an NI 43-101 technical report titled "*2015 Mineral Resource Update for the Renard Diamond Project, Québec, Canada*", with an effective date of September 24, 2015 (available on SEDAR at www.sedar.com). The work presented in the latter report was reviewed to determine the process for establishing the current Mineral Resource Estimate, as documented in Section 14 of the current technical report.

The 2015 Mineral Resource Estimate comprises the integration of kimberlite volumes, density, petrology and diamond content data derived from 101,078 m of diamond drilling (497 holes), 6,151 m of large-diameter reverse circulation (RC) drilling (36 holes), 23.7 t of samples submitted for microdiamond analysis, 196 cts of diamonds (3,107 stones) recovered from drill core, 605 cts of diamonds (7,181 stones) recovered from RC drilling, 4,404 cts of diamonds (40,521 stones) recovered from underground bulk sampling, and 5,219 cts of diamonds (52,474 stones) recovered from surface and trench sampling. The Mineral Resource Estimate also incorporates information derived from approximately 150 drill holes, 37 surface test pits and 12 trenches undertaken for geotechnical and hydrogeological purposes. Results are presented in Tables 1.1 and 1.2. The Mineral Resource Estimate is based on the continuity of geology between kimberlite at depth and kimberlite nearer surface, and the generally low variation in sample results for the different kimberlite phases with depth.

Table 1.1: September 2015[1] Indicated[2] Mineral Resources[3] – Renard Diamond Project

Body	Total Tonnes[4]	Total Carats[4]	Average cpht[5]	Average Dilution %
Renard 2 Total	25,696,000	21,578,000	84.0	54.7
Renard 2 w/o CRB[6]	*21,417,000*	*20,680,000*	*96.6*	*46.4*
Renard 2 CRB	*4,279,000*	*899,000*	*21.0*	*96.0*
Renard 3	1,820,000	1,859,000	102.2	33.5
Renard 4	7,246,000	4,437,000	61.2	48.9
Renard 65	7,865,000	2,300,000	29.2	42.8
Renard 9	0	0	0	n/a
Lynx	0	0	0	n/a
Hibou	0	0	0	n/a
Total	42,627,000	30,175,000	70.8	50.6

Notes
1) Effective date is September 24, 2015.
2) Classified according to CIM Definition Standards.
3) These Mineral Resources are not Mineral Reserves as they do not have demonstrated economic viability.
4) Figures may not add up to total due to rounding.
5) Carats per hundred tonnes. Estimated at a +1 DTC sieve size cut-off.
6) Excludes discrete, more dilute kimberlite facies not previously incorporated into the July 2013 Resource. Provided to facilitate more direct comparison with the 2013 Mineral Resource Estimate.

Table 1.2: September 2015[1] Inferred[2] Mineral Resources[3] – Renard Diamond Project

Body	Total Tonnes[4]	Total Carats[4]	Average cpht[5]	Average Dilution %
Renard 2 Total	6,589,000	3,883,000	58.9	72.8
Renard 2 w/o CRB[6]	*4,080,000*	*3,356,000*	*82.3*	*58.5*
Renard 2 CRB	*2,510,000*	*527,000*	*21.0*	*96.0*
Renard 3	542,000	609,000	112.3	39.4
Renard 4	4,750,000	2,455,000	51.7	56.3
Renard 65	4,928,000	1,181,000	24.0	56.5
Renard 9	5,704,000	3,040,000	53.3	63.6
Lynx	1,798,000	1,924,000	107	n/a
Hibou	178,000	256,000	144	n/a
Total	24,490,000	13,348,000	54.5	n/a

Notes:
1) Effective Date is September 24, 2015.
2) Classified according to CIM Definition Standards.
3) Mineral Resources are not Mineral Reserves and do not have demonstrated economic viability.
4) Figures may not add up to total due to rounding.
5) Carats per hundred tonnes. Estimated at a +1 DTC sieve size cut-off.
6) Excludes discrete, more dilute kimberlite facies not previously incorporated into the July 2013 Indicated Resource. Provided to facilitate a more direct comparison with the 2013 Mineral Resource Estimate.

There is additional potential for the Project because the geological models for Renard 2, Renard 3, Renard 4, Renard 65 and Renard 9 are based on conservative geometries for the kimberlites at depth and the models do not incorporate areas of limited drilling at depth. New work was also undertaken during 2013, 2014 and 2015 on the Renard 1, Renard 7, Renard 8 and Renard 10 kimberlite pipes, and various kimberlite dyke systems on the property. Table 1.3 provides details on the Target for Further Exploration (TFFE; previously known as Potential Mineral Deposit or PMD before the June 30, 2011 revisions to NI 43-101). Total TFFE was identified as representing between 76 and 113 Mt, containing between 33 and 71 million carats of diamonds at an average grade of 43 to 63 cpht. The potential quantity and grade of any TFFE is conceptual in nature, there is insufficient exploration to define a Mineral Resource, and it is uncertain if further exploration will result in the target being delineated as a Mineral Resource.

Table 1.3: September 2015 Target for Further Exploration[1] – Renard Diamond Project

	Low Range		High Range	
Body	**Total tonnes**	**Average cpht**	**Total tonnes**	**Average cpht**
Renard 2	6,138,000	60	15,472,000	100
Renard 3	3,352,000	105	3,773,000	168
Renard 4	11,120,000	50	15,358,000	77
Renard 65	29,026,000	25	40,926,000	33
Renard 9	3,858,000	52	6,327,000	68
Lynx	3,089,000	96	3,199,000	120
Hibou	3,469,000	104	4,028,000	151
Renard 1	8,620,000	20	12,983,000	30
Renard 7	6,342,000	30	9,431,000	40
Renard 10	1,217,000	60	1,730,000	120
Total[2]	76,232,000	43	113,227,000	63

Notes: 1) Previously known as Potential Mineral Deposit before the June 30, 2011 revisions to NI 43-101.
2) Figures may not add up to total due to rounding.

1.7 Mineral Reserves

A mine plan has been developed to extract the Indicated Mineral Resources of the Renard Project. The Renard 2 and Renard 3 kimberlite pipes will be mined through a combination of open pit and underground mining methods, whereas the Renard 65 kimberlite pipe will be mined by open pit mining method only and the Renard 4 pipe will be mined by underground method only. Open pit and underground Mineral Reserves were estimated independently based on criteria specific to each method. For both underground and open pit methods, a detailed mine plan was developed, and dilution and recovery modifying factors applied. All Mineral Reserves are classified as Probable Mineral Reserves.

A consolidated summary of open pit and underground Probable Mineral Reserves, by mining method and pipe, effective as of December 31, 2015, is presented in table 1.4. All Mineral Reserves are in the “Probable” category.

Table 1.4: Probable Mineral Reserve Summary – Open Pit and Underground

Mine	Tonnes (k)	Grade (cpht)	Carats (k)	% Tonnes Total	% Carats Total	% Tonnes Method	% Carats Method
OPEN PIT							
R2	1,489	92.7	1,381	4.5%	6.2%	16.7%	34.9%
CRB2A	475	31.4	149	1.4%	0.7%	5.3%	3.8%
CRB	1,575	20.2	319	4.7%	1.4%	17.7%	8.1%
R2 Subtotal	**3,539**	**52.2**	**1,849**	**10.6%**	**8.4%**	**39.7%**	**46.7%**
R3 Subtotal	**794**	**92.3**	**733**	**2.4%**	**3.3%**	**8.9%**	**18.5%**
R65 Subtotal	**4,579**	**30.1**	**1,376**	**13.8%**	**6.2%**	**51.4%**	**34.8%**
TOTAL OP	**8,912**	**44.4**	**3,958**	**26.8%**	**17.9%**	**100%**	**100%**
UNDERGROUND							
R2-290	5,111	63.3	3,236	15.4%	14.6%	21.0%	17.8%
R2-470	4,744	84.7	4,017	14.3%	18.1%	19.5%	22.1%
R2-590	4,750	89.9	4,270	14.3%	19.3%	19.5%	23.5%
R2-710	5,073	81.4	4,132	15.2%	18.7%	20.8%	22.7%
R2 Subtotal	**19,679**	**79.6**	**15,655**	**59.1%**	**70.7%**	**80.8%**	**86.1%**
R3 Subtotal	**1,223**	**70.2**	**858**	**3.7%**	3.9%	**5.0%**	**4.7%**
R4 Subtotal	**3,458**	**48.3**	**1,671**	**10.4%**	7.5%	**14.2%**	**9.2%**
TOTAL UG	**24,360**	**74.6**	**18,184**	**73.2%**	**82.1%**	**100%**	**100%**
STOCKPILE	**153**	**73.5**	**113**				
TOTAL OP, UG & Stockpile	**33,424**	**66.5**	**22,255**	**100%**	**100%**		

Notes to accompany Probable Mineral Reserves Table

1) Probable Mineral Reserves have an effective date of December 31, 2015. The Probable Mineral Reserves were prepared under the supervision of Patrick Godin, Eng., Chief Operating Officer and Director of Stornoway Diamond Corporation. Mr. Godin is a Qualified Person within the meaning of NI 43-101.
2) Probable Mineral Reserves are reported on a 100% basis.
3) The reference point for the definition of Probable Mineral Reserves is at the point of delivery to the process plant.
4) Probable Mineral Reserves are reported at +1.0 mm diamond sized (effective cut-off of 1.0 mm).
5) Probable Mineral Reserves that will be or are mined using open pit methods include Renard 2, Renard 3 and Renard 65. Mineral Reserves are estimated using the following assumptions: Renard 2 and Renard 3 open pit designs assuming external dilution of 4.3% and mining recovery of 98%; Renard 65 open pit design assuming external dilution of 3.5% and mining recovery of 98%.
6) Renard 2, Renard 3 and Renard 4 Probable Mineral Reserves are mined using underground mining methods. The Renard 2 Mineral Reserves estimate assumed an external dilution of 20% and mining recovery of 82%. The Renard 3 Mineral Reserves estimate assumed an external dilution of 14% and mining recovery of 85%. The Renard 4 Mineral Reserves estimate assumed an external dilution of 14% and mining recovery of 78%.
7) Tonnes are reported as thousand metric tonnes, diamond grades as carats per hundred tonnes, and contained diamond carats as thousands of contained carats.
8) Tables may not sum as totals have been rounded in accordance with reporting guidelines.

1.8 Status of Project Development & Project Infrastructure

The Project is currently at the construction and mining development stage and Mineral Reserves have been prepared based on the mining, processing and infrastructure already in place or on designs detailed in this report and incorporating current geological and diamond revenue data.

First ore delivery to the Renard diamond process plant has been scheduled for the end of September 2016, with commercial production (60% of plant capacity achieved over 30 days) to be achieved by December 31, 2016.

The Project Execution Plan proposes concepts and practices that are consistent with those in the Canadian diamond mining industry. The Project design includes open pit mining of the Renard 2, Renard 3 and Renard 65 kimberlite pipes, followed by underground mining of the Renard 2, Renard 3 and Renard 4 kimberlite pipes by the blasthole shrinkage method. The mine plans propose the extraction of 17.5 Mt of waste and 33.4 Mt of ore over a 14 year period using standard equipment and mine designs. Ore will be processed initially at the process plant nameplate capacity of 2.16 Mt/a, and subsequently expanded to 2.5 Mt/a. Process plant designs and equipment selections use proven suppliers and are based on experience from other operations. Security designs are industry standard practices.

Supporting infrastructure includes an effluent treatment plant, potable water treatment plant, a power plant (comprised of seven natural gas generators of 2,050 KW each and three diesel-powered generators of 1,800 KW each), LNG and diesel tank farms, LNG regasification plant, 370 bed accommodation complex, a six-bay maintenance facility, mine offices, an explosive storage and handling area, a 1,494 meter gravel airstrip and telecommunication systems.

Cost estimates were established from first principles where appropriate, or from benchmarking against comparable projects, or derived from actual costing incurred during the first year of the Project's mining operations. Appropriate contingencies and mitigation allowances have been applied, and risk management and peer reviews were held during the construction and development process.

Processed kimberlite will be disposed of through a dry stack facility currently under construction and the management of surface runoff water is facilitated through a system of peripheral drainage ditches designed to direct runoff water to an excavated catch basin for treatment, if required, before release to the environment. Suspended solids are determined to be the main concern for water treatment. Storage capacity of the catch basin accommodates spring runoff and a 100 year return storm event. Mine operations incorporate a program of progressive reclamation that minimizes costs and allows timely monitoring of performance. Waste rock generated by mining will be re-introduced to the underground underground as backfill.

Risks for the Project have been identified and are considered reasonable for the Project's development status.

1.9 Financial Analysis

The project financial analysis incorporates the estimated LOM diamond production profile based on the updated Mineral Reserves, diamond valuation and escalation estimates, CAPEX, OPEX, Sustaining Capital Costs, salvage value, working capital, closure and reclamation costs, taxation, costs under the Mecheshoo Agreement, royalties, the Renard diamond streaming agreement, and the project's financial parameter assumptions. After-tax and pre-tax net present value (NPV) evaluations have been undertaken in real dollar terms (or constant dollars) utilizing an effective date of January 1, 2016. Valuations are presented on an unlevered basis.

LOM gross revenue from diamond sales is estimated at $5,565 million in real dollar terms. LOM OPEX is estimated at $1,878 million in real terms to process 33.4 Mt of ore and produce 22.3 million diamond carats.

The LOM average operating costs of $56.20/tonne, or $84.37/carat produced with the average annual profile presented.

Unlevered after-Tax NPV (7%) is estimated at $974 million, and $1,349 million on a pre-tax basis, in real dollar terms. Project NPV is most sensitive to revenue, specifically the diamond price assumption, followed by operating costs and finally by diamond ore grade and exchange rate.

1.10 Conclusions and Recommendations

There is sufficient technical and scientific information and data to support the updated mine plan and Probable Mineral Reserves and initiate production in 2016.

2. INTRODUCTION

The Renard Diamond Project is located in north-central Québec. Stornoway Diamond Corporation, a company listed on the TSX, holds 100% interest in the Renard Diamond Project through its wholly-owned subsidiary, Stornoway Diamonds (Canada) Inc./Les Diamants Stornoway (Canada) inc. (SDCI). In this report, Stornoway may be used to refer either to Stornoway Diamond Corporation or to Stornoway Diamonds (Canada) Inc.

In 2011, Stornoway engaged a third party engineering firm to complete an independent feasibility study on the Renard Diamond Project in Québec, referred to in this document as the Renard Project or the Project, which Stornoway wholly owns through its subsidiary SDCI. The results of the feasibility study were disclosed in a press release dated November 16, 2011. An NI 43-101 technical report in support of the November 16, 2011 disclosure was filed on January 3, 2012.

In 2012, Stornoway completed an optimized feasibility study, described in a press release dated January 28, 2013, and a NI 43-101 technical report titled “Renard Diamond Project – Feasibility Study Update”, dated February 28, 2013.

On July 8, 2014, Stornoway announced that it had entered into a series of transactions intended to provide a comprehensive funding package for the construction of the Renard Diamond Project. In total, gross proceeds of $946 million (assuming a conversion rate of 1 USD = 1.10 CAD) has been funded or committed for funding through the combination of senior and subordinated debt facilities, equity issuances, a purchase and sale agreement (or streaming agreement) for the forward sale of diamonds, and an equipment facility (collectively referred to as the Financing Transactions). The construction of the Renard Diamond Project commenced on July 10, 2014, immediately following the closing of the Financing Transactions.

Stornoway filed a NI 43-101 technical report updating the Mineral Resource estimate of the Project dated January 11, 2016, titled “2015 Mineral Resource Update”.

This NI 43-101 Technical Report presents an updated mine plan as well as background and supporting information on the Mineral Reserves estimate at the Renard Diamond Project, as of December 31, 2015.

The general list of abbreviations is found in Table 2.1. Lithological abbreviations are found in Table 7.2.

Table 2.1: List of Abbreviations

Abbreviation	Description
2011 Feasibility Study	Refers to the report entitled " The Renard Diamond Project, Québec, Canada, Feasibility Study, NI 43-101 Technical Report" dated December 29, 2011"
2013 Optimization Study	Refers to the report entitled "The Renard Diamond Project, Québec, Canada, Feasibility Study Update, NI 43-101 Technical Report" dated February 28, 2013.
2015 Mineral Resource Estimate	Refers to the 2015 Mineral Resource estimate reported in "2015 Mineral Resource Update for the Renard Diamond Project, Québec, Canada, National Instrument 43-101 Technical Report" dated January 11, 2016"
$ or C$ or CAD	Canadian dollars
US$ or USD	United States dollars
AMEC	AMEC Americas Limited
Ashton	Ashton Mining Canada Inc.
BHS	Blasthole shrinkage
BHSP	Blasthole shrinkage with pillars
ca	Circa
Canadian Environmental Assessment Act or CEAA	*Canadian Environmental Assessment Act* (S.C., c.37)
CAPEX	Capital expenditures
CCR	Cracked country rock
CDPQ	Caisse de Dépot et Placement du Québec
CIM	Canadian Institute of Mining and Metallurgy
CK	Coherent kimberlite
cm	Centimetre
COF	Refers to the unsecured cost overrun facility agreement entered into between Stornoway and CDPQ on July 8, 2014.
Commencement of Commercial Production	means the first day of the month immediately following the month in which the Renard Project's processing plant first processes ore at an average rate of 3,550.7 tons per day.

COMEV or Evaluating Committee	Evaluating Committee of the MDDELCC
COMEX or Provincial Review Committee	Review Committee of the MDDELCC
cpht	Carats per hundred tonnes
cpt	Carats per tonne
CR	Country rock
CRB	Country rock breccia
cts	Carats
DFO or Department of Fisheries and Oceans	Department of Fisheries and Oceans (Canada)
DIAQUEM	DIAQUEM Inc.
DICAN	Diamonds International Canada
DMS	Dense media separation
El.	Elevation
Environment Quality Act or EQA	*Environment Quality Act* (RSQ c. Q-2)
EPCM	Engineering Procurement Construction and Management
Equipment Facility	Refers to the master lease agreement, dated July 25, 2014, entered amongst SDCI (as lessee), Ashton and Stornoway (as guarantors) and Caterpillar Financial Services Limited (as lessor) for the acquisition and financing of certain mine equipment items manufactured by Caterpillar and others, including the Renard Project's mobile mining fleet.
ESIA	*Environmental and Social Impact Assessment*
Explosives Act	*Explosives Act* (R.S.C., 1985 c. E-17)
FCDC	FCDC Sales and Marketing Inc., a wholly-owned subsidiary of SDCI
Financing Transactions	Series of financing transactions intended to provide a comprehensive funding package, in total gross proceeds of $946 million (assuming a conversion rate of 1 USD = 1.10 CAD), for the construction of the Renard Diamond Project, including senior and subordinated debt facilities, equity issuances, a purchase and sale agreement of diamonds, and an equipment financing facility.

Fisheries Act	*Fisheries Act* (R.S.C., 1985 c. F-14)
Foxtrot Property	Refers to 650 claims, mining lease BM 1021 and surface lease 1303 10 000
g	Gram
Ga	Giga annum (billion years)
GCC or GCC-EI or Grand Council of the Crees	Grand Council of the Cree of Québec (Eeyou Istchee)
GDP	Gross domestic product
GeoStrat	GeoStrat Consulting Inc.
GMining	G Mining Services Inc.
GN	Fresh, competent, fine-grained, banded quartz-feldspar-biotite gneiss
Golder	Golder Associates Ltd
GR	Fresh, competent equigranular granite
ha	Hectare
HADD	Harmful alteration, disruption or destruction of fish habitat
HK	Hypabyssal kimberlite
HKt	Transitional hypabyssal kimberlite
HPGR	High-pressure grinding roll
Hz	Hertz
ISRM	International Society for Rock Mechanics
Itasca	Itasca Consulting Canada Inc.
ITH	In-the-hole
JBNQA	James Bay and Northern Québec Agreement 1975
k	thousand
kg	Kilogram
km	Kilometre
km²	Square kilometre
l	Litre

L	Mine level (depth below surface in meters)
LNG	Liquefied natural gas
LNG Feasibility Study	Refers to the feasibility study undertaken during 2013 on the viability of a LNG fuelled power plant for the Renard Diamond Project, the results of which were announced by the Stornoway in November 2013.
kV	Kilovolt
Lakefield	SGS Lakefield Research Ltd
LOM	Life-of-mine
LDR	Large diamond recovery
m	Metre
m^2	Square metre
m^3	Cubic metre
Ma	Mega annum (million years)
Majescor	Majescor Resources Inc.
masl	Meters above sea level
MBJ	Municipality of James Bay
MDDELCC	Ministère du Développement durable, de l'Environnement et de la Lutte contre les changements climatiques (Ministry of Sustainable Development, the Environment and the Fight Against Climate Change) (formerly the MDDEP)
MERN	Ministère de l'Énergie et des Ressources naturelles (Ministry of Energy and Natural Resources) (formerly the MRNF)
MFFP	Ministère de la Faune, des Forêts et des Parcs (Ministry of Wildlife, Forests and Parks)
mm	Millimetre
MRNF	Ministère des Ressources naturelles et de la Faune (now the MERN)
Mt	Million tonnes (metric tons)
Mt/a	Million tonnes per annum
MTQ	Ministère des Transports du Québec
MVK	Massive volcaniclastic kimberlite

MW	Megawatts
NG	Natural gas
NI	National Instrument
Norton Rose	Norton Rose OR LLP
NPV	Net present value
NRCAN	National Resources Canada
NTS	National topographic sheet
OK	Ordinary kriging
OP	Open Pit
OPEX	Operating expenses
PDA	Pre-development agreement
PEA	Preliminary economic assessment
PEG	Fresh, competent, equigranular pegmatite
PEM	Mining exploration licence
PK	Processed kimberlite
PKC	Processed kimberlite containment
PMD	Potential mineral deposit
QC	Quality control
QP	Qualified person
Québec Explosives Act	*An Act respecting explosives (Québec)* (RSQ, c. E-22)
Québec Land use Act	*An Act respecting Land use planning and development* (RSQ, c. A-19.1)
Québec Mining Act or Mining Act	*Mining Act (Québec)* (RSQ, c. M-13.1)
RC	Reverse circulation
Regulation respecting mineral substances	*Regulation respecting mineral substances other than petroleum, natural gas and brine* (R.Q., c. M-13.1, r.2)
Regulation respecting the application of the Environment Quality Act	*Regulation respecting the application of the Environment Quality Act* (c. Q-2, r.3)

Renard Kimberlite Pipes	*Refers to the kimberlites known as Renard 1, 2, 3, 4, 65, 7, 8, 9 and 10 located on the Foxtrot Property*
Renard Project or Project	Renard Diamond Project
Roche	Roche Ltd, Consulting Group
ROM	Run-of-mine
Royalty Agreement	Restated Royalty Agreement between DIAQUEM and SDCI dated April 1, 2011.
Saskatchewan Research Council	Saskatchewan Research Council Geoanalytical Laboratories
SDCI	Stornoway Diamonds (Canada) Inc./Les Diamants Stornoway (Canada) Inc.
Senior Secured Loan	Refers to the credit agreement entered into between SDCI and DIAQUEM on July 8, 2014.
SNC-Lavalin or SLI	SNC-Lavalin Inc.
SOQUEM	SOQUEM Inc.
spt	Stones per tonne
Stornoway	May be used to designate Stornoway Diamond Corporation or its wholly owned subsidiary, SDCI, depending on the context.
Streamers	Refers to the purchasers of the Subject Diamond Interest under the Stream Agreement.
Streaming Agreement or Purchase and Sale Agreement	Refers to a streaming agreement for the forward sale of the Subject Diamond Interest entered into between FCDC and the Streamers, dated July 8, 2014.
Subject Diamond Interest	Refers to a 20% undivided interest in each of the run of mine diamonds produced from the Renard Project, until the Streamers have been delivered the subject diamonds interest in 30 million carats of diamonds, and following such time, from all run of mine diamonds derived from Renard 2, Renard 3, Renard 4, Renard 65 and Renard 9.
SWRPA	Scott Wilson Roscoe Postle Associates Inc.
t	Metric ton (tonne)
TFFE	Target for further exploration
Thunder Bay Laboratory	Thunder Bay Mineral Processing Laboratory
TK	Volcaniclastic kimberlite

TKB	Tuffisitic kimberlitic breccia
TKt	Transitional volcaniclastic kimberlite
tpa	Metric ton per annum
tpd	Metric ton per day
tph	Metric ton per hour
WWW	WWW International Diamond Consultants
V	Volts
yr	Year

2.1 Qualified Persons and Site Visits

Qualified Persons contributing to this report are listed in Table 2.2.

Table 2.2: Areas of Responsibility

Name of QP	Company	Site Visit	QP Sections
Patrick Godin, Eng.	Stornoway Diamond Corporation	Last visit March 22, 2016	Sections 1, 2, 3, 4, 5, 6, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 (at the exception of 18.4.1 and 18.4.2) , 25, 26 and 27
Robin Hopkins, P.Geol. (NT/NU)	Stornoway Diamond Corporation	Last visit October 22, 2014	Sections 7, 8, 9, 10, 11, 12, 13 and 14
Paul Bedell, P.Eng.	Golder Associates Ltd	From September 29 to October 1, 2010, inclusive	Sections 18.4.1 and 18.4.2

2.2 Previous Technical Reports and References

Below is a list of previous technical reports and references in reverse chronological order.

Farrow, D.F. and Hopkins, R. (2016): 2015 Mineral Resource Update for the Renard Diamond Project, Québec, Canada, NI 43-101 Technical Report. Prepared for Stornoway Diamond Corporation by GeoStrat Consulting Services Inc. Available at www.Sedar.com.

Bagnell, W., Bedell, P., Bertrand, V., Brummer, R., Farrow, D.F., Gagnon, C., Gignac, LP, Gormely L., Magnan, M, St-Onge, JF (2013):The Renard Diamond Project, Québec, Canada, Feasibility Study Update, NI 43-101 Technical Report February 28, 2013.

Farrow, D.F. and Farrow, D.G. (2013): 2013 Mineral Resource Update for the Renard Diamond Project, Québec, Canada, NI 43-101 Technical Report. Prepared for Stornoway Diamond Corporation by GeoStrat Consulting Services Inc. Available at www.Sedar.com.

Bedell, P., Bertrand, V., Brummer, R., Farrow, D.F., Gignac, LP., Gormely L., Kroon, A., Magnan, M, Taylor, G.,Therrien, P. (2011): The Renard Diamond Project, Québec, Canada, Feasibility Study, NI 43-101 Technical Report December 29, 2011.

Lecuyer, N.L., Roscoe, W.E., Farrow, D., L'Ecuyer, M., Kozak, A., 2010. Updated Technical Report on the Preliminary Assessment of the Renard Project, Québec, Canada. Published by Scott Wilson Roscoe Postle Associates Inc. Available at www.Sedar.com

Farrow, D.F. and Farrow, D.G. (2011): 2010 Mineral Resource Update for the Renard Diamond Project. Prepared for Stornoway Diamond Corporation by GeoStrat Consulting Services Inc. Available at www.Sedar.com.

Farrow, D.F. (2010): 2009 Mineral Resource Update, Renard Diamond Project, Northern Québec, Canada. Prepared for Stornoway Diamond Corporation by Golder Associates. Available at www.Sedar.com.

Lecuyer, N.L., Roscoe, W.E., Cullen, R., Kozak, A. And Wiatzka, G. 2009. Technical report on the Preliminary Assessment of the Renard Project, Québec, Canada. NI-43101 Report. Published by Scott Wilson Roscoe Postle Associates Inc. Effective Date: December 12, 2008). Available at www.Sedar.com.

2.3 Effective Date

The effective date of the report is December 31, 2015.

3. RELIANCE ON OTHER EXPERTS

The Qualified Persons (QPs) involved in this report are QPs for the sections identified in the certificates of QPs. The QPs have relied on expert opinions and information pertaining to diamond pricing.

Diamond Valuation

Patrick Godin and Robin Hopkins fully relied on WWW International Diamond Consultants (WWW) and disclaim responsibility for the diamond price information presented in sections 14.8 (Diamond Price Estimates), 19.1 (Diamond Pricing and Market Studies) and 22 (Economic Analysis). It is reasonable for the Qualified Persons to rely on WWW for this information because WWW is an internationally recognized independent diamond valuation and advisory service to diamond mining and exploration companies, governments of diamond producing countries and private diamond companies. WWW, through Diamonds International Canada (DICAN) Ltd, serves as the valuator for the Federal Government of Canada and the Ontario Government.

In May 2011, WWW supervised an open market valuation of parcels of diamonds from the Renard Project documented in a report entitled "Valuation and Modelling of the Average Price of Diamonds from the Renard Kimberlite Cluster - May 2011". In October 2011, WWW provided a marketing report titled "Rough Diamond Supply and Demand Forecast 2011 to 2025 Plus Cost Comparison". Between late February and early March 2013, a valuation of the Renard Diamonds was performed under the supervision of WWW in Antwerp, Belgium, and the Diamonds were repriced by WWW as of March 2014. The March 2014 diamond pricing was used to establish the "Reasonable Prospects of Economic Extraction" in the 2015 Mineral Resource Update. The March 2014 diamond pricing, adjusted with publically available rough market pricing data, was used to establish the Probably Mineral Reserves in this report and in the financial analysis.

The diamond price information could not be verified by the QPs due to the proprietary nature of the diamond price book used for the valuation. Risk exists that the diamond price values obtained from WWW may differ from those achieved during commercial production since the valuation was based on an exploration sized sample.

The diamond supply and demand forecasts of Bain & Company (2015) were referenced in the Market Studies section.

4. PROPERTY DESCRIPTION AND LOCATION

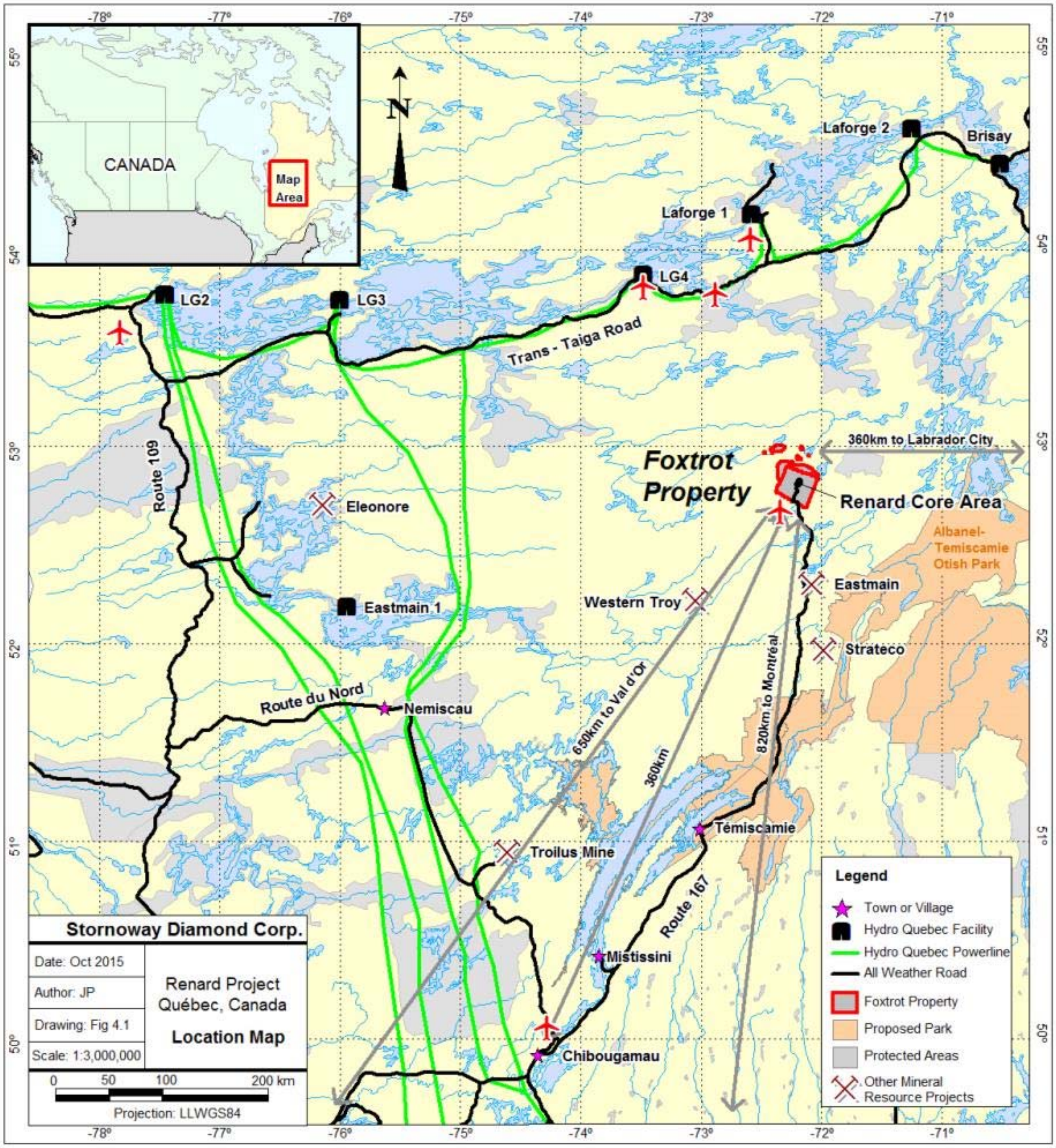
4.1 Location

The Renard Diamond Project is part of the Foxtrot Property situated in the Monts Otish (Otish Mountains) region of the province of Québec, Canada, 820 km north of the city of Montreal and 360 km north-northeast of the mining town of Chibougamau (Figure 4.1). The closest settlement is Témiscamie, situated on Lac Albanel, approximately 210 km to the south of the Project, which is connected by Route 167, all-weather provincial road, to the Cree community of Mistissini. From Témiscamie, the Project is connected by an all-weather access gravel road known as Route 167 Extension and by the Renard Mining Road completed in 2014.

The Foxtrot Property covers a total area 33,825.66 ha, much of which is encompassed within National Topographic Sheet (NTS) 033A16, although there are also smaller portions in NTS 033A09 and 033H01 (Figure 4.2).

The Renard Project surface and underground infrastructure are currently under construction. The Renard Project centroids are at approximately 52°49’N and 72°12’W.

Figure 4.1: Location Map



4.2 Tenure

The Foxtrot Property comprises 650 claims (33,629.95ha) in four blocks (one large contiguous landholding of 630 claims plus three smaller blocks), mining lease BM 1021 (143.71 ha) and surface lease number 1303 10 000 (199.85 ha). The claims, mining lease and surface lease are 100% owned by SDCI, a wholly owned subsidiary of Stornoway Diamond Corporation.

4.2.1 Claims

Of the 630 contiguous claims, 608 have anniversary dates of August 15, 2016 or August 16, 2016, and can be identified by the following claim numbers in the public register of real and immovable mining rights kept at the MERN: CDC numbers 2245719 to 2245734 (inclusive), 2245737 to 2245968 (inclusive), 2245973 to 2246148 (inclusive), 2246150, 2246152, 2246154, 2246156, 2246158, 2246160, 2246162, 2246164, 2246166, 2246168, 2246170, 2246295 to 2246314 (inclusive), 2246321 to 2246377 (inclusive), 2246381 to 2246389 (inclusive), 2246393 to 2246401 (inclusive), 2246405 to 2246473 (inclusive), 2246493 to 2246499, and 2246572 to 2246573 (inclusive). All of these claims have sufficient banked expenditures to allow them to be renewed for a minimum of two additional years (see Table 4.1)

Ten mineral claims (CDC numbers 22446378 to 2246380 (inclusive), 2246390 to 2246392 (inclusive), 2246402 to 2246404 (inclusive) and 2388520) have anniversary dates of January 16, 2017.

The remaining 32 claims are in good standing until April of 2017: CDC numbers 1006892, 1007101, 1007102, 1007123, 1007124, 1007418 to 1007424 (inclusive), 1007426 to 1007429 (inclusive), 1009047, 1009057, 1009366, 1009385 to 1009387 (inclusive), 1009409, 1009410, 1009414, 1009415, 1009441 to 1009443 (inclusive), and 1009897 to 1009899 (inclusive).

The claim holder has the exclusive right to search for all mineral substances in the public domain, with the exception of petroleum, natural gas, brine and loose surface materials. Claims are valid for a two-year period and can be renewed every two (2) years. Renewal fees are fixed by regulation and are currently set at $ 138.24 per claim if renewed at least 60 days before the expiry date and doubled if renewed within 60 days. In order to maintain tenure, exploration work equivalent ranging from $135 to $2500 per claim are required depending on the number of terms of renewal a claim has undergone. Table 4.1 details the work requirements per renewal period north of the 52 degree of latitude. When the work carried out is insufficient, or if work was not carried out, the titleholder may pay an amount equivalent to the required amount in lieu of work. Alternatively, work expenditures that are in excess of the amount required for the term on a claim can be transferred to other contiguous claims that are within a 4.5 km radius, or can be credited towards future renewals.

Table 4.1: Work Requirement per Renewal Period

Term	Surface area of claim		
	Less than 25 ha	25 to 45 ha	More than 45 ha
1	$48	$120	$135
2	$160	$400	$450
3	$320	$800	$900
4	$480	$1,200	$1,350
5	$640	$1,600	$1,800
6	$750	$1,800	$1,800
7 or more	$1,000	$2,500	$2,500

4.2.2 Mining Lease

Mining Lease BM 1021, totaling 143.71 ha in size, was granted to SDCI on October 16, 2012. BM 1021 encompasses the Renard Project mine site and surface operations (excluding the Processed Kimberlite Containment area). Mining Leases are surveyed, valid for 20 years (renewable for three 10-year periods and following that for 5-year periods) and have annual payments currently set at $46.50/ha.

4.2.3 Surface Rights

The Foxtrot Property is located entirely in Cree Territory, or *Eeyou Istchee*, on category III lands belonging to the Québec government and subject to the *James Bay and Northern Québec Agreement 1975*, as amended (JBNQA).

The JBNQA provides for three categories of land, Categories I to III, each with specifically defined rights. Category III lands are public lands where Cree communities have certain rights, particularly in regard to trapping, hunting, fishing and the development of outfitter operations. Members of the CNM undertake hunting, fishing and trapping activities within the Foxtrot Property, with the Renard Kimberlite Pipes occurring in an area known to them as “yuus-kanchiisu-saakahiikan” (mild rock Ptarmigan Lake). More specifically, the Renard Kimberlite Pipes lie within the CNM trapline area designated as M-11, used by Sydney and Emerson Swallow (known as the ‘tallyman’) and their respective families.

Under the terms of the JBNQA, governments have the right to develop Category III lands. Surface and mineral rights on Category III lands reside with the Government of Québec and are governed by the applicable land use laws and regulations, implemented by the relevant regulatory authorities. Infrastructure for the Renard Project required the acquisition of surface leases issued by the Ministry of Natural Resources (MNR), including a surface lease encompassing the Process Kimberlite Containment area which was granted on October 16, 2012 under #1303 10 000 for an area of 199.85 ha. Yearly renewal fees for surface leases that are used to store mine residues are $94/ha.

All surface leases required for the infrastructure of the Renard Project are listed in Table 4.2 and borrow pits in Table 4.3.

Figure 4.2: Landholdings, Mineralization and Local Infrastructure

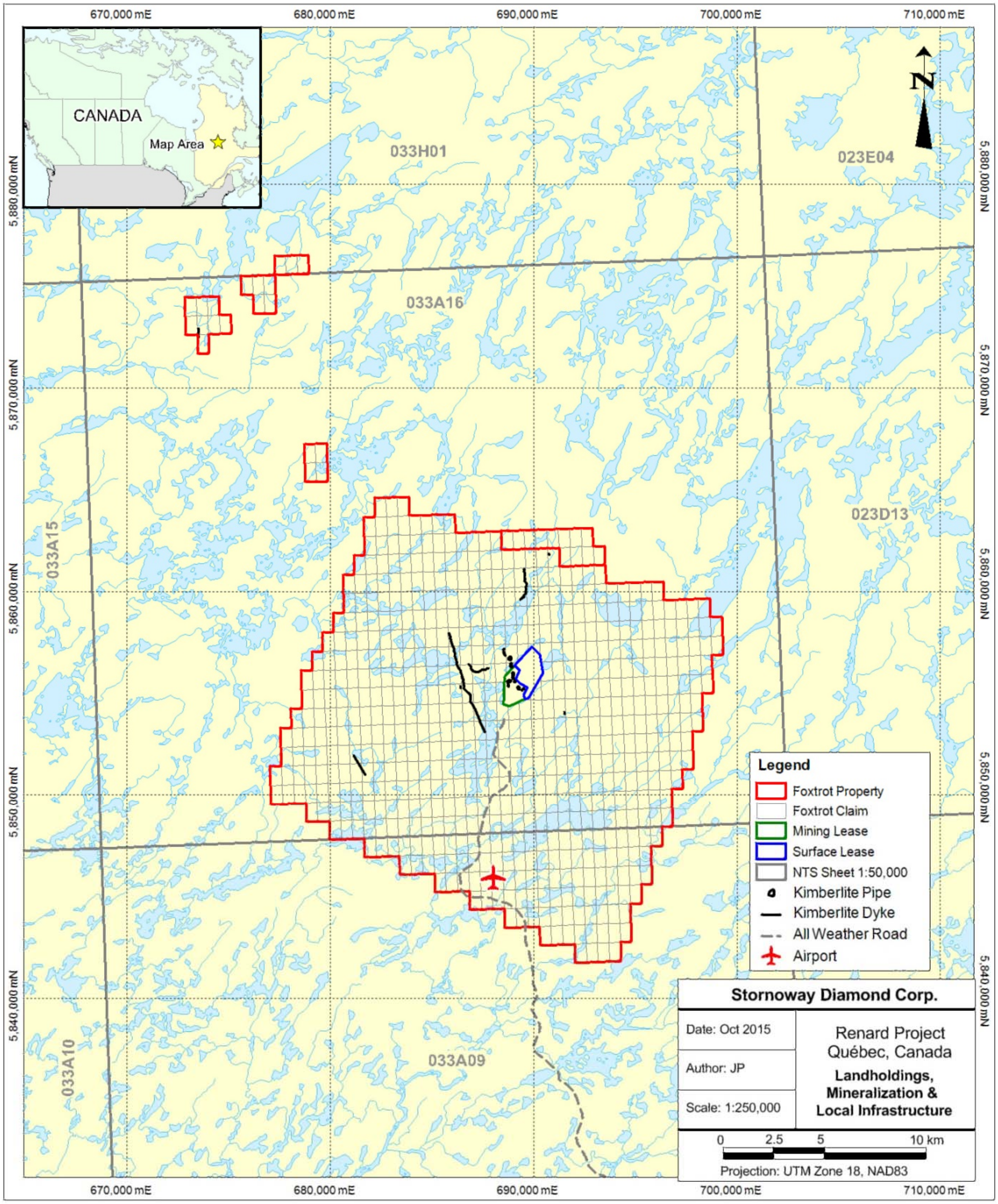


Table 4.2: Surface Leases

Lease number#	Description	Date obtained
1303 10 000	Tailings and waste rock accumulation areas	October 16, 2012
1349 10 000	Renard Project Airport	January 23, 2013
1352 10 000	Explosive storage and handling area	January 23, 2013
1350 10 000	Domestic wastewater treatment plant	January 23, 2013
1351 10 000	Trench landfill	January 23, 2013
1901 10 000	LNG storage	December 10,2015
1897 10 000	Mine waste water treatment plant	December 10,2015
1998 10 000	Cultural Cree center	December 10,2015

Table 4.3: Borrow Pits

Lease number	Location
BNE 40154	33A16-6
BNE 40153	33A16-5
BNE 40152	33A16-4
BNE 40151	33A09-5
BNE 40150	33A09-12
BNE 40149	33A09-14
BNE 40148	33A09-15
BNE 40147	33A08-15
BNE 40146	33A08-13
BNE 40145	33A08-11
BNE 40144	33A08-12
BNE 40143	33A08-7
BNE 40142	33A01-23
BNE 40141	33A01-24
BNE 40140	33A01-18
BNE 40139	33A01-19
BNE 40138	33A01-11
BNE 40137	33A01-17
BNE 3895	33A16-7

4.3 Agreements

4.3.1 Royalites

SDCI holds a 100% interest in the Property, subject to a 2% direct gross revenue royalty on future life of mine diamond production in favour of DIAQUEM as documented in the Restated Royalty Agreement between DIAQUEM and SDCI dated April 1, 2011 (the Royalty Agreement).

4.3.2 Encumbrances

The Renard Diamond Project is encumbered by the following:

The obligations of SDCI under the Royalty Agreement are secured by a hypothec and a security agreement both dated April 1, 2011 on all of SDCI's mining titles related to the Renard Project, including the mining lease BM 1021.

SDCI's obligations under the Mecheshoo Agreement have been secured by a hypothec registered against the mining lease BM 1021.

On May 3, 2012, Stornoway entered into a $20 million unsecured debt facility with Fonds de solidarité des travailleurs du Québec (FTQ), Fonds Régional de solidarité FTQ. Nord-du-Québec, s.e.c. and DIAQUEM (the Fonds Loan). In connection with the Fonds Loan, SDCI granted the lenders a 1% contingent secured royalty interest in the Renard Project which is only triggered upon the occurrence of certain specified events, such as a payment default or a default following a change of control of Stornoway, in each case capped at an amount equal to the aggregate value of the principal and interest then outstanding on the Fonds Loan. The obligations of Stornoway under the Fonds Loan have also been secured by a hypothec registered on all of SDCI's mining titles on the Renard Project including the mining lease BM 1021.

On July 8, 2014, SDCI entered into a credit agreement with DIAQUEM for an amount not to exceed $120 million plus an amount equal to the full outstanding principal amount of all loans under a loan agreement between DIAQUEM and SDCI dated October 1, 2013 (the Senior Secured Loan). The obligations under the Senior Secured Loan have been secured by a security interest and hypothec on the universality of property and assets constituting the Renard Diamond Project.

Under the Streaming Agreement for the forward sale of diamonds (the Forward Sale of Diamonds) with certain purchasers (the Streamers), FCDC agreed to sell to the Streamers a 20% undivided interest (the Subject Diamonds Interest) in each of the run of mine diamonds produced from the Renard Project, until the Streamers have been delivered the Subject Diamonds Interest in 30 million carats of diamonds, and following such time, from all run of mine diamonds derived from Renard 2, Renard 3, Renard 4, Renard 65 and Renard 9 (Subject Diamond Interest). The obligations under the Streaming Agreement have been secured by a security interest and hypothec on the universality of the property and assets constituting the Renard Project until such time as the Deposit is fully offset in accordance with the terms of the Streaming Agreement, following which the security granted to the Purchasers is limited to a security interest and hypothec on the mining lease BM 1021 and the Subject Diamond Interest.

On July 25, 2014, SDCI (as lessee), Ashton and Stornoway (as guarantors) entered into a master lease agreement with Caterpillar Financial Services Limited (as lessor) for the acquisition and financing of certain mine equipment items manufactured by Caterpillar and others, including the Renard Project's mobile mining fleet (the Equipment Facility). The Equipment Facility is available to SDCI in the maximum aggregate amount of US$75 million of acquired equipment value (less the applicable upfront payments due by the lessee). The obligations of SDCI under the Equipment Facility have been secured by a movable hypothec on the movable Caterpillar equipment, and like equipment, present and future, wherever situated, together with all parts and accessories thereof, including the equipment used in connection with the Renard Diamond Project.

4.4 Permits

The current permitting status of the Project and the permits remaining to be required are discussed in Section 20.

4.5 Environment

The current status of environmental liabilities and other environmental obligations are discussed in Section 20.

4.6 Social Licence

The Foxtrot Property is located within the region of Northern Québec governed by the JBNQA, a land claims agreement executed by the government of Québec, the government of Canada, the Grand Council of the Cree of Québec ("**GCC**") and the Northern Québec Inuit Association, among others. This agreement defines the social and environmental protection regimes for the regions of James Bay and Nunavik. The JBNQA provides for three categories of land, Categories I to III, each with specifically defined rights. Category III lands are public lands where Cree communities have certain rights, particularly in regard to trapping, hunting, fishing and the development of outfitter operations. Members of the CNM undertake hunting, fishing and trapping activities within the Foxtrot Property, with the Renard Kimberlite Pipes occurring in an area known to them as "yuus-kanchiisu-saakahiikan" (mild rock Ptarmigan Lake). More specifically, the Renard Kimberlite Pipes lie within the CNM trapline area designated as M-11.

In February 2002, Québec and the Cree Nation signed a fifty year political and economic agreement known as *La Paix des Braves*. The GCC is the political body that represents approximately 14,500 Crees of Eastern James Bay and Southern Hudson Bay in Northern Québec. The GCC has twenty members: a Grand Chief and Deputy-Grand Chief elected at large, the chiefs elected by each of the nine Cree communities of the territory, and one other representative from each community. The CNM is the largest Cree community with 3,500 residents.

On July 24, 2012, the Crees of Eeyou Istchee and the Gouvernement du Québec signed the Agreement on Governance in the Eeyou Istchee James Bay Territory. This agreement provides for the creation of a joint Regional Government composed of Crees and Jamésiens (the "James Bay Regional Government") replacing the Municipalité de Baie-James. The James Bay Regional Government is responsible for the management of Category III lands and exercises the same jurisdictions, functions, and authority on Category III lands in the Eeyou Istchee James Bay Territory as those formerly attributed to the Municipalité de Baie-James. The James Bay Regional Government is directed by a Council composed of 11 Cree representatives (the Grand Chief of the Cree Nation Government along with ten other designates from elected members of the Council of the Cree Nation Government), 11 Jamésien representatives (designated from among elected members of the councils of the enclave municipalities and the non-Crees residing within the James Bay Regional Government's territory), and one non-voting representative of the Québec government. Authorizations and permits required for the development of all industrial projects are provided by the James Bay Regional Government.

Since the early stages of the Renard Diamond Project, Stornoway has developed and maintained significant communications and relations with stakeholders including: the CNM, the Grand Council of the Crees (Eeyou Istchee), tallymen of Trapline M11 and the towns of Chibougamau, Chapais and the Municipality of James Bay. Public meetings, field visits, meetings with chiefs or mayors, environmental exchange group meetings, and business meetings with contractors and suppliers are amongst the many activities held by Stornoway on a regular basis. Stornoway has opened offices in Mistissini and Chibougamau to facilitate communications with these parties.

In March 2012, Stornoway completed negotiations with the CNM, the Grand Council of the Crees (Eeyou Istchee) and the Cree Regional Authority on the Mecheshoo Agreeement. The Mecheshoo Agreement is a binding agreement that will govern the long-term working relationship between Stornoway and the Cree parties during all phases of the Renard Diamond Project. It provides for training, employment and business opportunities for the Crees during project construction, operation and closure, and sets out the principles of social, cultural and environmental respect under which the project will be managed. The Mecheshoo Agreement includes a mechanism by which the Cree parties will benefit financially from the success of the project on a long term basis, consistent with the mining industry's best practices for engagement with First Nations communities.

In July 2012, Stornoway executed a Declaration of Partnership (the "Declaration") with the communities of Chibougamau and Chapais in the James Bay Region of Québec. The Declaration is a statement of cooperation between the partners for the responsible development of the Renard Diamond Project based on the principles of environmental protection, social responsibility and economic viability. The Declaration includes provisions to set up a Renard Liaison Committee that will address issues of mutual interest such as communication, employment, and the economic diversification of local communities. In particular, the committee will oversee initiatives to attract and retain new residents to the towns of Chibougamau and Chapais.

The environmental licence for the Renard Project is discussed in Section 20.

4.7 Comments on Section 4

The responsible QP is of the opinion that:

- Stornoway, through its wholly-owned subsidiary SDCI, holds 100% of the Renard Project;
- The Renard Project comprises 650 claims (33,629.95ha), one mining lease BM 1021 (143.71 ha), a surface lease for the processed kimberlite area # 1303 10 000 (199.85 ha) and several other surface leases for infrastructure related to the Project (Table 4.2), all of which are expected to be sufficicent for the operations of the Renard Project;
- A 2% direct gross revenue royalty is payable to DIAQUEM on future life of mine diamond production;
- A 20% undivided interest in each of the run of mine diamonds produced from the Renard Project, until the Streamers have been delivered the Subject Diamonds Interest in 30 million carats of diamonds, and following such time, from all run of mine diamonds derived from Renard 2, Renard 3, Renard 4, Renard 65 and Renard 9, has been sold to Streamers for a fixed price;
- SDCI has entered into the Mecheshoo Agreement; and
- Permits obtained, or to be obtained in the normal course of the development of the Renard Project, are expected to be sufficient to ensure that activities are conducted within the regulatory framework required by local, provincial and federal governmants.

5. ACCESSIBILITY, CLIMATE, LOCAL RESOURCES, INFRASTRUCTURE & PHYSIOGRAPHY

5.1 Accessibility

5.1.1 Air

In September 2014, the Renard airport, specifically built for the use of the Renard Project and located along the Renard Project access road 10 km away from the Renard Project gate, was completed and inaugurated as the Clarence & Abel Swallow Airport. The Clarence and Abel Swallow Airport comprises a 1,497m long year-round gravel landing strip with associated instrumentation, maintenance facilities and a passenger terminal.

Charter flights transport workers to the Renard Project. The airport accommodates Dash-8 aircraft and there are regular flights to the Renard Project departing from St-Hubert and Chapais-Chibougamau. Smaller aircraft are chartered from time to time to answer specific transportation demands.

5.1.2 Roads

The Renard Project benefits from an all-weather gravel road access. Provincial highway 167 affords paved access to the city of Chibougamau and the village of Mistissini, and links these communities to the provincial road network. A 105-km, all-weather gravel road proceeds north from the junction of Mistissini to Témiscamie. From Témiscamie, the extension of Road 167 and the Renard mining road, which were completed in fall 2014, provide access to the Renard Project. The length of the road between Chibougamau and the Renard Project is 420 km.

All the material, supplies and food for construction and operational phases of the Renard Project are transported along this access route. A transhipment and consolidation center is operated out of Chibougamau.

5.2 Climate / Operating Seasons

Construction activities are, and mining activitites will be, conducted year-round. Climate at the Renard site is characterized by long winters and short summers. Temperatures range from summer maximums of +35°C to winter minimums of -45°C. Lakes freeze over in late October, thawing in late April–May. Abundant precipitation falls in the form of rain and snow. Total annual precipitation averages around 80–100 cm. The Otish Mountains form a local topographic high that affects both precipitation and fog formation and distribution in the area.

Forest fires are common in the area during the spring and summer months, but to date have not adversely affected the Project.

5.3 Local Resources and Infrastructure

5.3.1 Local Resources

The nearest communities to the Renard Project are Chibougamau (population about 8,000), Mistissini (population 3,500), and Chapais (population 1,300). Chibougameau offers extensive community, health and transportations services. It also serves as the major supply centre for regional industries. Since March 2015, the Renard mine accommodation complex is fully operational. This accommodation complex, located on the mine site, includes 370 rooms (private bathroom, TV screen, phone and wireless internet connection), cafeteria, fitness room, recreation room, locker room, training room, reception services and security offices. In addition, for the construction period, there is an adjacent temporary camp that can accommodate an additional 310 workers. These latter temporary units will be dismantled after the construction period.

5.3.2 Power

During construction, electrical power is supplied to the mine site by three diesel powered generators (1800 KW each). These generators are in place to support the buildings' needs and the construction works. The diesel fuel is transported on site by truck and stored in one of the eight 50,000 l tankers set-up beside the entrance of the mine site.

During the operational phase, power will be generated by a combination of diesel and LNG.

An LNG storage and regasification facility was constructed on the Renard Project site to power the main generators in the power plant. This LNG facility consists of a truck offloading station, six storage tanks and a regasification station with its ancilliary. Daily deliveries of LNG by tanker truck from Montreal are made possible by the all-season Route 167 access road to the Project. On site LNG storage capacity is rated at 10 days of supply at full Project production.

5.3.3 Water

Water is required for the process plant and accommodation complex. During the operation, the process water will be sourced from Lagopede Lake or will be recirculated water from surface facilities such as the R65 runoff water collecting pond and/or from underground mine water.

Raw water is sourced from the Lagopede Lake and treated by a water treatment plant on the Renard Project, having a capacity to supply potable water for up to 800 workers.

5.4 Physiography

Topographic relief within the Foxtrot Property consists of steep-sided hills with rounded tops separated by muskeg-covered valleys. Elevations range between 450 masl and 550 masl. Lakes, ponds and small rivers are common.

5.5 Flora and Fauna

The Foxtrot Property is located in the Taïga belt of Northern Québec. Vegetation consists largely of immature to mature black spruce, poplar, alders and muskeg, with increasing proportions of muskeg and black spruce toward the north. Animals such as bear, fox, moose, marten and caribou, as well as various species of birds, are present in the vicinity of the Foxtrot Property.

5.6 Sufficiency of Surface Rights for Mining Operations

The sufficiency of surface rights for mining operations is discussed in Section 4.2.3.

6. HISTORY

Diamond exploration commenced in the Foxtrot Property area in 1996, with the formation of a 50:50 joint venture between Ashton Mining Canada Inc. (Ashton) and SOQUEM (SOQUEM). Prior to that time, regional exploration had been undertaken for gold and base metals by a number of parties using prospecting and geochemical techniques. These activities were limited in scope due to the location of the current property between two Archean volcanic belts, an area that was considered to be non-prospective for traditional gold and base metal targets. Both BHP Billiton and De Beers Canada Exploration have conducted regional diamond exploration programs in the general area of the current Foxtrot Property, but results are not publicly available. Although there had been scattered small-scale claim blocks throughout the area explored by the Ashton-SOQUEM joint venture, no large-scale land acquisition had occurred prior to the August 2000 grant of Mineral Exploration Licences PEM 1555 and 1556.

Initial stage exploration by the joint venture comprised heavy mineral sampling that defined a number of areas with anomalous indicator mineral counts. Geophysical surveys over the geochemical anomalies were completed, using fixed-wing airborne, magnetic surveys, and ground geophysical surveys.

Core drill testing of these targets led to the discovery of the Renard 1 and Renard 2 kimberlites in September 2001, the Renard 3, Renard 4, Renard 5 and Renard 6 kimberlites in March and April 2002, Renard 7 in August 2002, Renard 8 in September 2002, Renard 9, Renard 10 and the 4.2 km long Lynx kimberlitic dyke occurrences in 2003, and the 850 m long Hibou kimberlite dyke in 2005. The Renard 5 and Renard 6 kimberlites were subsequently determined to be one kimberlitic body, and the occurrence was renamed Renard 65.

Mini-bulk sampling using trenching and core drilling commenced in 2002 on the Renard 2, Renard 3 and Renard 4 kimberlites. Additional minibulk sampling since that date has been undertaken on the Renard 1, Renard 65, Renard 7, Renard 9, Renard 10, Lynx, Lynx South and Hibou kimberlites. Mini-bulk samples of kimberlite boulders collected down-ice of the Lynx and Hibou kimberlites have also been completed.

During 2004, a conceptual tonnage and grade estimate was prepared for the combined Renard 2, Renard 3, Renard 4 and Renard 65 kimberlites, by Wardrop Engineering Inc. of Vancouver, British Columbia (Maunala, 2004). This estimate included valuation of Renard 2, Renard 3, Renard 4 and Renard 65 diamonds recovered through RC sampling.

Additional exploration resulted in the discovery of the North Anomaly kimberlite dyke in 2005 and the Southeast Anomaly and G04-296 kimberlite dykes in 2006. An occurrence of kimberlitic float material was also discovered north of the Renard pipe cluster.

Bulk sampling commenced in 2006, testing the Renard 2, Renard 3, Renard 4, Renard 65 and Renard 9 kimberlites. Samples were sourced from trench, drill core, RC drill chips and underground. The underground exploration workings comprised a portal and the excavation of an inclined ramp to a depth of approximately 55 m below surface. Horizontal drifts were driven to access the Renard 2 and Renard 3 kimberlites. A modular 10 tph dense media separation (DMS) test facility was erected on the Foxtrot Property close to Renard 2 and Renard 3, to treat the material from the bulk sampling.

In December 2008, an economic assessment study jointly undertaken by Agnico-Eagle Mines Limited (Agnico-Eagle) and AMEC Americas Limited (AMEC), was completed and formed the 2008 NI 43-101 Technical Report on the Preliminary Assessment of the Renard Project authored by Scott Wilson Roscoe Postle Associates Inc. (SWRPA).

Agnico-Eagle was responsible for mining, cost estimating and financial analysis. Elements of the Agnico-Eagle mining study, such as cost estimation, were completed to a pre-feasibility standard. AMEC designed the DMS facility, and prepared the first Mineral Resource Estimate. Given the more conceptual nature of the mine plan, and inclusion of Inferred Resources from AMEC's Mineral Resource Estimate used in the study, the two studies combined comprise a Preliminary Assessment under the definitions contained within NI 43-101. An audit of the Agnico-Eagle study and economic analysis was completed by SWRPA, as independent Qualified Persons for the 2008 NI 43-101 report.

The December 12, 2008 Technical Report was revised on March 25, 2009 to clarify that the QPs were taking responsibility for the WWW International Diamond Consultants valuation and that diamond valuations are not amenable to data verification procedures that can be performed by a QP. None of the scientific and technical information in the report changed.

In 2009, Golder Associates Ltd (Golder) completed updated NI 43-101 Mineral Resource estimate for the Renard Project (Farrow, 2010). The effective date of the report was December 8, 2009. The initial report was dated January 21, 2010, and revised on January 27, 2010. The revision did not make any material changes to the initial report.

SWRPA was then retained by Stornoway to prepare an update of the March 25, 2009 Independent Technical Report on the Project. The purpose of the report was to provide an updated document for the disclosure of a Preliminary Assessment of a potential mining and processing operation at the Project.

A Preliminary Assessment of the Project was completed on March 22, 2010. This study comprised a conceptual mine plan, capital and operating cost estimates, and cash flow model prepared by SWRPA; a diamond processing plant design, with capital and operating cost estimates, prepared by AMEC; and social, environmental and permitting aspects contributed by Stantec. The conceptual mine plan was based on the NI 43-101 Mineral Resource Estimate prepared by Golder. Golder, AMEC and Stantec prepared or contributed to sections of the technical report, which was dated May 5, 2010 (Lecuyer et al., 2010). The report conformed to NI 43-101 Standards of Disclosure for Mineral Projects.

Stornoway retained GeoStrat Consulting Services Inc., (GeoStrat) to provide an independent Mineral Resource estimate update for the Renard Project (Farrow and Farrow, 2011). This estimate represented the third reporting of Mineral Resources from the Renard Project. Despite 7,654 m of additional core drilling and 3.6 t of sampling since the last disclosure, the Indicated Resource remained relatively unchanged. The increase in Inferred Resource was primarily due to the upgrade of a portion of the Renard 65 kimberlite from potential mineral deposit (PMD) to Inferred Resource. The subsequent National Instrument 43-101 technical report was filed on February 3, 2011.

In 2011, Stornoway completed a feasibility study for the Project. The lead consultant for the 2011 Feasibility Study was SNC-Lavalin Inc. The study and subsequent National Instrument 43-101 technical report were prepared by the following independent consultants: SNC-Lavalin Inc., AMEC Americas Limited, Golder Associates Limited, G Mining Services Inc., GeoStrat Consulting Inc., Itasca Consulting Canada Inc. and Roche Ltd, Consulting Group. The technical report summarizing the 2011 Feasibility Study was filed on January 3, 2012, although its effective date is December 29, 2011.

Subsequent to the 2011 Feasibility Study, an updated feasibility study, also referred to as the 2013 Optimization Study, was completed February 28, 2013 and voluntarily filed March 27, 2013 on SEDAR as a National Instrument 43-101 technical report (Bagnell et al., 2013). This report summarizes the revised mining costs estimated in the initial 2011 Feasibility Study.

During 2012, Stornoway enlarged an existing surface trench at the Renard 65 pipe, and collected/processed approximately 5,080 tonnes of kimberlite, returning 962.8 cts of diamonds. The largest diamond was a 9.78 carat white octahedral gem. A valuation by WWW using their March 2013 price book returned an average price of US$250 per carat on the total Renard 65 parcel due to prices of US$8,500 per carat and US$5,900 per carat on the two largest stones. The WWW diamond price model for Renard 65 of US$180 per carat had a “High” sensitivity of US$203 per carat and a “Minimum” sensitivity of US$169 carat.

In July 2013, GeoStrat completed an update on the Renard Mineral Resource Estimate, which incorporated results from processing of the Renard 65 bulk sample, recent geotechnical and hydrogeological drilling that intersected the ore bodies, and a refined method for calculating country rock dilution within each kimberlite. This resulted in changes to the diamond content estimates of Renard 2, Renard 3, Renard 4 and Renard 9, and a substantial quantity of Country Rock Breccia ("CRB") within Renard 2 being added to the Mineral Resource Estimate, including 0.87 Mt at 32 cpht (0.28 million carats) in the Indicated category and 6.54 Mt at 19 cpht (1.24 million carats) in the Inferred category.

In late 2013 Stornoway announced that a $10 million exploration program would be undertaken on the Project, consisting primarily of deep directional drilling at Renard 2, but also including work on the other project pipes and dykes. Field aspects of this program were completed in November of 2014, followed by detailed geological logging of drill core, the preparation and collection of hundreds of samples, processing by various laboratory facilities and the compilation/interpretation of results in 2015. The 2014/2105 work program allowed the Project's Mineral Resource Estimate to be updated, as documented in an independent 43-101 Technical Report authoured by GeoStrat (effective date of September 24, 2015), and as confimed in this current NI 43-101 Technical Report.

On October 21, 2013 Stornoway announced the results of a feasibility study on the viability of a Liquefied Natural Gas ("LNG") fuelled power plant for the Renard Diamond Project. The study was authored by SNC-Lavalin Inc. and AMEC America Ltd under the Renard Project EPCM joint venture, and demonstrates substantial benefits to the project in terms of annual operating cost and environmental emissions compared to the planned diesel gen-set option.

On July 8, 2014 Stornoway announced the closing of a comprehensive $946 million funding package for construction of the Renard Diamond Project through a combination of senior and subordinated debt facilities, equity issuances, equipment financing facility, and the forward sale of diamonds. Following a formal production decision, mine construction activities commenced with a ground breaking ceremony on July 10, 2014. Mine construction is still underway at the present time. In March 2015, mining activities commenced in the Renard 2 & 3 open pit with the removal of overburden and the prestripping of waste rock. Ore extracted during these early mining operations is being stockpiled pending start-up of the diamond processing plant.

No commercial production has been carried out on the property to date.

7. GEOLOGICAL SETTING AND MINERALIZATION

7.1 Regional Geology

The Project area is located within the eastern portion of the Superior Craton (Superior Structural Province) as shown in Figure 7.1. The Superior Province forms the Archean core of the Canadian Shield, and is an amalgamation of small continental fragments of Meso-Archean age and Neo-Archean oceanic plates, with a complex history of aggregation between 2.72 Ga and 2.68 Ga. Since about 2.6 Ga, the Province has been tectonically stable (Percival, 2007).

The Superior Province is surrounded by provinces of Paleo-Proterozoic age on the west, north and east (Churchill Province), and Meso-Proterozoic age (Grenville Province) on the southeast. Margins of the Superior Province were affected during Paleo-Proterozoic and Meso-Proterozoic tectonism. Proterozoic and younger activity is limited to rifting of the margins (Mid Continent Rift System), emplacement of numerous mafic dyke swarms, compressional reactivation and large-scale rotation at ca. 1.9 Ga, and failed rifting at ca. 1.1 Ga (Percival, 2007). Figure 7.2 shows the Archean and Proterozoic rocks in the general region of the Foxtrot Property.

Quaternary glacial cover in the area was controlled by the New Québec Ice Divide. From the divide, ice flowed north and northeast toward Ungava Bay and west to southwest toward Hudson Bay. Glacial lineaments are well developed and widespread. Eskers and hummocky to discontinuous, unmoulded, ground moraine deposits are also common.

There are five known episodes of kimberlitic volcanism in Québec (Moorhead et al., 2003) as shown in Figure 7.1 and summarized below from south to north:

- Témiscamingue: six diatreme facies pipes intruding the Pontiac Subprovince. Two age dates, 125 Ma (Rb–Sr) and 142 Ma (U–Pb) have been obtained. Kimberlites are hosted in the northwest-trending Témiscaming structural zone.
- Desmaraisville: five hypabyssal pipes and numerous dykes located in the central portion of the Abitibi Subprovince. Age date of 1104 Ma (Rb–Sr from phlogopite). Hosted in the Waswanipi–Saguenay Tectonic Zone; pipes are in close proximity to northeast-trending Proterozoic dykes.
- Otish: at least 12 pipes intruding the northeast portion of Opatica and Opinaca Subprovinces. Age dates range from 550.0 +/- 3.5 Ma at Beaver Lake to 640.5 +/- 28 Ma at Renard. The kimberlite field is associated with the southern end of the Mistassini-Lemoyne structural zone, and near-northwest, and northeast-trending Proterozoic diabase dykes.
- Wemindji: kimberlitic sills intruding Archean-age gneisses of La Grande Subprovince, located at the western end of the Wemindji–Caniapiscau structural zone where it intersects the northeasterly projection of the Kapuskasing zone.
- Torngat: diamond-bearing dykes recognized in the Paleo-Proterozoic Rae Province near the Archean Nain Craton. These dykes were classified as carbonatized ultramafic lamprophyres and dated at 550 Ma.

The Renard Cluster is considered to be part of the Otish kimberlitic volcanic event.

Figure 7.1: Regional Geology – Superior Craton

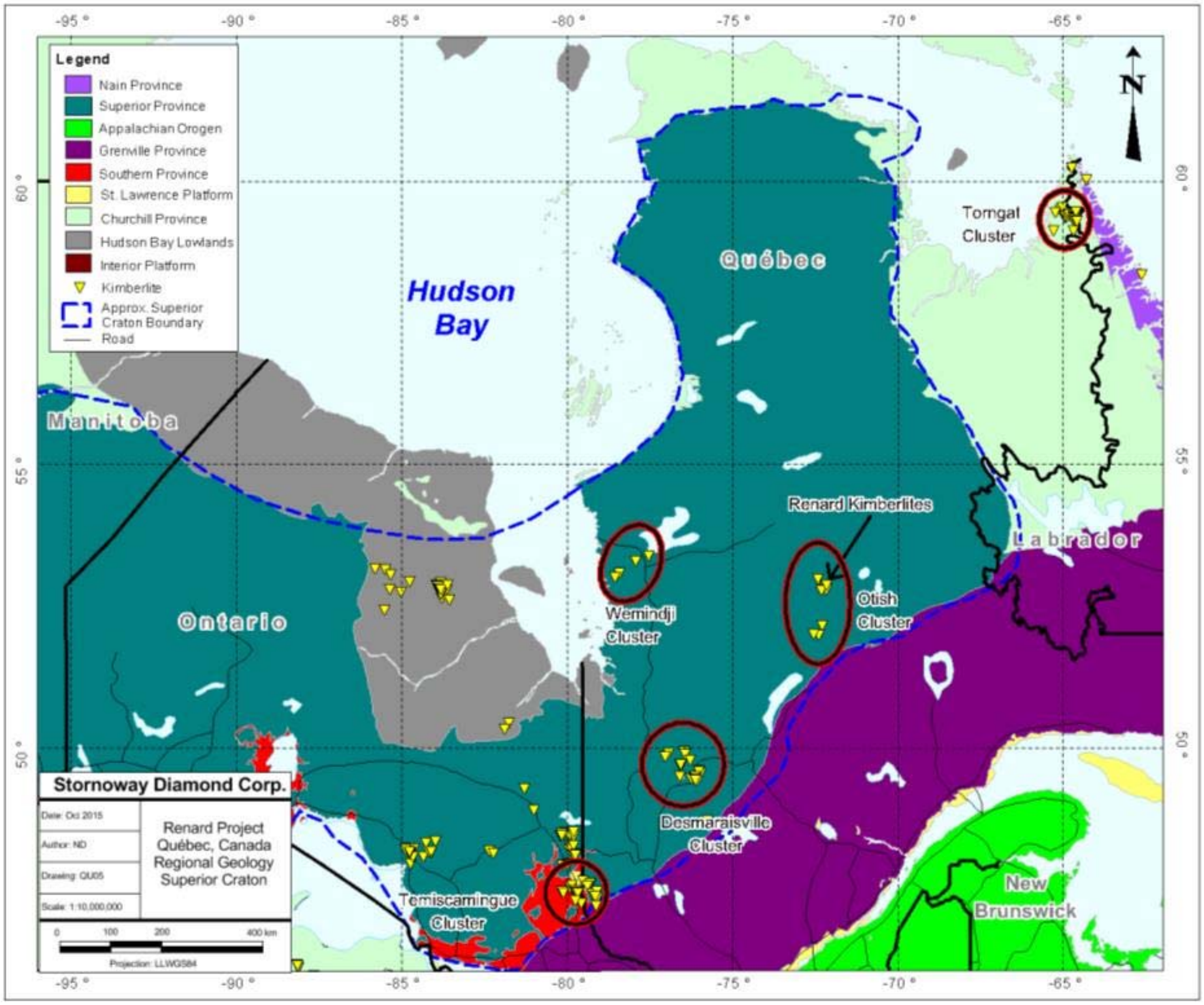
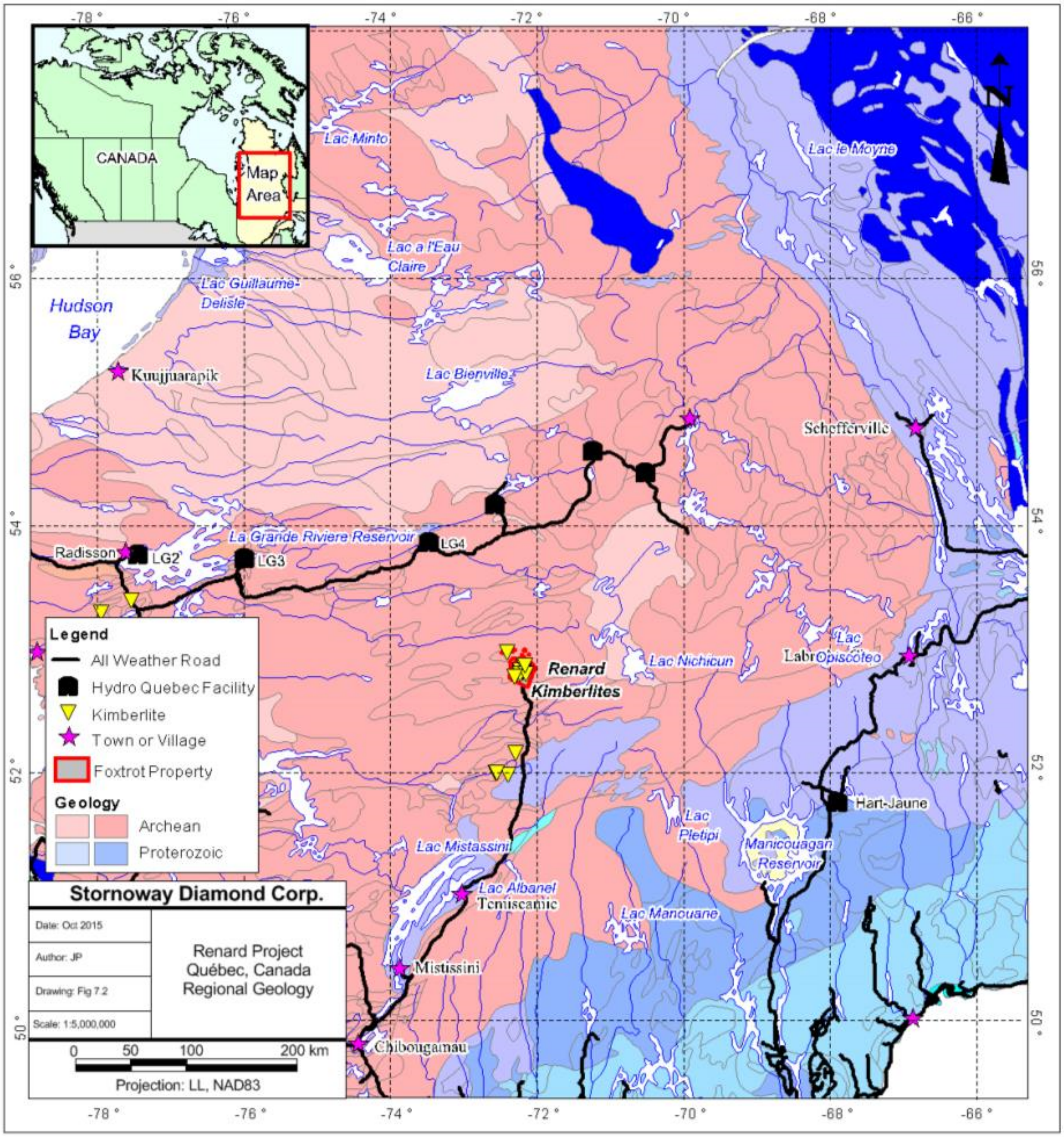


Figure 7.2: Regional Geology – Foxtrot Area



7.2 Project Geology

The Project area is located on the south-eastern portion of the Superior Structural Province, bordered by Proterozoic rocks of the Labrador Trough in the east and the Grenville Province in the south (Figure 7.1). This portion of the Superior Craton corresponds to the Opinaca subprovince, and is sometimes referred to as the “Ungava Craton”. Metagreywacke, derived migmatite and granite characterize the Opinaca subprovince. Polydeformed schists occur at the belt margins, whereas the interior portions are metamorphosed to amphibolite and granulite facies (Percival, 2007).

Proterozoic rocks of the Labrador Fold Belt in the east, the Cape Smith Fold Belt in the north and the Grenville Province in the south surround the Project area. Northern portions of the Project area consist of north-northwest trending, plutonic and gneissic terranes. Based on metamorphic grade, mineralogy, lithology and aeromagnetic observations, the terranes appear to vary in width from 70 km to 150 km (Percival et al., 1994).

The Foxtrot Property is situated between the La Grande greenstone (volcanic) belt to the north and the Eastmain greenstone (volcanic) belt to the south. Granite-gneiss and retrograde granulite gneiss are the predominant lithologies, with lesser amounts of granite and granodiorite. Contained within the gneiss are relict metasedimentary and metavolcanic rock assemblages along with associated mafic and ultramafic intrusive rocks. The Otish Mountain and Mistassini groups of Proterozoic, clastic, metasedimentary rocks overlie the Archean lithologies, marginal to the Grenville Province. Mafic and ultramafic intrusive rocks of variable affinities are more common in the southeast than in the southwest (O’Connor and Lépine, 2006).

Granite–gneiss and retrograde granulite gneisses of sedimentary origin are the predominant lithologies in the Property area; however, lesser granite and granodiorite may also be present. The gneisses may contain relict metasedimentary and metavolcanic rock assemblages, as well as associated mafic and ultramafic intrusive rocks. Minor linear belts of supracrustal metavolcanic rocks occur throughout the area, generally trending east-west or west-northwest. Northwest-trending, Proterozoic Mistassini Swarm diabase and gabbro dykes up to 30 m wide cross-cut all lithologies. Isolated outliers of Proterozoic clastic metasedimentary rocks are present in the area (O’Connor and Lépine, 2006).

Metamorphic grade within the Foxtrot area is primarily amphibolite facies with local granulite facies being reported near Lac Minto (Percival et al., 1994). Higher-grade lithologies in the north are interpreted as supracrustal relicts dating to 3.1 Ga. Granite and granite gneiss are dated at 2.7 Ga and local felsic and intermediate intrusive rocks are dated at 2.5 Ga.

Glacial overburden within the Foxtrot Property can be up to 34 m thick, but is on average 10 m thick in the area of the Renard Cluster. Glacial deposits consist of till, eskers, moraine and post-glacial sediments, and their orientation reflects ice transport from the north-northeast.

In this report, reference is made to the Renard Core Area, which is defined as a 37 ha area that contains all of the known kimberlite pipes as well as the Lynx and Hibou dyke systems. Kimberlites are discussed individually in the following sections.

7.3 Kimberlite Mineralization at Renard

The Renard kimberlites were emplaced into granitic and gneissic host rocks, and contain diamonds of economic interest. The bodies comprise a late Neo-Proterozoic to Cambrian kimberlite field in Québec (Girard, 2001; Moorhead et al., 2002; Letendre et al., 2003).

To date, nine kimberlite pipes have been identified over a 2 km^2 area in the Renard Cluster (Renard 1 to Renard 10; with Renard 5 and Renard 6 forming one body, referred to as Renard 65). The kimberlite pipes are typically spaced between 50 m and 500 m from each other (Figure 7.3). Geophysical data and drill information from delineation and bulk sampling programs indicate that, in general, most of the Renard kimberlites are irregular and elliptical in plan view. Surface areas of the kimberlite portion of the pipes range from 0.1 ha to 2.0 ha. A summary of each pipe is presented in Table 7.1. The Mineral Resource Estimate in Section 14 describes Indicated/Inferred resources for the Renard 2, Renard 3, Renard 4, Renard 9 and Renard 65 pipes. The Renard 1, Renard 7 and Renard 10 pipes may have economic potential and are classified as TFFE. Two laterally extensive kimberlite dyke systems, known as the Lynx and Hibou dykes, have been identified to the west and northwest of the pipe cluster, respectively (Figure 7.3). Portions of both dykes are included in the mineral resource estimation. Additional dyke-like kimberlites have been discovered elsewhere on the property. These are not included in the mineral resource estimation but may warrant additional work at a later date.

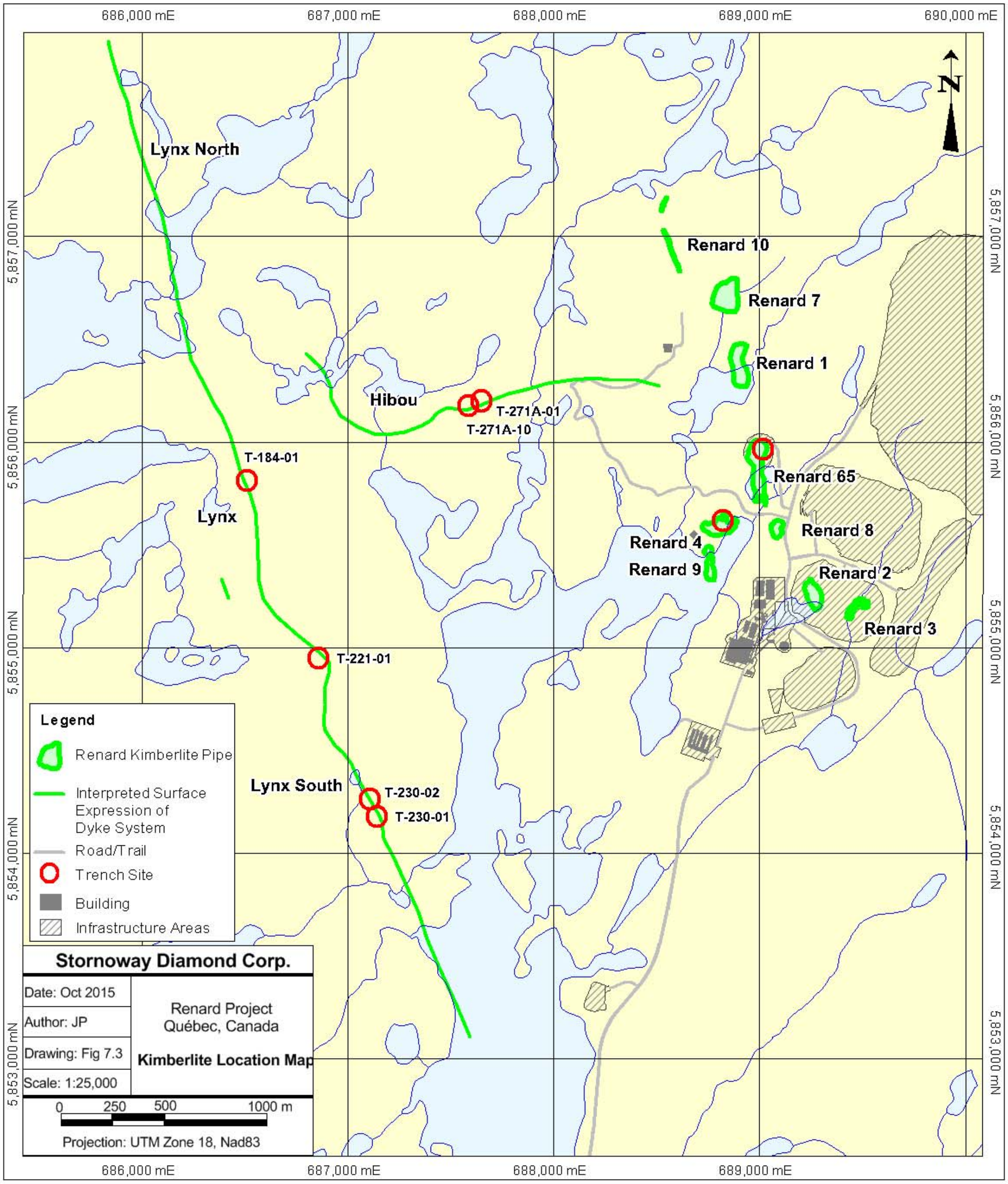
The following sections focus on the internal geology of the pipes which is fundamental to the resource estimation process.

7.3.1 General Geology

Kimberlite nomenclature has evolved several times throughout the work carried out on the Renard kimberlites. The terminology used at this time to describe the rock types in these kimberlites is in accordance with that used in most scientific literature (Field and Scott Smith, 1999; Hetman et al., 2008; Sparks et al., 2006; Cas et al., 2009; Webb, 2006). Within this report the following terms and definitions are used:

- Massive Volcaniclastic Kimberlite (MVK): a general term that refers to kimberlite that has been fragmented (i.e., the magma broken apart as a result of emplacement processes) and includes kimberlite classified texturally as tuffisitic kimberlite breccia (TKB).
- Coherent Kimberlite (CK): a general term that refers to kimberlite that has not been fragmented and includes kimberlite classified texturally as hypabyssal kimberlite (HK). In general, the term coherent kimberlite is used to refer to large, pipe-infilling events of this nature.
- Hypabyssal Kimberlite (HK): a more specific historical textural term for CK. Typically used here to describe the detailed texture of a CK rock and commonly used when referring to dykes or irregular intrusions.
- Tuffisitic Kimberlite Breccia (TKB): a more specific textural term of an MVK. Characterized by microlitic clinopyroxene in the matrix of the rock.
- Transitional Kimberlite: this refers to kimberlite that shows textures of both MVK and CK. A small “t” denotes a transitional textured HK or TK when describing rock-types (e.g., HKt or TKt).

Figure 7.3: Kimberlite Location Map



The Renard kimberlite pipes comprise diatreme-zone to root-zone kimberlites, with overall similar internal geologies. These pipes can be classified as “typical” South-African-style kimberlites (Hawthorne, 1973, Clement and Skinner, 1979, Clement, 1982 and Field and Scott Smith, 1999). In most pipes, with the exception of Renard 3 and Renard 10, the dominant phase is a MVK that can be classified as TKB. In general, these TKBs are extensively altered and have a massive texture. They consist of varying amounts of olivine, juvenile clasts and country rock xenoliths that are poorly sorted, typically loosely packed and less commonly clast supported, all set within a highly altered interclast matrix. In many pipes an additional pipe-filling phase is present that is typically a more coherent or transitional kimberlite characterized by lower country rock xenolith content and higher olivine content set within a crystalline to semi-crystalline groundmass. In all bodies, HK is present as both dykes and irregularly shaped intrusions that are found within each pipe infilling phase, between contacts of phases and along pipe margins. These are typically considered later stage intrusions. The HK intrusions can vary in thickness from a few centimeters to several meters and, in the case of the Lynx and Hibou dyke system, for example, can be laterally extensive.

The Renard pipe-like bodies are all associated with extensive cracked country rock (CCR) created during the emplacement event and, with the exception of Renard 3 and Renard 8, have a significant marginal country rock breccia (CRB). The CCR consists of both broken and solid country rock with small amounts of HK dykes and veins throughout, and minor zones containing kimberlite-derived constituents. The CRB typically lies between the main kimberlite units and the CCR, and is characterized by dominantly broken and pulverized clast-supported country rock, with an overall dilution of 95% or greater. CRB contains up to 5% of kimberlitic components, present as olivine, rare altered magmaclasts and very rare garnet xenocrysts in the breccia matrix. The CRB contains a significant amount of additional diamond-bearing kimberlitic material, in the form of late-stage, cross-cutting HK dykes, and helps to define the pipe shape.

Each Renard kimberlite contains a variety of phases that are distinguishable from one another by differing macroscopic and microscopic properties as well as diamond grades. A summary of the various kimberlite lithologies present in the Renard bodies with Indicated or Inferred resources is provided in Table 7.2.

Previous U-Pb dating of groundmass perovskite in HK dykes within Renard 1 suggested an emplacement age of 631.6 +/- 3.5 Ma (Birkett et al., 2004). Recent data obtained for the main rock-types in Renard 2 and Renard 3 using the same method suggest an emplacement age of 640.5 +/- 2.8 Ma.

Table 7.1: Renard Kimberlite Pipes and Significant Dyke Systems

Pipe (year of discovery)	Surface Area of Kimberlite* (ha)	Dimensions of Kimberlite at Surface* (m)	Vertical Extent of Kimberlite Modelled (m)	Depth of Cover (m)	Major Kimberlite Units	Exploration Potential	Comments
Renard 1 (2001)	2.3	250 x 100	325	6	MVK, transitional CK and large HK intrusion	Additional drilling and collection of a large tonnage DMS sample	Diatreme-zone kimberlite with overall regular shape. Slightly elongate north-south, sub-circular shape at surface. Large body with multiple kimberlite facies including discrete HK.
Renard 2 (2001)	1.9	220x120	850	22	MVK and transitional CK	Remains open at depth	Diatreme-zone kimberlite with irregular shape near surface. Slightly elongate north-south at surface. Partly covered by lake.
Renard 3 (2002)	0.5	160 x 40	442	12	Transitional CK, MVK, large HK intrusions	Remains open at depth	Deep diatreme to root-zone kimberlite with overall irregular shape. May have larger depth extent than originally anticipated.
Renard 4 (2002)	1.6	190 x 125	642	11	MVK, transitional CK	Remains open at depth	Partly covered by shallow lake (about 0.4 ha). Diatreme-zone kimberlite with overall regular shape. Slightly elongate, east-west trending subcircular shape at surface.
Renard 65 (2002)	3.1	375 x 100	585	8	Transitional CK, MVK	Remains open at depth	Diatreme zone kimberlite. Slightly elongate north-south trending irregular shape at surface.

Table 7.1: Renard Kimberlite Pipes and Significant Dyke Systems (cont'd)

Pipe (year of discovery)	Surface Area of Kimberlite* (ha)	Dimensions of Kimberlite at Surface* (m)	Vertical Extent of Kimberlite Modelled (m)	Depth of Cover (m)	Major Kimberlite Units	Exploration Potential	Comments
Renard 7 (2002)	1.6	160x120	240	5	MVK	Additional drilling and collection of a large tonnage DMS sample	Diatreme-zone kimberlite with overall regular shape and subcircular shape at surface. Potentially extensive and discrete HK unit near surface.
Renard 8 (2002)	0.3	80 x 50	240	8	MVK, large HK intrusion	No additional work planned	Diatreme-zone kimberlite with overall regular shape and subcircular shape at surface. Proximal to mine infrastructure.
Renard 9 (2003)	0.7	195 x 50	377	10	MVK, transitional CK	Remains open at depth	Diatreme zone kimberlite. Elongate, north-south trending, elliptical shape at surface. Body shape dips to the east with depth. Entirely covered by 2–4 m deep lake.
Renard 10 (2003)	1.5	370 x 50	210	3	CK	Additional drilling and collection of a large tonnage DMS sample	North-northwest trending dyke-like kimberlite. Possible blind intrusion suggested by large hanging wall CRB.
Lynx (2003)	n/a	4230 (strike length only)	95	5	HK	Remains open along strike and down dip	North-northwest trending dyke, variable dip from 10° to 50° to the east. Geologically and geochemically similar to Hibou.
Hibou (2005)	n/a	1940 (strike length only)	92	8	HK	Remains open along strike and down dip	East-west trending dyke; shallow dip approximately 10° to the north-northeast. Similar geologically and geochemically to Lynx.

* Based on 3D geological model; excludes CCR but includes CRB as an integral part of pipe.

Table 7.2: Major Geological Units

Kimberlite	Major Geological Unit	Dominant Colour	Textural Classification	Textural Classification Codes	Distinguishing Characteristics
Renard 2	Kimb2a	blue to blue green	volcaniclastic kimberlite	TK	high abundance and size of country rock xenoliths (CRx), blue clay matrix
	Kimb2b	brown	magmatic > volcaniclastic kimberlite	HK-HKt to TKt	coarse olivine sizes, lower CRx% and size, abundant perovskite
	Kimb2c	dark green to black	magmatic kimberlite	HK	uniform distribution of crystalline groundmass, low dilution
	CRB-2a	white with blue matrix	volcaniclastic kimberlite	TK	highly diluted Kimb2a characterized by very large, tightly packed country rock xenoliths with minor amounts of blue kimberlite matrix
Renard 3	Kimb3b	blue to blue green	volcaniclastic kimberlite	TK	high abundance and size of CRx, abundant juvenile clasts
	Kimb3c	dark green to grey/black	magmatic kimberlite	HK	uniform distribution of crystalline groundmass, low dilution
	Kimb3d	black to dark brown	magmatic kimberlite	HK-HKt	strongly altered CRx with black to green alteration rims and bleached centres
	Kimb3g	mottled green-brown to dark brown	magmatic > volcaniclastic kimberlite	HKt-TKt	texturally complex, similar to kimb3f but has lower abundance of HK autoliths and mantle nodules
	Kimb3f	light-dark brown with mottled blue-green zones	volcaniclastic > magmatic kimberlite	TKt-HKt	higher abundance of HK autoliths and higher% of larger CRx than in 3d, mantle nodules present
	Kimb3h	black	magmatic kimberlite	HK	uniform distribution of crystalline groundmass, 15% - 25% commonly bleached CRx
	Kimb3i	black	magmatic kimberlite	HK	uniform distribution of coarse crystalline groundmass, indicator minerals more common

Table 7.2: Major Geological Units (cont'd)

Kimberlite	Major Geological Unit	Dominant Colour	Textural Classification	Textural Classification Codes	Distinguishing Characteristics
Renard 4	Kimb4a	blue grey to grey green	volcaniclastic kimberlite	TK	high abundance and size of white-pink CRx, common carbonatized olivines
	Kimb4b	brown	magmatic > volcaniclastic kimberlite	HK-HKt to TKt	coarse olivine sizes, lower CRx% and size, common mantle nodules, abundant perovskite
	Kimb4c	black to dark green	magmatic kimberlite	HK (to HKt)	uniform distribution of crystalline groundmass, low dilution
	Kimb4d	dark blue	highly variable - volcaniclastic > magmatic kimberlite	TK-TKt with HKt-HK zones	dark blue clay matrix, common indicator minerals, higher abundance of HK autoliths and HK dykes
Renard 65	Kimb65a	pale blue-grey to dark grey-green	volcaniclastic kimberlite	TK-TKt	high abundance and size of CRx, abundant juvenile clasts
	Kimb65b	black to dark brown	magmatic kimberlite	HK-HKt	strongly altered CRx dark green to partially bleached centres, common mantle nodules
	Kimb65c	dark green to grey/black	magmatic kimberlite	HK	uniform distribution of crystalline groundmass, low dilution, common flow banding of olivines
	Kimb65d	light-dark brown with mottled blue-green zones	volcaniclastic > magmatic kimberlite	TKt-HKt	dirty brown appearance, creamy green and yellow rimmed CRx, common HK autoliths
	Kimb65e	dark brown	magmatic > volcaniclastic kimberlite	HKt	complex magmatic transitional unit in the southern part of the body

Table 7.2: Major Geological Units (cont'd)

Kimberlite	Major Geological Unit	Dominant Colour	Textural Classification	Textural Classification Codes	Distinguishing Characteristics
Renard 9	Kimb9a	greyish green	volcaniclastic kimberlite	TK	high abundance and size of CRx, abundant juvenile clasts, pyrite replaces olivines
	Kimb9b	black to brownish black	magmatic > volcaniclastic kimberlite	HK-HKt (rare TKt)	varies in texture from east (HK) to west (TKt)
	Kimb9c	dark green to grey/black	magmatic kimberlite	HK	uniform distribution of crystalline groundmass, two types – non-magnetic carbonate rich and magnetic
Lynx		dark green to black	magmatic kimberlite	HK	uniform distribution of crystalline groundmass, coarse olivines, abundant ilmenite
Hibou		dark green to black	magmatic kimberlite	HK	uniform distribution of crystalline groundmass, very coarse olivines common, abundant ilmenite and chrome diopside
All	CR	variable	n/a	n/a	country rock
	CCR	variable	n/a	n/a	large country rock blocks separated by thin kimberlite-filled fractures and HK dykes situated around the main kimberlite phases
	CRB	variable	n/a	breccia	brecciated country rock cut by HK dykes
	CRB+K	variable	n/a	breccia	brecciated country rock with up to 5% kimberlite material usually identified as traces of kimb2a, olivine macrocrysts and very rare magmaclasts

Notes:
HK = hypabyssal kimberlite
HKt = transitional hypabyssal kimberlite
HK-HKt = hypabyssal to transitional kimberlite
HKt-TKt = magmatic to tuffisitic kimberlite
TK = tuffisitic kimberlite
TK-TKt = tuffisitic to transitional tuffisitic kimberlite

7.3.2 Renard 2

The Renard 2 kimberlite is a mid-sized kimberlite pipe within the Renard Cluster. It is interpreted as a diatreme-zone kimberlite with irregularities in the external shape of the kimberlite near the surface, but an overall smooth and tapering shape when considering the emplacement envelope (including the CRB and CCR). The internal geology of Renard 2 (Figure 7.4 and Figure 7.5) was established using geological logs of both core and reverse circulation drill holes combined with detailed mapping of the underground drift and petrographic and geochemical studies. A detailed description of the geology and emplacement history of Renard 2 is described in Fitzgerald et al. (2009).

Renard 2 consists of two main pipe-infills: an MVK referred to as Kimb2a and a transitional CK referred to as Kimb2b. The two main infills exhibit contrasting primary textures: olivine abundances and populations, country rock xenolith abundances and populations, and diamond contents. The kimberlite is surrounded by extensive marginal CRB and CCR. In addition to these, HK dykes and intrusions (referred to as Kimb2c) of varying thickness are found throughout the body, along the pipe contacts, within the marginal breccia and in the cracked country rock (Fitzgerald et al., 2009).

Each kimberlite unit can be described as follows:

Kimb2a is volumetrically the most significant kimberlite rock type infilling the pipe, accounting for 70% of total kimberlite by volume. It is extensively altered, massive and can be texturally classified as a TKB (Clement and Skinner, 1979; Clement, 1982; Clement and Reid, 1989; Field and Scott Smith, 1999; Hetman, 2008). This blue to blue-green rock consists of olivine, juvenile clasts and country rock xenoliths that are poorly sorted, typically loosely packed and less commonly clast supported and set within a highly altered interclast matrix. Olivine macrocrysts comprise 5-15% of the rock and are typically medium-grained to rarely coarse-grained. Juvenile clasts are common, have sharp margins and are of both cored and uncored varieties (Webb, 2006). Primary groundmass minerals in juvenile clasts include phlogopite, spinel and perovskite. This unit commonly contains 40% - 65% fresh to moderately altered granitoid and lesser gneissic country rock xenoliths (granitoid dominate). These are set within a non-crystalline matrix that comprise clays, serpentine and microlitic clinopyroxene. This rock can be classified mineralogically as a phlogopite kimberlite.

Kimb2b is the second most volumetrically significant kimberlite rock type, accounting for 30% of total kimberlite by volume. This rock is a moderately altered, massive and texturally variable CK that displays both coherent and rarer volcaniclastic textures classified as HK-HKt to TKt. This brown rock consists of olivine, country rock xenoliths and rare juvenile clasts that are poorly sorted and set within a crystalline to semi-crystalline groundmass. Olivine macrocrysts comprise 10% - 25% of the rock and are medium- to coarse-grained with common very coarse grains. Juvenile clasts are relatively rare and, where observed, typically have diffuse margins. Primary groundmass minerals include phlogopite, perovskite, spinel, carbonate and rare monticellite. Secondary minerals in the groundmass include serpentine, clays and microlitic clinopyroxene. This unit commonly contains 20% - 50% moderately to strongly altered granitoid and gneissic country rock xenoliths (gneissic appear to dominate). This rock can be classified mineralogically as a monticellite-phlogopite kimberlite.

Contacts between these two rock types are often sharp and commonly highlighted by HK dyke intrusions. The contrasting textural and component features described above and the nature of the contacts, combined with the difference in diamond grade, supports interpretation of the two pipe-infills as two distinct rock-types emplaced during two separate volcanic events. The emplacement of Kimb2b is thought to have preceded that of Kimb2a based on the contact relationships between these two phases, the relative distribution of these within the pipe and age dating.

Also modelled in Renard 2 is the CRB-2a unit, accounting for 3% of the pipe (Kimb2a, Kimb2b, CRB2a and CRB units) by volume. This internal unit was identified during mapping of the underground exposure and drill core logging, and consists of large fresh blocks of white granitoid and gneiss with a matrix component manifested as Kimb2a intersections < 10 cm up to several meters wide. Country rock xenolith content is around 95%. Intersections of Kimb2c are also common throughout, similar to other units within Renard 2. The unit has a vertical extent of 230 m from surface and is nestled between both the Kimb2a and Kimb2b units. The presence of the Kimb2a intersections, overall blue colour of the matrix, similarities in appearance, type and alteration state of the country rock xenoliths, presence of Kimb2c intersection as well as its spatial location suggest that this country rock dominated unit represents a highly diluted zone of Kimb2a. Petrographic examination of matrix material supports this interpretation.

Coherent kimberlite (Kimb2c) occurs in Renard 2 in the form of late-stage HK dykes and irregular intrusions, ranging in thickness from a few centimeters up to 15 m and contributing 17.4% by relative proportion of drill intercepts (as calculated within the Kimb2a and Kimb2b units). Kimb2c is black to dark-green coloured with medium- to coarse-grained and, commonly, very coarse-grained macrocrystic olivine, comprising 25% to 30% of the rock. The various HK intrusions include a spectrum of primary groundmass mineral assemblages, ranging from monticellite dominated to phlogopite dominated, each additionally consisting of common carbonate, spinel and perovskite. This unit contains approximately 12% country rock xenoliths which are typically <5 cm in size, irregular in shape and highly altered.

The kimberlite in Renard 2 is surrounded entirely by extensive marginal CRB and CCR. The CRB has a maximum width of approximately 100 m near surface and extends the full depth of the body on the north side of the pipe. Due to drilling constraints, the full extent of the CCR has not been delineated, and so it is not shown in Figures 7.4 and 7.5. HK dykes are present within the CRB and CCR units. CCR units were identified in drill core, however, there was not enough external contact information to warrant modelling it in 3D in the updated 2015 model.

Figure 7.4: Plan View – Renard 2, Renard 3, Renard 4 and Renard 9

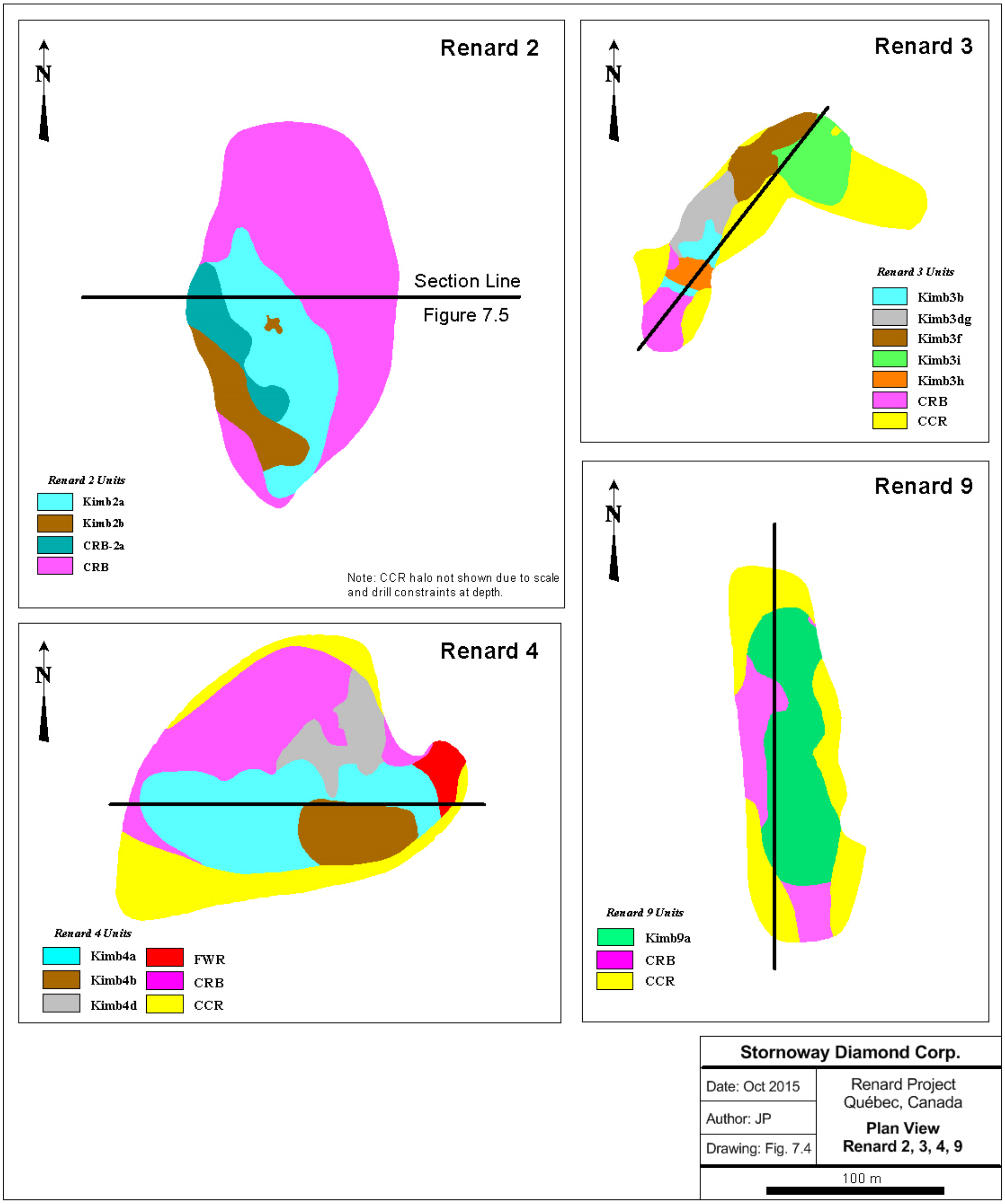
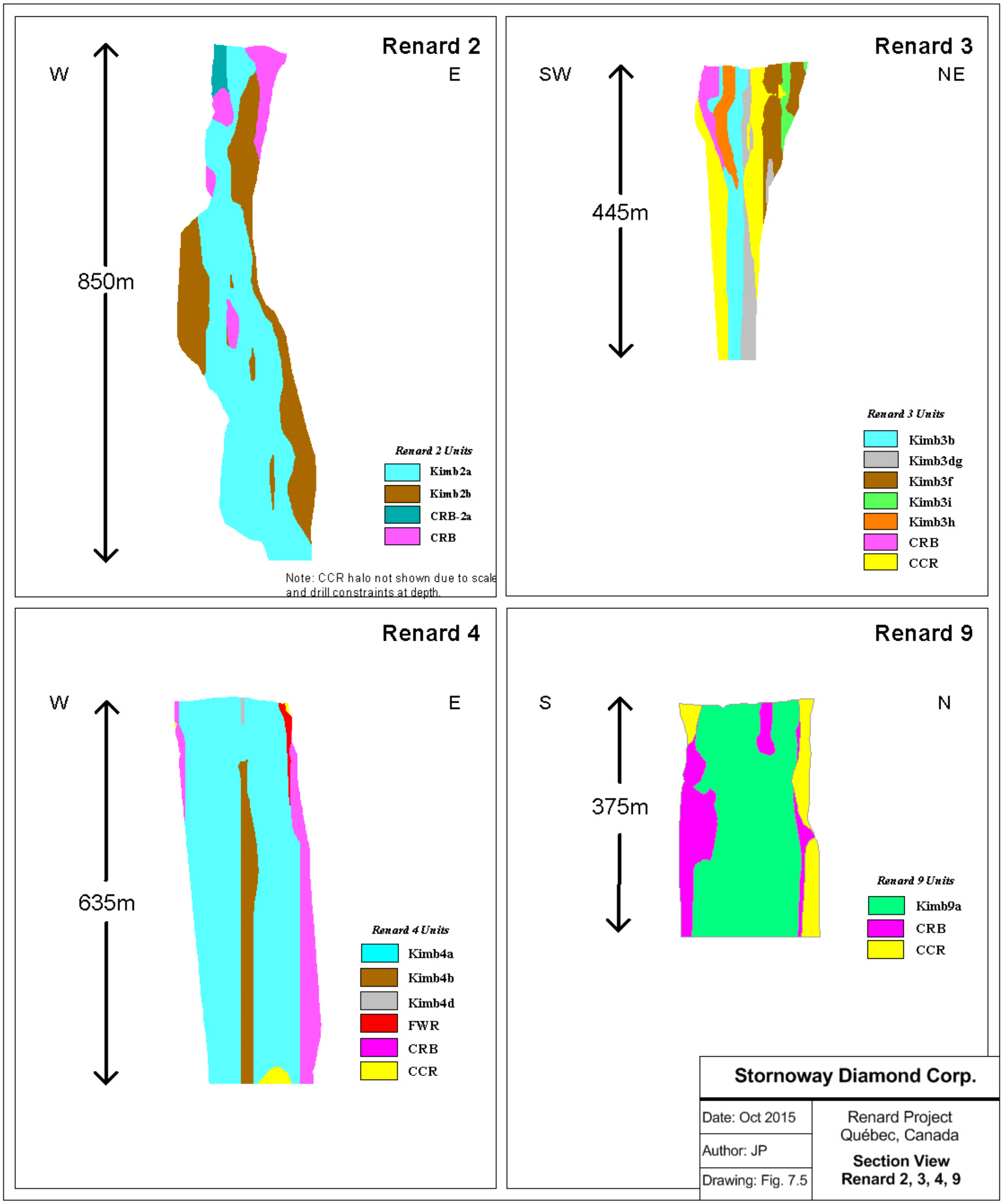


Figure 7.5: Section View – Renard 2, Renard 3, Renard 4 and Renard 9



7.3.3 Renard 3

The Renard 3 kimberlite is one of the smallest kimberlite pipes within the Renard Cluster. It is interpreted as a steep-sided, deep diatreme to root-zone kimberlite, with an irregular in external pipe shape. The geology of this kimberlite has been determined using detailed logging of drill core, mapping of the underground decline walls, and results of petrographic and geochemical studies (Figure 7.4 and Figure 7.5). A detailed description of the geology and emplacement history of Renard 3 is described in Muntener and Scott Smith (2013).

The current three-dimensional (3D) model of Renard 3 has two possible feeder zones and consists of six kimberlite rock types in order of volumetric significance: Kimb3d, Kimb3g, Kimb3i, Kimb3b, Kimb3h, and Kimb3f. Currently, the Kimb3d and Kimb3g units are modelled together as one 3D solid, based on their comparative mineralogy and abundances of groundmass minerals observed through thin section analysis, as well as similar grade and diamond counts (stones per tonne) obtained from underground sampling. A small CRB zone exists on the upper, southern portion of the pipe, in association with the Kimb3b phase. More extensive CCR discontinuously surrounds the majority of the kimberlite pipe margin. In addition to the six main kimberlite rock types infilling the pipe, a number of HK dykes and intrusions (Kimb3c) occur throughout the body, along the pipe contacts, within the marginal breccia and in the cracked country rock.

Each kimberlite unit can be described as follows:

Kimb3d is a predominantly coherent, black to dark brown kimberlite with medium- to coarse-grained and common very coarse-grained macrocrystic olivine. This unit commonly contains 20% - 45% strongly altered, dark green to black-rimmed granitoid and lesser gneissic country rock xenoliths with bleached centres. The matrix is crystalline and dominated by primary groundmass minerals consisting of phlogopite, carbonate, perovskite, spinel and rare monticellite. Texturally this rock is classified as an HK-HKt and mineralogically as a phlogopite kimberlite.

Kimb3g is a coherent to volcaniclastic, dark brown to mottled-brown-green kimberlite with medium- to coarse-grained and rare very coarse-grained macrocrystic olivine. This unit contains approximately 30% - 50% granitoid and gneissic country rock xenoliths moderately altered with creamy green and yellow rims. Juvenile clast abundance is variable and exhibits common thick selvages. The matrix is non-crystalline to crystalline and contains both primary groundmass minerals and patchy zones of late-stage reaction minerals. Texturally this rock is classified as an HKt-TKt and mineralogically as a phlogopite kimberlite.

Kimb3d and Kimb3g are modelled as one unit and together account for 46% of the kimberlite by volume.

Kimb3b is a blue to blue-grey MVK with fine- to medium-grained and less commonly coarse-grained macrocrystic olivine. This unit contains fresh to altered pink to patchy green granitoid xenoliths, with lesser gneissic xenoliths that are fresh to moderately altered green. Xenolith content varies from 55% - 75%. Juvenile clasts are common and are both cored and uncored. The matrix is dominated by clinopyroxene, minor phlogopite, serpentine and clays, contributing to its blue-grey colour. Texturally this rock can be classified as a TKB and mineralogically as a phlogopite kimberlite. Kimb3b represents 17% of total kimberlite by volume.

Kimb3i is a coherent, black kimberlite with medium- to coarse-grained and common very coarse-grained macrocrystic olivine. This unit contains on average 10% - 20% strongly altered granitoid and gneissic country rock xenoliths, with alteration style varying from entirely bleached and dark-green coloured to having black rims with bleached centres. The matrix is uniform, crystalline and dominated by primary groundmass minerals consisting of carbonate, phlogopite, perovskite, spinel. Texturally this rock is classified as a HK and mineralogically as a phlogopite carbonate kimberlite. Kimb3i represents 16% of total kimberlite by volume.

Kimb3h is a coherent black kimberlite with medium- to coarse-grained and rare, very coarse-grained macrocrystic olivine. This unit contains 15% - 25% strongly altered, dark green to black-rimmed granitoid and gneissic country rock xenoliths with bleached centres. The matrix is crystalline with primary groundmass minerals consisting of carbonate, phlogopite, spinel, perovskite and rare monticellite and apatite. Texturally this rock is classified as a HK and mineralogically as a carbonate phlogopite kimberlite. Kimb3h represents 10% of total kimberlite by volume.

Kimb3f is a volcaniclastic brown to mottled green-blue kimberlite with medium-grained and common coarse-grained macrocrystic olivine. This unit contains 35% - 50% gneissic and granitoid country rock xenoliths, moderately altered with creamy yellow to green rims. Juvenile clasts with thick selvages are abundant, however, are commonly diffuse and are not always easily observed, particularly in darker, more magmatic zones. Autoliths of HK are abundant and, in general, this unit contains the highest percentage of autoliths compared to other units in the Renard 3 body. The matrix is dominated by primary groundmass minerals consisting of phlogopite, perovskite, spinel as well as late-stage clinopyroxene and richterite. Texturally this rock is classified as a TKt-HKt and mineralogically as a phlogopite kimberlite. Kimb3f represents 10% of total kimberlite by volume.

Contacts between these main rock types are sharp and commonly highlighted by HK dyke intrusions. Contrasting textural and component features described above and the sharp nature of the contacts, combined with the difference in diamond grades, supports interpretation of Renard 3 containing five distinct phases of kimberlite, each the product of distinct and separate batches of mantle-derived magma. Based on the current understanding of Renard 3, the emplacement history is complex and interpreted as at least a nine stage process involving five main pipe-infilling events. Kimb3b emplaced first, contemporaneously with the CRB, followed by Kimb3f, Kimb3d/g, Kimb3h, and Kimb3i (Muntener and Scott Smith, 2013).

Coherent kimberlite (Kimb3c) occurs in Renard 3 in the form of late-stage HK dykes and irregular intrusions contributing in the order of 14% by relative proportion of drill intercepts (as calculated within the main Renard 3 units). They range in thickness from a few centimeters up to 20 m. Kimb3c is a dark-green to black HK with medium- to coarse-grained, and commonly, very coarse-grained macrocrystic olivine. This unit does not commonly contain country rock xenoliths, but where present, they comprise approximately 10% of the rock.

CRB is found only in the southern, uppermost part of the pipe and therefore comprises only a minor portion of the entire kimberlite pipe. It has a maximum width of 45 m, wrapping around the southern end of the pipe, and extends vertically for almost 175 m. The kimberlite is surrounded from top to bottom by discontinuous CCR.

7.3.4 Renard 4

The Renard 4 kimberlite is one of the larger kimberlite pipes within the Renard Cluster. It is interpreted as a diatreme-zone kimberlite with a relatively regular external pipe shape. The geology of this kimberlite has been determined using detailed logging of drill core, mapping of the surface trench and petrographic studies (Figure 7.4 and Figure 7.5).

The Renard 4 pipe contains three kimberlite geological units: an MVK, referred to as Kimb4a, a transitional CK, referred to as Kimb4b, and a texturally variable MVK, referred to as Kimb4d. In the current 3D model there is also a unit referred to as FWR (further work required) on the eastern edge of the body that is currently unclassified kimberlite. It may represent a more highly diluted Kimb4a or Kimb4d. A significant marginal CRB and CCR surround the main kimberlite. The Renard 4 CCR adjoins at depth with the CCR of Renard 9, along the southern margin of the pipe. In addition to the three main pipe infills, a number of HK dykes and irregular intrusions occur throughout the body, along the pipe contacts, within the marginal breccia and in the cracked country rock.

Each kimberlite unit can be described as follows:

Kimb4a is an MVK that can be texturally further classified as a TKB. It is blue to green coloured with medium-grained and, less commonly, fine-grained macrocrystic olivine. A characteristic feature of this unit is that the olivine macrocrysts are commonly carbonatized, which is not typical of other Renard TKB's. Juvenile clasts are common and are of both cored and uncored varieties. The unit contains 50% - 80% granitoid and gneiss country rock xenoliths that are fresh to moderately altered. The matrix is dominated by clay, serpentine carbonate and clinopyroxene, contributing to its blue-green colour. Kimb4a has been divided into two subtypes based on subtle differences in xenolith character and mineralogy: Kimb4a-1 is a blue-grey rock with fresh xenoliths and is mineralogically classified as a phlogopite monticellite kimberlite; Kimb4a-2 is a green rock with dark pink, hematized xenoliths that can be classified mineralogically as a phlogopite spinel monticellite kimberlite. Further work is required in order to determine if in fact these are two separate phases, or if the differences in the rock characteristics are due to variations within one phase. Kimb4a represents 69% of total kimberlite by volume.

Kimb4b is a mottled light to dark brown coherent to volcaniclastic kimberlite, with medium- to coarse-grained macrocrystic olivine. Juvenile clasts are present and typically appear as diffuse magmatic domains within a more altered groundmass. Their abundance is variable and they exhibit thick selvages. Country rock xenoliths are fresh to more commonly, strongly altered granitoid and gneiss, with content ranging from 25% - 50%. The groundmass varies from crystalline to non-crystalline. Mantle nodules are very common and characterize this unit. Texturally this rock is classified as a HK-HKt to TKt and mineralogically as a perovskite monticellite phlogopite kimberlite. Kimb4b represents 25% of total kimberlite by volume.

Kimb4d is a mostly volcaniclastic, dark blue kimberlite with medium-grained and, less commonly, fine- and coarse-grained macrocrystic olivine. It is a highly variable kimberlite texturally that contains common more coherent intervals within an overall MVK texture. These more coherent zones occur as both dykes and possible autoliths of undiluted Kimb4d. Juvenile clasts are both uncored and cored varieties, some with diffuse margins. Country rock xenoliths are fresh to moderately altered granitoid and gneiss and represent 40% to 75% of the rock. The matrix is dominated by serpentine, carbonate, clay minerals, clinopyroxene microlites and mica. Mantle indicator minerals are common and notably more abundant than in the other units. Texturally this rock is classified as a TK-TKt with zones of HKt-HK. Mineralogically this rock can be classified as a phlogopite spinel monticellite kimberlite. Kimb4d represents 3% of total kimberlite by volume.

Contacts between these main rock types are sharp and commonly highlighted by HK dyke intrusions. Contrasting textural and component features described above and the nature of the contacts, combined with the difference in diamond grade, supports interpretation of the three pipe infills as distinct rock-types emplaced during three separate volcanic events. The emplacement of Kimb4d is thought to have preceded that of Kimb4a based on the contact relationships between these two phases and the fact that Kimb4d appears as a remnant block of kimberlite that does not extend to depth. A second kimberlite emplacement event resulted in the removal of a significant volume of Kimb4d and infilling of the diatreme by Kimb4a. Finally, the Kimb4b unit removed a portion of the Kimb4a unit and infilled the south eastern portion of the pipe. Emplacement of Kimb4a is thought to have preceded formation of Kimb4b based on the presence of a remnant block of Kimb4a within the Kimb4b unit.

Coherent kimberlite (Kimb4c) occurs in Renard 4 in the form of late-stage HK dykes and irregular intrusions contributing in the order of 12% by relative proportion of drill intercepts (as calculated from within the main Renard 4 units). These HK dykes and intrusions range in thickness from a few centimeters up to 34 m. Kimb4c is a black to dark-green HK with medium- to coarse-grained and, commonly, very coarse-grained macrocrystic olivine. This unit does not commonly contain country rock xenoliths but, where present, they comprise approximately 11% of the unit and are strongly altered.

CRB is present dominantly in the north and west portion of the body, with smaller blocks in the south and east. The marginal breccia is both vertically and horizontally extensive, reaching a maximum width of 60 m adjacent to the body near surface and extending almost 650 m vertically. CCR surrounds the kimberlite from top to bottom discontinuously, however it is more extensive on the southern side of the body. The southern part of the Renard 4 CCR merges with the northern side of the Renard 9 CCR at approximately 210 m depth below surface and was modelled as one continuous unit.

7.3.5 Renard 9

The Renard 9 kimberlite is one of the smaller kimberlite pipes within the Renard Cluster. It is interpreted as a lower diatreme to root-zone kimberlite with an irregular external pipe shape that dips to the east with depth. The internal geology of Renard 9 has been established using geological logs of both drill core and reverse circulation drill holes and petrographic studies (Figure 7.4 and Figure 7.5).

Renard 9 consists of two main pipe-infills: an MVK referred to as Kimb9a and a volumetrically minor (2%) texturally variable CK referred to as Kimb9b. In general, the texture of these kimberlite phases change from east to west across the body: Kimb9b changes from more HK-like in the east to more TK-like to the west; and the dilution of Kimb9a increases significantly from east to west. An extensive CRB is present on the western side of the body, across both the length and height of the body. The CRB is spatially associated with the Kimb9a unit. Also peripheral to the kimberlite is significant CCR, which joins with that of Renard 4 at depth. In addition to these units, hypabyssal kimberlite (HK) dykes and intrusions (referred to as Kimb9c) of varying thickness are found throughout the body, along the pipe contacts, within the marginal breccia and in the cracked country rock.

Each kimberlite unit can be described as follows:

Kimb9a is volumetrically the most significant kimberlite rock type infilling the pipe, accounting for 98% of total kimberlite by volume. It is extensively altered, generally massive and can be texturally classified as a TKB. This pale-green to grey-green rock consists of olivine, juvenile clasts and country rock xenoliths that are poorly sorted, typically loosely packed and less commonly clast, supported and set within a highly altered interclast matrix. Olivine macrocrysts are typically fine- to medium-grained with rare coarse grains and are commonly altered to serpentine, carbonate and rare pyrite. Two mineralogical types of juvenile clasts are observed in Kimb9a. Both have sharp margins and include both cored and uncored varieties. Type one is classified as a spinel phlogopite kimberlite and type two as a spinel phlogopite monticellite kimberlite. This unit contains on average 60% - 75% fresh to moderately altered granitoid and gneissic country rock xenoliths set within a non-crystalline matrix comprising clays, serpentine and microlitic clinopyroxene.

Kimb9b accounts for only 2% of total kimberlite by volume. The relative scarcity of this rock type in the body, position and contact relationships suggest that Kimb9b is a remnant kimberlite phase emplaced before Kimb9a. This rock is a moderately altered, massive and texturally variable CK that displays both coherent and rare volcaniclastic textures. This brown kimberlite consists of olivine, country rock xenoliths and rare juvenile clasts that are poorly sorted and set within a dominantly crystalline groundmass. Olivine macrocrysts are medium- to coarse-grained. Juvenile clasts are relatively rare and, where observed, typically have diffuse margins. Groundmass minerals consist of perovskite, monticellite and phlogopite, set within an inhomogeneous, variably crystallized, interclast matrix dominated by carbonate, clays and microlitic clinopyroxene. This unit contains 20% - 50% moderately to strongly altered granitoid and gneissic country rock xenoliths. This rock exhibits textures of both MVK and a CK and can be classified texturally as a HK-HKt and mineralogically as a perovskite monticellite phlogopite kimberlite. The abundance and character of perovskite and phlogopite are diagnostic for this unit.

Coherent kimberlite (Kimb9c) occurs in Renard 9 in the form of late-stage HK dykes and irregular intrusions or autoliths. These range from black to dark green to grey coloured and are typically less than 1 m thick in drill core, contributing in the order of 14% by overall volume to the pipe (as calculated within the main Renard 9 units). HK dykes can be classified mineralogically as phlogopite monticellite kimberlites which are strongly magnetic. The more irregular intrusions can be classified as phlogopite spinel monticellite kimberlites with extensive carbonate throughout and are non-magnetic. These may in fact represent undiluted Kimb9a, occurring as autoliths throughout the TKB. Both subtypes of Kimb9c do not commonly contain country rock xenoliths, but where present, they comprise approximately 9% of the unit and are strongly altered.

The Renard 9 kimberlite has an extensive marginal CRB on the western and southern sides of the body, spatially associated with Kimb9a. It has a surface width of approximately 30 m and extends the full depth of the body. Contacts between this and the Kimb9a unit are typically gradational. Peripheral CCR with rare HK dykes discontinuously surrounds the kimberlite and joins with the CCR of Renard 4 at depth.

7.3.6 Renard 65

The Renard 65 kimberlite is the largest kimberlite in the Renard Cluster by surface area. It is interpreted as a diatreme-zone kimberlite with a slightly irregular shape at surface, and dipping gradually to the east. The internal geology of this kimberlite has been determined using geological logs of both core and reverse circulation drill holes, mapping of the surface trench and petrographic studies (Figure 7.6).

Renard 65 consists of four main pipe-infilling kimberlite units: Kimb65a, Kimb65b, Kimb65d and Kimb65e. In addition, CRB and CCR surround the main kimberlite pipe infills. In the current 3D geological model, Kimb65e was modelled with Kimb65a as its distribution is discontinuous and small, suggesting it is a remnant unit within Kimb65a. In addition, a number of HK dykes and irregular intrusions (Kimb65c) occur throughout the body as late-stage intrusions, and within the CRB and CCR.

Each kimberlite unit can be described as follows:

Kimb65a is a pale blue-grey to green-grey volcaniclastic kimberlite breccia accounting for 68% of total kimberlite by volume. It contains fine- to medium-grained and, less commonly, coarse-grained macrocrystic olivine and 50% - 75% fresh to altered, pink to patchy-green-coloured granitoid xenoliths, with lesser amounts of gneiss. Juvenile clasts are common and are both cored and uncored. Texturally this rock can be classified as a TK-TKt, mineralogically as a phlogopite kimberlite.

Kimb65b is a coherent black to dark brown kimberlite accounting for 12% of the kimberlite by volume. It contains coarse to very coarse-grained macrocrystic olivine with 5% - 35% strongly altered, dark-green to black-rimmed gneissic and lesser granitoid country rock xenoliths. Mantle nodules are abundant and include harzburgite, peridotite and possible minor dunite types. The matrix is crystalline with minor patches of alteration from the digestion of country rock xenoliths. Texturally this rock can be classified as HK-HKt.

Kimb65d is a volcaniclastic light to dark brown, occasionally mottled green kimberlite accounting for 19% of total kimberlite by volume. It contains medium- to coarse-grained and, rarely, very coarse-grained macrocrystic olivine and 30% - 50% (may reach up to 70%) granitoid and gneissic country rock xenoliths. These are moderately altered with pale green and yellow rims. Juvenile clasts are present, are both cored and uncored, and are typically < 3 cm in size. The matrix varies from crystalline to non-crystalline, with patchy zones of late-stage mineral alteration. Texturally this rock can be classified as TKt-HKt.

Kimb65e is a coherent to volcaniclastic dark-brown kimberlite with medium- to coarse-grained macrocrystic olivine. This unit contains 25% - 55% gneissic and granitoid country rock xenoliths (granite may dominate), fresh to moderately altered pale green, with stronger alteration in smaller xenocrysts. The matrix is crystalline to patchy crystalline with alteration minerals from xenolith digestion. The primary groundmass mineralogy includes phlogopite, carbonate, spinel and perovskite and can texturally be classified as HKt. Due to the small and discontinuous nature of the contacts, Kimb65e is currently modelled in Kimb65a.

Contacts between these main rock types are sharp and commonly highlighted by HK dyke intrusions. At the present time, more work needs to be carried out to define the relationship between Kimb65d and

Kimb65b, however based on the current understanding of Renard 65, the emplacement of Kimb65d is thought to have preceded that of Kimb65b. It is also thought that the emplacement of Kimb65a precedes both Kimb65d and Kimb65b.

Coherent kimberlite (Kimb65c) occurs in Renard 65 in the form of late-stage HK dykes and irregular intrusions contributing in the order of 9% by relative proportion of drill intercepts (as calculated within the main units). They range in thickness from a few centimeters up to 35 m. Kimb65c is a coherent black to dark-green kimberlite with medium- to coarse-grained and, commonly, very coarse-grained macrocrystic olivine. This unit does not commonly contain country rock xenoliths, but where present, they comprise approximately 5% of the unit.

Renard 65 kimberlite is surrounded by extensive CRB and CCR units that extend to depth but are discontinuous around the pipe. CRB is dominant on the western side of the body whereas CCR dominates on the east and north sides.

7.3.7 Other Renard Kimberlite Pipes

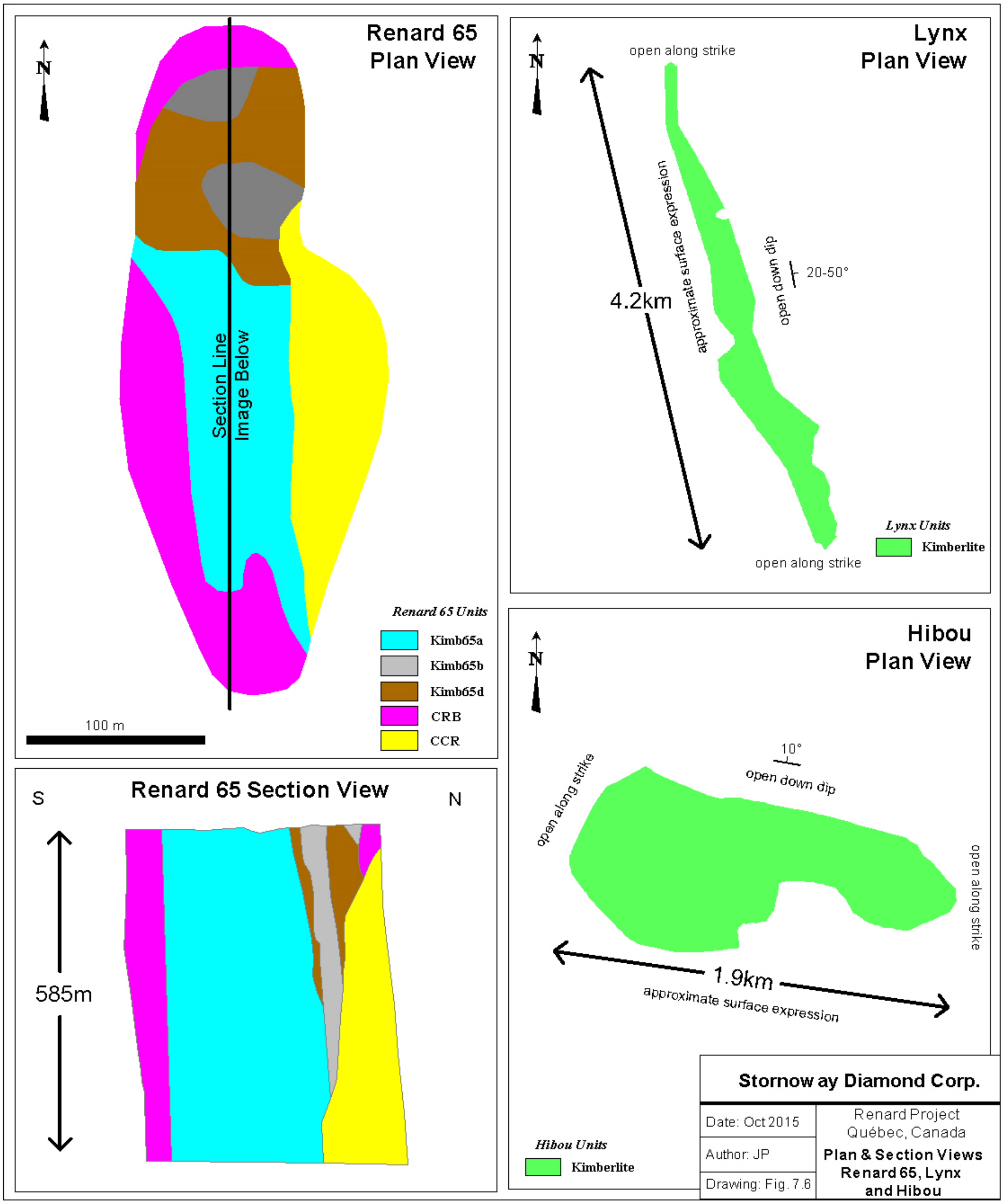
The remaining Renard kimberlite pipes (Renard 1, Renard 7, Renard 8 and Renard 10) are discussed briefly in the following summary as they represent non resource bodies in the project. Plan and Section views are shown in Figure 7.7 and Figure 7.8, respectively.

Renard 1

The Renard 1 kimberlite is the second largest kimberlite in the Renard Cluster, by surface area. It is interpreted as a diatreme-zone kimberlite that is oval shaped at surface with a small indent on its eastern side. It tapers at approximately 85° with depth. The geology of this kimberlite has been determined by petrology logging and petrographic work of 12 drill cores. The deepest hole (537 m), completed in 2010, crossed the body from south to north at an inclination of -58° and intersected kimberlite, CRB and CCR from 195 to 508 m downhole.

Renard 1 consists of three main pipe-infilling kimberlite units: a blue-grey MVK referred to as Kimb1a (comprising 26% of total kimberlite) and is classified as TK; a transitional CK referred to as Kimb1b (comprising 49% of total kimberlite) and a large, irregular HK intrusion referred to as Kimb1d (comprising 25% of total kimberlite). It is unclear at this time how these kimberlites relate temporally, but petrographic work indicates they are three distinct and separate phases. In addition to these main infills, a CRB unit surrounds the main kimberlite pipe infills on all sides but the eastern side. CCR units were identified in drill core, however there was not enough external contact information to warrant modelling the CCR in 3D. A number of cross cutting HK dykes and irregular intrusions occur throughout the main phases of kimberlite, along the pipe contacts and within the marginal CRB and CCR units. These are referred to as Kimb1c.

Figure 7.6: Plan and Section Views: Renard 65, Lynx and Hibou



Renard 7

The Renard 7 kimberlite is a moderate to large sized kimberlite in the Renard Cluster. It is interpreted as a diatreme-zone kimberlite with a slightly irregular shape at surface, and tapering gradually with depth. The geology of this kimberlite has been determined with petrology logging and petrographic work of seven drill holes.

The current 3D geological model of Renard 7 consists of at least one main pipe-infilling kimberlite unit: a diluted grey-green MVK that is referred to as Kimb7a and is classified as TK. An extensive amount of HK is localized in the upper southern portion of the body and is referred to as Kimb7c. It is unclear at this time if this HK represents a main pipe infilling event or is simply a larger cross cutting sheet. Due to its spatial occurrence in drill holes, it was not modelled separately in the current 3D model, therefore relative abundances are unknown. Total kimberlite represents 76% of the modelled body by volume. CRB partially surrounds the kimberlite pipe along the northern, southeastern and southwestern margins, along with one large internal block located in the north. CCR units were identified in drill core, however there was not enough external contact information to warrant modelling the CCR in 3D. A number of HK dykes and irregular intrusions occur throughout the body, along the pipe contacts, within the marginal breccia and in the cracked country rock. These are also referred to as Kimb7c because at this time they cannot be distinguished geologically from the large intrusion of HK in the upper southern portion of the body.

Renard 8

The Renard 8 kimberlite is the smallest kimberlite pipe within the Renard Cluster and is situated between Renard 2 and Renard 65, very proximal to ongoing infrastructure development. It is interpreted as a diatreme-zone kimberlite and is oval shaped at surface with a small indent on its eastern side. The geology of this kimberlite has been determined with petrology logging and petrographic work of ten drill holes.

The current 3D model of Renard 8 consists of one main pipe infill kimberlite unit: a diluted grey-green MVK that is classified as TKB, and referred to as Kimb8a. Total kimberlite represents 91% of the modelled body by volume. CCR units were identified in drill core, however there was not enough external contact information to warrant modelling the CCR in 3D. CRB is located in the south as an isolated block which penetrates the body rather than being solely marginal to the pipe. In addition, a number of HK dykes and intrusions referred to as Kimb8c occur throughout the body, along the pipe contacts, within the marginal breccia and in the cracked country rock.

Renard 10

The Renard 10 kimberlite is one of the smallest “pipes” within the Renard Cluster. It does not have a traditional pipe shape but is elongate and narrow in nature. The kimberlite (20% of the body by volume) in this body is surrounded by a significant marginal CRB (80% of the body by volume). This is a feature that has not been noted to be associated with other large dykes on the Foxtrot property. The geology of this kimberlite has been determined with petrology logging and petrographic work of ten drill holes.

Figure 7.7: Plan View – Renard 1, Renard 7, Renard 8 and Renard 10

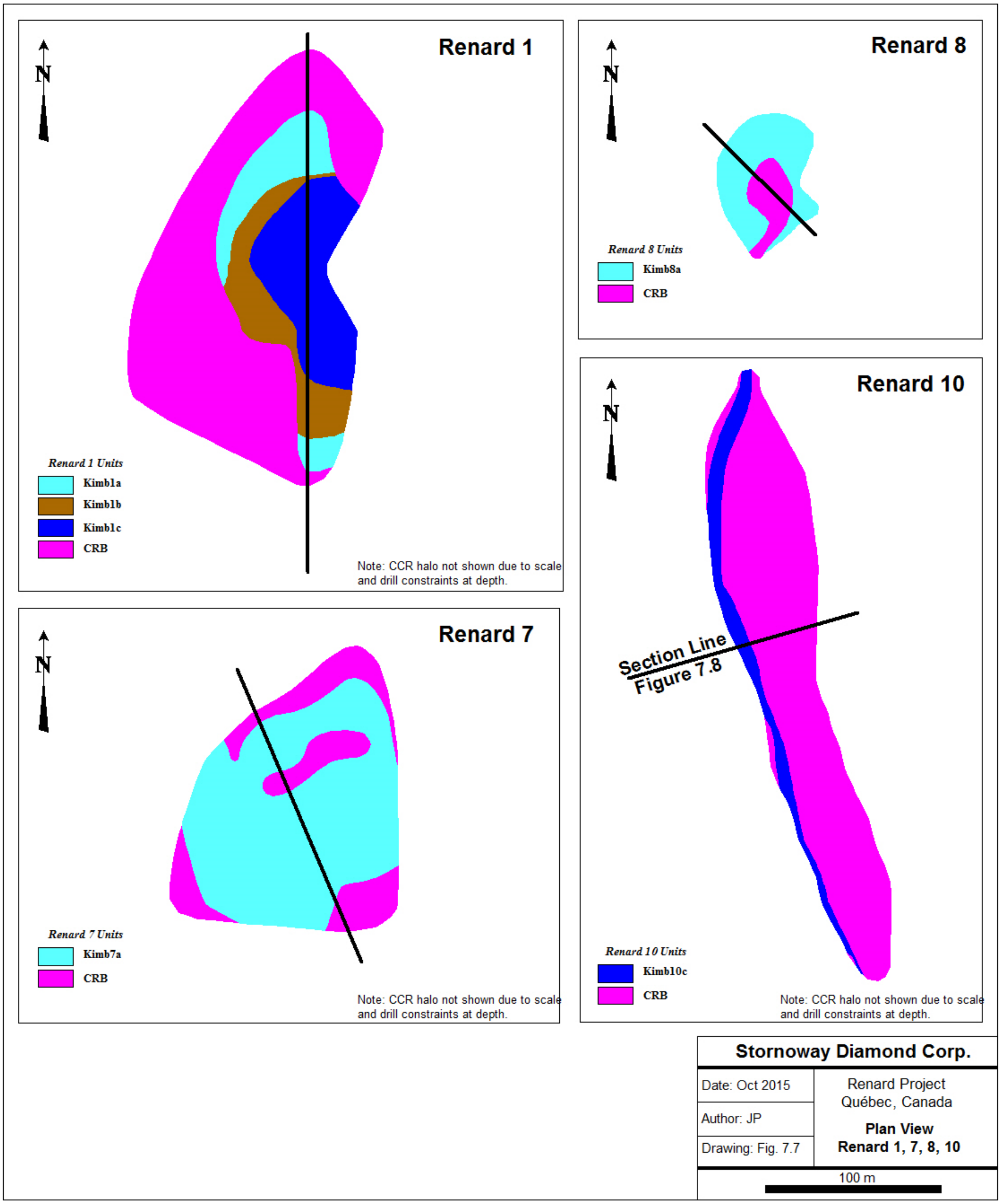
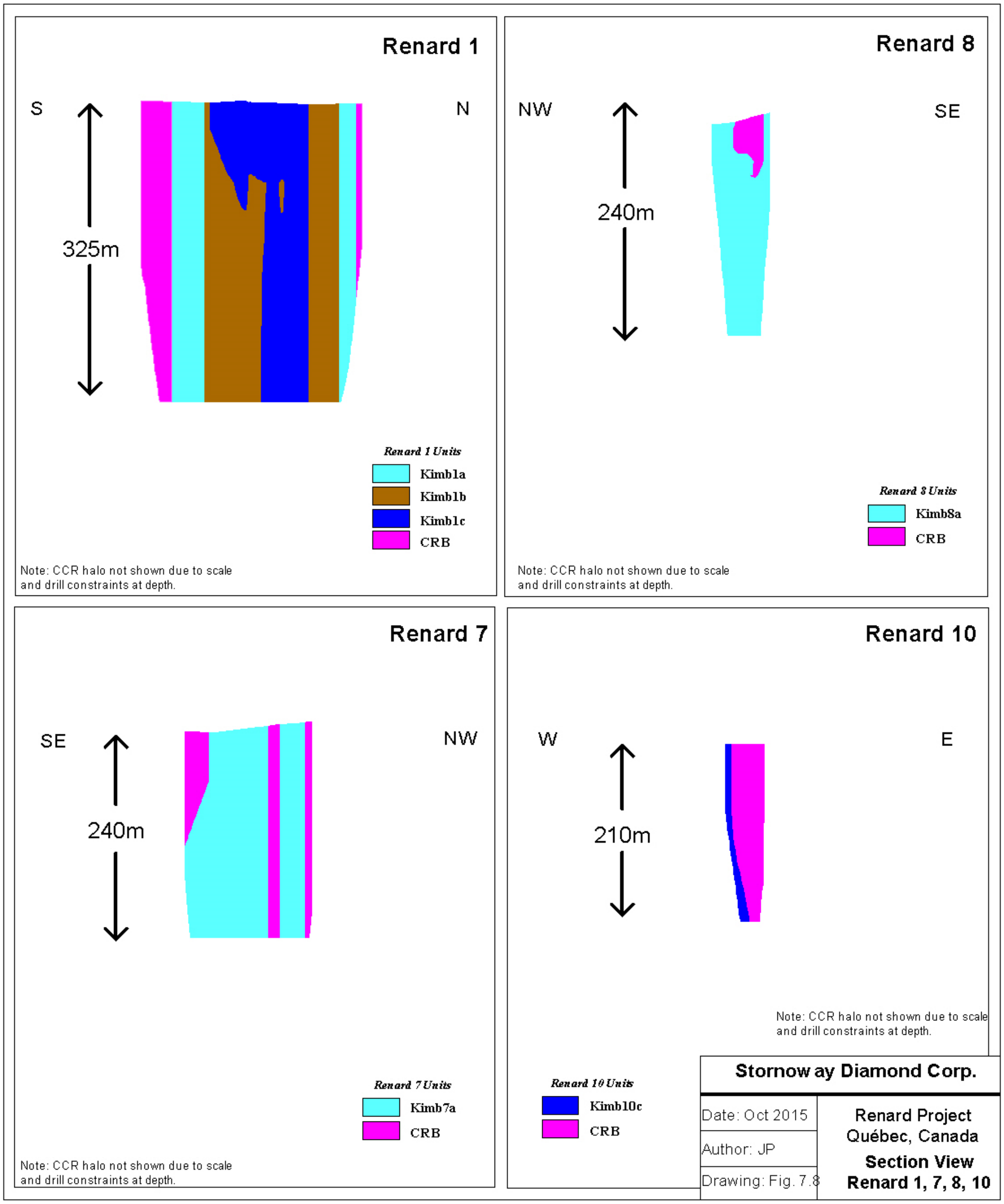


Figure 7.8: Section View – Renard 1, Renard 7, Renard 8 and Renard 10



The main kimberlite in Renard 10 is classified as HK with a dyke-like morphology and is referred to as Kimb10d. Small intersections of both an MVK and transitional CK unit are present within drill core (referred to as Kimb10a and Kimb10b respectively), but their spatial distribution does not allow for them to be modelled separately. Total kimberlite represents 20% of the modelled body by volume and is typically undiluted. Contacts between this HK and the adjacent CRB are sharp. A number of cross cutting HK dykes and irregular intrusions occur throughout the entire kimberlite body, along the pipe contacts and within the marginal CRB unit. These are referred to as Kimb10c. CCR units were identified in drill core, however there was not enough external contact information to warrant modelling the CCR in 3D. The CRB can be divided into two distinct units; one has an overall appearance and texture similar to that of the other Renard bodies, and the second, which is generally more proximal to the main kimberlite, contains a higher amount of kimberlitic fluid material and a higher degree of rounding of CR clasts. Together this suggests that the main HK is a later intrusion which eradicated the majority of any previous MVK and CK material near surface, and remobilized portions of the previous CRB material. In turn this implies that MVK and/or CK material could be found at depth.

7.3.8 Lynx and Hibou Kimberlite Dykes

Lynx Kimberlite Dyke

The Lynx dyke consists of CK that can be further classified as HK. Trenching and mapping of Lynx and the logging of 73 drill holes reveals that the dyke comprises semi-continuous thick intersections of kimberlite with many thin (from < 1 cm to 40 cm) discrete sheets adjacent to it. Lynx extends for a minimum of 4.2 km along strike and has a variable dip (10º to 50º) to the east. The dyke may reach cumulative thicknesses of up to 3 m, however the average thickness is 1.8 m. The main kimberlite intersection appears to pinch and swell and, where thin, is strongly altered and replaced by clays. Overall, Lynx displays varying levels of alteration, often showing zonation with portions of the main intersection being highly altered and friable on one half, but massive and relatively unaltered in the other. Fracturing is present throughout the dykes and varies in intensity (Figure 7.6).

Macroscopically, the Lynx HK is dark grey to green coloured, strongly magnetic and consists of two generations of olivine (macrocrysts and phenocrysts) set in a carbonate rich groundmass. Olivine macrocrysts are medium to coarse-grained, with common very coarse grains, and are completely serpentinized. They comprise on average 25% of the rock, however local variations range from 0% to 50%. Olivine megacrysts and olivine dominated mantle nodules larger than 10 cm are present. Mantle indicator minerals are very common throughout this HK, representing up to 3% of the rock by volume and include coarse- to very coarse-grained picroilmenite, chrome diopside and pyrope garnet. Country rock xenoliths are rare, representing less than 2% of the rock, and are rounded to subrounded, strongly altered granitoid or gneiss fragments. Contacts with the surrounding country rock are marked in the kimberlite by local minor fracturing and sometimes extensive carbonate veining. Chilled margins with flow aligned olivine macrocrysts are common and typically less than 5 mm in width. Fracturing parallel to the contact, and joints perpendicular to the contact, are present within the country rock and decrease in intensity away from the contact. The Lynx dyke typically cuts the foliation of the country rock.

Microscopically, the Lynx HK consists of olivine macrocrysts and phenocrysts within an evenly distributed crystalline groundmass dominated by carbonate, serpentine and phlogopite with a carbonate content of 15% to 35%. Olivine macrocrysts and phenocrysts are completely pseudomorphed to pale-yellow green serpentine (rarely carbonatized) and commonly show a preferred orientation with their long axes subparallel to the dyke margins. Groundmass minerals consist of carbonate, phlogopite, monticellite, serpentine and ilmenite with minor dolomite. Accessory minerals (5% modal) include perovskite, spinel and apatite. Distinctive magmatic textures are observed, where calcite occurs as millimetre scale oikocrysts that enclose small olivine and other matrix minerals (Patterson et al., 2009).

Hibou Kimberlite Dyke

The Hibou dyke is a CK that can be further classified as an HK. Hibou consists of one main kimberlite intersection with thin HK veins and dykes (from 1 cm to 30 cm) adjacent to the main intersection. Trenching and mapping of Hibou and the logging of 41 drill holes reveals that the HK may be up to 3.5 m thick, but is on average 2 m thick. The dyke has a westerly strike extent of at least 1,900m with a shallow dip of approximately 10º to the north. It is open down dip to the north, and along strike in both directions. The HK pinches and swells and, where thin, is strongly altered to clay. Overall Hibou displays varying levels of alteration and/or weathering that are zoned. Fracturing is present throughout the dyke and varies in intensity (Figure 7.6).

Macroscopically, Hibou HK is grey-green to green and strongly magnetic. Olivine macrocrysts are medium to coarse-grained with very common coarse grains set in a carbonate rich groundmass. Olivine macrocrysts comprise approximately 25% of the rock but can vary from 0% - 45% locally. Olivine size frequently continues beyond the size range of macrocryst classification with individual olivine grains measuring more than 3 cm in length. Olivine dominated mantle nodules are also present and reach up to 10 cm in size. Olivines are typically completely serpentinized. Mantle indicator minerals such as pyrope garnet, chrome diopside and picroilmenite are readily observable, representing up to 3% of the rock by volume. Local calcite segregations are observed and calcite veining is present through the dyke and can reach up to 2 mm in thickness. Country rock xenoliths are relatively rare, representing < 2% of the rock by volume. Xenoliths are strongly altered, are subrounded to rounded and consist of gneiss or granitoid fragments that measure up to 7 cm in diameter. Contacts with the surrounding country rock are sharp and marked in the kimberlite by local minor fracturing and carbonate veining. Chilled margins with flow-aligned olivine macrocrysts are common and typically less than 5 mm in width. Fracturing parallel to the contacts, and joints perpendicular to the contacts, are present within the country rock. Hibou is cut by rare black hematite veins less than 5 mm thick that appear to follow the jointing in the dyke. Flow banding of groundmass minerals is evident and tends to be somewhat erratic in orientation.

Microscopically, Hibou HK consists of olivine macrocrysts and phenocrysts set within an evenly distributed, crystalline groundmass dominated by carbonate, serpentine and phlogopite with an average modal carbonate estimate of 15% to 35%. Olivine macrocrysts are fresh to completely pseudomorphed to a dark green to black serpentine and are rarely carbonatized. They commonly display a preferred orientation with their long axes subparallel to the dyke margins. Unlike Lynx, fresh olivine macrocrysts are more common in Hibou. Groundmass minerals consist of carbonate, phlogopite, monticellite, serpentine and ilmenite with minor dolomite. Accessory minerals (5% modal) include perovskite, spinel and apatite. Calcite oikocrysts (as in Lynx) and dolomite rhombs occur in fresh samples of Hibou (Patterson et al., 2009).

8. DEPOSIT TYPES

There are two types of diamond deposits: primary and secondary. Primary deposits are those in which the diamonds remain inside the original host rock (usually kimberlite) that conveyed them to the surface. Secondary deposits are formed when the diamonds are eroded from the host rock and concentrated by the action of water into alluvial deposits (in rivers) or marine deposits (in beaches). The Renard kimberlites are primary deposits.

8.1 Overview of Primary Diamond Deposits

Primary diamond deposits such as kimberlites and lamproites have produced over 50% of the world's diamonds. The remainder was derived from recent to ancient placer deposits that originated from the erosion of kimberlite and/or lamproite. Although diamondiferous kimberlite and lamproite comprise most of the economic diamond deposits, other diamond-bearing rocks have also been discovered and are the subject of numerous academic papers. Such diamond-bearing rocks include ultramafic lamprophyres (aillikites) in Canada and volcaniclastic komatiites in French Guiana (Capdevila et al., 1999). It has been established by the scientific community that diamonds are not genetically related to kimberlite or lamproite but that kimberlite and lamproite intrusives serve as a transport mechanism for bringing diamonds to surface (Kirkley et al., 1991) from the mantle.

Clifford (1966) and Janse (1991) stated that a majority of economic diamondiferous kimberlites occur in stable Archean age cratonic material that has not undergone any thermal or deformational event since 2.5 Ga. Such Archean age cratons include the Kaapvaal, Congo and West African cratons (Africa), Superior and Slave Provinces (Canada), East European Craton (Russia, Finland), and the West, North and South Australian cratons. The only exceptions to date are the Argyle and Ellendale mines of Australia, which occurred in Proterozoic-age, remobilized, cratonic material.

To date, over 6,000 known kimberlite and lamproite occurrences have been discovered, of which over 1,000 are diamondiferous. Some of the well-known diamondiferous kimberlites/lamproites currently being mined include Argyle (lamproite) in Australia; Orapa and Jwaneng (kimberlites) in Botswana; Jubilee, Udachnaya and Mir (kimberlites) in Russia; Venetia (kimberlite) in South Africa, and the Ekati and Diavik clusters (kimberlites) in Canada.

Economic diamond kimberlite and/or lamproite pipes generally range from less than 0.4 ha to 146 ha in surface area, with the maximum size being more than 200 ha (for example, Catoca, Angola). Economic diamond grades can range from 3.5 cpht to 600 cpht.

8.2 Kimberlite-Hosted Deposits

Kimberlites remain the principal source of primary diamonds despite the discovery of high-grade deposits in lamproites. Recent mineralogical and Rb–Sr isotopic studies have shown that two varieties of kimberlite exist (Mitchell, 1991).

- Group 1, or olivine-rich monticellite serpentine calcite kimberlites; and
- Group 2, or micaceous kimberlites (predominantly occur in southern Africa).

“Group 1” kimberlites are complex, hybrid rocks consisting of minerals that may be derived from:

- Fragmentation of upper mantle xenoliths (including diamond);
- Megacryst or discrete nodule suite; or
- Primary phenocrysts and groundmass minerals.

The contribution to the overall mineralogy from each source varies widely and significantly influences the petrographic character of the rocks. Consequently, Group 1 kimberlites comprise a petrological clan of rocks that exhibit wide differences in appearance and mineralogy as a consequence of the above variation, coupled with differentiation and diverse styles of emplacement of the magma. Currently, three textural-genetic groups of kimberlite are recognized, each being associated with a particular style of magmatic activity in such a system. These are:

- Crater facies;
- Diatreme facies; and
- Root Zone facies.

Rocks belonging to each facies differ in their petrology and primary mineralogy, but may contain similar xenocrystal and megacrystal assemblages.

With a few exceptions, such as the Finsch Kimberlite Mine in the Republic of South Africa and the Dokolwayo Kimberlite Mine in Swaziland, most of the well-known diamondiferous kimberlites in South Africa and elsewhere are Group 1 kimberlites. The Renard kimberlites are considered to be Group 1 kimberlites.

The Renard kimberlites are interpreted to be steep-sided, pipe-like structures with irregular to elongate shapes in plan view. Surface expressions of the kimberlite vary between 0.3 ha and 3.1 ha although there are larger haloes of broken country rock. The Renard Cluster is mainly composed of diatreme-like kimberlitic breccia lithologies and hypabyssal kimberlitic material. No crater material is noted in the Renards.

The Lynx, Hibou, North Anomaly, Southeast Anomaly and G04-296 hypabyssal dykes are interpreted to be intrusions of kimberlitic material that did not vent to the earth’s surface at the time of emplacement.

9. EXPLORATION

9.1 Geological Mapping

Structural geological mapping was undertaken in the area of the Lynx dyke in 2004, and over the Renard bodies in 2006 and 2010. The objective of the 2004 and 2006 mapping programs was to identify structural controls to locate more kimberlitic intrusions or dykes. Results were inconclusive but it was noted that rock fracture intensity increased near the kimberlitic bodies. The objective of the 2010 geological mapping was to identify large-scale structural features (ie. faults, shear zones) inferred from remote sensing data. Overburden was cleared to expose the bedrock for mapping, and several faults were identified within the proposed mine site.

Sampling trenches excavated on Renard 4, Renard 65, Lynx, Hibou, and the North Anomaly were mapped in detail prior to sampling. Mapping was undertaken with grid control and reference points surveyed by a registered surveyor. Mapping notes included the mineralogy, relationship of contacts, and structural measurements in the host rocks.

During the underground bulk sampling program at Renard 2 and Renard 3, geological mapping was completed on all workings. The face was mapped after each round during the development of the ramp and drifts and, after the underground excavation was completed, the ramp and drift walls were mapped in the kimberlites.

9.2 Heavy Mineral Sampling

Since inception of the Foxtrot Project, approximately 12,000 heavy mineral samples have been collected over a 400,000 km^2 area of which some 8,140 lie within the current property. Heavy mineral samples collected to date are summarized by year in Table 9.1. This work has been reported in previous technical reports (Clements and O'Connor, 2001, 2002; Lucas et al., 2003; Lépine and O'Connor, 2004; O'Connor and Lépine, 2005, 2006).

Table 9.1: Heavy Mineral Sampling

Year	Number of Samples	Comments
1996–1999	13	Regional samples within what became the Foxtrot Property
2000	48	Highly anomalous indicator mineral counts in 1 sample
2001	252	Prioritizing geophysical anomalies for drilling
2002	785	Increase sample density within the property
2003	914	Detailed grids and reconnaissance samples
2004	2,000	Detailed grids and reconnaissance samples
2005	1,412	Detailed grids and reconnaissance samples
2006	1,203	Detailed grids and follow-up samples
2007	959	Detailed grids and follow-up samples
2008	554	Detailed grids and follow-up samples
Total	**8,140**	

9.3 Geophysical Surveys

Since 2000, several ground and airborne geophysical surveys were completed on the Foxtrot Property. This work has been reported in previous technical reports (Clements and O'Connor, 2001, 2002; Lucas et al., 2003; Lépine and O'Connor, 2004; O'Connor and Lépine, 2005, 2006). Geophysical surveys that have been completed to the effective date of this report are summarized in Table 9.2.

Table 9.2: Geophysical Surveys

Year	Airborne Geophysical Surveys (total line km)	Number of Ground Magnetic Surveys (total line km)	Number of Electromagnetic Surveys (total line km)
2000	1,419	0	0
2001	0	5 (38.4)	0
2002	900	21 (140.5)	14 (20.9)
2003	8,900	52 (452.3)	10 (19.4)
2004	4,778	58 (505.4)	33 (67.7)
2005	19,491	25 (215.2)	41 (84.2)
2006	0	29 (308.0)	51 (81.6)
2007	0	52 (441.5)	43 (41.9)
2008	1,969	32 (385.0)	12 (11.8)
Totals	37,457	274 (2,486.3)	204 (327.5)

9.4 Drilling

Drilling is discussed in Section 10.

9.5 Sampling for Diamonds

Three basic levels of progressively larger diamond sampling procedures are summarized below (caustic fusion sampling, minibulk sampling and bulk sampling), followed by descriptions of the comparable core, reverse circulation, trenching and underground sample programs.

9.5.1 Caustic Fusion Sampling

The caustic fusion process is used to evaluate, characterize and correlate the diamond potential of individual kimberlite lithologies, and to provide data to facilitate the grade estimation process. The objective of this type of test is to extract all diamonds greater than 0.1 mm in size, through chemical dissolution of the host rock sample. Individual samples may vary in size from a few kilograms to hundreds of kilograms, depending on the available material and the specific purpose of the testing. Kimberlite may be collected from drill core, float boulders, subcrop, outcrop, underground exposures and subsamples of material in a process facility or a combination thereof. Individual sample results from comparable kimberlite units may be merged together to provide larger, statistically more representative, samples.

Kimberlite is collected, described and recorded by the site geologists following protocols in place at the time. Samples are individually numbered, weighted, sealed in a tamper-resistant container appropriate for the volume of material, and transported to the test facility by a combination of charter aircraft and commercial couriers.

9.5.2 Mini-Bulk Sampling

Although there is no formal industry-accepted definition of a 'minibulk' sample, many companies would agree that the term is generally used to refer to the processing of kimberlite material up to several tens of tonnes. This material may be derived from drill core, RC chips, boulders, subcrop, outcrop, trenches or underground workings. Mini-bulk samples are usually processed through Dense Media Separation (DMS) equipment that, depending on specifications and diamond recovery objectives of a particular program, may be configured to recover diamonds of greater than 0.3mm, 0.5 mm, 0.85 mm or 1.18 mm on square-mesh screens. In some cases caustic dissolution or other extraction techniques may be used for final diamond recovery. All of Stornoway's minibulk samples reported herein were processed through DMS equipment, and the reported diamond content is based upon stones retained on either 1.18 mm square-mesh screens or +1 DTC screens.

Stornoway's minibulk sampling programs reported herein have used drill core, RC chips, boulders, and surface trenches to source kimberlite material. Drill core was collected, described and recorded by the geologists following protocols in place at the time. Prior to the 2013 work, core boxes were sealed, weighted and secured to pallets for shipping, then forwarded to the North Vancouver laboratory facilities. Upon receipt samples were verified, then composited and processed through an onsite DMS plant. From 2013 onwards core boxes were sealed and shipped to North Vancouver for detailed geological logging and sample preparation, before dispatching to Microlithics Laboratories in Thunder Bay for DMS processing.

During the various RC drill programs (see Section 10), screen openings of either 0.98 mm or 1.18 mm were used to dewater the RC kimberlite chip recovery flow and to remove the undersize fraction. Recovered kimberlite was separated at regular downhole intervals into individual samples and collected in either large 1.5 t bulk sample bags or individually lined 205 l steel drums. Bulk bags and/or drums were clearly identified with a unique numbering system, fastened with tamper-proof seals and transported to secured Stornoway storage facilities. Charter aircraft and bonded freight services were organized as required to transport sample materials to the process facility. In 2004, RC minibulk samples were dispatched to the external Thunder Bay Mineral Processing Laboratory (TBMPL) in Ontario for DMS processing, whereas RC minibulk samples from 2006 were sent to Stornoway's North Vancouver laboratory and those from 2007 were processed by the Lagopede-based DMS. Mini-bulk samples collected and processed between 2013 and 2015 used Stornoway's 1.5 tph DMS plant situated in Thunder Bay, Ontario, and operated by Microlithics Laboratories.

9.5.3 Bulk Sampling

Although there is no formal industry-accepted definition of a 'bulk' sample, many companies would agree that the term is generally used to refer to the processing of kimberlite material exceeding several tens of tonnes. This material may be derived from drill core, RC chips, boulders, subcrop, outcrop, trenches or underground workings. Bulk samples are usually processed through DMS equipment that, depending on specifications and diamond recovery objectives of a particular program, may be configured to recover diamonds of greater than 0.85 mm or 1.18 mm on square-mesh screens. In some cases larger screen sizes or other extraction techniques may be used for diamond recovery. All of Stornoway's bulk samples reported herein comprise either surface trench or underground sample material, and were processed through DMS equipment. The reported diamond content is based upon stones retained on either 1.18 mm square mesh or +1 DTC screens.

To maintain the integrity of the bulk samples, a sampling protocol was established before the exploration work took place in order to ensure that the final diamond recovery data can be linked to the correct kimberlite, sample location within the body and rock type. Each surface or underground round was assigned a sample number and a colour code by the mine geologist. For surface samples, individual loads were directed to specific dump piles in a secure, access-controlled, storage area located near the DMS facility.

The storage area was monitored 24 hours per day by an independent security force (Groupe Canaprobe of Montreal, Québec) and by CCTV surveillance.

For underground samples, a sample number placard was posted in the remuck bay and on the dump truck so the underground operators and process plant staff were constantly aware of which sample was being handled. On surface, a member of the DMS staff at the controlled access storage site directed the loads to the correct stockpile locations and kept a log of loads that were delivered from underground. Once all loads were hauled to surface, the mine geologist checked that the remuck bay was cleaned out in preparation for the next sample. Sample number placards were taken to the surface with the final dump truck load and placed on the appropriate stockpile. A location plan of the surface or underground samples was updated each time a new pile was initiated, in order to easily identify sample location.

9.5.4 Core Sampling

Certain drill core collected during historical drill programs (discussed in Section 10) was composited and treated for macrodiamond recovery. Results are summarized in Table 9.3.

Table 9.3: Summary of Macrodiamond Sampling Results

Kimberlite Body	Sample Type	Year	Number of Samples	Weight (dry t)	Total Carats (+1 DTC)
Renard 1	Drill Core	2002	1	0.3	0.00
	Drill Core	2003	11	10.0	0.73
	Drill Core	2014/2015	4	2.9	0.02
Renard 2 (continued below)	Drill Core	2002	7	5.0	3.29
	Drill Core	2003	8	8.6	5.24
	Drill Core	2004	9	12.5	13.45
	Drill Core	2005	16	6.7	4.96
	Drill Core	2006	7	2.8	2.71
	RC Chips	2004	12	171.2	146.96
	RC Chips	2007	15	86.8	70.95

Table 9.3: Summary of Macrodiamond Sampling Results (cont'd)

Kimberlite Body	Sample Type	Year	Number of Samples	Weight (dry t)	Total Carats (+1 DTC)
Renard 2 (continued)	Underground	2006/2007	15	2448.8	1601.94
	Underground (drums)	2014/2015	7	1.4	1.77
	Drill Core	2014/2015	71	54.6	44.65
Renard 3	Drill Core	2002	5	4.9	6.47
	Drill Core	2004	13	13.8	13.66
	RC Chips	2004	10	157.0	185.11
	RC Chips	2007	13	59.4	34.86
	Underground	2006/2007	13	2113.7	2799.85
Renard 4	Drill Core	2002	6	4.8	2.94
	Drill Core	2003	15	12.4	5.32
	Drill Core	2004	36	32.4	13.62
	Drill Core	2005	1	0.5	0.48
	RC Chips	2004	17	141.8	53.23

	RC Chips	2006	14	41.4	33.21
	Surface Sample	2004	2	1.8	3.09
	Surface Sample	2005	6	9.8	17.76
	Surface Sample	2006/2007	7	2104.2	2721.88
Renard 65	Drill Core	2002	2	0.8	1.19
	Drill Core	2003	23	19.8	8.47
	Drill Core	2004	22	17.9	4.05
	RC Chips	2004	18	149.6	32.50
	Surface Sample	2007	2	266.0	51.77
	Surface Sample	2012	1	5080.8	963.38
Renard 7	Drill Core	2005	4	4.1	0.10
	Drill Core	2014/2015	3	2.3	0.00
Renard 8	Drill Core	2005	4	6.1	0.47
	Drill Core	2014/2015	2	1.4	0.03
Renard 9	Drill Core	2004	6	6.0	5.65
	Drill Core	2005	4	6.2	6.38
	RC Chips	2006	19	70.3	35.84
	RC Chips	2007	5	27.3	11.97
Renard 10	Drill Core	2014/2015	3	1.7	0.00
Hibou	Surface Sample	2005	5	19.8	4.68
	Surface Sample	2006	2	31.4	39.53
	Surface Sample	2008	1	543.9	781.41
Lynx	Surface Sample	2003	3	3.9	4.46
	Surface Sample	2004	2	10.3	14.92
	Surface Sample	2005	6	34.7	42.33
	Surface Sample	2007	3	494.3	528.93
North Anomaly	Surface Sample	2006/2008	3	46.4	44.90

9.5.5 Reverse Circulation (RC) Sampling

RC chip sampling programs on the Renard Project were undertaken with objectives that varied over time from simply collecting a large amount of kimberlite to create representative samples, to characterizing the grade over various depth intervals, to regular sampling intervals. Early work on the Project also considered that some minimum quantity of kimberlite should comprise each sample to accommodate processing. The data set of sampling results represents:

- Sampling intervals from 10 m to more than 50 m;
- Combination of chips from adjacent holes over similar intervals in order to double the volume of kimberlite from that interval;
- In some areas, regular sampling at uniform downhole spacing; and
- Sampling governed by internal geological contacts, completed during the 2007 program.
- Results are summarized in Table 9.3.

9.5.6 Trench Sampling

Since 2005, several thousand tonnes of kimberlitic material have been excavated from trenches on the Renard 4 and Renard 65 pipes and the Lynx, Hibou and North Anomaly dykes. Trench locations are shown in Figure 7.3. Trench details are summarized in Table 9.4 and macrodiamond results in Table 9.3.

Table 9.4: Trenching

Area	Trench Name	Date of sampling	Dimension (m)	Sample Collected (dry t)
Renard 4	Renard 4	2004	5 x 5	1.8
	Renard 4	2005	25 x 25	9.8
	Renard 4	2006	35 x 27	2345.0
Renard 65	Renard 65	2007	20 x 16	266.0
	Renard 65	2012	26 x 32	5080.8
Lynx Dyke	T-184-01	2005	8 x 35	6.2
	T-221-01	2005	6 x 25	12.9
	T-221-01	2007	8 x 40	364.1
	T-230-01	2005	12 x 10	12.7
	T-230-02	2005	10 x 25	3.0
	T-230-02	2007	10-15x 106	130.2
Hibou	T-271A-01	2005	10 x 6	19.8
	T-271A-10	2006	31 x 10	31.4
	T-271A-10	2008	40 x 40	543.9
North Anomaly	T-222-01	2006	1.5-3.5 x 13	46.4

9.5.7 Underground Bulk Sampling

The underground exploration program at the Renard Project was designed to extract a minimum of 2,000 t of kimberlite from each of Renard 2 and Renard 3 at a depth of approximately 55 m below ground surface. Supplies and equipment were mobilized from Chibougamau via Buffalo aircraft to the ice airstrip at Camp Lagopède in March and April 2006.

Preparatory work included establishing powder magazines, office and dry facilities, a maintenance shop and generator shed, a mine water sump and a waterline. Equipment had to be assembled, and included a two-boom jumbo, one haul truck, two scoops, a scissorlift truck, and a crew transport vehicle. Genivar Inc. supervised all aspects of the operations, while mining contractor Monterie Expert Inc. was engaged to provide equipment and personnel to complete the work.

The portal was located on the southwest side of a hill immediately north of Renard 2. An 8 m, near-vertical outcrop face was present at the location, reducing the need for an extensive portal trench. Following mobilization and assembly of the equipment, cleaning of the portal area commenced at the end of June to expose the bedrock. A small airtrack drill was used to bore the portal blast holes, and the first blast occurred on July 16, 2006.

The first round in the ramp was taken August 1, 2006, and work was completed on February 22, 2007. The ramp dimensions are 3.8 m by 4.2 m, with a maximum grade of 15%. The back was screened and bolted, generally using a 1.2 m by 1.2 m pattern with Swellex bolts or resin rebar and galvanized, welded wire mesh. A total of 749.1 m of underground workings was completed, comprising a portal, ramp, ore drifts, safety bays, sump pump stations, an electrical substation and a powder magazine. Excavations were completed to a depth of approximately 55 m below surface. The main ramp length totalled 435.6 m with lateral excavation in waste of 111 m (from base of ramp to Renard bodies) and lateral excavation in ore of 202 m. Horizontal drifts to access the Renard 2 and Renard 3 kimberlites were each about 100 m in length.

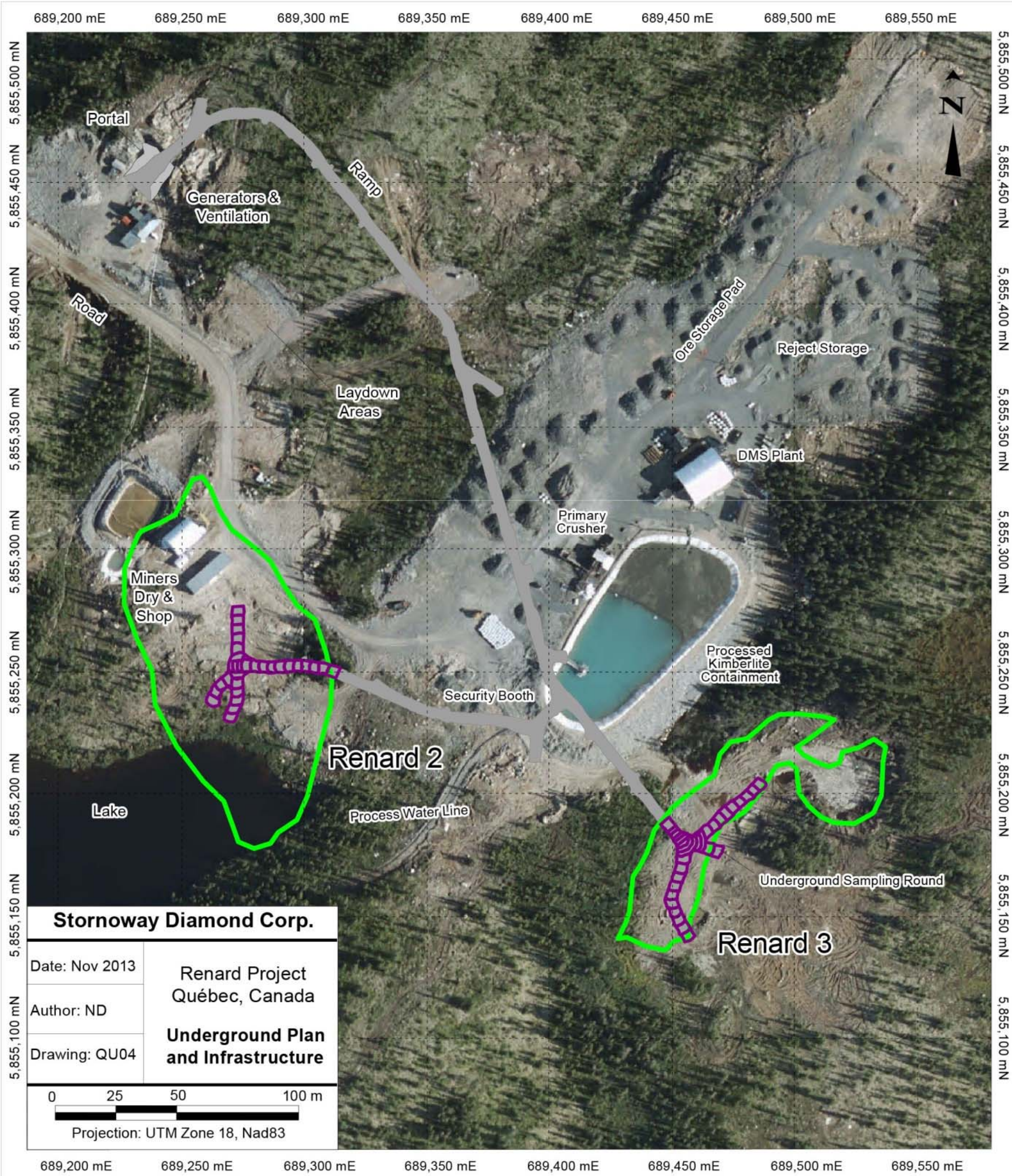
Production rates averaged 3.59 m per day; but varied depending upon whether single or double faces were being advanced.

Underground, the faces were surveyed using a Leica total station instrument equipped with a reflectorless measuring device. Details of each side of the gallery and the front face were surveyed with a variable number of points depending on surface regularity. Each survey comprised five different surfaces: floor, back, left and right walls, and front face. These surveys were used to measure the rate of advance and to calculate the volume of material removed.

Sampling consisted of collecting the kimberlite material blasted from each round. After each blast, material was mucked from the face with scoops and collected in a remuck bay. From the remuck bay, the ore was loaded into a haul truck and transported to the secure ore stockpile area located adjacent to the DMS facility on the surface, some 200 m southeast of the portal (Figure 9.1). A typical round was three to five meters in length, or approximately 150 t. A total of 60 individual samples were collected: 29 from Renard 2 (5,010 calculated tonnes) and 31 from Renard 3 (5,161 calculated tonnes). A representative subsample from each of Renard 2 (15 rounds; 2,449 t) and Renard 3 (13 rounds; 2,114 t) were processed through the Lagopede DMS for macrodiamond recovery (Table 9.3). The remaining unprocessed rounds are stored for future DMS plant commissioning.

The underground workings were allowed to flood late in 2007, and the portal is currently closed.

Figure 9.1: Underground Plan and Infrastructure circa 2012



9.6 Bulk Density Determinations

The dry bulk density database for the Foxtrot Property comprises 2,127 bulk density records, consisting of 1,672 measurements from drill core and 455 from bulk sampling. When multiple measurements from the same sample, and multiple subsamples from the same rock are averaged, and the laboratory Quality Control (QC) checks removed, there are 1,770 spatially discrete density samples (increased from 1,284 by new work undertaken from 2013 to 2015).

Morris Magnetics Inc. performed the first density tests in 2004 on Renard 2, Renard 3 and Renard 4 using the water displacement method. Between 2004 and 2015, additional bulk density samples were measured by Morris Magnetics Inc., SGS Canada Inc. and Acme Analytical Laboratories (now Bureau Veritas Commodities Canada Ltd). Some of the more porous samples were treated by wax immersion, but the large majority were through the immersion/water displacement method, as follows:

1. Dry the sample in the oven at 49ºC overnight.
2. Weigh each sample in air.
3. Weigh each sample suspended in water.
4. Calculate the displaced volume of the dry rock in water.
5. The difference between the dry rock weight and the water displaced gives the calculated bulk density.

Quality control checks comprising duplicate measurements with water and wax immersion techniques, repeat density determinations, blind checks between laboratories, and multiple subsamples of the same rock demonstrate small density variances on a sample-by-sample, measurement-by-measurement and process-by-process basis, but there does not appear to be a consistent shift from laboratory to laboratory. Variability is generally in the order of 0.00 g/cm^3 to 0.05 g/cm^3 (or from 0% - 2% assuming an average bulk density of 2.60 g/cm^3), and should not be a concern for data interpretation. Density variations did not show a correlation with country rock dilution, nor was there a clearly demonstrable change of density with increasing depth in the kimberlite pipes.

Country rock (CR) consists of predominantly gneissic, lesser granitic and minor pegmatitic phases. The relationship between granitic and gneissic units is commonly cyclic, such that large intersections of drill core are best described as mixed granite-gneiss. Given the proportion of mixed units, and the overlapping density ranges for granite and gneiss, a calculated average bulk density of 2.71 g/cm^3 has been applied to the country rock of all bodies.

Bulk density ranges for both the country rock breccia (CRB) and country rock breccia with minor kimberlite as thin matrix/vein fillings (CRB+K) are a little lower than the country rock (CR) units, likely as a function of fracturing and alteration associated with brecciation, with overlapping ranges. An average bulk density of 2.60 g/cm^3 for both CRB and CRB+K units has been applied.

Table 14.7 shows the average bulk density values for the various geological units within the Renard kimberlites.

9.7 Moisture Content

Determining the moisture content of each sample prior to caustic fusion, DMS or bulk density work is necessary to allow an accurate dry weight of the kimberlite to be calculated. Drying of bulk density samples is part of the standard laboratory routine, as documented above.

Drill core samples processed through the North Vancouver laboratory during historical minibulk programs were stored inside for extended periods of time prior to DMS processing (23 days to 598 days; average 141 days), considered as dry, and no moisture contents were measured.

Moisture content tests were performed on the RC chip samples to allow an estimate of percentage chip recovery during drilling. These moisture contents were not used in the determination of RC minibulk sample masses, or during RC diamond grade calculations, as tonnages were based upon theoretical sample volumes of each contributing geological unit and the appropriate density.

During bulk sampling programs, moisture content tests were completed on kimberlite from both the surface and underground samples. As each individual sample was being run through the DMS plant, between five and 15 aliquots of raw feed (one to two kilograms each) were taken from the feed conveyor. Each aliquot was weighed, then dried in the oven at 105ºC for two hours and reweighed. Results for each aliquot were recorded and averaged to represent the assumed moisture content for the complete subsample. Average moisture values of 5.8% (range 3.0% - 8.5%) and 7.8% (range 3.9% - 15.4%) were obtained for the underground samples at Renard 2 and Renard 3, respectively. The large tonnage surface samples varied from an average of 7.1% (range 6.5% - 7.7%) at Lynx to an average of 10.8% (range 5.4% - 21.1%) at Renard 4. Average moisture content at Renard 65 was 6.0% (range of 4.2% - 7.8%) and the Hibou sample was 9.8% (treated as one large sample). Moisture contents for recent minibulk DMS samples collected from drill core and processed between 2013 and 2015 were determined in a similar manner, and varied from 0 to 3.66%, averaging 0.60%.

9.8 Petrography, Mineralogy and Other Research Studies

Since 2001, petrographical, mineralogical and other research studies have been undertaken on the Renard kimberlitic bodies and the Lynx, Hibou and North Anomaly dykes. These studies include:

- Detailed trench mapping;
- Detailed core logging in the Renard Core Area including the various dyke systems;
- Petrographic descriptions of thin sections and polished slabs;
- Indicator mineral and geochemical studies on the various kimberlite phases in the pipes and dykes;
- Diamond morphology, chemistry and inclusion studies
- Age dating of various kimberlite phases from the pipes and dykes; and
- Characterization of geophysical signatures on a unit by unit basis.

The various studies were used to help distinguish the different lithological units within the kimberlite pipes, and were important for construction of 3D geological models.

9.9 Further Exploration

Several unexplained kimberlite indicator mineral trains and geophysical anomalies remain within the Foxtrot Property. To explore for additional kimberlite occurrences, Stornoway may collect additional heavy mineral samples, undertake ground and airborne geophysical surveys, prospect areas of anomalous indicator mineral results and drill exploration targets generated from the heavy mineral sample, geophysical and prospecting work. No work is planned or budgeted at the present time.

10. DRILLING

10.1 Background and Summary

All drilling has been carried out under the control of Stornoway and predecessor company Ashton. A total of 900 drill holes (133,089m) has been drilled on the Property since 2001, comprising 36 RC holes (6,151m) and 631 exploration core holes (120,994m), 35 geomechanical holes (3,471 m), 133 geotechnical holes (1,219m) and 64 hydrogeological holes (884m). Total surface drilling on a year-to-year basis for exploration work is summarized in Table 10.1. Total drilling for geomechanical, geotechnical and hydrological purposes is summarized in Table 10.2. During 2007, and as part of the underground bulk sample work, 22 holes were drilled from underground on Renard 2 (1,508 m) and 21 holes from underground on Renard 3 (874 m). These 43 holes are included in the totals above, and in Table 10.1.

Vertical and angled holes were drilled through the kimberlite bodies, from which three dimensional geological models were constructed for resource estimation. Drilling intersections are therefore not related to true thickness of mineralization.

Between 2001 and 2002, drilling was completed for early-stage, exploration-focused programs for all the bodies except for Renard 9 and Renard 10, which were discovered in 2003 and 2005, respectively. From 2003, drilling was used primarily to support advanced-stage project evaluation and deposit delineation by providing bulk and minibulk samples. Target exploration drilling was undertaken between 2001 and 2010. Drilling in 2011 and 2012 was focussed on collecting data to support the proposed mine plan and infrastructure design, and drilling in 2014 concentrated on the Renard 2 kimberlite.

Table 10.1: Summary of Exploration Drill Programs

Program by Year	Number of Core Holes	Number of Extended Holes	Drilled Meters (core)	Number of RC Holes	Drilled Meters (RC)
2001	6	0	554	0	0
2002	33	0	4,688	0	0
2003	71	0	12,642	0	0
2004	104	0	17,699	23	4,157
2005	137	3	25,914	0	0
2006	90	1	11,343	5	805
2007	95	3	12,243	8	1,189
2008	16	0	2,160	0	0
2009	29	5	16,506	0	0
2010	12	2	5,209	0	0
2014	38	0	12,036	0	0
Total	**631**	**14**	**120,994**	**36**	**6,151**

Of the exploration drill holes listed above, 497 holes totalling 101,078m were used in the 2015 mineral resource estimate (totals therefore exclude work on Renard 1, Renard 7, Renard 8 and Renard 10 as they do not contain 'resources' as defined by NI 43-101 and related documents). Collar locations and surface traces for these 497 holes are shown in Figure 14.1.

Table 10.2: Summary of Geomechanical, Geotechnical and Hydrogeological Drill Programs

Year	Purpose	Number of Drill Holes	Drilled Meters
2006	Hydrogeological	8	123
2010	Geomechanical	26	2,445
2010	Hydrogeological	12	106
2010	Geotechnical	39	203
2011	Geotechnical	23	499
2012	Geomechanical	7	696
2012	Geotechnical	49	423
2012	Hydrogeological	26	413
2014	Geotechnical	22	94
2014	Hydrogeological	18	242
2015	Geomechanical	2	330
Total		**233**	**5,944**

10.2 Reverse Circulation Drilling

Foundex Exploration of Surrey, British Columbia, was the drilling contractor for all reverse circulation (RC) work on the Renard Project, employing an HT-2000 drilling system rig. All RC boreholes were drilled vertically, ranged from 88 m to 211 m depth, and measured 27.7 cm to 28.8 cm diameter. A two part 23-hole, RC drilling program was completed in 2004 on the Foxtrot Property, for 4,157 m of drilling. Another five and eight RC holes were drilled in 2006 and 2007, for a total of 805 m and 1,189 m, respectively. Due to the complex geology of the Renard bodies and the reduced geological information gathered by using RC drilling methods, all RC borehole locations were twinned by core drill holes.

The winter 2004 RC drilling program was undertaken using a mix of water and air to bring the kimberlitic chips to the surface, whereas the summer 2004 program used a “reverse flood” technique. Reverse flood drilling uses much larger volumes of water, which is thought to reduce diamond breakage by flushing the cuttings away from the drill head in a timely manner, and facilitating a more rapid, less aggressive transport to surface. In the 2006 and 2007 drill programs, the reverse flood technique was used for all RC drilling, with the exception of one hole on Renard 4, where the air and water method was required due to technical difficulties.

A downhole calliper survey was performed when each RC hole was completed. The goal of the RC calliper survey was to measure the diameter of the hole for the calculation of volume (in cubic meters) of material drilled along the length of the RC hole for diamond grade estimation.

The calliper system comprises a mechanical three-arm calliper with a winch and cable system. Each arm of the calliper can extend to a maximum distance of 0.2 m in length. The survey methodology consists of lowering the calliper to the bottom of the RC hole, extending the arms until they contact the RC hole wall, and then raising the instrument at a constant rate so that the calliper arms can measure the RC hole profile in real time. Each RC hole was surveyed twice so that the repeatability of the data is consistent and reliable. The information is recorded on a laptop computer and subsequently processed and interpreted. The data were presented as a graphic 3D downhole log and a downhole Excel spreadsheet.

RC drilling recovers kimberlite as a continuous stream of rock chips ranging in size from a few millimeters to several centimeters depending on the characteristics of the rock being drilled and the type of bit being used. Since a certain percentage of rock material is discarded as under-sized material, or lost in fractures in an RC borehole, the weight of the sample is calculated rather than measured directly. The weight equals the volume of the hole multiplied by the bulk density of the kimberlitic material. Volumes were determined mathematically as a function of hole diameter and sample length. Tonnages were determined on a sample-by-sample basis using the proportion of each geological unit and the measured density as determined from the same interval in the twinned core hole (refer to Section 9).

Rock chips are collected at regular depth intervals, logged geologically, and archived for later use. Rock chip size, morphology, and granulometry are monitored constantly and recorded in drill logs. The Foundex drillers also recorded technical information during the drilling, including rotary pressure, bit pressure, revolutions per minute, air pressure and production rates. This RC data is used to compare with diamond recovery data and to determine the best bits for particular rock types.

RC chip recovery is monitored on a sample-by-sample basis by estimating the theoretical tonnage (calliper volume times sample length times density) minus the approximate dry weight of the samples processed. RC recoveries typically vary from 51% - 78%. Recoveries are lower when the rock is strongly diluted by country rock xenoliths. Stornoway observed that certain RC bit types and drilling techniques did not give good recoveries.

10.3 Core Drilling

Chibougamau Diamond Drilling of Chibougamau, Québec, was the drill contractor for both surface and underground exploration on the Renard Project from 2001 until 2013. Diesel-powered unitized HC150's (essentially equivalent to a Boyles 44) drill rigs used for the majority of the surface work were housed in metal shacks, skid-mounted and moved by bulldozer. Drill rods and ancillary equipment were transported by skid-mounted sloop (or sled). Prior to the 2009 summer drill program, the aluminum tower on one of the rigs was replaced with a stronger steel tower to allow deeper drilling. When required to test outlying exploration targets, the drill rig was configured for helicopter transport by removing it from the metal shack. Surface holes ranged from 12 m to 843.5 m in length, and were drilled at a variety of orientations with dips from -45º to -90º. The lightweight drill rig used for underground core holes was manufactured by Chibougamau Diamond Drilling and driven by air pressure. It was mounted on a steel frame and drilled ATW-size (3.0 cm) core. The shortest underground hole was 7.5 m in length, and the longest 118 m in length.

During 2014, Forages Rouillier (Amos, Québec) provided specialized deep directional drilling services, supported by Tech Directional Drilling (Millertown, NL) for downhole survey control. Fordia Group Inc. (Val d'Or, Québec) provided water treatment services initially, but were replaced by ASDR Industries (Malartic, Québec) early in the program. The first drill rig mobilized to Renard on March 27, 2014, and the last drilling was completed on November 11, 2014. Three drill rigs each established a single pilot hole, from which a series of tightly constrained branches were cut using directional drilling equipment. Two drills were situated within the outline of Renard 2 and drilled subvertically within the kimberlite, while a third drill was positioned outside the body and focused on inclined holes. A total of 33 branches were cut from the three pilot holes, ranging in length from 5.3 m to 724.8 m. Two separate shorter holes were also drilled from surface within the Renard 2 body. Total meterage drilled during the program was 12,145 m and comprised a mixture of BQ, NQ and HQ diameter core. The longest continuous drill hole trace ran 1,086 m from surface to the end of the hole. The deepest vertical depth tested from surface was 1,056 m, and the deepest vertical depth of kimberlite intersected was 1,012 m.

From 2001 to 2006, geologists provided by SOQUEM and Stornoway logged core using a method termed by Stornoway as "lithological" logging. During this period, all core logging was completed at the secure core shack at the Lagopède camp prior to being shipped to North Vancouver for diamond analysis. In 2006, the logging style changed to a more detailed procedure termed by Stornoway as "petrological" logging. This method involves laying out one (or more) drill hole(s) in its (or their) entirety to allow the logger to compare geology between sections within an individual drill hole or across several drill holes more efficiently and effectively. In addition, with the petrological logging method, more emphasis is placed on petrographic and thin section work to identify and correlate geological units. Beginning in 2006, all drill cores containing or suspected to contain kimberlite were logged using the lithological technique in camp and then shipped from the Project site to Stornoway's North Vancouver facilities for petrological logging. In addition, core remaining from drill programs completed prior to 2006 was re-logged using the petrological logging technique. In 2009 and 2010, core was logged using the petrological method in the field. To allow core holes to be laid out in their entirety, core logging was completed in a larger facility adjacent to the core shack.

In addition to a traditional drill core log which noted rock type, mineralogy, colour, texture, structure, and alteration, a modal analysis spreadsheet was completed for holes drilled between the summer of 2003 and the end of 2008. Estimates of the country rock xenolith, kimberlitic matrix, olivine macrocryst, magmaclast, and autolith content, in percent, were noted for each 3 m core interval, and used to produce a visual representation of the estimated percent kimberlite content versus depth. Beginning in 2009, the percentage of country rock xenoliths within the kimberlite units was measured directly using a line scan method. Line scans are completed by measuring all country rock xenoliths greater than 0.5 cm along a line drawn on the core. The country rock xenolith percentage is calculated by dividing the total of the country rock xenolith measurements by the length of the section measured, which is typically 1 m. Line scans have also been undertaken on archived drill core.

Core recovery and other geotechnical parameters are collected for all drill cores on the Project kimberlites. Core boxes were labelled with aluminium embossing tape and stacked on wooden pallets. Core recovery in the area of the Renard Cluster, Lynx, Hibou and North Anomaly dykes is typically greater than 90%. Each box of core is photographed with a digital camera at the Lagopède core shack. Photographs are then identified with the drill hole and core box numbers and stored on Stornoway's network in the North Vancouver office. A portion of the core was labelled as “waste” and remains stored onsite.

10.4 Geotechnical, Geomechanical and Hydrogeological Drilling

While drilling for delineation or minibulk samples in the Core Area, detailed geotechnical observations have been recorded from exploration drill core. All holes are logged for geotechnical parameters such as total core recovery (TCR), rock quality designation (RQD), intact rock strength, weathering/alteration, joint orientation, joint condition rating and fracture frequency in order to obtain rock mass quality values. Point load and Schmidt Hammer measurements have been completed on selected holes for Renard 2, Renard 3, Renard 4, Renard 9 and Renard 65. Beginning in 2009, holes have been drilled to produce oriented core for the purpose of obtaining orientation data from the core. Azimuth (Alpha) and inclination (Beta) measurements for all fractures in the oriented drill core were recorded to aid in the development of a geotechnical model of the Renard mine site.

Eight holes (123 m) were drilled for hydrological studies in the vicinity of Renard 2 and Renard 3 before the beginning of the underground operations (Barbeau, 2006). Seven of these drill holes were subsequently converted to water wells. The holes were advanced through the overburden down to the bedrock. Soil sampling and soil identification were performed every 1.5 m. Two to three permeability tests were carried out in each borehole. A pumping test was performed to evaluate the hydraulic conductivity of the ground. This drilling was done using a regular diamond drill with PQ diameter drill rods, allowing the construction of observation wells with 50 mm PVC pipes and screen.

A total of 26 geomechanical drill holes to test the proposed open-pit walls and deep underground at Renard 2, Renard 3, and Renard 4 were completed in March 2010 for a total of 2,445 m. These core holes were also used for geological modelling purposes.

Packer testing was carried out in March 2010 on two historical boreholes to test the hydraulic conductivity of both country rock and kimberlite at depth at Renard 2 and Renard 4.

Twelve boreholes (106 m) were drilled in 2010 for hydrological studies in the areas for future Processed Kimberlite Containment (PKC) facility, Plant, Airstrip and Landfill sites. The holes were advanced through the overburden and a short distance into bedrock to verify competency. Soil sampling and soil identification were performed at regular intervals during drilling, and hydraulic conductivity tests were also carried out within the wells. Drilling was done using a NQ-size (4.75 cm) split-spoon-hammer setup to advance through the overburden. Observation wells were then constructed using 50 mm PVC pipes and screens, sand filters, with bentonite plugs at the base of the well.

Thirty-nine overburden “profiling” boreholes were drilled in October 2010 totalling 203 m. The objective of this drill program was to map overburden thicknesses in proposed areas for future PKC facility, process plant, airstrip and landfill sites. Drilling was done using a standard diamond drill rig to advance through the overburden down to bedrock.

Twenty-three boreholes totalling 499 m were drilled in winter 2011 to characterize sediment and bedrock conditions along a proposed dyke that was considered at one point in the study; the final Project design does not incorporate the proposed dewatering dyke. A triple tube core rig was mobilized to site to conduct the drilling campaign. Twelve of the holes were for triple-tube soil coring and were advanced 3 m into bedrock, six holes were advanced 15 m to 50 m into bedrock, and two holes were drilled to define bedrock profile with soil samples recovered. Four of the holes were equipped to perform hydraulic conductivity testing and static water level measurements. One of the holes was equipped with a Casagrande-type piezometer.

During the summer of 2012, a series of geotechnical and geomechanical investigations were performed in order to obtain additional information on soil and bedrock for the detailed engineering design. Seven geomechanical holes (696 m) were drilled in the bedrock to obtain structural data in the vicinities of the portal and underground ramp. Two holes (69 m) were drilled to sample the bedrock in the vicinity of R65 in order to perform tests to determine the cement aggregate potential of this rock. Three holes (159 m) were drilled to perform packer tests, to assess the hydraulic conductivity of the rock. The hydraulic conductivity has also been verified at different areas along fault 28 using packer testing inside two existing holes and without packer testing inside another three existing holes. A total of 49 boreholes have been drilled in soil in order to determine the properties (soil type, thickness, etc.) in areas where infrastructure will be built, such as the airstrip, mine roads, buildings, etc. Monitoring wells have been installed in 26 of the boreholes to monitor water levels through the seasons. Some boreholes in soil have been extended below the rock contact to determine the rock unit type and its potential bearing capacity for infrastructures.

In 2014, two geomechanical holes were completed to investigate ground conditions in the future fresh air raise and return air raise vicinities (370 m). Another hole was drilled vertically in the CRB (135 m) to verify ground conditions, but also the hydraulic conductivity of this unit. Other geotechnical (94 m) and hydrogeological holes (107 m) were completed at various locations to generally verify soil thickness and water levels (monitoring wells). The main areas targeted by these holes are the Renard 2/Renard 3 and Renard 65 open pits, and underneath the future waste pile.

During 2015, one geomechanical hole was completed to verify the ground conditions in the fresh air raise area. The hole completed in the same area in 2014 was stopped at 80 m, but the mine design has changed since then and another hole was required to investigate the ground conditions down to 330 m. This hole will also be used as a pilot hole during the raise excavation, and packer testing was performed in early 2016 to verify the potential for water infiltration in this main excavation. A hydrogeological/geomechanical campaign was initiated at the end of 2015 to investigate a structure providing a significant amount of water at the face of the ramp at that time. This campaign is not yet complete, and meterages have not been compiled into Table 10.2.

10.5 Collar Surveys and Down Hole Surveys

Before the fall of 2010, all positional work on the Property was carried out using a transformed UTM 18 NAD 27 local coordinate system (Canada Mean). Since then the transformed UTM 18 NAD 83 local coordinate system (Canada Mean) has been the reference system on the Foxtrot Property.

Exploratory drill holes prior to 2004 were located relative to marked grids constructed using either Global Positioning System (GPS) units or chain and compass. In addition to using the marked grid, all exploratory borehole locations were verified by hand-held GPS units with no differential correction. Exploration drill hole azimuths and inclinations are set using a compass and protractor, respectively. Final exploration borehole inclinations are surveyed using an “acid test” system.

To position delineation and minibulk hole collar location pickets and front sights for holes drilled within the Renard Cluster, a registered surveyor uses GPS equipment with sub-centimetre accuracy. Technicians use a theodolite to align the drill rig with the collar picket and front sight pickets to ensure the holes are started accurately. After the drill programs are completed, the surveyor returns to locate the drill collar positions accurately by surveying the casings for locations and orientations. During 2003 and 2004, Corriveau J.L. and Associates Inc. of Val d’Or, Québec, surveyed all the delineation and minibulk (as well as several of the exploratory) borehole collars using a Leica System 500 RTK real-time, differentially corrected GPS system (DGPS) with +/- one centimetre accuracy. In addition, cut-and-chained stations on the ground grids covering Renard 2, Renard 3, Renard 4, Renard 65 and Renard 8 were accurately located using the same survey equipment, and a series of control points were established on the Property.

From the winter of 2005 until 2012, Paul Roy Surveyor of Chibougamau, Québec confirmed locations of all survey collars. The downhole track of core holes drilled at Renard 2, Renard 3, Renard 4, Renard 65, Renard 7, Renard 8, Renard 9, and Renard 10 and several exploratory holes drilled outside the Renard Cluster, were surveyed using the FlexIt borehole survey instrument to determine the azimuths and inclinations more accurately. The FlexIt instrument is an electronic downhole compass tool that measures several parameters, such as azimuth, dip, magnetic field, and dip of the magnetic field. Once the data are downloaded into a computer, a program corrects for magnetic declination and rejects readings above the average magnetic field (56,000 nanoTeslas) for the area of the Renard bodies. The readings above average generally correspond to areas of magnetic rock. In addition to the FlexIt system, a gyroscope system from Gyrosmart was used for the summer 2007 drilling campaign. This particular survey tool (IBG10), known as Imego's digital Butterfly Gyroscope, uses a MEMS (Micro-Electro-Mechanical System) gyro sensor and allows the downhole orientation survey to be carried out inside the drill string. For the drill programs completed during 2009 and 2010, in addition to using the FlexIt tool, a DeviFlex system from Devico A.S. was used to determine borehole geometry. The DeviFlex is a non-magnetic, electronic, multishot tool that uses accelerometers and strain gauges to calculate changes in inclination and azimuth. It is not affected by magnetic fields and is used for surveying inside casings and drill strings.

During the 2014 deep directional drilling program at Renard 2, three different downhole survey tools operated were used by two companies. The primary tool was the north-seeking, non-magnetic SPT Gyro Tracer, supported by the magnetic multishot PeeWee (DeviTool) survey tool, both operated by Tech Directional Drilling (Millertown, NL). For QA/QC purposes, downhole audits were performed on five holes by Halliburton Sperry Directional Services (North Bay, ON) using a north seeking non-magnetic Memory Keeper Gyro manufactured by Applied Technologies. Data quality was generally good and can be used for 3D modelling. Technical problems with drilling prevented some of the drill branches from being completely surveyed, however efforts were made to survey each branch whenever a kimberlite contact was encountered. Drill collar locations from 2014 were established by Stornoway's internal mine construction survey department.

All the core holes drilled on the Property are surveyed for magnetic susceptibility. From 2001 to 2006, an Exploranium KT-9 hand held instrument was used to manually measure the core at select intervals. Since 2006, magnetic susceptibility readings have been taken using a continuous reading, multi-parameter probe (MPP) from Instrumentation GDD.

10.6 Drill Programs

Between 2001 and 2002, drilling in the Renard Core Area was undertaken primarily as early-stage, exploration-focused programs. From 2003, most of the drilling was used to support delineation work and advanced-stage project evaluation through the collection of minibulk and bulk samples for all bodies except Renard 9 and Renard 10, which were discovered in 2003. Additional target exploration drilling has been ongoing since 2001 on the larger Foxtrot Property. These drilling programs have been discussed in detail in earlier technical reports (Clements and O'Connor, 2001, 2002; Lucas et al., 2003; Lépine and O'Connor, 2004; O'Connor and Lépine, 2005, 2006). Location plans for drilling not used to support mineral resource estimation can be found in these reports. Details of the 2009 program can be found in the May 2010 Preliminary Assessment (Lecuyer et al., 2010) while the 2010 program is discussed in the technical report for the 2011 Feasibility Study filed in 2011 (Bedell, et al., 2011). Drilling in 2011 and 2012 was conducted for geomechanical, geotechnical and hydrogeological purposes and is described in Section 10.4 of this report. The 2014 directional drill program targeted Renard 2 with more details provided in Section 10.3.

Tables 10.1 and 10.2 summarize the drilling on the Foxtrot Property. Drill hole collar locations and outlines of the kimberlite bodies contributing to the mineral resource estimate are discussed in Section 14.

11. SAMPLE PREPARATION, ANALYSES AND SECURITY

11.1 Laboratories

Stornoway has used a combination of internal and external facilities for the processing and extraction of both macrodiamonds and microdiamonds from the Renard Project.

Four laboratories have been used to provide primary macrodiamond extraction through the Dense Media Separation (DMS) process: an external unrelated commercial facility owned and operated by Kennecott Canada Exploration Inc. (Thunder Bay Mineral Processing Laboratory - TBMPL); an external unrelated commercial facility owned and operated by Microltihics Laboratories (also in Thunder Bay, Ontario); an internal facility situated in North Vancouver, British Columbia (owned and operated by Stornoway); and, an internal facility formerly situated at the Renard Project in Québec (owned and operated by Stornoway). The TBMPL is accredited by the Standards Council of Canada to the ISO/IEC 17025 standard as a testing laboratory for specific tests. Neither Microlithics Laboratories Inc. nor the two internal Stornoway facilities are accredited, but have been independently audited and are subjected to ongoing QA/QC testing.

During the Renard exploration programs, microdiamonds were recovered by one internal facility situated in North Vancouver, British Columbia (owned and operated by Stornoway) and four external unrelated commercial facilities: Microlithics Laboratories Inc. located in Thunder Bay, ON; Saskatchewan Research Council Geoanalytical Laboratories (SRC), Saskatoon, SK; TBMPL, Thunder Bay; ON; and, SGS Lakefield Research Ltd (Lakefield) in Lakefield, ON. Neither Stornoway's internal facility nor Microlithics Laboratories are accredited, but have been independently audited and are subjected to ongoing QA/QC testing. Both Lakefield and SRC are accredited by the Standards Council of Canada under ISO/IEC 17025 "General Requirements for the Accreditation of Calibration and Testing Laboratories (CAN-P-4D)". The TBMPL is accredited by the Standards Council of Canada to the ISO/IEC 17025 standard as a testing laboratory for specific tests. Microlithics, SRC and Lakefield were further used as secondary laboratories to cross-check and verify recoveries from the other microdiamond facilities.

TBMPL, Microlithics Laboratories, SRC and Lakefield are commerical laboratory facilities independant of Stornoway, and have no interest in the Renard Diamond Project.

11.2 Dense Media Separation (DMS) Facilities

Dense media separation is a standard industry process for the liberation and extraction of macrodiamonds from large volumes of sample material (commonly tens to thousands of tonnes). Rock samples are progressively crushed and the disaggregated material passed over a series of size sorting screens before being mixed with a slurry of ferrosilicon and water. A cyclone is used to split the heavy minerals, including diamonds, from the lighter waste rock. The heavy mineral concentrate is removed from the DMS plant and stored under secure conditions until the diamonds can be extracted. Waste material is recycled through the plant and re-crushed to liberate finer and finer diamonds. The minimum and maximum diamond size that can be recovered by the process is determined by the plant configuration. For the Renard Project, all DMS plants targeted stones of +1 DTC screen size.

The non-accredited, internal Stornoway laboratory in North Vancouver provides mineralogical and geochemical analyses for diamond exploration in support of Stornoway's exploration projects. The laboratory previously included a 5 tph Bateman/Van Eck & Lurie DMS facility for the recovery of commercial-sized diamonds from core, minibulk and bulk samples. The diamond recovery circuit currently includes a sizing circuit, an X-ray flow-sorting machine, grease table and magnetic separation equipment as well as heavy liquids.

A 10 tph Bond Equipment DMS plant was mobilized, assembled and commissioned on site at Renard during 2006 and 2007. This internal facility was used in 2007, 2008 and 2012 to process material from minibulk and bulk sampling programs, but has now been decommissioned.

Kennecott's TBMPL owns and operates a 10 tph Bateman/Van Eck & Lurie DMS facility, initially constructed in 1993 and rebuilt during 2003. This facility was used during 2004 to process RC chip samples collected from the Renard Project.

Microlithics operates a 1.5 tph Dowding, Reynard and Associates (DRA) DMS plant delivered new in November of 2005. This facility was used as a secondary facility for auditing tailings from minibulk and bulk samples prior to 2013. After 2013 the DMS plant was the primary treatment facility for minibulk samples.

Kimberlite concentrate generated by the TBMPL and Lagopede DMS facilities was shipped to the North Vancouver laboratory for final treatment and recovery of diamonds. Concentrates from Microlithics were either finished onsite or shipped to the North Vancouver laboratory for final treatment and recovery of diamonds.

11.3 Caustic Fusion Sampling

The caustic fusion technique, more properly known as caustic dissolution, uses chemical processing to provide total liberation of all diamonds within a given sample in order that an accurate diamond distribution can be determined. Caustic dissolution processes are usually applied to recover microdiamonds from relatively small volume samples (tens to hundreds of kilograms). Rock samples are loaded into large steel pots and caustic soda is added to dissolve the mineral matrix hosting the diamonds. Dissolution takes place over an extended period of time in temperature-controlled kilns. Once the reaction is complete, the residue is cooled and poured through stainless steel wire mesh screens at the required size to avoid loss of small diamonds. Depending on the size of the residue, further standard dissolution may be required. In cases where abundant oxides remain in the residue, a variety of other chemicals may be used to reduce the size of the concentrate, without harming the diamonds. Residues are then observed under microscopes by trained personnel, and the diamonds recovered, counted, sized and weighed.

To assure the integrity of the process, a chain of custody is established between the customer and the laboratory. Customer samples are processed in a controlled environment to ensure that confidentiality is maintained at all times. All samples are handled with due diligence during processing stages, according to previously defined protocols. Quality control grains are added to each aliquot undergoing the caustic dissolution process to monitor recovery.

Similar caustic dissolution processes are used by all four external unrelated commercial facilities, however, historically Stornoway's internal laboratory facility (as operated by Ashton Mining) applied a process of attrition milling, followed by heavy mineral separation and caustic fusion dissolution, as the initial method of testing kimberlite discoveries for microdiamonds. Processing was a complex process of controlled and iterative crushing followed by fractionation, magnetic separation techniques and heavy liquid separation to produce diamond-enriched concentrates. Caustic reagents are mixed with a portion of the concentrate and melted by heating in a muffle furnace at high temperature. Diamonds are then recovered from the resulting fusion residue.

11.4 Database

Data collected from the various exploration, minibulk, and sampling programs were collated into a SQL Server relational database. Data requests are processed through the database administrator. Access to the data in SQL Server is restricted to the database administrator only.

The database is stored on the server in the North Vancouver office, with backups being performed every day. One copy of the database is removed from site on a regular basis. Hard copies of processing and diamond data are stored in fire resistant filing cabinets in the North Vancouver office, as are hard copy data of the Renard core field logging. In addition, these hardcopies have been scanned as digital PDF files which are stored on the servers.

11.5 Sample Security

All sample processing is undertaken by qualified operators who conduct their work in secure laboratory areas with restricted access and following strict sample handling protocols. Diamonds recovered from the caustic dissolution process are generally very small (< 0.85 mm), have limited commercial value and the focus is to extract and retain all available stones. Security, while present, does not form as large a component as it does for the DMS work which can recover hundreds of carats of diamonds, many of which may be potentially attractive for theft.

DMS operations, post-processing treatment of DMS concentrates, and handling of concentrates and diamonds, from 2004 to the effective date of this report, were conducted under approved security protocols and procedures, which include but are not limited to:

- Chain of Custody documentation;
- Dual locking containers;
- Uniquely numbered, single use, tamper resistant seals;
- Monitoring and control of sample weights;
- Limited access or dual access to certain laboratory premises;
- GPS tracking of concentrates;
- Closed-circuit TV surveillance; and
- External (third party) security guards as required.

Comparative analysis of diamond size distribution is checked against historical and external laboratory results.

11.6 Drill Core

Once a full core tray has been reviewed by a geologist at the drill rig, the core boxes are screwed closed for ground transport to the core logging facility, situated approximately 500 m west of the Renard kimberlite cluster. This facility has been used for all of the initial Renard core work, but has subsequently been decommissioned. The core facility was a secure, locked facility without windows. Only authorized personnel were allowed inside.

Drill core sampled during the 2013 to 2015 time frame included both archived core stored in a secured facility in North Vancouver and new core collected onsite at the Renard Project. Core from the latter was briefly logged in an onsite core shack, photographed and measured for geotechnical purposes. Telescopic samples were removed at 3 m intervals systematically throughout each hole and stored separately. Core boxes were labeled with aluminum tags, their lids screwed down and then strapped onto pallets with ¾” metal banding in a minimum of three places. Pallets and contents were numbered, recorded and consigned to trucks travelling south on the Route 167 Extension/Renard Mine Road. Pallets were checked in and marshalled at a yard in Chibougamau belonging to Durocher Transport, before being loaded onto dedicated flatbed trucks, strapped again, covered with heavy rubberized tarpaulins and strapped once more. The loaded trucks were dispatched to Stornoway's North Vancouver facility and detailed waybills submitted in advance of delivery. The telescopic samples were shipped separately from the drill core to provide an additional level of comfort.

Trucks were unloaded under CCTV camera coverage by Stornoway employees, contents of the pallets verified, and stored within locked, access controlled warehouse facilities protected by 10-ft-high steel fencing and crash bollards. No concerns with sample tampering during shipping were identified.

Detailed geological logging took place under the same access controlled conditions and, once complete, a sampling plan was developed on a hole by hole basis. In certain cases diamond spikes were added to the core samples to assist in the sample tracking and QA/QC verification procedures. Essentially all core was consumed through either caustic dissolution, minibulk DMS processing or geochemical analysis, although a partial core record does exist in the form of both telescopic and representative samples collected from each hole. Material collected for sampling were stored in either uniquely numbered, sealed heavy plastic bags in 20 l white plastic pails with a single use locking lid (microdiamond and geochemical samples) or double lined 90 cm x 90 cm x 90 cm white ricine megabags, each of which were labelled with a unique number and sealed (minibulk samples). Locking single use uniquely numbered wire cable security seals were also applied to the megabags. Both the white plastic pails and the megabags were strapped and shrink wrapped to pallets before being labelled for shipping to the laboratories. Shipment weights, seal numbers and documentation were exchanged with the contract analytical facilities. Although at the start of the program the internal inventories were occasionally found to be inaccurate, this was corrected later in the program, and no serious shipping concerns were identified.

In the case of drill core collected for microdiamond analyses (i.e., caustic dissolution) during the 2009 and 2010 drill programs (handled almost entirely onsite at Renard), once logging was complete and sample intervals had been established in the field, core was extracted from the core boxes and packed into 20 l, white plastic pails lined with heavy polyethylene sample bags. Bar coded sample tags with unique identification numbers were inserted into each sample bag and the bags sealed. Pails were clearly labeled inside and out with the same sample number, and then sealed with tamper resistant single use lids. Individual pail weights were recorded and this information forwarded to the company's expeditor in Chibougamau and the office in North Vancouver. The expeditor was notified once the pails left camp on a charter aircraft, met the aircraft at the landing site and transferred the sample pails to a secure location. The pails were then consigned to a commercial courier service for delivery to either the contract caustic fusion laboratory or to North Vancouver. Upon receipt of the samples the caustic fusion facility would take possession of the samples, document the number and weight of the pails and notify both the camp and the North Vancouver office. In certain cases diamond spikes were added to the core samples to assist in the sample tracking and QA/QC verification procedures.

Historically, during drilling operations for minibulk sampling, specific security measures were instituted to minimize the potential for tampering and to maintain the integrity of the drill core from initial coring on site to delivery of the core to the North Vancouver laboratory.

11.7 Reverse Circulation (RC) Chips

Reverse Circulation sample drilling operations occur under strict security measures to minimize potential tampering and to maintain the integrity of the samples while they were being shipped from the drill site to the North Vancouver or commercial laboratory. A fenced area with controlled access at the Renard camp and/or the LG-4 site, was used for temporary storage of RC chip samples during the summer 2004 and 2006 drilling programs, prior to shipping the samples south. For the winter and fall 2007 RC drilling program, RC bags were stored in a restricted access area adjacent to the Renard DMS facility, before being processed on site.

Joint venture geologists supervised the sample collection. During drilling, access to kimberlitic chips was limited to authorized joint venture staff and drillers. Once filled, 1.5 t sample bags or 205 l steel drums were sealed immediately with cable ties and a security seal was applied to deter opening the bags or the drums. The bags and the barrels were loaded on pallets and transported by forklift to the camp's restricted loading area until loaded into aircraft.

During the 2004 and 2006 winter drilling program, large capacity aircraft were used to transport the material to Chibougamau or Mirage. Once at the destination, the bags were unloaded from the aircraft for storage at a secure location where only authorized personnel were permitted, before being shipped to the Thunder Bay DMS facility or the North Vancouver facilities. During the summer of 2004, float-equipped airplanes were used to carry secured 205 l barrels to Mirage, a tourist outfitter lodge near LG-4, before being shipped to the Thunder Bay DMS facility. Seal number, trailer and container numbers and date of departure were recorded on the chain of custody and were transmitted to the Vancouver office. Once the shipment reached its destination, the seal was verified and the van was off-loaded by a technician who verified the contents against the shipping papers and checked sample weights.

11.8 Bulk Sampling

11.8.1 Underground Bulk Samples

Access to the underground workings was restricted to ensure safety of personnel, as well as security and integrity of the samples. Only personnel involved in the underground program had access to the kimberlite, such as geologists, surveyors, equipment operators and drillers. On surface, the sample storage areas, treatment compound and DMS facility were considered as restricted areas and subject to access control, camera surveillance and security monitoring. An independent Québec-based security firm, Canaprobe, was mandated to implement site security protocols and provided bonded security officers. During bulk sample processing, only authorized personnel had access to the restricted areas. Security personnel escorted all approved visitors. All personnel involved in the operations at the DMS facility were subject to criminal record checks and random daily searches. Surveillance by the security force included use of overlapping and task specific CCTV coverage, as well as a continuous physical presence within the DMS plant. Locked dual custody provisions were in force in the vicinity of the DMS concentrate.

The concentrate was discharged from the DMS into 205 l steel drums lined with a heavy polyethylene sample bag. Once sample processing was complete, or the drum was full, the sample bag was sealed with a numbered tamper-resistant security tag by both a processing representative and a security representative. Each drum was mechanically sealed, locked and secured with two additional numbered tags. Sealed concentrate drums were weighted, labelled and moved to a secure, access controlled, storage site under constant camera surveillance and chain of custody documentation. Once enough concentrate drums had been collected, they were removed from site on a charter aircraft and transported under chain of custody to Chibougamau or Mirage, where they were temporarily stored in locked containers or secure vans before being shipped to North Vancouver with a bonded trucking company. Concentrate was transported in a locked, sealed and documented tractor-trailer on a "non-stop, 24/7" continuous basis from

Chibougamau or Mirage directly to North Vancouver. Concentrate reception in North Vancouver was under dual custody conditions and included monitoring by external security officers, drum/external security seal number verification, integrity inspections and weight control. Concentrate drums were then placed in a locked, access-controlled storage cage under CCTV surveillance until they could be processed. Paper records of access/transfer points were also maintained.

11.8.2 Renard 4, Renard 65, Lynx and Hibou Trench Bulk Samples

To ensure the security and integrity of the large tonnage surface samples, and the sample process itself, access to the trenches and the DMS stockpiles was limited to personnel specifically involved in the sampling, such as geologists, field assistants and heavy equipment operators. A log of personnel accessing the sites was recorded daily by Security Personnel. Detailed records of the sample collection were kept including times of various activities and any incidents such as spillages. In addition, records were kept of all sample transfers from the trenching site to the DMS stockpile.

11.9 Final Diamond Treatment and Recovery

Diamond bearing concentrates generated by DMS processing of underground bulk samples, large tonnage trench samples and RC chip samples from the Renard Project were all subjected to final processing at Stornoway's North Vancouver laboratory facilities. The diamond recovery circuit includes a sizing circuit, an X-ray Flow-Sort machine and grease table equipment.

All processing of concentrates was undertaken in secured, controlled access, CCTV monitored areas of the North Vancouver facilities. An independent, external and bonded security force (FBIG Investigations) monitored CCTV equipment, provided a physical presence at critical points of the diamond extraction process and recorded both routine activities and any abnormal incidents (sample spillage, etc.) during the large tonnage trench and underground diamond extraction programs. These security personnel also checked sample seals, sample weights and provided key control services for dual-locked storage areas, concentrate canisters and restricted areas. Processing personnel were subject to random searches at various times.

For the 2013–2015 work, the size and scope of the North Vancouver facility had been significantly reduced, allowing security functions to be undertaken by internal personnel using personal visits and CCTV based coverage.

11.10 Stornoway Quality Assurance and Quality Control Programs

Stornoway has a series of QA/QC programs in place that are applied to samples submitted to either external laboratories or the Company's internal facilities. Historically, QA/QC testing was conducted on 5% of all samples passing through the North Vancouver microdiamond circuit and the Dense Media Separation (DMS) circuit (Lecuyer et al., 2009). Checking associated with the 2013 to 2015 internal macrodiamond sample processing was conducted at a more frequent rate.

QA/QC check programs conducted by Stornoway include:

- Regular testing of all machines and equipment;
- Calibration and verification procedures;
- Use of unique sample numbers on preprinted cards to avoid duplication;
- Weighting of raw samples prior to submission, and after receipt at laboratory, to check for sample switching;

- Submission of blank samples (i.e., samples that should not return mineral grains or diamonds), spiked blank samples and samples of a known chemical composition;
- Blind spiking of samples in processing, using “known” natural or synthetic diamonds and/or manufactured density tracers prior to processing with recovery determined at the observation stage;
- Blind spiking of samples in observing, using “known” diamonds prior to observation;
- Second passing/audits of mineral concentrates by different personnel;
- Routine audits of non-observable fractions and reject materials;
- Routine re-processing of both random and targeted samples to verify recovery
- Use of internal standards and reference materials which are calibrated to provide traceability and reproducibility;
- A record-keeping system of documentation, which retains in archives all original records and data, with all amendments clearly marked, initialled and dated for reference;
- Vetting of compiled digital databases against original hard copy and digital records;
- Corrective actions which are implemented immediately when any aspect of laboratory analysis, or chain of custody documentation does not conform to procedural standards;
- The investigation and verification of any result which appears to be a potential statistical anomaly, to ensure laboratory results fit within the geological context; and
- Use of external laboratories for check samples. Up to 5% of all samples are routinely sent for external analysis.

The caustic fusion and diamond recovery facilities are governed by a series of detailed procedures that are appropriate to ensure the security and integrity of samples and the final results. All samples received in the laboratory are accompanied by a chain of custody document and with security seals that must be verified prior to processing any sample. Upon receipt, the samples are stored in a secure facility with restricted access. The diamond recovery circuits are in restricted areas and all samples, concentrates, diamonds and data are locked in safes, cabinets, drying ovens, or secure rooms when not being handled.

11.11 Conclusions

The responsible QP is of the opinion that the core sampling, RC chip sampling, minibulk and bulk sampling, sample preparation and analysis, and the QA/QC and security protocols used for the Renard Diamond Project are adequate, and that the data is valid and of sufficient quality to be used for Mineral Resource estimation. The results of Stornoway’s QA/QC programs did not identify any significant analytical issues in any data used for the 2015 Mineral Resource Estimate.

12. DATA VERIFICATION

12.1 GeoStrat Verification

As part of the independent expert review undertaken in conjunction with the 2015 Mineral Resource Estimate (Farrow and Hopkins, 2015), David Farrow and Darrell Farrow of GeoStrat conducted the following verification checks on the Foxtrot Property during the most recent project work:

- Site visits to the Renard Project by Darrell Farrow from July 29, to July 31, 2014 and from April 20, to April 22, 2015;
- Site visit to Microlithics Laboratories in Thunder Bay by Darrell Farrow from November 25, to November 26, 2014.
- Site visits to Stornoway's internal diamond recovery facilities throughout the 2013 to 2015 work programs
- Review of the surface and underground geological and mineralization interpretations (Section 7);
- Review of the historic and current exploration programs (Sections 6 and 9);
- Review of the deposit model (Section 8);
- Review of data that are supporting mineral resource models (Sections 9, 10, and 11). The review covered drill core inspection, core logging, sampling and assay protocols and methods, and sample security measures and sample storage;
- Review of QA/QC data protocols and methods, data integrity and validation historical RC, drill core and underground data (Sections 11 and 12; see also Bangnell et al., 2013), and
- Review of diamond content modelling and valuation methodologies.

GeoStrat has visited the Project site and the North Vancouver offices in order to audit procedures at Stornoway. Independent samples were not collected and treated by GeoStrat since this is not practical for diamond sampling.

The audit process requires matching of raw data from field copies for the various data collection areas to final copies of data to be used in public reporting and resource estimation.

GeoStrat has further reviewed documentation of procedures and verified that activities in the field conform to Stornoway's published internal procedures for those activities.

GeoStrat is of the opinion that Stornoway's published and practiced procedures for collection of data in the field and transposition of these data into data 'products' to support resource evaluation work and initial costing exercises meet industry best practice guidelines (Farrow and Hopkins, 2015).

12.2 Special Considerations for Diamond Resource Determination

Unlike commodities such as gold or base metals, diamonds do not have a standard value per unit weight that can be used to calculate the value of a deposit. A one carat diamond can be worth less than one dollar to tens of thousands of dollars, depending on the shape, colour and quality. A parcel of diamonds must be individually examined to establish an average value. Diamond values also change with the mix of diamonds over time; however, as a whole, diamond values have tended to increase with time. As this can be a somewhat subjective exercise, multiple valuations from different professional diamond valuers, or diamantaires, are necessary, and are usually averaged to give an estimate of the probable true price of the goods in question. Diamond price estimates can differ between valuers by as much as ±20%, especially on smaller parcels of diamonds. These differences are simply due to the fact that different diamantaires will perceive the value of a stone or parcel of stones differently. Their price guidelines will differ somewhat as well.

In a valuation exercise, it is necessary to involve a number of diamantaires to obtain a range of valuations that can be averaged to get an accurate price estimate and to use these data to model an average price. Often, in early stage evaluations of diamond projects, diamond price modelling is undertaken. In price modelling, the small sample size is compensated for by estimation of what the diamond population in a larger sample would be. By doing this, the valuer attempts to predict the likelihood of finding larger stones and what their effect on the overall value of the parcel would be and, as such, estimate more closely what the run-of-mine value would be. Modelling involves study of the diamond parcel on hand, including size distributions and valuations, to statistically estimate the upper and lower limits of a production parcel at certain confidence levels based upon the small parcel on hand. To accomplish this, Stornoway contracted WWW Diamond Consultants International (WWW) of London to obtain valuations and perform price modelling. WWW are recognized international leaders in this field.

Small parcels of diamonds are difficult and time consuming to value, so individual sample goods are generally combined on the basis of geology or some other parameter. Valuation parcels are generally sieved into DTC sieve classes (+1, +3, +5, +7, +9, +11), and grainer and carater categories. A cumulative 9,312.22 carat diamond parcel acquired by bulk sampling underground, trench sampling and RC drilling completed by Stornoway between 2003 and 2013 was used for value modelling.

12.3 Conclusions

The responsible QP is of the opinion that the data verification process demonstrated the validity of the data and protocols for the Renard Diamond Project. The responsible QP considers the Renard Project database valid and of sufficient quality to be used for the 2015 Mineral Resource Estimate.

13. MINERAL PROCESSING AND METALLURGICAL TESTING

13.1 Introduction

Dense media separation is a standard industry process for the liberation and extraction of commercial-sized macrodiamonds from large volumes of core, minibulk and bulk sample material. Rock samples are progressively crushed and the disaggregated material passed over a series of size-sorting screens before being mixed with a slurry of ferrosilicon and water. A cyclone is used to separate the heavy minerals, including diamonds, from the lighter waste rock. The heavy mineral concentrate is removed from the DMS plant and the diamonds are extracted. Waste material is recycled through the plant and re-crushed to liberate finer and finer diamonds. The minimum and maximum diamond size that can be recovered by the process is determined by the plant configuration. For the Renard Project, all DMS plants targeted stones of +1 DTC screen size or above, although smaller stones have been recovered.

Three DMS process facilities were used as primary macrodiamond extraction laboratories during the Renard exploration programs up to 2012: an internal 10 tph facility previously situated onsite at the Renard Project in Québec (owned and operated by Stornoway); an external unrelated 10 tph commercial facility at Thunder Bay, Ontario (owned and operated by Kennecott Canada Exploration Inc. and doing business as the Thunder Bay Mineral Processing Laboratory (TBMPL)); and an internal 5 tph facility situated in North Vancouver, British Columbia (owned and operated by Stornoway).

A fourth facility (a 1.5 tph DMS plant located in Thunder Bay, Ontario and operated by Microlithics Laboratories) was used for audit work during the 2007 to 2012 programs, and from 2013 to 2015 provided the primary mineral concentration process for the numerous minibulk samples. The largest proportion of DMS processing, by mass, has been through the Renard DMS plant, as shown by facility and sample type in Table 13.1. Table 13.2 shows the sample breakdown for the individual Renard kimberlite bodies comprising the current resource calculations.

Table 13.1: DMS Facilities – Sample Processing Breakdown

Sample Type	Renard		TBMPL		North Vancouver		Microlithics	
	# Samples	Weight (t)	# Samples	Weight (t)	# Samples	Weight (t)	# Samples	Weight (t)
RC	33	173.5	57	617.3	33	111.7	0	0
Surface	16	8538.2	0	0	27	109.0	0	0
Core	0	0	0	0	200	174.5	85	63.9
Underground	28	4562.5	0	0	0	0	7	1.4
Totals	**77**	**13274.2**	**57**	**617.3**	**260**	**395.2**	**92**	**65.3**

Table 13.2: DMS Facilities – Allocation of Sample Processing for Resource Bodies

Resource Body	DMS Plant			
	Total Weight Processed (tonnes)			
	Renard	TBMPL	North Vancouver	Microlithics
Renard 2	2535.6	171.2	34.6	56.0
Renard 3	2173.1	157	18.6	0
Renard 4	2104.2	141.8	103	0
Renard 65	5346.8	147.2	38.4	0
Renard 9	27.3	0	82.5	0
Lynx	494.3	0	48.9	0
Hibou	564.9	0	30.1	0
Totals	**13246.2**	**617.2**	**356.1**	**56.0**

The basic flow sheet used for each of the four mineral processing facilities is described in the following subsections, as well as an outline of the QA/QC procedures.

13.2 DMS Processing - Renard

A 10 tph Bond Equipment DMS facility was mobilized, assembled and commissioned on site at Renard in Québec during 2006 and 2007. Stornoway operated this facility in 2007, 2008 and 2012 to process material from minibulk and bulk sampling programs. Trained DMS process operators supervised each shift in a secure building. Sample processing followed procedures similar to those of the Thunder Bay and North Vancouver facilities, with some exceptions due to the nature of remote field operations, as noted below.

After excavation, kimberlite samples were stockpiled outside the facility in a secure compound with restricted access. Individual samples were prepared through a primary jaw crusher, and the 50.0 mm crushed product fed directly into the DMS facility where a scrubber trommel unit removes the +20.0 mm oversize. The +20.0 mm oversize material was reduced using a secondary cone crusher set at 10.0 mm. Coarse rejects were re-crushed to -6.0 mm and re-circulated through the facility. The first seven subsamples of the 2007 bulk sample program used a 1.2 mm by 13.0 mm slotted screen. Subsequent samples employed a 1.0 mm by 12.0 mm slotted screen to increase the recovery of smaller diamonds. Contract security services were provided on site by an independent third party (Groupe Canaprobe; Montreal, Québec).

DMS concentrates generated by the Renard facility were sealed in drums and dispatched in a locked container under chain of custody protocols to Stornoway's internal laboratory in North Vancouver for final treatment and recovery of diamonds. The laboratory owns and operates integrated diamond recovery equipment that includes a sizing circuit, X-ray flow-sort machine and grease table. Upon arrival at the North Vancouver diamond recovery facility, concentrates are screened into four size fractions, passed through an X-ray sorter twice, and the tailings diverted to a grease table circuit to generate a final diamond concentrate. Diamond-enriched concentrates are stored in sealed lock boxes within high-security cages before being transferred to the observation laboratory for final diamond recovery and reporting.

A team of trained mineral observers and mineralogists undertakes final diamond recovery using a combination of hand-sorting and binocular microscopy techniques. All diamonds recovered are routinely verified, described, weighed, photographed and recorded by trained mineralogists. Results are reported as diamonds retained on a +1 DTC sieve, or as stones greater than 1.18 mm using Tyler square mesh sieve classes when required. All diamond recovery is carried out with dual custody handling provisions under video surveillance in restricted areas. An independent external security team monitored and recorded all operations, transfers and seal controls. FBIG Investigations (Vancouver, British Columbia) provided these contract services from 2004 to 2011. Securiguard Services Ltd (Vancouver, British Columbia) provided these services during the 2012/2013 operations. No third party security was engaged for the 2014 and 2015 operations due to the much more limited scope of diamond recovery associated with minibulk sampling, as opposed to bulk sampling activities.

Neither the internal DMS facility nor the diamond recovery circuit are accredited, but both have been independently audited and are subjected to ongoing QA/QC testing. The Renard 10 tph DMS plant has been decommissioned to allow construction associated with mine development.

13.3 DMS Processing – Thunder Bay Mineral Processing Laboratory (TBMPL)

Kennecott's TBMPL owns and operates a 10 tph Bateman/Van Eck & Lurie DMS facility, initially constructed in 1993 and rebuilt in Thunder Bay during 2002 and 2003. This facility was used on two occasions in 2004 to process RC chip samples collected from the Renard Project. Operators from Stornoway's North Vancouver facility supervised contract DMS processing in Thunder Bay to verify the quality and continuity of work being performed. The TBMPL is accredited by the Standards Council of Canada to the ISO/IEC 17025 standard as a testing laboratory for specific tests.

RC chip samples were sent to the facility in sealed pails, drums, or bulk bags, each with an individual sample number and security seal. Security seals were verified as being intact upon arrival at the laboratory, recorded in a chain of custody document and isolated in a secure storage area with controlled access overseen by an external independent security service (Apex Security, Thunder Bay). The same security team also monitored DMS processing and maintained video surveillance in restricted areas.

Since the kimberlite material was already broken into chips from the RC drilling process, it was directly fed to the DMS scrubber unit to remove the +16.0 mm oversize fraction, which was sent to a secondary rolls crusher set at 8.0 mm. Kimberlite was processed through the facility, then the tails screened at 6.0 mm. Tails greater than 6.0 mm were re-crushed in a roll crusher set at 3.8 mm and repassed through the plant. Finer tails were discarded. A 0.85 mm by 14.0 mm slotted product screen was used within the Thunder Bay DMS facility.

DMS concentrates were dried, and then screened into six size fractions before being sent to the North Vancouver laboratory in secure shipments for further processing by heavy liquid and magnetic separation techniques. Final diamond recovery, classification and reporting were also carried out in North Vancouver under secure, independently monitored conditions FGIB Investigations (pre-2012) and Securigard Services Ltd (2012–2013), both based in Vancouver, British Columbia).

13.4 DMS Processing – North Vancouver Facility

Stornoway's internal laboratory in North Vancouver provides mineralogical and geochemical analyses for diamond exploration in support of Stornoway's exploration programs. The laboratory included a 5 tph Bateman/Van Eck & Lurie DMS facility and a diamond recovery circuit. Kimberlite samples were received at the laboratory as small diameter drill core in sealed core boxes; reverse circulation drill chips in sealed pails, bags or drums; or as surface material from outcrop or boulders packed in drums or bulk bags. Each box, pail, bag or drum was given an individual sample number and sealed in the field prior to shipment. Bags and drums were identified with security seals that were verified as being intact upon arrival at the laboratory in a locked and sealed shipment. Once the chain of custody documents sample numbers, and security seals were checked and verified, the sample was isolated in a secure storage area with restricted access.

Trained operators conducted DMS processing in a secure portion of the laboratory. Samples were initially reduced through a primary jaw crusher at a nominal gap of 20.0 mm to 25.0 mm. This material was then passed through a combination of jaw and roll crushing at a nominal gap of 10.0 mm to 14.0 mm. The crushed product was fed into the 5 tph DMS facility where a scrubber trommel unit removed the +14.0 mm oversize for re-crushing.

After primary feeding and sizing, all sample material is processed through the cyclone. DMS floats are screened at 4.0 mm with the undersize going to tails, and the oversize to iterative re-crushing and repassing at 4.0 mm and 2.0 mm to minimize the potential for diamond breakage. On early samples, a 1.0 mm by 12.0 mm slotted screen was employed within the DMS facility. In 2006, this was changed to a 0.85 mm by 14.0 mm slotted screen to better recover small diamonds, ensure a bottom diamond cut-point of 1.18 mm using a Tyler square mesh screen, and produce a minimum stone size smaller than that generally retained on a +1 DTC sieve.

Resultant DMS concentrates are screened into four or six size fractions, passed through an X-ray sorter twice, and the final tailings diverted to a grease table circuit for final recovery. Heavy liquid and magnetic separation recovery methods were employed on samples prior to 2006. Diamond-enriched concentrates are stored in sealed lock boxes within high-security cages before being transferred to the observation laboratory for final diamond recovery and reporting.

Final diamond recovery, classification and reporting were also carried out in North Vancouver under secure, independently monitored conditions (FBIG Investigations and Securiguard Services Ltd, both in Vancouver, British Columbia).

Neither the internal DMS facility nor the diamond recovery circuit are accredited, but both have been independently audited and are subjected to QA/QC testing. The 5 tph DMS plant has been decommissioned and was not used for the 2014–2015 work. The remaining part of the diamond recovery circuit, as described in Section 13.2, remains in operation.

13.5 DMS Processing – Microlithics Laboratories Facility

Stornoway owns a 1.5 tph DRA DMS facility situated in Thunder Bay, Ontario, which is operated and maintained by Microlithics Laboratories. Kimberlite samples from 2013 to 2015 were received at the laboratory as small diameter drill core in 90 cm x 90 cm x 90 cm ricine bulk bags containing up to 1,500 kg of kimberlite. The bulk bags were double-lined, individually numbered and security-sealed. Each of these minibulk samples was prepared in North Vancouver under the supervision of company geologists, following detailed geological logging and the collection of representative subsamples for future reference. When a sample arrives, it is checked into the LIMS (laboratory information management system), recording all relevant information including dates, security seals and condition upon arrival. The sample is isolated in a secure storage area with restricted access.

Trained operators and support staff conduct DMS processing in an access-controlled part of the laboratory under CCTV coverage. Immediately prior to processing, each sample bag is weighed on a pallet scale, and the weight of the wet material along with tare weights of the bag and or pallet are recorded into the LIMS and to the data sheet. At this time, a moisture sample is taken from the sample to represent the contents of the bag.

Once moisture content has been determined, the sample is lightly crushed using a 10 in by 7 in jaw crusher with a closed side setting of between 40 and 75 mm, and added to the main feed belt. The feed conveyor sends all material into the scrubber where it is mixed with water. The scrubber is lined with rubber and fitted with 2-in paddles to agitate the material, and is designed to maximize sample retention to better perform autogenous milling. Material that exits the scrubber via the overflow passes through a 12.7 mm square perforated trommel screen where any +10 mm oversize material gravitates to the primary jaw crusher. This equipment reduces oversize to -10 mm and returns it by conveyor to the feed conveyor for reprocessing. Product from the primary jaw crusher is routinely tested using sieve analysis to check for proper crusher operation.

All -10 mm material that passes through the trommel screen gravitates to the feed preparation screen where it is washed to remove undersize. The undersize is pumped to the de-gritting circuit for water recovery and fine tails disposal. This material is routinely tested using sieve analysis to check for any oversized material. Washed and sized -10 mm material reports to the mixing box where it is mixed with ferrosilicon (FeSi) before being pumped to the 150 mm DMS cyclone. A separation is effected under pressure. The slurry density is monitored using an inline Dense Medium Controller. The density is also monitored and calibrated using a pulp density scale. The quality of separation is monitored using a series of density tracer tests as well as quality control grains on a regular basis. Tracer tests are performed at the start of each shift as well as periodically throughout the shift. The resultant products are first drained and then washed to recover adhering FeSi. The screen washings are pumped to the wet drum magnetic separator for medium recovery and densification.

Floats report to the DMS tailings conveyor and are further screened and crushed in a 10-in Cone Crusher Circuit. All tailing material larger than 6 mm is crushed to -6+4 mm and all material -6+4 mm is then crushed to -4 mm to provide further diamond liberation and returned to the scrubber via conveyor to re-enter the separation process. The material is cycled in this manner until all tailing material is -4 mm at which time it is stored in cataloged mega bags.

The sinks (DMS concentrates) are collected in lined steel drums. The drums are sealed using metal security seals, weighed and shipped to Stornoway's North Vancouver facility where they are screened into four or six size fractions, passed through an X-ray sorter twice, and the final tailings diverted to a grease table circuit for final recovery. Heavy liquid and magnetic separation recovery methods were also employed. Diamond-enriched concentrates are stored in sealed lock boxes within high-security cages before being transferred to the observation laboratory under secure CCTV monitored conditions for final diamond recovery, classification and reporting.

Neither the Microlithics DMS facility nor Stornoway's internal diamond recovery circuit are accredited, but both have been independently audited and are subjected to ongoing QA/QC testing.

13.6 QA/QC

Quality control testing was routinely conducted on drill core, RC chips, surface and underground samples processed through the four DMS facilities. Tailings and various processing residues are also subject to audit. QA/QC measures include but are not limited to:

- Adherence to established/documented processing and handling protocols;
- Systematic density bead tracer testing prior to operating plant to ensure efficiency and diamond recovery;
- Addition of luminescent density tracers prior to DMS to monitor processing and X-ray sorter efficiency. Recovery is measured after X-ray sorter recovery;
- Addition of identifiable natural diamond spikes prior to any DMS processing; recovery is measured at the diamond sorting stage;
- Addition of natural diamonds prior to observation to determine the efficiency of diamond sorting and for security control purposes;
- Audit of representative coarse DMS tailings from select samples;

- Re-crush of DMS tailings and concentrate tailings to monitor processing efficiency, X-ray/grease recovery, and diamond liberation;
- Regular DMS feed size analysis;
- Testing of processing circuit clean-out residues;
- Audits of processing rejects using secondary laboratories and DMS facilities;
- Monitoring of diamond recovery statistics, including size frequency analyses;
- Independent third party audit processes (spiking, process review, etc.); and
- Review and audit of DMS and diamond data, operating procedures and QA/QC programs.

13.7 Metallurgical Testing

13.7.1 Mini Bulk Sample Processing

During advanced-stage evaluation of the Project, minibulk samples from drill core and RC chips (Renard 2, Renard 3 and Renard 4) were processed using small-scale dense media separation processing plants located in Stornoway's facilities at North Vancouver and Kennecott's TBMPL in Thunder Bay. Some useful DMS concentrate yield data were generated. The information is based on dry concentrate weight and dry plant feed weight expressed as a percentage and summarized in Tables 13.3 and 13.4.

Table 13.3: DMS Yield Data – North Vancouver Plants

	1 tph Plant DMS Yield			5 tph Plant DMS Yield		
	Max %	Ave %	Min %	Max %	Ave %	Min %
Renard 2	3.07	1.59	0.22	1.63	0.61	0.12
Renard 3	3.46	2.38	1.47	2.14	0.72	0.25
Renard 4	5.78	2.3	0.46	1.06	0.23	0.05

Table 13.4: DMS Yield Data – Kennecott TBMPL Plant

	10 tph Plant DMS Yield		
	Max %	Ave %	Min %
Renard 2	0.2	0.13	0.06
Renard 3	0.67	0.34	0.04
Renard 4	0.24	0.14	0.08

The 1 tph plant was operated to recover diamonds greater than 0.5 mm and DMS rejects were recycled and re-crushed to liberate additional diamonds and produce final plant rejects of less than 2 mm. Compared to a commercial plant where recovery of diamonds greater than 1 mm is usually targeted and final plant rejects are crushed to less than 6 mm, or in some cases higher, the DMS yields generated in the 1 tph plant are unrealistically high for commercial plant design.

The 5 tph plant was operated to recover diamonds greater than 1 mm and to produce final plant rejects less than 4 mm. Considering that the commercial plant will employ a high-pressure grinding roll unit, which is proven to liberate diamonds and hence heavy minerals better than conventional cone crushing, it is reasonable to assume DMS yields achieved in the 5 tph plant will equate to the expected higher yields with the inclusion of a HPGR (Bagnell et al., 2013).

The 10 tph TBMPL plant was operated with commercial plant settings: recovery of diamonds greater than 1 mm and production of final plant rejects less than 6 mm. However, because initial liberation of diamonds occurred during RC drilling, and up to 50% of the kimberlite was removed by screening out fines as part of the RC sample collection process, the DMS yields are understated in comparison if these data are applied to a circuit with a HPGR.

13.7.2 Bulk Sample Processing – 2007 and 2012

Bulk samples from Renard 2 (2,449 t), Renard 3 (2,114 t), Renard 4 (1,659 t) and Renard 65 (199 t) were processed on site at Renard during 2007, using the Lagopede 10 tph DMS plant. An additional 5,081 t of Renard 65 were completed in 2012. The 10 tph Lagopede plant was operated with commercial plant settings; recovery of diamonds greater than 1 mm and production of final plant rejects less than 6 mm. However, because liberation was achieved with conventional cone crushing the DMS yields will be understated if these data are applied to a circuit with a HPGR (Bagnell et al., 2013).

DMS concentrate was shipped off-site to Stornoway's recovery plant in North Vancouver. The recovered diamonds were used for diamond price estimation and grade determinations through the analysis of Size Frequency Distribution models of the Renard diamond population. The data listed below was collected during bulk sample processing to support future process design.

- Plant feed particle size distribution;
- DMS concentrate yield (summarized in Table 13.5);
- Diamond recovery;
- Diamond size frequency distributions;
- Granulometry (rejects and DMS concentrate);
- Plant mass balance and operating data; and
- X-ray sorter concentrate yields.

Table 13.5: DMS Yield Data – Lagopede 10 tph Plant

	10 tph Plant DMS Yield		
	Max %	**Ave %**	**Min %**
Renard 2	0.34	0.17	0.06
Renard 3	0.47	0.32	0.10
Renard 4	0.21	0.12	0.09
Renard 65[1]	-	0.22	-

Note: 1) The 5,081t of Renard 65 was processed as a single sample, therefore 'Max' and 'Min' are not available.

Data from the bulk sample processing indicated the following (Bagnell et al., 2013):

- The production of fines (-1 mm), a predictor of ore hardness, ranged from hardest to softest as follows: Renard 3 at 26%, Renard 2 at 31% and Renard 4 at 39%;
- The kimberlites have a low DMS yield using conventional cone crushing for liberation;
- Approximately 95% of the diamonds by weight were recovered after two stages of x-ray sorting. The remainder were recovered using a grease table;
- Grease scavenging of x-ray rejects will be worthwhile in a commercial plant; and
- Renard 2 exhibits potential to host larger diamonds than the standard commercial plant top size cut-off of 25 mm.

Consideration of the DMS concentrate yield from bulk sampling generated the plant design data presented in Table 13.6 (Bagnell et al., 2013).

Table 13.6: Design SMA Yields

	Max %	Ave %	Min %
Design DMS Yield	2.14	1.63	1.06

13.7.3 Additional Testwork Completed to Support Process Design

The testwork shown in Table 13.7 was completed to support the process design.

Table 13.7: Summary of Testing Completed during the 2011 Feasibility Study and after the 2011 Feasibility Study to Support the Process Design

Tests Undertaken / Data Generated	Test Location
HPGR bench-scale tests to product fineness; results were modelled to predict full-scale performance	SGS, Ontario
HPGR pilot-scale tests to determine product fineness, influence of moisture in feed and expected operating gap	BC Mining, British Columbia
Scrubber tests to determine resistance to abrasion breakage	SGS, Ontario
Grindability tests to characterize ore hardness	SGS, Ontario
Settling tests to permit thickener sizing, select suitable flocculant and estimate flocculant consumption	Outotec, Ontario
Filtration tests to determine dewatering characteristics of fine kimberlite particles	Outotec, Ontario
Centrifuge tests to determine dewatering characteristics of fine kimberlite particles	Decanter, Tennessee
NFTR, gangue & diamond LI testwork and magnetic separation tests using DMS concentrate	DebTech, South Africa
Flow property test results for PK	Jenike & Johanson, Ontario

Testwork findings were as follows:

- Renard 2 kimberlites exhibit similar impact breakage as country rock breccia containing significant granite xenoliths. The crusher work index for these samples indicates moderately hard ore (Bagnell et al., 2013).
- Renard 2 kimberlites and country rock breccia are resistant to comminution by abrasion as indicated by t_a values of 0.37 to 0.55, where t_a is the measure of resistance to abrasion breakage. This indicates that a scrubber or a mill could be used for washing rather than comminution (Bagnell et al., 2013).
- Laboratory scrubbing results indicate that comminution attributed to scrubbing would generate limited additional fines: less than 1% for Renard 2 kimberlite and less than 5.5% for Renard 3 (Bagnell et al., 2013).

- BC Mining generated process design data for a HPGR unit (750 mm diameter x 220 mm wide) in a tertiary crushing and recrush mode with a full feed size distribution. In the commercial plant design, the required product size for treating a mixture of fresh feed (-50 mm, 135 tph) and DMS rejects (-30+6 mm, 100 tph) will be achieved with a specific pressing force of 2.5 N/mm^2 and with a net specific energy consumption of about 2 kWh/t. For these operating conditions, a large operating gap of 34 mm is predicted to protect against possible diamond damage or breakage. The HPGR product flake was found to be soft and easily broken (Bagnell et al., 2013).
- Outotec generated a settling test and scale-up data for kimberlite slimes (-0.25 mm) thickening. Samples from Renard 2 settled easily, requiring low flocculant dosage rates. In the commercial plant design, using a high-rate thickener, overflow clarities of 100 ppm will be achieved with a flocculant dosage of 30 g/t or slightly higher and a high-density underflow of 45% w/w solids will easily be achieved and maintained (Bagnell et al., 2013).
- Outotec performed filtration evaluation tests using -0.25 mm processed kimberlite. The material proved difficult to filter such that filtration of fine processed kimberlite for the purpose of dry stacking is unlikely to be successful (Bagnell et al., 2013).
- Decanter Machine Inc. performed pilot-scale testing to determine the dewatering capabilities of an 18 x 60 solid bowl centrifuge with -1 mm processed kimberlite as feed. The tests proved that this material can be successfully dewatered for the purpose of dry stacking, thus eliminating the traditional processed kimberlite containment dam with a central water pond (Bagnell et al., 2013).
- The HPGR product generated from Renard 2 samples was processed in Stornoway's 5 tph plant to produce concentrate. This material was tested in a dual wavelength x-ray machine to determine probable yields that could be obtained under production conditions. The tests demonstrated that satisfactory yields are possible with the dual wavelength feature (Bagnell et al., 2013).
- Material flow property tests were conducted at Jenike & Johanson (J&J), and the final PK loadout system was designed based on the test results.
- To optimize the revised recovery design for mass reduction and X-ray yield recovery, further testwork was conducted at DebTech. Test results confirmed mass reduction and design. On the middle stream, one additional Raven machine was included due to the higher expected yield from the X-ray machines.

13.8 Basis for Recovery Estimates

DMS rejects (-6+1 mm) were collected after the treatment of subsamples from Renard 2, Renard 3 and Renard 4, and reprocessed in a small-scale DMS plant and in the recovery plant at Stornoway's North Vancouver facility. From this audit exercise, the lowest recovery efficiency of liberated diamonds in the Lagopede bulk sample plant was estimated to be 97.5% by carat.

Further plant efficiency audits were completed by adding 30 identifiable natural diamonds to the bulk sample plant feed and in a separate audit, 20 identifiable natural diamonds were added. In the first audit, 29 out of 30 diamonds were recovered giving a recovery efficiency of 96.7% by stone and 98.2% by carat. In the second audit, 100% recovery was achieved.

Also in accordance with established QA/QC procedures, 10 diamond simulants were added to each sample. A total of 149 simulants from 150 simulants introduced into 15 Renard 2 samples were recovered for an indicated recovery efficiency of 99.3%. All 120 simulants in the 12 Renard 3 samples and 70 simulants in the 7 Renard 4 samples were recovered for an indicated recovery of 100% in both cases.

The recovery efficiency for the commercial plant is estimated to be not less than 98% by carat weight.

13.9 Metallurgical Variability

Renard 2 will account for 79.1% of diamond production, Renard 3, 7.2%, Renard 4, 7.5% and Renard 65 6.2% All major geological units from each deposit were processed during the bulk sampling campaign at Lagopede. Also the major geological units from Renard 2, the dominant kimberlite source, were tested as described in Section 13.1 above.

14. MINERAL RESOURCE ESTIMATES

14.1 Introduction

This section documents the process by which the updated 2015 Mineral Resource Estimate was established for the Renard 2, Renard 3, Renard 4, Renard 65 and Renard 9 kimberlites and the Lynx and Hibou dykes (Figure 7.3).

The 2015 Renard Mineral Resource Estimate was prepared by an independent Qualified Person, Darrell Farrow, Pr.Sci.Nat. P.Geo. (British Columbia), Ordre des geologues du Québec (Special Authorization #332), of GeoStrat, in accordance with the Canadian Institute of Mining (CIM) Definition Standards for Mineral Resources and Mineral Reserves as incorporated by National Instrument 43-101, Standards of Disclosure for Mineral Projects. The 2015 Mineral Resource Estimate comprises the integration of kimberlite volumes, bulk density, petrology and diamond content data derived from 101,078 m of diamond drilling (497 holes), 6,151 m of reverse circulation (RC) drilling (36 holes), 23.7 t of samples submitted for microdiamond analysis, 196 cts of diamonds (3,107 stones) recovered from drill core, 605 cts of diamonds (7,181 stones) recovered from RC drilling, 4,404 cts of diamonds (40,521 stones) recovered from underground bulk sampling and 5,219 cts of diamonds (52,474 stones) recovered from surface and trench sampling. The estimate also incorporates information derived from approximately 150 drill holes, 37 surface test pits and 12 trenches undertaken for geotechnical and hydrogeological purposes. Drill collars and surface traces of the holes used to estimate the 2015 Mineral Resources are shown in Figure 14.1.

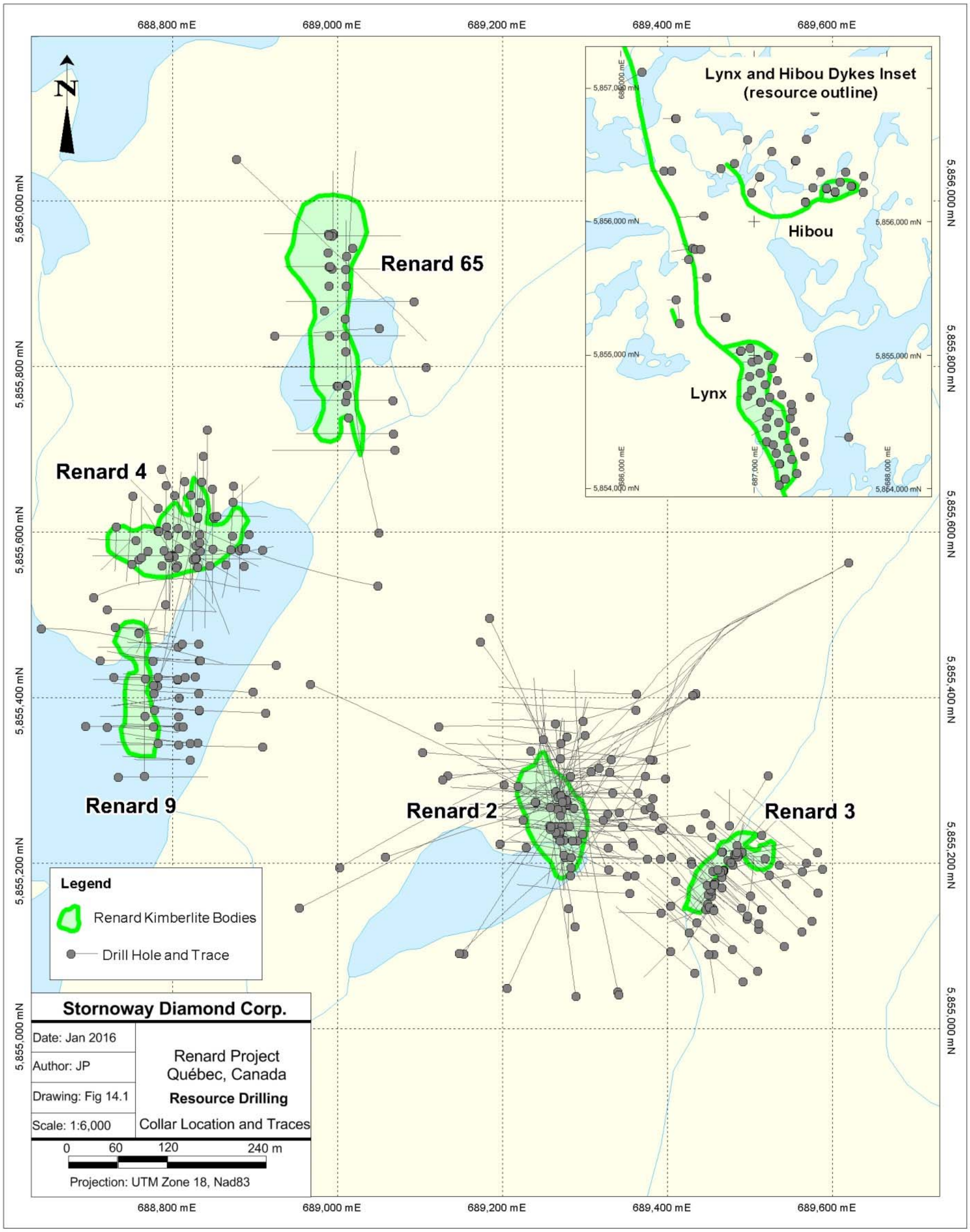
The methodology of the 2015 Mineral Resource Estimate has been reviewed by David Farrow of GeoStrat as documented in a separate NI 43-101 Technical Report (Farrow and Hopkins, 2015). Information provided below has been reviewed and is valid as of the effective date of the current technical report.

14.2 Previous Work

Prior to the 2015 Mineral Resource Estimate, a previous Mineral Ressource estimate was produced in July of 2013 by GeoStrat Consulting Services Inc., and is summarized in Table 14.1.

The July 2013 Mineral Resource Estimate was completed in accordance with the Canadian Institute of Mining (CIM) Definition Standards for Mineral Resources and Mineral Reserves as incorporated by National Instrument (NI) 43-101, Standards of Disclosure for Mineral Projects. It comprised the integration of kimberlite volumes, density, petrology and diamond content data derived from 88,887 m of diamond drilling (457 holes), 6,151 m of large diameter reverse circulation (RC) drilling (36 holes), 18.3 t of samples submitted for microdiamond analysis, 600.8 cts of diamonds (6,457 stones) recovered from RC drilling, and 9,575.0 cts of diamonds (92,940 stones) recovered from surface trenching and underground bulk sampling. The estimate also incorporated information derived from approximately 135 drill holes, 37 surface test pits and 12 trenches undertaken for geotechnical and hydrogeological purposes.

Figure 14.1: Drill Collar Locations and Traces for 2015 Mineral Resources



In addition to incorporating the results of the Renard 65 bulk sampling completed in February 2013, the July 2013 Mineral Resource Estimate incorporated the results of geotechnical and hydrogeological drilling at Renard, where such drilling intersected the ore bodies, and used a refined method for calculating country rock dilution within each kimberlite. The grade contribution provided by the abundant occurrences of hypabyssal kimberlite observed within each body's eruptive envelope has also been re-assessed. This resulted in changes to the diamond content estimates of Renard 2, 3, 4 and 9, as well as a substantial quantity of Country Rock Breccia ("CRB") within Renard 2 being added to the Mineral Resource Estimate, including 0.87 Mt at 32 cpht (0.28 million carats) in the Indicated category and 6.54 Mt at 19 cpht (1.24 million carats) in the Inferred Mineral Resource category. In previous Mineral Resource estimates, this CRB material was assigned a zero grade, and was treated within the earlier Renard underground mine plan as waste or mining dilution.

Table 14.1: Historical Mineral Resource as of July 2013

Mineralization Category	Renard 2	Renard 3	Renard 4	Renard 9	Renard 65	Lynx	Hibou	Total
Indicated								
Tonnage	18,580,000	1,760,000	7,250,000	0	7,870,000	0	0	35,460,000
Total Carats	18,660,000	1,820,000	4,310,000	0	2,300,000	0	0	27,090,000
Average cpht	100	103	60	0	29	0	0	76.5
Inferred								
Tonnage	11,770,000	540,000	4,750,000	5,700,000	4,930,000	1,800,000	180,000	29,670,000
Total Carats	7,470,000	610,000	2,370,000	3,040,000	1,180,000	1,920,000	260,000	16,850,000
Average cpht	64	112	50	53	24	107	144	56.8

Subsequent to the July 2013 Mineral Resource Estimate, more drilling has been added and more geological information has become available. The July 2013 Mineral Resource Estimate cannot be considered as current; it is included in this section for illustrative purposes and should not be taken out of context.

14.3 Methodology

The 2015 Mineral Resource Estimate for Renard kimberlite pipes 2, 3, 4, 9 and 65 is based upon the determination of undiluted grade models for each kimberlite lithofacies, as derived from diamond sampling, and their application to a dilution block model derived from diamond drilling. This methodology has been selected as appropriate based on the geology of the Renard kimberlites ("transitional" TK/HKt-TKt kimberlites), the recognition that country rock dilution is the principal driver of grade variability in the Renard kimberlites, and the observed stability of the diamond population on an undiluted basis within each kimberlite lithofacies. Resource block models created in this manner can then be tested against observed diamond sampling data, which includes RC drilling and diamond drilling in Renard 2, Renard 3, Renard 4, Renard 9 and Renard 65, surface trenching in Renard 4 and Renard 65, and underground bulk sampling in Renard 2 and Renard 3.

The Mineral Resource estimation process comprised the following steps:

1) Establish geological models for each kimberlite pipe using diamond drill logs, petrographic analysis, and emplacement modelling.
2) Analyze the diamond populations:
   a) Review macrodiamond populations to ensure they are robust for mineral resource valuation;
   b) Develop the relationship between macrodiamond and microdiamond populations; and
   c) Establish that the microdiamond population is stable with depth.
3) Establish undiluted grade models for each lithofacies:
   a) Undertake diamond size frequency analysis using combined macrodiamond and/or microdiamond data from representative samples;
   b) Where appropriate, establish individual population grades for mixed samples; and
   c) Determine a factor to account for the presence of discontinuous, but mappable, hypabyssal units present within each lithofacies (units Kimb2c, Kimb3c, Kimb4c and Kimb65c).
4) Establish the parameters of the block model.
5) Establish the dilution model:
   a) Establish distributions for country rock dilution in the various bodies and lithofacies based on line scan and/or modal means; and
   b) Undertake geostatistical analysis and estimation of the dilution per rock type, per kimberlite.
6) Undertake bulk density analysis:
   a) Establish bulk density per lithofacies; and
   b) Demonstrate that bulk density is constant with depth and dilution.
7) Integrate undiluted grade models and bulk density values with country rock dilution in the block model.
8) Validations:
   a) Calculated dilution in the block model against measured dilution in drilling; and
   b) Calculated grade in the block model against observed diamond recoveries in sampling.

The 2015 Mineral Resource Estimate for the Lynx and Hibou dykes, given the different emplacement model and simpler internal composition of kimberlite dykes, was established by applying average grades from surface trench samples within a geological model established by drill hole data.

14.4 Geological Models

The geological investigation of the Renard kimberlites is based on the megascopic observation (logging/mapping) of drill core, reverse circulation drill holes, underground drifting, magnetic susceptibility and the macroscopic and/or microscopic observation of representative petrographic samples of kimberlite. The combined data were used to differentiate geological units and in turn identify distinct internal phases of kimberlite, required for any economically relevant three dimensional (3D) geological model.

The Renard 3D geological models were constructed using Dassault Systémes GEOVOA Gems™ (Gems) and GEOVIA Surpac™ (Surpac) software. An initial 3D solid was constructed for the entire kimberlite body (including the CRB and, where applicable, the CCR) which was clipped by the topographic and overburden surfaces. Following this, 3D clipping solids were created for each geological unit and phase of kimberlite and systematically intersected or clipped with the initial kimberlite pipe, clipping from the outside units inward. Each phase of kimberlite is typically separated by steep, near-vertical and predominantly sharp, cross-cutting internal contacts.

Geological solid models were originally created by Stornoway for each of the Renard bodies included in this study: Renard 2, Renard 3, Renard 4, Renard 9, Renard 65, Lynx and Hibou. In 2015, GeoStrat updated the geological model for Renard 2 based on the results of the 2014 deep directional drilling program at Renard 2, and modified the shape of Renard 3 at surface where overburden stripping activities associated with mine development had exposed additional kimberlite.

14.5 Diamond Sampling Analysis and Grade Models

Diamond sampling of the Renard Project kimberlites includes core sampling, RC chip sampling, small tonnage trench sampling, and bulk sampling from both surface trenches and the underground exploration workings. These data have been collected, compiled and analyzed by a combination of methodologies in order to cater to the spatial distribution of sampling and the amount of information available for each kimberlite domain in each body.

14.5.1 Macrodiamond Sample Analysis

Table 14.2 summarizes diamond recoveries from bodies that were used for size frequency analysis and estimation of average stone size.

Table 14.2: Recovery Data Used in Diamond Size Frequency Analysis

Kimberlite	Weight (t)	Total Stones (+3 DTC)	Carat Weight (+3 DTC)	Sieve
Renard 2 UG	2,448.8	13,246	1,590.58	DTC (+3)
Renard 3 UG	2,113.7	24,407	2,753.995	DTC (+3)
Renard 4 Trench	2,104.2	25,480	2,688.71	DTC (+3)
Renard 65 Trench	5,346.8	8,489	1,003.2	DTC (+3)
Lynx Trench	494.3	5,271	515.32	DTC (+3)
Hibou Trench	543.9	8,401	781.49	DTC (+3)

These data and results for size frequency and carats per stone are considered representative of each body for the purposes of resource estimation.

14.5.2 Diamond Grade Models

The relationship between microdiamonds and macrodiamonds can be used to establish diamond content for a kimberlite through the creation of a grade model (Chapman and Boxer, 2004). Diamond grade models were created for each lithofacies in each kimberlite pipe where representative macro and micro diamond sampling was available. For Renard 2, representative samples weighing approximately 200 kg were collected for microdiamond recovery from each underground bulk sample round and placed in sealed 205 l drums for storage. Archived material from specific underground rounds, representative of specific kimberlite facies, were submitted for standard caustic fusion microdiamond recovery. Dilution estimations were made for each caustic sample and were complemented by measured line scan data for each underground round. Dilution estimates were combined with microdiamond and macrodiamond results to determine undiluted stones per 10 kg, undiluted stones greater than 0.6 mm per 10 kg, undiluted total stones and an undiluted model grade (Table 14.3). The model grade was adjusted to reflect non-recovery of small stones due to commercial DMS plant inefficiencies (i.e., the adjusted model grade is lower than the unadjusted model grade). No attempt has been made to account for potential higher stone recoveries in a more efficient commercial plant setting, or grade loss during sampling through diamond breakage.

The grade for the discontinuous hypabyssal units referred to as “c” units was determined in a similar manner as the other diamond grade models. Due to the discontinuous nature of the “c” units, macrodiamond data were not always available for modelling with microdiamond data. In such instances, the microdiamond curves were used in conjunction with selected macrodiamond data from more magmatic samples to provide guidance in the larger stone sizes. Macrodiamond data from unit Kimb2b, in conjunction with UG rounds that host appreciable mappable contributions from unit Kimb2c, were used to assist in forming the shape of the distribution curve. Kimb4c used 791.8 kilograms of caustic fusion data from drill core with a dilution value of 3.9% for microdiamond data, and a 509.6 t macrodiamond sample with 4.0% dilution from a surface exposure, to define the curve. No macrodiamond sample is available for Kimb65c, therefore the coarse portion of the Kimb65b macrodiamond distribution was used to assist in defining the macro portion of the Kimb65c curve. A total of 256 kg of drill core caustic fusion data, assigned a dilution value of 5%, comprises the microdiamond dataset.

Table 14.3: Macrodiamond and Microdiamond Samples Used to Determine the Micro/Macro Relationship in Renard 2, Renard 3, Renard 4 and Renard 65

	Kimb 2a	Kimb 2b	Kimb 2c	Kimb 3d/g	Kimb 3f	Kimb 3h	Kimb 4a	Kimb 4b	Kimb 4c	Kimb 4d	Kimb 65a/e	Kimb 65b	Kimb 65c	Kimb 65d
Microdiamond Sample Weight (kg)	4141.4	3844.3	1981.5	594.5	222.6	217.8	869.4	594.1	791.8	221.9	710.0	868.5	256.0	607.4
Microdiamond Dilution%	53	29	12	18	25	7	63	30	4	51	55	22	5	34
Microdiamond Total Stones	1496	3117	2815	365	80	203	194	291	1092	117	79	179	105	143
Undiluted Stones per 10 kg	7.6	11.4	16.2	7.5	4.8	10.0	6.1	7.0	14.4	10.8	2.5	2.6	4.3	3.6
Undiluted Stones >0.6mm per 10 kg	0.8	1.1	1.3	0.6	0.1	0.8	0.5	0.7	1.1	1.4	0.2	0.2	0.4	0.4
Macrodiamond Sample Weight (t)	140.4	515.7	7.4	384.2	317.9	143.4	158.6	158.6	509.6	1504.0	185.6	5080.0	-	185.6
Macrodiamond Dilution %	35	16	11	14	25	3	51	37	4	60	49	24	-	49
Macrodiamond Total Carats (+1 DTC)	58.2	646.7	19.3	559.1	278.6	274.8	61.5	61.5	938.2	1672.1	45.6	950.0	-	45.6
Macrodiamond Total Stones	512	5646	31.2	5207	2197	2931	927	927	9708	16376	639	8375	-	639
Model Grade Adjusted (cpht)	83	181	261	168	119	214	81	67	202	250	42	30	85	51

14.5.3 Microdiamond Sample Analysis

To test for the continuity of mineralization and stability of the diamond population with depth in the Renard 2 body, sampling for microdiamond analysis was undertaken on 50 m levels (horizontal slices) from surface to the base of the body at 1,000 m below the land surface (-500 masl). From each 50 m level, approximately 200 kg of Kimb2a, Kimb2b and Kimb2c was collected from drill core and treated for microdiamond recovery. A total of 4,141 kg of Kimb2a, 3,844 kg of Kimb2b and 1,982 kg of Kimb2c were analyzed by standard caustic dissolution.

The results for Kimb2a, Kimb2b and Kimb2c are presented in Figures 14.2, 14.3 and 14.4, respectively, as plots of stones per hundred tonnes per unit interval, which demonstrate that grades, on an undiluted basis, remain consistent with depth.

Figure 14.2: Kimb2a Microdiamond Sampling

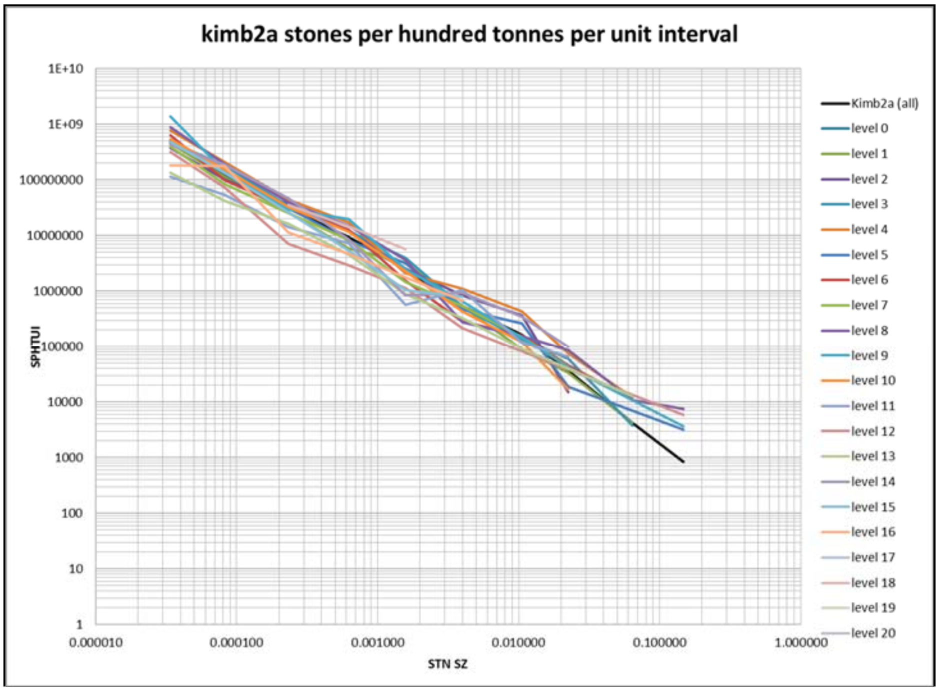


Figure 14.3: Kimb2b Microdiamond Sampling

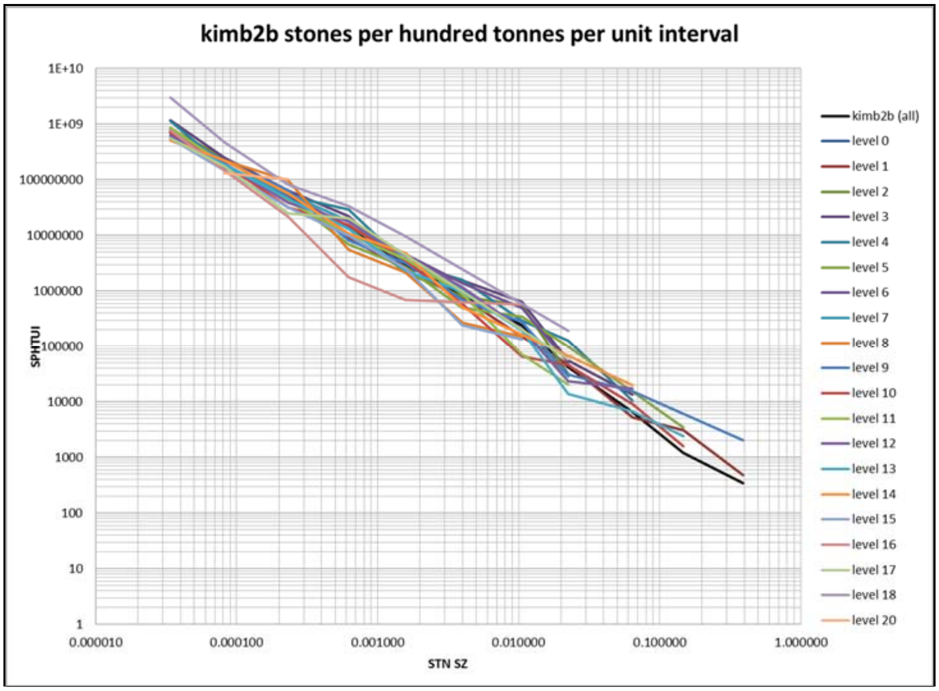
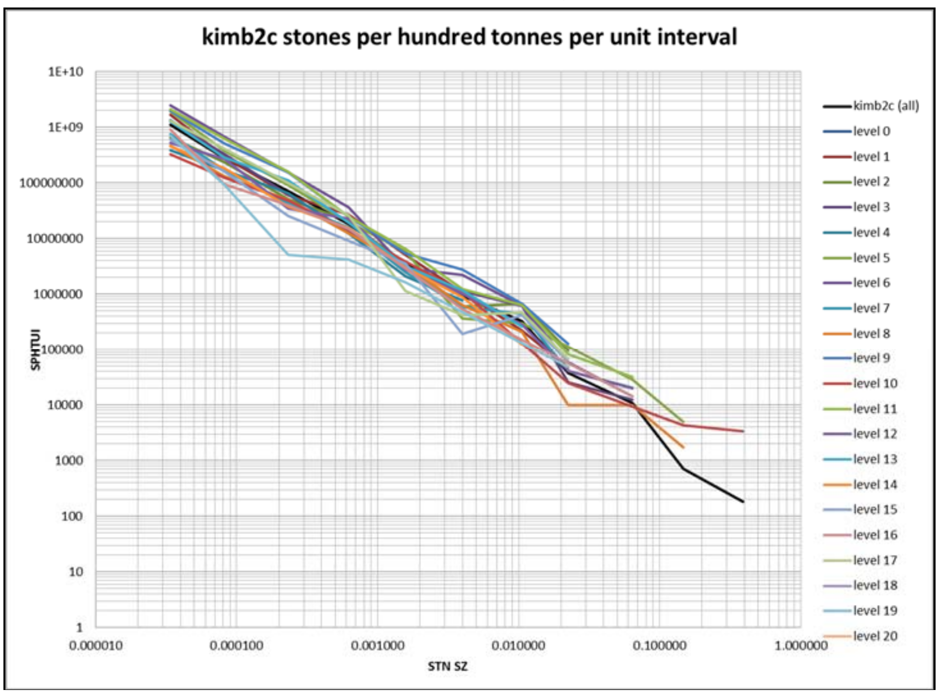


Figure 14.4: Kimb2c Microdiamond Sampling



14.5.4 Macrodiamond Size Frequency Modelling

Diamond size frequency modelling was undertaken for all remaining kimberlite lithologies to determine undiluted grades for all units (Table 14.4). This was carried out primarily using the macrodiamond samples and fitting the curves to remain parallel to the microdiamond curves established for each pipe.

Table 14.4: Summary of Macrodiamond Data Modelling

	Kimb3i	Kimb9a	Kimb9b
Macrodiamond Sample wt (t)	28.5	27.6	n/a
Macrodiamond Dilution %	42	59	n/a
Macrodiamond Total Carats	9.39	20.23	n/a
Macrodiamond Total Stones	77	150	n/a
Model Grade (cpht)	45	136	136

14.5.5 Mixed Sample Deconvolution

Deconvolution is the process of calculating individual lithofacies grades from mixed samples. Representative microdiamond and macrodiamond data for essentially pure Kimb3b and Kimb3c units were not available for grade modelling. Undiluted grades have been estimated through a back calculation methodology. Based on volume calculations from underground survey and mapping data, the proportion of undiluted Kimb3h, Kimb3d/g and Kimb3c have been calculated for underground rounds 3107, 3006, and 3008. The proportional contribution of total carats attributable to unit Kimb3d/g and Kimb3h was calculated based on pure grade calculations (see Table 14.3). This carat contribution was removed from the overall recovered carat total for each bulk sample round and the remainder attributed to the Kimb3c unit. The resulting weighted average grade for the Kimb3c unit is 230 cpht (Table 14.5).

A similar exercise was conducted for the Kimb3b unit and involved bulk sample rounds 3104 and 3105, which are mixtures of Kimb3h, Kimb3d/g and Kimb3b. The undiluted proportion of carats was calculated for Kimb3h, and Kimb3d/g and removed from the total carats recovered from the bulk sample rounds. The remaining carats were assigned to unit Kimb3b and the weighted average grade for Kimb3b is 101 cpht (Table 14.5).

Table 14.5: Calculated Macrodiamond Grades

	Kimb3b*	Kimb3c*
Macrodiamond Sample wt (t)	348.3	521.7
Macrodiamond Dilution %	50.61	11.72
Macrodiamond Total Carats	244.98	938.19
Proportional Carat Contribution (Kimb3h, Kimb3d/g)	136.06	476.57
Proportional Weight (kg)	108.2	201.1
Proportional Carat Contribution	108.92	461.62
Model Grade (cpht)	101	230

* Grades back calculated by stripping out contributions from Kimb3h and Kimb3d/g

14.5.6 Intrusions of Hypabyssal (Coherent) Kimberlite

Hypabyssal (Coherent) Kimberlite is present in each kimberlite pipe as coherent but discontinuous units (e.g., Kimb2c, Kimb4c, Kimb65c and Kimb3c). Where encountered in the underground decline on Renard 2 and Renard 3, the hypabyssal kimberlite units tend to be irregular in shape, orientation and both lateral and vertical continuity. Analysis of drill core shows a consistent proportion of hypabyssal kimberlite in each lithofacies unit of each body. In Renard 2, the average contribution of Kimb2c to the Kimb2a and Kimb2b units is 17.4%. For Renard 3, the percentage of Kimb3c present in Kimb3b, Kimb3d/g, Kimb3f, Kimb3h and Kimb3i is 14.3%. For Renard 4, the percentage of Kimb4c present in all kimberlite units is 12.3%. For Renard 65, the percentage of Kim65c is 9%. In Renard 9, the percentage of Kimb9c is 13.8%. Hypabyssal kimberlite is also observed within the peripheral Country Rock Breccia (“CRB”) and to a lesser extent in the Cracked Country Rock (“CCR”) units, outside of the Mineral Resource. The CRB in Renard 2 contains 9.47% Kimb2c.

14.6 Dilution Models

14.6.1 Dilution Sampling Analysis

Table 14.6 lists the sampling undertaken to establish the models of country rock dilutions in each kimberlite body. Different methods of dilution determination were employed by Stornoway over time. The most recent method used being “Line Scans”, where the amount of dilution in each metre of core was measured in a quantitative and consistent manner. Where available, both line scan and modal data were used to establish the dilution models. The older method, known as “Modal”, employed a visual estimate per 3-m geological unit. Whilst coarser in resolution, where sufficient data coverage exists, modal data yields similar overall dilution figures to the line scan data. The newer line scan data were averaged to mimic modal data and improved both representative and spatial coverage. Intersections of CR, CRB and CCR greater than a metre and less than 5 m within the kimberlite units were identified and flagged as xenoliths and included in the dilution data set. Xenoliths greater than 5 m were modelled in three dimensions, included in the geological and block models, and consequently removed from the dilution data set.

Table 14.6: Line Scan and Modal Dilution Data

Lithology	Line Scan			Modal		
	Count	Mean	Std Dev	Count	Mean	Standard Deviation
Kimb2a	9196	50.4	29.1	2161	48.3	29.4
Kimb2b	5337	29.2	23.2	407	32.1	20.7
Kimb3b	-	-	-	405	54.1	32.2
Kimb3d/g	162	33.9	17.4	541	37.2	24.8
Kimb3f	-	-	-	187	36.3	23.5
Kimb3h	-	-	-	117	24.8	22.9
Kimb3i	-	-	-	243	25.3	29.4
Kimb4a	330	59.6	20.3	1307	48.1	29.1
Kimb4b	131	32.5	17.5	703	37.9	20.9
Kimb4d	53	62.9	20.1	302	45.0	34.8
Kimb65a	413	56.5	23.0	625	50.7	27.5
Kimb65b	76	30.8	25.9	104	38.1	12.6
Kimb65d	223	38.4	15.7	132	52.7	14.7
Kimb9a	48	70.6	17.3	1553	59.0	31.4
Kimb9b	-	-	-	57	22.8	17.6

14.6.2 Dry Bulk Density

Table 14.7 summarizes the bulk density results. A single bulk density per lithofacies type is used in this resource evaluation.

Table 14.7: Summarized Density Data

Body	Lithology	Number of Samples	Average Bulk Density	Standard Deviation
Renard 2	Kimb2a	158	2.61	0.07
	Kimb2b	123	2.73	0.09
	Kimb2c	110	2.68	0.09
Renard 3	Kimb3b	11	2.61	0.04
	Kimb3c	27	2.66	0.05
	Kimb3d+Kimb3g	77	2.74	0.06
	Kimb3f	11	2.67	0.06
	Kimb3h	13	2.69	0.06
	Kimb3i	12	2.74	0.06
Renard 4	Kimb4a	136	2.59	0.05
	Kimb4b	45	2.72	0.06
	Kimb4c	35	2.61	0.06
	Kimb4d	25	2.59	0.06
Renard 65	Kimb65a+Kimb65e	67	2.58	0.07
	Kimb65b	39	2.78	0.05
	Kimb65c	26	2.61	0.09
	Kimb65d	39	2.7	0.09
Renard 9	Kimb9a	131	2.56	0.06
	Kimb9b	8	2.64	0.05
	Kimb9c	20	2.61	0.03
Lynx	HK	20	2.54	0.05
Hibou	HK	20	2.53	0.08
CR all bodies	GN, GN-GR, GR-GN, GR, PEG, CCR	170	2.71	0.07
CRB & CRB+K all bodies	CRB, CRB-2a, CRB-3, CRB+K, CRB+/-K	84	2.6	0.06

No correlation is found between variations in average bulk density within a kimberlite lithology and depth; moreover, for the most part, the samples for each kimberlite lithology tend to cluster in a narrow density range.

Much of the Renard kimberlite material contains a significant contribution of country rock xenoliths. These xenoliths vary in percentage and in degree of alteration (fresh to strongly altered), and could be expected to have an impact on the density of the samples; for example, if the density of fresh country rock is greater than that of the host kimberlite, or alternatively, if a strongly altered country rock xenolith is lower than the overall bulk density of the kimberlite. Bulk density samples from recent core drilling in and around the five main Renard bodies were line scanned prior to density determinations in an effort to establish a quantitative dilution estimate. There is minimal correlation between the sample results, suggesting that country rock dilution has a minimal effect on density.

14.6.3 Block Model Setup

A block model size of 5 m x 5 m x 5 m was selected as most appropriate to represent the small size of the kimberlite pipes, and as best suited to accommodate open pit mine planning, which is currently based on increments of 5-m bench heights (e.g., 10 m, 15 m). The block model is not rotated, and incorporates Renard 2, Renard 3, Renard 4, Renard 65 and Renard 9.

14.6.4 Estimation Process

The Ordinary Kriging (OK) interpolation method was used to estimate the country rock dilution into the block model using variography parameters defined from the geostatistical analysis and summarized in Table 14.8. Inverse distance squared (ID2) and nearest neighbour (NN) models were run to validate the kriging interpolation.

Table 14.8: Kriging Parameters

			Range				Range		
	Nugget	**x1**	**y1**	**z1**	**sill1**	**x2**	**y2**	**z2**	**sill2**
Kimb2a	260	20	20	25	78	75	75	155	160
Kimb2b	130	30	30	15	45	65	65	100	103
Kimb3b	500	5	5	10	400	10	10	30	150
Kimb3d/g	200	4	4	4	150	8	8	20	200
Kimb3f	250	5	5	5	100	10	10	20	200
Kimb3h	200	5	5	5	50	15	15	25	275
Kimb3I	400	20	20	20	450	-	-	-	-
Kimb4a	600	25	25	50	250	-	-	-	-
Kimb4b	225	12	12	35	190	45	45	80	90
Kimb4d	950	20	20	40	175	-	-	-	-
Kimb65a	400	30	30	50	275	-	-	-	-
Kimb65b	40	35	35	110	155	-	-	-	-
Kimb65d	50	18	18	55	230	-	-	-	-
Kimb9a	500	15	15	15	200	25	25	50	275
Kimb9b	250	10	10	25	60	-	-	-	-

14.6.5 Integration Process

For each kimberlite lithology, the undiluted grade for the block was multiplied by the estimate of the dilution for the block. For each rock type, if a factor was required to account for the Kimb2c/Kimb3c/Kimb4c/Kimb65c/Kimb9c component, this was applied. In the block model, the resulting grade was multiplied by the proportion of the rock type in the block, the relevant density was applied to each rock type and the results summed to establish the total carats and tonnes in the block.

Available information for the Lynx and Hibou dykes was examined, and average grades from the bulk samples were assigned as appropriate grades for the bodies.

14.7 Validation

14.7.1 Dilution Validation

Proper estimation of country rock dilution is a key determinant of the Mineral Resource confidence. Amongst many issues checked, dilution is the crucial key to this estimation. In Figures 14.5 and 14.6, estimates of dilution are shown against the measured dilution.

Figure 14.5: Estimated vs. Actual Dilution for Kimb2a

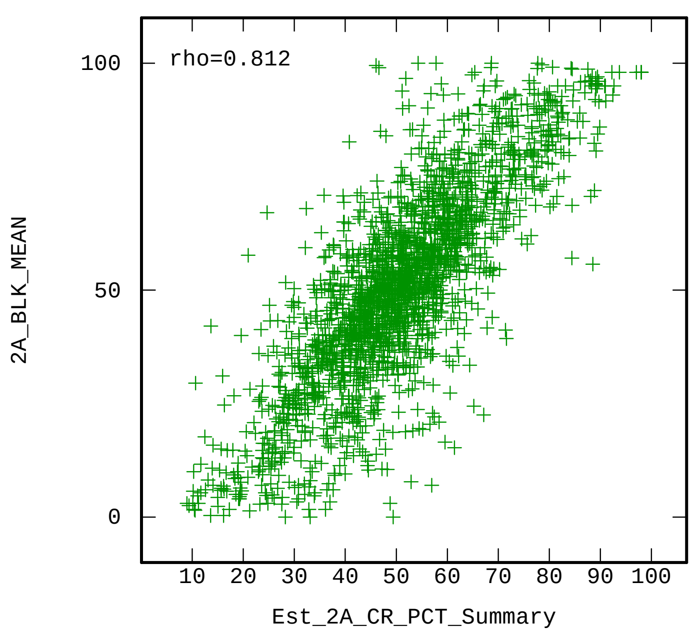
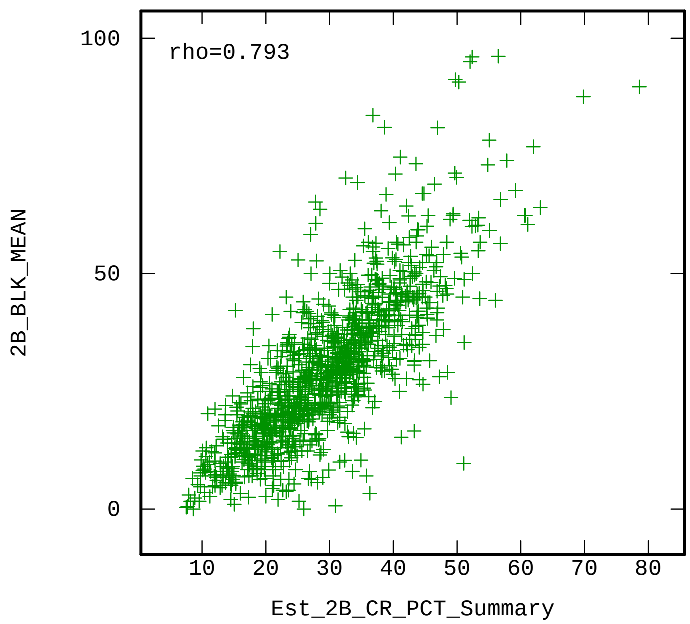


Figure 14.6: Estimated vs. Actual Dilution for Kimb2b



14.8 Diamond Price Estimates

Between May 9th and May 13th 2011, parcels of diamonds from the Renard 2, 3 and 4 kimberlite pipes were valued in an "open market" valuation in Antwerp, Belgium by WWW International Diamond Consultants Ltd (WWW) and four additional independent valuators. Diamond price models were developed by WWW based on the average of the five valuations, with "High" and "Minimum" sensitivities. In addition, WWW also valued diamond parcels from the Lynx and Hibou kimberlite dykes. In March 2013 WWW valued a parcel of diamonds recovered from the Renard 65 kimberlite pipe, and re-assessed their diamond price models for the Renard 2, 3 and 4 samples based on market price movements in the intervening period and a re-assessment of each samples diamond size frequency model. This exercise was repeated again in March 2014.

A description of the methodology of these successive valuations is given in Section 19.2. Diamond price estimates for the purposes of determination of the Renard Mineral Resources are derived from the March 2014 re-valuation exercise for Renards 2, 3, 4 and 9, and the May 2011 valuation exercise for the Lynx and Hibou kimberlite dykes, as illustrated in Table 14.9.

Table 14.9: Diamond Price Estimates for Resource Determination

Body	Valuation Sample (carats)	May 2011 Open Market Valuation[1]		March 2013 Valuation[1]		March 2014 Valuation[1]		Diamond Price Estimate for Resource Determination (US$/carat)[4]
		Observed Average Diamond Price (US$/carat)[2]	Diamond Price Model (US$/carat)[2]	Observed Average Diamond Price[3]	Diamond Price Model (US$/carat)[3]	Observed Average Diamond Price[3]	Diamond Price Model (US$/carat)[3]	
Renard 2	1,580	$173	**$182** (Sens. $163 to $236)	$161	**$190** (Sens. $151 to $214)	$167	**$197** (Sens. $178 to $222)	**$197**
Renard 3	2,753	$171	**$182** (Sens. $153 to $205)	$157	**$151** (Sens. $141 to $185)	$163	**$157** (Sens. $146 to $192)	**$157**
Renard 4[5]	2,674	$100	**$112 *($164)*** (Sens. $105 to $185)	$93	**$104 *($150)*** (Sens. $98 to $168)	$96	**$106 *($155)*** (Sens. $100 to $174)	**$106 *($155)***
Renard 65	997	*n/a*		$250	**$180** (Sens. $169 to $203)	$262	**$187** (Sens. $160 to $190)	**$187**
Renard 9[6]	*n/a*							**$155**
Lynx[7]	537	$99	**$119** (Sens. $99 to $144)	*n/a*				**$119**
Hibou[7]	771	$71	**$118** (Sens. $88 to $136)	*n/a*				**$119**

Notes:

1) *As determined by WWW International Diamond Consultants Ltd. at a +1 DTC sieve size cut off.*
2) *Based on the average of five independent valuations for Renard 2, 3 and 4 and a WWW valuation alone for Lynx and Hibou.*
3) *Based on the adjusted average of five independent valuations for Renard 2, 3 and 4 and WWW alone for Renard 65.*
4) *As recommended; see Section 19.2 for additional discussion.*
5) *Should the Renard 4 diamond population prove to have a size distribution equal to the average of Renard 2 and 3, WWW estimated that a base case diamond price model of US$164 per carat would apply based on May 2011 pricing, US$150 per carat on March 2013 pricing and US$155 per carat on March 2014 pricing.*
6) *Size frequency analysis of Renard 9 diamond samples indicates an overall similarity with other kimberlite pipes of the Renard cluster. The recommended price for Renard 9 for Mineral Resource determination purposes is based on a size distribution model equal to the average of Renard 2 and 3 and the valuation model of Renard 4.*
7) *The proposed mine plan for the Renard Diamond Project presently does not include a plan to mine the Lynx and Hibou dykes.*

14.9 Mineral Resource Classification

The reported Mineral Resources for the Renard Project kimberlites were constrained using drill density and sample distribution. Consideration was also made of the confidence with which diamond sampling results and the location of drill hole intercepts could be forecast. Classification of the resources in diamonds is an inclusive process, looking at aspects of geology, grade, revenue, density, diamond size frequency and continuity.

For the Renard kimberlite pipes, the Indicated Resource classification incorporates areas where substantial sampling has been undertaken. For Renard 2 this includes the RC, underground bulk samples and microdiamond sampling to 1,000 m depth. For the other pipes, it is limited to the RC hole depths that indicate grade continuity. These depths are supported by the more extensive core drilling, which is sufficient to delineate the resources at an Indicated level from the geological/lithological/volumetric/density aspect.

Sufficient carats have been recovered for revenue analysis, and support that the revenue is at the Indicated level of confidence for Renard 2, Renard 3, Renard 4 and Renard 65.

14.9.1 Reasonable Prospects of Economic Extraction

Canadian Institute of Mining and Metallurgy (CIM) standards and Securities Commission disclosure regulations require that a mineral resource can only be declared on a mineral deposit that has “reasonable prospects of economic extraction”. Work completed for the 2013 Optimization Study (Bagnell et al., 2013) supports GeoStrat’s opinion that the 2015 Mineral Resource has reasonable prospects of economic extraction, and is further buttressed by Stornoway’s successful $946 million mine financing and ongoing construction/development of the Renard Diamond Mine. Nothing in this NI 43-101 report refutes that opinion.

14.9.2 Uncertainty of Mineral Resources

Mineral Resources for the Renard Project documented in Section 14 are estimates and no assurance can be given that the anticipated tonnages and grades will be achieved or that the indicated level of recovery will be realized. Market fluctuations and diamond prices may render the mineral resources uneconomic. Moreover, short-term operating factors relating to the diamond deposits, such as the need for orderly development of the deposits or the processing of new or different grades of diamonds, may cause any mining operation to be unprofitable in any particular accounting period. Until they are categorized as “Mineral Reserves” under NI 43-101, the new Indicated mineralization outlined at Renard by this report is not determined to be economic ore. Stornoway is currently developing the Renard Project, with commercial production scheduled for 2017; however, profitability may be affected by ongoing engineering studies, the potential need for additional permits and general economic conditions.

14.10 Mineral Resource Statement

The 2015 Mineral Resource Estimate for the Renard Project is summarized in Tables 14.10 and 14.11.

The 2015 Mineral Resource Estimate takes into account geological, mining, processing and economic constraints, and is classified in accordance with the 2005 CIM Definition Standards for Mineral Resources and Mineral Reserves. The Mineral Resource Estimate is based on the continuity of geology between kimberlite at depth and kimberlite nearer surface, and the generally low variation in sample results for the different kimberlite phases with depth. There is potential for additional volume at depth for the Renard Project, as the geological models have been constructed conservatively in areas of limited drilling.

The Indicated tonnage reported in Table 14.10 and the Inferred tonnage in Table 14.11 lie within the solid model shells. Tonnage and grade estimates include estimation of internal dilution (barren material) within each body.

Table 14.10: September 2015[1] Indicated[2] Mineral Resources[3] – Renard Diamond Project

Body	Total Tonnes[4]	Total Carats[4]	Average cpht[5]	Average Dilution %
Renard 2 Total	25,696,000	21,578,000	84.0	54.7
Renard 2 w/o CRB[6]	*21,417,000*	*20,680,000*	*96.6*	*46.4*
Renard 2 CRB	*4,279,000*	*899,000*	*21.0*	*96.0*
Renard 3	1,820,000	1,859,000	102.2	33.5
Renard 4	7,246,000	4,437,000	61.2	48.9
Renard 65	7,865,000	2,300,000	29.2	42.8
Renard 9	0	0	0	n/a
Lynx	0	0	0	n/a
Hibou	0	0	0	n/a
Total	42,627,000	30,175,000	70.8	50.6

Notes:

1) Effective Date is September 24, 2015
2) Classified according to CIM Definition Standards.
3) The Mineral Resources have reasonable prospects of economic extraction.
4) Totals may not add due to rounding.
5) Carats per hundred tonnes. The diamond size cut-off was +1 DTC.
6) Excludes discrete more dilute kimberlite facies not previously incorporated into the July 2013 resource. Provided to facilitate a more direct comparison with the 2013 Mineral Resource Estimate.

Table 14.11: September 2015[1] Inferred[2] Mineral Resources[3] – Renard Diamond Project

Body	Total Tonnes[4]	Total Carats[4]	Average cpht[5]	Average Dilution%
Renard 2 Total	6,589,000	3,883,000	58.9	72.8
Renard 2 w/o CRB[6]	*4,080,000*	*3,356,000*	*82.3*	*58.5*
Renard 2 CRB	*2,510,000*	*527,000*	*21.0*	*96.0*
Renard 3	542,000	609,000	112.3	39.4
Renard 4	4,750,000	2,455,000	51.7	56.3
Renard 65	4,928,000	1,181,000	24.0	56.5
Renard 9	5,704,000	3,040,000	53.3	63.6
Lynx	1,798,000	1,924,000	107	n/a
Hibou	178,000	256,000	144	n/a
Total	24,490,000	13,348,000	54.5	n/a

Notes:

1) Effective Date is September 24, 2015
2) Classified according to CIM Definition Standards.
3) The Mineral Resources have reasonable prospects of economic extraction.
4) Totals may not add due to rounding.
5) Carats per hundred tonnes. The diamond size cut-off was +1 DTC.
6) Excludes discrete more dilute kimberlite facies not previously incorporated into the July 2013 Resource. Provided to facilitate a more direct comparison with the 2013 Mineral Resource Estimate.

14.11 Target for Further Exploration

In addition, Targets for Further Exploration (TFFE) were identified on the various kimberlites. *[Note: before changes to NI43-101 came into effect on June 30, 2011, TFFE were known as Potential Mineral Deposits (PMD)].* TFFE are derived from geological volumes based on projection at a standard 85° of the pipe margin in pipes, or to within 50 m of known borehole intersections in the case of the Lynx dyke. These are detailed in Table 14.12 as low and high ranges, both of which are considered to be geologically realistic. Total TFFE was identified as being between 76 and 113 Mt, at overall average diamond contents of 43 cpht to 63 cpht.

The potential quantity and grade of any target for further exploration is conceptual in nature; there is insufficient exploration to define a mineral resource and it is uncertain if further exploration will result in the target being delineated as a mineral resource. The 2015 TFFE includes TFFE contributions from Renard 1, Renard 7 and Renard 10 because sufficient data in the form of diamond drilling and sampling exists from these bodies to warrant additional exploration work.

Table 14.12: September 2015 Target for Further Exploration[1] Renard Diamond Project

	Low Range		High Range	
Body	**Total Tonnes**	**Average cpht**	**Total Tonnes**	**Average cpht**
Renard 2	6,138,000	60	15,472,000	100
Renard 3	3,352,000	105	3,773,000	168
Renard 4	11,120,000	50	15,358,000	77
Renard 65	29,026,000	25	40,926,000	33
Renard 9	3,858,000	52	6,327,000	68
Lynx	3,089,000	96	3,199,000	120
Hibou	3,469,000	104	4,028,000	151
Renard 1	8,620,000	20	12,983,000	30
Renard 7	6,342,000	30	9,431,000	40
Renard 10	1,217,000	60	1,730,000	120
Total[2]	76,232,000	43	113,227,000	63

Notes
1) Previously known as Potential Mineral Deposit prior to the June 30, 2011 changes to NI 43-101.
2) Totals may not equal the sum of the individual figures due to rounding.

15. MINERAL RESERVE ESTIMATE

15.1 Basis of Estimate

The requirements for Mineral Resources to be converted to Mineral Reserves are that:

- Only Measured and Indicated Resources can be included;
- The resources are part of a mining plan;
- Modifying factors of dilution and mining recovery are applied; and
- The mine plan passes an economic test.

Each of these requirements was addressed in establishing the Mineral Reserves for the Renard Project.

A detailed mine plan was developed to extract the Indicated Mineral Resources of the Renard Project. Dilution and recovery assumptions were applied, and cut-off grades were calculated using preliminary costs and diamond valuations.

Open pit and underground Mineral Reserves were estimated independently based on criteria specific to each method. A reconciliation of the Mineral Resources included in the open pit and underground mine plans was completed, confirming that all available resources were included.

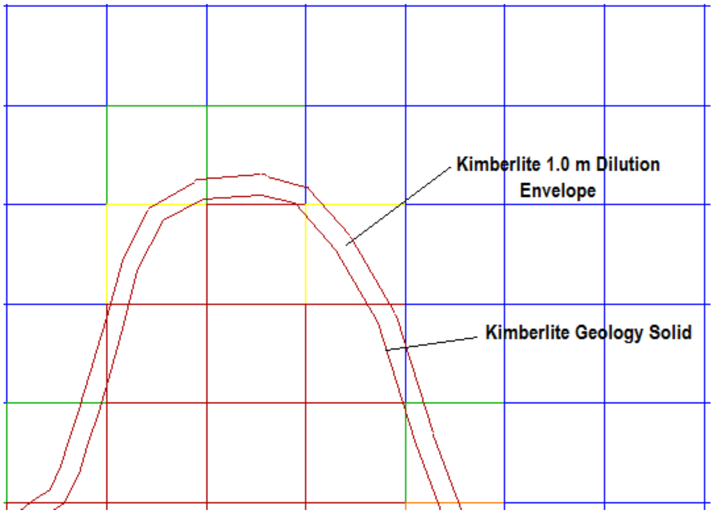
15.2 Open Pit

The Mineral Reserves for the surface mine design are reported according to the Canadian Institute of Mining, Metallurgy and Petroleum's (CIM) standards. According to these standards, Mineral Resource model blocks classified as Measured and Indicated are reported as Proven and Probable Reserves, respectively. Owing to the above reporting standards, Inferred Resources cannot be included as Mineral Reserves and so have not been included in the life of mine schedule. No Mineral Resource blocks are classified as Measured and therefore no part of the Mineral Reserve classifies as Proven.

The total Probable Mineral Reserve includes an ore dilution factor and an ore recovery factor to estimate a recoverable mining reserve.

The ore-outline mining shapes consisted of expanding the modelled kimberlite solids to incorporate a 1-m dilution envelope (Figure 15.1). Specific dilution factors were estimated for each kimberlite orebody through compilations performed on a bench by bench basis. The dilution envelope is assumed not to contain diamonds, and is therefore considered to be zero grade.

Figure 15.1: Illustration of Dilution Envelope



The dilution factors by bench per kimberlite pipe are shown in Tables 15.1 and 15.2. For mineral reserve estimation, a mine recovery factor of 98% is applied to R2/R3 and 98% to R65. The cut-off grades used for the Mineral Reserve estimation are shown in Table 15.3

Table 15.1: R2 & R3 & R65 Dilution by Bench

Bench	Bench Dilution (%)				
	R2	R3	CRB	CRB2a	R65
500	-	10.1	-	-	-
490	4.4	10.5	4.4	3.4	-
480	1.6	12.9	3.3	2.5	4.2
470	0.8	13.6	3.4	2.3	3.8
460	0.9	16.6	3.5	2.1	3.6
450	1.5	12.6	3.3	1.9	3.7
440	1.7	10.8	4.6	2.0	3.6
430	2.0	9.8	5.3	1.7	4.1
420	2.6	9.1	5.3	1.3	3.6
410	2.4	9.3	6.7	1.1	3.8
400	1.3	5.5	6.2	0.6	3.6
390	0.5	-	-	-	3.6
380	0.2	-	-	-	3.5
370	-	-	-	-	3.5
360	-	-	-	-	2.6
350	-	-	-	-	2.1
Average	1.72	11.44	4.32	1.75	3.51

Table 15.2: Average Dilution Factors

Dilution	R2 (Kimb2a; Kimb2b; CRB2a; CRB)	R3	R65
Diluted Ore Tonnage	4,333,020	793,719	4,578,679
Dilution Factor	2.73%	11.44%	3.51%

Table 15.3: Open Pit Cut-off grade (cpht)

R2 (Kimb2a; Kimb2b; CRB2a; CRB)	R3	R65
16.2	22.1	17.0

15.3 Underground

15.3.1 Estimation Procedure

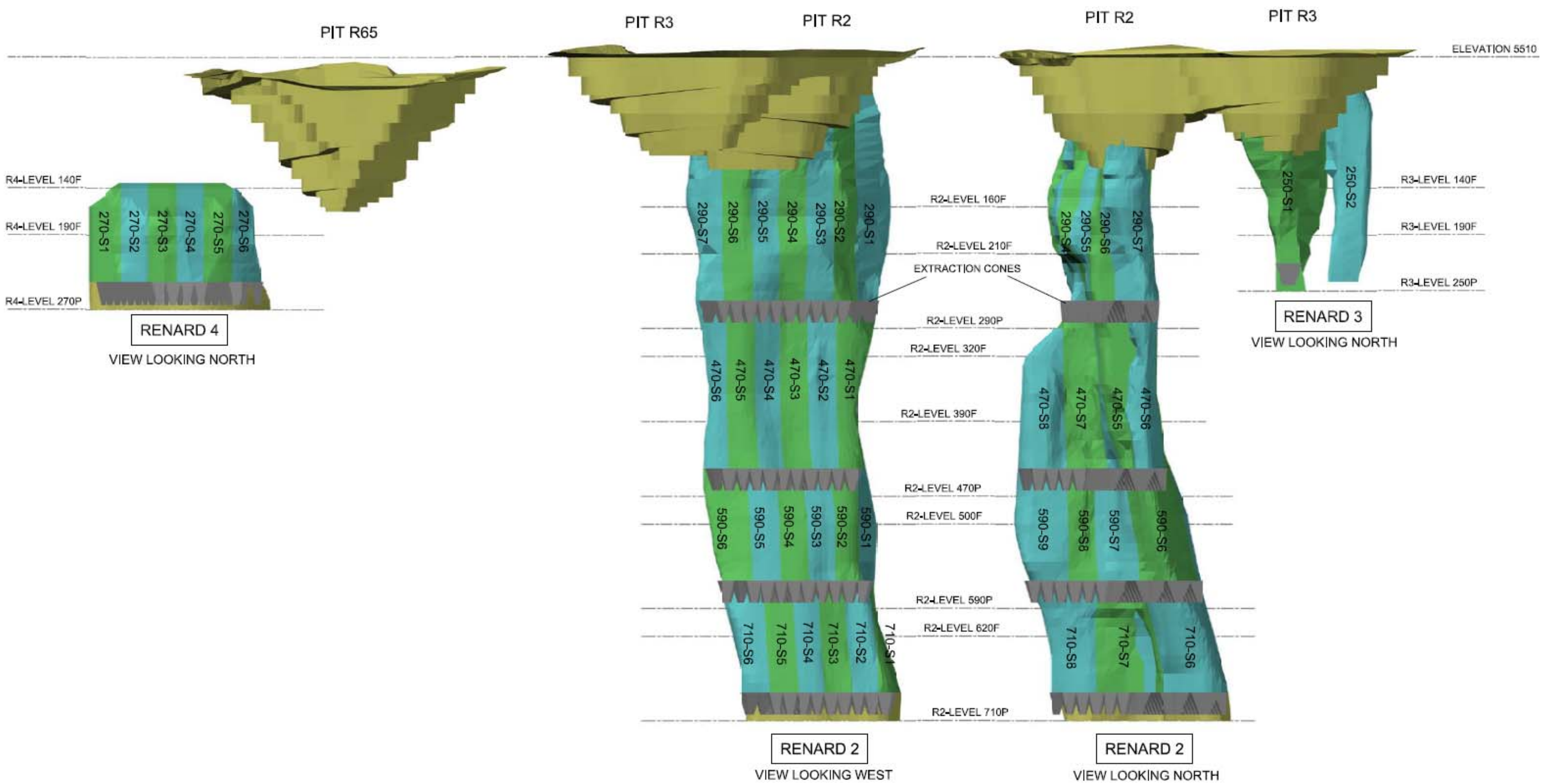
The first step in the estimation of the Mineral Reserves was to create 3D shapes for each of the proposed stopes based on the Indicated Mineral Resource model and the planned mining method. Practical shapes compatible with the planned development of the drawpoint levels and drill drifts, available drill patterns, the type of selected drilling equipment, and likely post-blast outlines, were created. The pipes outlines in some areas are irregular; in these areas, practical (smoothed) mining outlines were created, which typically encapsulated some waste rock and occasionally excluded some kimberlite.

The stope outlines were created by cutting a series of Mineral Resource plans every 10 m vertically, and then planning and digitizing stope outlines for each elevation. The individual stope outlines created over the full height of each pipe were created in Deswik software. Attributes were then assigned to each outline so that the software could create 3D solids by regrouping many outlines with same attributes. These attributes include the pipe number, level, workplace, drill direction, financial case, cost type, etc. At the end of the process, mining volumes were created for each drawcone and stopes, which are also subdivided in three or four volumes. Each of these stope volumes represents a portion of the main stope between two levels, sometimes with different drilling directions (up hole and down hole). For example, stope 3 on 290L includes eight drawcones and four stope volumes.

The planned development excavations were completed by first drawing the centerlines for the ramps, raises, drill levels and production levels. This was done with respect to the selected mining method and also to the limitations imposed by various geomechanical studies (e.g., subsidence model) performed to date. Once completed, each centerline segment was assigned attributes (pipe, level, etc.) and a tunnel type (main ramp, muck bay, refuge station, etc.). Drift dimensions and shapes were previously assigned to each drift type so that Deswik software would recognize which design to assign to each centerline segment during the 3D volume creation process.

Once all volumes were created, the tonnage and grade of all the Mineral Resources contained within each stope shape were queried using the resource block model, Deswik software and excel spreadsheet for the Mineral Resource to Mineral Reserve conversion. The stopes included in the underground Mineral Reserves are shown in Figure 15.2. As part of this process, the block model grades within the pipes were examined and, while there was some variation in grade across the levels, there was no real opportunity to eliminate local areas of lower grade kimberlite from the stope shapes given that BHS is a bulk mining method.

Figure 15.2: Stopes Included in the Underground Mineral Reserves



15.3.2 Modifying Factors

As discussed in Section 15.3.1, the stope shapes include any waste contained within them as internal dilution. The modifying factors of mining recovery and external dilution have been applied to the contained Mineral Resource and waste in the stope shapes.

The assumptions of recovery and dilution for BHS mining used for this study are listed in Table 15.4. Itasca's REBOP kinematic flow simulation software was used to provide more accurate values of these factors for the R2 kimberlite pipe. For the extraction of the R3 and R4 pipes, they were estimated based on benchmarking of other similar mining operations and on the experience of involved experts. Additionnal simulations with REBOP will be performed to optimise these factors as the mining strategy is being refined for R2, and new simulations will be done for R3 and R4 to validate current assumptions.

Table 15.4: Assumptions of Recovery and Dilution of BHS Mining

Modifying Factor	Pipe		
	R2	R3	R4
Mining Dilution	20%	14%	14%
Mining Recovery	82%	85%	78%

Considerations for recovery and dilution estimate assumptions for BHS mining are:

- With BHS, the first 35% of the production is swell, which is expected to have very low dilution. When the stope is full of blasted ore, the first 50% to 60% of the drawdown is also expected to have very low dilution and high recovery. When backfill waste starts mixing with the ore as it is drawn down, it is expected that dilution for each tonne drawn will start increasing significantly. The build-up of the dilution and recovery assumptions are shown in Table 15.5.

Table 15.5: Assumptions for Dilution and Recovery of BHS Mining for R2 pipe

Drawdown Stage in Stope	Portion of Total Ore in Stope	Recovery Estimates	Dilution Estimates
Shrinkage Blasting	35%	98%	5%
Middle Drawdown	32% (50% of ore remaining after blasting)	85%	12%
Final Drawdown	31% (47% of ore remaining after blasting)	71%	40%
Final Clean-up	2% (3% of ore remaining after blasting)	40%	105%
Subtotal	**100%**	**84%**	**20%**
Mining Losses[1]		2.0%	-
Total		**82%**	**20%**

Note: 1) Mining losses due to: (a) irregular geometry of the walls; (b) miss-holes and blasting problems; (c) ore stranded above arches in bottom drawpoint level; and (d) failure of brows in drawpoints.

- The placement of drawpoints and drawdown planning are critical to achieve mass flow and minimize "rat-holing" of the blasted ore in the stope during drawdown.
- The grade of all external dilution has been assumed to be zero, even if it is known that the CRB units, which surround the pipes in many locations, contain diamonds.

- For internal design dilution, a grade of 21 cpht has been assumed for the CRB above the 300 m elevation (Indicated Resources). The grade of the CRB below this elevation (Inferred Resources) was assumed at 0 cpht, the same for all other waste rock units.
- It is expected that much of the dilution will come from the open pit waste rock, to be placed on top of the blasted kimberlite, which will contain no grade.
- Material from drawcones located mostly in waste will be screened according to the following guidelines:
  - waste/ore percentage < 75% = 0%
  - 250% > waste/ore percentage > 75% = 33%
  - 750% > waste/ore percentage > 250% = 67%
  - 750% < waste/ore percentage = 100%
- Where a significant amount of blasted waste is included in the stope design, it was considered that it will either be left in the stopes (if on top of the ore) or screened (if below the ore) following the assumption that 33% will be mixed with the blasted ore and recovered, and 67% will be left in place at the end of the stope extraction. This was done on a stope by stope basis, making sure that it is operationally feasible.

At the very bottom level (710L for Renard 2 and 270L for Renard 4), the kimberlite remaining on the underside of the drawpoint troughs will not be recovered as part of BHS mining. To recover this ore, sublevel retreat mining (SLR) will be used at the very end of the mine life. Up holes will be drilled from all the drawpoints and scram drifts located below the troughs and cones, and then the holes will be blasted and the ore mucked in a retreat fashion.

For this Technical Report, the mine-life estimated average recovery and dilution factors have been applied to each individual mined block (drawcone, stopes and sills) as not enough information is available to perform this exercise on a stope by stope basis. These factors will certainly vary for each mined block.

The grade of all the stopes was compared to the cut-off grades shown in Table 15.6, and all stopes were found to exceed this cut-off.

Table 15.6: Underground Cut-off grade (cpht)

R2 (Kimb2a; Kimb2b; CRB2a; CRB)	R3	R4
29.2	36.6	37.1

15.4 Mineral Reserves

15.4.1 Open Pit Mineral Reserve Statement

The total in-situ open pit Probable Mineral Reserve is estimated at 8.91 Mt of ore at an average diluted grade of 44.4 cpht for 3.96 million carats, as at December 31, 2015. The Renard 3 pipe has the highest average diamond grade at 92.3 cpht, followed by Renard 2 at 59.6 cpht. Renard 65 is a lower grade pit containing 4.58 Mt of ore at an average grade of 30.1 cpht for 1.38 million carats. As of December 31, 2015, 152 kt of ore have been mined and stockpiled containing an estimated 112 thousand carats.

The open pit Mineral Reserves are summarized by bench in Table 15.7 and 15.9 and by kimberlite pipe in Table 15.8 and 15.10

Table 15.7: R2/R3 Open Pit Probable Mineral Reserves by Bench

Bench Level	Ore Tonnes (x 1000)	Carats (x 1000)	Avg. grade (cpht)
510	-	-	-
500	2.3	1.0	43.5
490	158.4	39.0	24.6
480	580.6	288.9	49.8
470	565.0	286.6	50.7
460	518.1	268.2	51.2
450	518.0	279.4	53.9
440	479.6	285.3	59.5
430	458.8	267.9	58.4
420	341.0	233.7	68.5
410	325.3	206.3	63.4
400	209.7	134.2	64.0
390	144.2	89.2	61.9
380	32.0	20.7	64.7
Total	4,333	2,581	59.6

Table 15.8: Open Pit Probable Mineral Reserves by Category by Kimberlite Pipe

Pipe	Classification	Tonnes (x 1000)	Grade (cpht)	Carats (x 1000)
R2	Probable	3,539	52.2	1,849
R3	Probable	794	92.3	733
Total	**Probable**	**4,333**	**59.5**	**2,581**

Table 15.9: R65 Open Pit Probable Mineral Reserves by Bench

Bench Level	Ore Tonnes (x 1000)	Carats (x 1000)	Avg. grade (cpht)
510	-	-	-
500	-	-	-
490	6.4	2.	31.3
480	213.5	66	30.9
470	411.6	136	33.0
460	560.6	144	25.7
450	401.1	130	32.4
440	423.1	128	30.3
430	419.0	123	29.4
420	405.6	119	29.3
410	381.3	114	29.9

400	358.1	107	29.9
390	281.7	87	30.9
380	270.8	84	31.0
370	200.7	63	31.4
360	153.2	48	31.3
350	75.2	22	29.3
340	17.0	4	23.5
330			
Total	4,579	1,376	30.1

Table 15.10: Open Pit Probable Mineral Reserves by Category by Kimberlite Pipe

Pipe	Classification	Tonnes (x 1000)	Grade (cpht)	Carats (x 1000)
R65	Probable	4,579	30.1	1,376
Total	**Probable**	**4,579**	**30.1**	**1,376**

15.4.2 Underground Mineral Reserve Statement

The underground Probable Mineral Reserves are summarized by pipe and mining zone in Table 15.11. All Mineral Reserves are in the "Probable" category. Of the three underground pipes in the Mineral Reserve, Renard 2 contains 81% of the tonnes and 86% of the carats, Renard 4 contains 14% of the tonnes and 9% of the carats, and Renard 3 contains 5% of the tonnes and carats.

Table 15.11: Underground Probable Mineral Reserve Statement

Zone/Level	Mining Parameters		Proven Reserve			Probable Reserve		
	Mining Recovery (%)	External Dilution (%)	Tonnes (,000 t)	Grade (cpht)	Carats (,000 c)	Tonnes (,000 t)	Grade (cpht)	Carats (,000 c)
R2 Pipe								
R2-290	82%	20%	-	-	-	5,111	63.3	3,235
R2-470	82%	20%	-	-	-	4,744	84.7	4,016
R2-590	82%	20%	-	-	-	4,750	89.9	4,270
R2-710	82%	20%	-	-	-	5,073	81.4	4,131
Sub total R2	82%	20%	-	-	-	19,679	79.6	15,654
R3 Pipe								
R3-250	85%	14%	-	-	-	1,222	70.2	858
R4 Pipe								
R4-270	78%	14%	-	-	-	3,458	48.3	1,671
TOTAL	**82%**	**19%**	**-**	**-**	**-**	**24,360**	**74.6**	**18,184**

15.4.3 Consolidated Mineral Reserve Statement

A consolidated summary of the Probable Mineral Reserves for the Renard Project including both underground and open pit mining is presented in Table 15.12. The table includes a breakdown of the Probable Mineral Reserves by pipe and mining method, as well as totals by pipe and mining method. In summary, 82.1% of the diamonds are mined by underground methods and overall 79.1% of the diamonds are contained in the Renard 2 kimberlite pipe.

Table 15.12: Probable Mineral Reserves Summary – Open Pit and Underground

Mine	Tonnes (k)	Grade (cpht)	Carats (k)	% Tonnes Total	% Carats Total	% Tonnes Method	% Carats Method
OPEN PIT							
R2	1,489	92.7	1,381	4.5%	6.2%	16.7%	34.9%
CRB2A	475	31.4	149	1.4%	0.7%	5.3%	3.8%
CRB	1,575	20.2	319	4.7%	1.4%	17.7%	8.1%
R2 Subtotal	**3,539**	**52.2**	**1,849**	**10.6%**	**8.4%**	**39.7%**	**46.7%**
R3 Subtotal	**794**	**92.3**	**733**	**2.4%**	**3.3%**	**8.9%**	**18.5%**
R65 Subtotal	**4,579**	**30.1**	**1,376**	**13.8%**	**6.2%**	**51.4%**	**34.8%**
TOTAL OP	**8,912**	**44.4**	**3,958**	**26.8%**	**17.9%**	**100%**	**100%**
UNDERGROUND							
R2-290	5,111	63.3	3,236	15.4%	14.6%	21.0%	17.8%
R2-470	4,744	84.7	4,017	14.3%	18.1%	19.5%	22.1%
R2-590	4,750	89.9	4,270	14.3%	19.3%	19.5%	23.5%
R2-710	5,073	81.4	4,132	15.2%	18.7%	20.8%	22.7%
R2 Subtotal	**19,679**	**79.6**	**15,655**	**59.1%**	**70.7%**	**80.8%**	**86.1%**
R3 Subtotal	**1,223**	**70.2**	**858**	**3.7%**	3.9%	**5.0%**	**4.7%**
R4 Subtotal	**3,458**	**48.3**	**1,671**	**10.4%**	7.5%	**14.2%**	**9.2%**
TOTAL UG	**24,360**	**74.6**	**18,184**	**73.2%**	**82.1%**	**100%**	**100%**
STOCKPILE	**153**	**73.5**	**113**				
TOTAL OP, UG & Stockpile	**33,424**	**66.5**	**22,255**	**100%**	**100%**		

Table 15.12: Probable Mineral Reserves Summary – Open Pit and Underground (cont'd)

Mineral Reserves by Pipe

Pipe	Tonnes (k)	Grade (cpht)	Carats (k)	% Tonnes Total	% Carats Total
R2 Pipe	23,218	75.4	17,504	69.8%	79.1%
R3 Pipe	2,016	78.9	1,591	6.1%	7.2%
R4 Pipe	3,458	48.3	1,671	10.4%	7.5%
R 65 Pipe	4,579	30.1	1,376	13.8%	6.2%
TOTAL PIPE	**33,272**	**66.5**	**22,142**	**100%**	**100%**

Mineral Reserves by Mining Method

Method	Tonnes (k)	Grade (cpht)	Carats (k)	% Tonnes Total	% Carats Total
Open Pit	8,912	44.4	3,958	26.8%	17.9%
Underground	24,360	74.6	18,184	73.2%	82.1%
TOTAL	**33,272**	**66.5**	**22,142**	**100%**	**100%**

TOTAL Mineral Reserves OP, UG & Stockpile

	Tonnes (k)	Grade (cpht)	Carats (k)	% Tonnes Total	% Carats Total
Stockpile R3+R65	153	73,9	113	0,5%	0,5%
Total Op & UG	33,272	66,5	22,142	99,5%	99,5%
TOTAL	**33,424**	**66,6**	**22,255**	**100%**	**100%**

Notes to accompany Probable Mineral Reserves Table

9) Probable Mineral Reserves have an effective date of December 31, 2015. The Probable Mineral Reserves were prepared under the supervision of Patrick Godin, Eng., Chief Operating Officer and Director of Stornoway Diamond Corporation. Mr. Godin is a Qualified Person within the meaning of NI 43-101.
10) Probable Mineral Reserves are reported on a 100% basis.
11) The reference point for the definition of Probable Mineral Reserves is at the point of delivery to the process plant.
12) Probable Mineral Reserves are reported at +1.0 mm diamond sized (effective cut-off of 1.0 mm).
13) Probable Mineral Reserves that will be or are mined using open pit methods include Renard 2, Renard 3 and Renard 65. Mineral Reserves are estimated using the following assumptions: Renard 2 and Renard 3 open pit designs assuming external dilution of 4.3% and mining recovery of 98%; Renard 65 open pit design assuming external dilution of 3.5% and mining recovery of 98%.
14) Renard 2, Renard 3 and Renard 4 Probable Mineral Reserves are mined using underground mining methods. The Renard 2 Mineral Reserves estimate assumed an external dilution of 20% and mining recovery of 82%. The Renard 3 Mineral Reserves estimate assumed an external dilution of 14% and mining recovery of 85%. The Renard 4 Mineral Reserves estimate assumed an external dilution of 14% and mining recovery of 78%.
15) Tonnes are reported as thousand metric tonnes, diamond grades as carats per hundred tonnes, and contained diamond carats as thousands of contained carats.
16) Tables may not sum as totals have been rounded in accordance with reporting guidelines.

15.5 Factors that may affect the Probable Mineral Reserve Estimates

Factors that may affect the Probable Mineral Reserve estimates include:

- New data from ongoing and upcoming sampling programs;
- Updates to assumptions used in estimating diamond carat content, including bulk density, pipe geometry and dimension, and grade interpolation method;
- Geological interpretation of internal kimberlite units and/or domain boundaries;
- Changes to mine design and/or planning parameters;
- Unforeseen mine geotechnical and/or hydrological conditions;
- Depletion due to mining or sampling;
- Further improvement, or deterioration, of process plant recovery;
- External influences on operating and sustaining capital costs, including without being limited to, energy costs and escalation;
- Diamond price and valuation assumptions;
- Foreign exchange rates, especially Canadian versus US;
- Variations to the permitting, operating or social licence regime assumptions, in particular if permitting parameters are modified by regulatory authorities during permit renewals.

15.6 Comments on Mineral Reserve Estimates

The responsible QP is of the opinion that the Probable Mineral Reserves for the Renard Diamond Mine appropriately consider modifying factors, have been estimated using industry best practices, and conform to CIM definitions.

16. MINING METHODS

A mine plan has been developed to exploit the Indicated Mineral Resources in the Renard 2, Renard 3, Renard 4 and Renard 65 kimberlite pipes. The mine plan for Renard 2 and Renard 3 combines open pit and underground mining methods, with pit depth determined through a trade-off study between open pit and underground mining costs. The Renard 65 mine plan employs open pit method only, whereas the Renard 4 mine plan employs underground mining method only. Since the Renard 4 kimberlite is located below Lake Lagopede, a 100-m-thick crown pillar will be maintained throughout the life of mine.

The Renard 2 and Renard 3 open pit excavations result in a single pit at surface with two pit bottoms centered on the respective orebodies. The excavation is therefore treated as a single pit and is referred to as the R2/R3 OP.

Operation in the R2/R3 OP started in March 2015 and the pit will be depleted in April 2018. Until the process plant start-up, expected in October 2016, all the kimberlite produced will be stockpiled. This ore stockpile will be sufficient to feed the process plant at a rate of 6,000 tpd until the underground mine comes into full production (6,000 tpd) in June 2018. Another 1,000 tpd of ore feed will be added once the R65 OP operation starts in July 2018.

The underground mine will be developed while mining the R2/R3 OP, and following its depletion, the underground mine will be extended to surface through the pit floor.

Waste rock from the open pit will be placed into the Renard 2 and Renard 3 underground stopes via the open pit to provide support for the stope walls during mining. Waste material to support the Renard 4 underground stopes will be backfilled through backfill raises from surface.

The mine design was developed by Stornoway. The overburden slope and hydrogeology assessment was performed by Golder, and the rock mechanics elements of the mine design were supplied by Itasca.

16.1 Overburden Pit Slope Assessment

This section summarizes the overburden slope assessment used to provide guidelines on the overall slope design for the open pits. The open pit designs are discussed in Section 16.4.

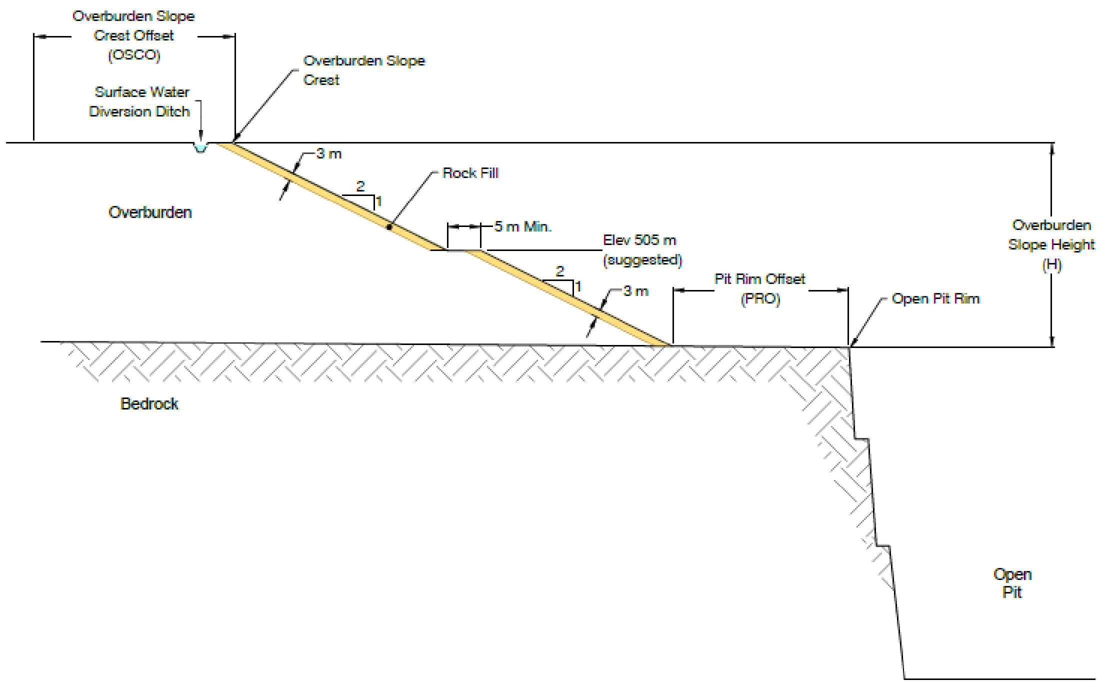
The geotechnical properties and parameters for the overburden used in the analyses were based on the geotechnical field investigations prior to 2014. Stability analysis of the overburden slopes for the R65 OP were performed in 2015 and were updated for the R2/R3 OP in 2015. The overburden thickness over the R2/R3 OP was determined with a model developed using the exploration drilling logs. The overburden thickness ranges from 0 m to 27 m with an average of about 10 m.

The analysis did not include the section of the pit perimeter within the R65 pound as no information was available on the soil conditions in the lake. A 2D stability assessment was performed. Selected cross sections of differing overburden thicknesses were analyzed. The position of the groundwater table was varied to assess its effect on slope stability. The groundwater level used in the assessment varied between El. 504 m and El. 514 m. The range of groundwater levels is based on results of site-specific monitoring.

Slope stability is affected by overburden conditions (thickness, stratigraphy, strength, etc.) and the position of the groundwater table. The typical overburden slope configuration is shown for the R2/R3 OP in Figure 16.1. The maximum slope angle is 2 horizontal to 1 vertical (2H:1V). Slope benches are limited to a maximum height of 10 m. The offset from the overburden slope crest in which no equipment or infrastructure is allowed depends on total overburden thickness. The pit rim offset (i.e., the distance from the toe of the overburden slope to the crest of the bedrock pit slope) also depends on overburden thickness.

A bench width of 5 m is to be provided. A 3-m-thick (perpendicular to the slope face) layer of waste rock or free-draining granular material is to be placed on the overburden slope face to provide additional stability against erosion and seepage from the overburden. The rockfill blanket thickness for the R65 OP can vary between 0 m, 1 m, 2 m and 3 m depending on which area the overburden is excavated. Golder has defined 4 areas with varying characteristics, all similar to the R2/R3 OP overburden recommendations. A surface water diversion ditch will be excavated along the crest of the overburden slope. A surface water diversion ditch is to be excavated in the bedrock near the crest of the bedrock slope to divert and collect water away from the open pit. Piezometric monitoring of the overburden slopes during excavation and throughout the life-of-mine is recommended. This monitoring will allow the determination of the phreatic surface to enable confirmation and assessment of the stability of the overburden slopes.

Figure 16.1: Typical Overburden Slope Configuration



16.2 Rock Mechanics

Most of the geomechanical empirical and modelling studies performed to date were completed by Itasca in collaboration with Stornoway. Rock mechanics laboratory testing was completed by the rock mechanics laboratories at the University of British Columbia and Université Laval (Québec), whereas Stornoway personnel were responsible for all on-site characterization, borehole logging, etc. Detailed individual reports have been completed on each of the aspects described below. A complete list of relevant reports prepared by Itasca has been included in the list of references in Section 27.

16.2.1 Site Investigations

Various field investigations for the feasibility-level studies have been conducted (by others) since 2005. These included on-site geological mapping (both at surface and underground), and several campaigns of geomechanical data collection, including joint orientation data from oriented drill holes.

During the 2009 drilling program, a total of 24 geomechanical/exploration boreholes were drilled between June and September for a total of 12,717 m. Of the 24 drill holes, 13 were oriented for further geomechanical investigation. During the 2010 field program, a total of 36 geomechanical/exploration boreholes were drilled between February and November for a total of 7367 m. All 36 boreholes were oriented for further geomechanical investigation. In 2014, a large drilling campaign (38 holes for a total of 12,036 m) was held to investigate the lower parts of Renard 2. During this campaign, all boreholes were oriented, and much emphasis was placed on the geomechanical logging of the core to compare and validate existing data from previous exploration campaigns. A select group of drill holes from past drill programs (prior to 2009) were chosen for supplementary geomechanical study, and these were updated to the same set of core logging standards as in recent drill programs. The updated drill holes were then added to the geomechanical database. By December 31, 2015, close to 37,000 m of core had been logged using Laubscher's Rock Mass Rating (RMR_{L90} – Laubscher, 1990), and approximately 31,000 m using Barton's Q system (Barton, 1994).

Five databases are being kept up-to-date as new data becomes available: the geomechanical database (RMRL90 and Q'), the joints database (stereonets), the laboratory testing database (UCS, triaxial, direct shear, etc.), the fault and in situ alteration database, and the RQD/FF database.

All recovered core was photographed and logged.

16.2.2 Laboratory Testing and Geomechanical Database

The descriptions, terms and standards used by Stornoway to construct the laboratory testing and geomechanical databases follow the terminology recommended by the International Society for Rock Mechanics (ISRM) and, in particular, the book *Rock Characterization and Testing: ISRM Suggested Methods* (1981).

The geomechanical field assessment identified the following geomechanical units for the purposes of mine design:

- ***Blue Massive Volcaniclastic Kimberlite (Kim2A)***, an extensively altered and generally massive. This blue to blue–green rock is comprised of olivine, juvenile clasts and country rock xenoliths that are poorly sorted, typically loosely packed and less commonly clast supported and set within a highly altered interclast matrix. Country rock xenoliths are comprised of locally derived granite and gneiss that are fresh to moderately altered.

- ***The Brown Coherent Kimberlite (Kim2B)***, a moderately altered, massive and texturally variable unit that displays both coherent and less common volcaniclastic (i.e., Tuffitic Kimberlite Breccia) textures. Macroscopically, the Brown CK is pale brown to black, and comprises two generations of olivine set within a crystalline groundmass containing common country rock xenoliths that typically show significant reaction to the host kimberlite. Country rock xenoliths include both granitoid and gneiss, although the latter predominates.
- ***Coherent Kimberlite (Kim2C)***, occurring in the form of late-stage dykes and irregular intrusions, and both are classified as Hypabyssal Kimberlite (HK). HK dykes are found throughout the body and typically occur along contacts between types of main pipe infill, along the pipe walls and throughout the marginal breccia. Dykes range in thickness from a few centimeters to 1 m, and contacts are often characterized by flow alignment of elongated minerals and carbonate veining. The HK is dark green to black and is comprised of two generations of olivine (macrocrysts and phenocrysts) that are evenly distributed and set within a homogeneous, crystalline groundmass.
- ***Country Rock Breccia (CRB)***, which can encompass both brecciated gneiss or brecciated granite or pegmatite. All types are present with the CRB of the Renard deposit. Granitic-type is most common. It is described as rock that consists of more than 95% country rock fragments with < 5% to no kimberlite component.
- ***Cracked Country Rock (CCR)***, a fractured country rock with varying degrees of fracturing. The breakage of this country rock can be related to kimberlite emplacement, whether as emplacement of the kimberlites or of a peripheral HK dyke.
- ***Gneiss (GN)***, consisting of fresh, competent, fine grained, banded, quartz-feldspar-biotite gneiss with less common coarse-grained sections displaying less banding.
- ***Granite (GR)***, consisting of fresh, competent, equigranular granite; primary mineralogy includes quartz, feldspar and biotite.
- ***Pegmatite (PEG)***, consisting of fresh, competent, equigranular pegmatite; primary mineralogy includes quartz, feldspar and biotite.

Samples from selected boreholes and rock types were collected for laboratory testing. All samples were tested at the UBC Rock Mechanics Laboratory in Vancouver and the Université Laval's Rock Mechanic Laboratory in the city of Québec. Laboratory testing included triaxial, uniaxial compression (including modulus of elasticity), direct shear and Brazilian indirect tension tests on intact rock samples. The laboratory tests were complemented by Schmidt hammer testing directly on core in the core shack at regular intervals.

As it is widely known that typical kimberlites are susceptible to weathering when exposed to humidity, a series of tests have been performed since 2012 to verify this possibility with the Renard kimberlites. These tests included three series of Slake Durability tests, two series of Field Slake Durability tests, X-ray diffraction and various checks of core exposed to outside weather conditions for long periods of time (months to years). Work is still in progress on this issue, but initial conclusions suggest that the weathering level varies within the various Renard's kimberlites, and that overall, they are less susceptible to weathering compared to other kimberlites around the world.

16.2.3 In Situ Stress

During the early stages of design, it is necessary to have reasonable estimates of in situ stress in order to provide the required input to numerical stress analysis models. Stresses were selected for the Renard Project based on information available from the World Stress Map. The current design considers a conservative range of stresses, within the range for comparable mines in the Canadian Shield.

16.2.4 Caveability Considerations

Caving methods were initially evaluated to determine whether they could be employed at the Renard Project. Prior to the analysis, Itasca reviewed the provided rock mass classification results. Based on the empirical and laboratory rock data, it was decided that all kimberlite units could be treated as equivalent in their geomechanical properties. Furthermore, there was no major variation in rock quality with depth within the kimberlites. The results of the weathering tests justified the assumption that only slight weathering was to be expected in the kimberlites. The Mine Rock Mass Rating (MRMR) empirical approach (Laubscher, 2000) was employed to determine whether the kimberlite domains are amenable to caving. Results from the various analyses concluded that in all cases considered for Renard, natural progressive caving was unlikely. It was concluded that the Renard orebodies will require some form of blast-assisted method.

16.2.5 Fault Analysis

Investigations into the presence of faults have been ongoing since 2010, including a campaign initiated at the end of 2015 to investigate a conductive structure intersected in the ramp vicinity. The fault analysis work included a review of prior work undertaken at Renard, orthophoto lineament analysis in July 2010, and groundtruthing and structural mapping from October 4 to 13, 2010. A first round of Datamine modelling of faults (as wireframe surfaces incorporating drill hole fault intercepts) was carried out in November 2010. Then, fault modelling was revised in May 2011 based on validations carried out by Stornoway. The most recent fault modelling work, conducted in 2012 and updated with all currently available information, included a detailed review of the previous model.

Presently, eight (8) brittle fault segments have been identified and modelled from drill core logging, surface and underground mapping and geophysical information. Modelled strike lengths range from 76 to 440 m, dip lengths are 116 to 736 m, and modelled thicknesses range from 0.3 to 18.4 m, based on the true thicknesses of the drill hole fault intercepts used to construct the fault wireframe solids. Two weak zones have been newly identified, and have been modelled based on 5- to 90-m-thick intervals of broken to rubbly material. The zones are shallow and have possibly been subject to near-surface weathering, and potentially influenced by forceful kimberlite emplacement. The focus for fault analysis has been on the Renard 2 and Renard 3 pipes given they will be mined first. Fault analyses will be ongoing as more data becomes available from the site.

16.2.6 Pit Slope Assessment

A series of open pit bedrock slope analyses for the Renard 2 and Renard 3 pits were carried out in 2010. Following a series of iterative analyses, the slope stability analysis was revised (in 2011) to take into consideration the updated and optimized final layout, geology and wall orientations of the pits.

The stability analysis using the standard approach, which includes stereonets and wedge analysis, was updated in 2013 using the most up-to-date information available in the pit areas. Also, the influence of the newly discovered weak zones (described in the previous section) was evaluated. In addition, modelling with FLAC3D software was performed to verify the influence of water on the overall slope stability and to address the presence of the newly identified weak Zone 1 in the pit. During the FLAC3D work, the influence of the geomechanical properties was tested by intentionally decreasing the original values until failure was obtained. Overall, the initial suggested design parameters from 2011 were confirmed following this phase of analysis.

In the fall of 2015, a feasibility-level stability analysis study was performed for the R65 pit by Golder. Initially, the R2/R3 pit design criterias were applied to the R65 OP assuming similar geomechanical properties for both areas. Using all data available from Renard 65, including geological/geomechanical core logs and field observations, Golder provided more reliable design parameters so that the Indicated Resources in this pit can be added to the mine reserves. Additional data from new boreholes will be required to confirm the suggested design criterias can be used for execution.

16.2.7 Subsidence Assessment

A subsidence assessment was carried out at Renard with the main goal of validating the proposed placement of surface infrastructure. Any critical life of mine or surface infrastructure should be located outside of the influence of mining-induced subsidence.

Subsidence limits were initially established using an empirical approach. A break angle of 85º was determined for the limit of discontinuous subsidence, and a limit angle of 80º was assumed as the limit of continuous subsidence. Detailed numerical analyses were completed in 2012 using *FLAC^{3D}*, which incorporated a more rigorous sensitivity analysis while considering a number of parameters; e.g., rock mass strength, influence of faulting, effect of backfill etc. The empirical limits were confirmed with the numerical models. Furthermore, analogues to similar kimberlite deposits were investigated. Based on the empirical and numerical analyses, as well as the analogue deposits, the subsidence limits determined at this stage of study appear to be adequate. As more information becomes available, the analyses should be revisited to ensure these limits will still be valid, particularly during the detailed engineering and execution phases.

16.2.8 Panel Retreating Sequence

The approach to mining at Renard is a hybrid method with two major phases: 1) a drilling and blasting phase, and; 2) an ore extraction phase. The drilling/blasting phase will be very similar to operations commonly undertaken in blast hole stoping mines. The ore extraction phase should be thought of in the same context as a caving style operation (minus the natural caving component), where draw control strategies are considered paramount for ensuring optimal dilution control to maximize recovery. In order to safely implement this method, a panel retreating sequence has been recommended whereby a mining horizon is subdivided into smaller panels, and mined in a retreating, pillarless fashion.

Only swell tonnes are to be pulled during the drilling/blasting process inside an active panel, in order to promote panel/wall stability. Blasted waste material is to be placed at the bottom of the pit on top of the blasted ore. The rationale behind this measure is based on the fact that broken rock (ore or waste) will:

- always provide some level of confinement to the pipe walls contributing to all aspects of stability for the mine;
- minimize the surface disturbance footprint due to subsidence;
- improve overall wall stability and minimize the risk of air blasts, which may possibly be caused by large falling blocks or hung-up material that suddenly falls.

16.2.9 Panel Dimensioning

Individual panels have been dimensioned with empirical stability methods (Laubsher's MRMR), using the geomechanical logging data. It was found that the panel dimension (footprint) must not exceed a maximum hydraulic radius of 12 m. Detailed 3D numerical modelling, incorporating the geometrical complexities and sequencing detail, is planned for the detailed engineering phase. Also, it has been recommended that a monitoring strategy be implemented to allow early detection of any movements occurring in the temporary sills while working in active areas and to serve as a means to monitor air gaps (beneath temporary sills) to keep these at a minimum.

16.2.10 Draw Level Design Guidelines

As part of ongoing optimization studies conducted in late 2011 and early 2012, the draw level design was updated to help improve recovery. A conventional caving draw level layout was proposed. This layout would lend itself to the panel retreat concept, meet operational and production requirements, and provide optimal recovery while minimizing waste entry.

In order to ensure the layout would provide adequate recoveries, kinematic flow modelling was conducted using Itasca's REBOP software. A parametric study was initiated, which considered two widely accepted variations on the Offset Herringbone layout, while testing sensitivity to various parameters; e.g., orebody width, mean fragment diameter etc. The two layouts only varied slightly, and it was decided to adopt a 15 m x 30 m (drawpoint spacing x haulage drift spacing) Offset Herringbone layout.

16.2.11 Fragmentation Considerations

Fragmentation is one of the key controlling parameters in terms of recovery and dilution for the planned mining method at Renard. Given the importance of fragmentation as a design parameter, empirical analyses were undertaken using the Kuz-Ram blast fragmentation model to help anticipate the fragmentation of the kimberlites resulting from the current blast design.

Based on successful R65 bulk sample trials, the Kuz-Ram model was used to anticipate the range of fragmentation that might be achieved during production blasting, with specific emphasis on R2. Sensitivity analyses were undertaken, with consideration given to explosive strength, drill hole spacing and to some degree, rock mass character. The results from this work were encouraging, and demonstrated that a fair amount of fragmentation control would be possible through blast design. The results of these analyses were used as a key input parameter for the global dilution and recovery analyses.

The blasting in the open pits is currently an opportunity to better understand the variations in the obtained fragmentation based on various drill/blast patterns and parameters. Fragmentation pictures are regularly being taken in the pits in order to perform size distribution charts of the blasted material. These will then be used to optimize the fragmentation of the waste backfill and to optimize the drill and blast strategy in the underground stopes. It is known that fragmentation is a key factor, with the current mining method, during the ore extraction sequence and also to allow a good control on the ore/backfill interaction at the top of the stopes. In situ fragmentation analysis will also be done by Itasca in 2016 to verify the natural fragmentation of the rock masses without adding explosives.

16.2.12 Global Recovery and Dilution Estimate for R2

Itasca's REBOP kinematic flow simulation software was used to simulate the recovery of the R2 kimberlite pipe. The REBOP simulations provided reasonable recovery values ranging from ~81% to ~87%. These simulations incorporated a number of the previously mentioned design elements giving due consideration to the anticipated fragmentation ranges for Renard; however, the values used must be verified once more fragmentation data becomes available from the site.

Additional REBOP modelling will be done as new fragmentation data becomes available and also as the mine design evolves. The draw strategy, consequently the recovery and the dilution factor, will be optimized with the use of this tool. Sensitivity analysis will also be performed with various ore/backfill fragmentation combinations and draw level/stopes designs.

16.2.13 Ground Support Standards

The design of ground support for Renard was mostly based on the empirical guidelines published by Barton (2002), and Grimstad and Barton (1994). These guidelines are based on rock mass quality, size of the opening, and intended use of the excavation. The prescribed ground support specifications were based on standards currently adopted in Canadian hard rock mines, and are in compliance with good engineering practice.

Current ground support standards includes 2.4 m to 3.7 m rebars with screen in the back in a 1.4 m x1.2 m pattern, with 0.9 m rockbolts and 1.8 m split set with screen in the walls. In the intersections or wider excavations, 6.1 m cablebolts in a 2.0 m x 2.0 m pattern are added to provide long-term stability. Other types of ground support, such as swellex and shotcrete, are available depending encountered ground conditions. Inspections are performed on a daily basis to verify ground conditions as the tunneling progresses and ground support requirements are adapted as required.

16.2.14 Crown Pillars

The Renard 4 kimberlite pipe is located below a lake. Consequently it is important to ensure the stability of the Renard 4 crown pillar since an uncontrolled failure could lead to water inflow into the mine during production. This would represent a significant risk to the safety of personnel working underground.

Empirical studies have been undertaken using the approach proposed by Carter (1992). This work suggested that a very thick crown pillar would be required for stability unless the spans could be reduced. Given the importance of crown pillar stability, Stornoway requested that 3D global stress numerical modelling be undertaken to assess the stability of a 100-m-thick crown pillar.

The modelling for the crown pillar in the Renard 4 pipe indicates that the crown pillar is stable under base case assumptions. It was further shown that within the range of reasonable assumptions with regards to material properties or stress, there are combinations in which the crown pillar is unstable. The greatest sensitivity in the modelling performed to date is the presence of a horizontal stress at surface.

The models suggest that if a flat-back profile was to be used, the depth of failure into the crown pillar will be between 25% and 35% of the span of the orebody. This coincides well with the general rule of thumb in which arching is assumed to occur at 1/3 of span. The back of the stope should, within reasonable bounds, be formed to create such an arch with the blasted profile to limit the ability of the rockmass to define its own stable arch (which can be quite different depending on local jointing). The peak of the arch should not be above the 100-m limit defined by the crown pillar. Additional work will be performed prior to mine R4 in order to better assess the crown pillar situation and the associated risks.

16.2.15 Limitations

For any mining project, there are always unknown rock mechanics and hydrogeological conditions that cannot be predicted ahead of actual mining. These unknown conditions, such as faulting, zones of weak rock, or zones of unanticipated water inflow, may only be discovered during mining and may require significant changes to the mining plan. While additional laboratory testing and characterization work should always be ongoing at every stage of project development to reduce uncertainty, it is never possible to carry out enough drilling/characterization work ahead of time to identify all of these potential risks. This may in turn have significant effects on cost, as relates for example to slope angles, dewatering and/or underground development, excavation size, etc.

It must be understood that there are limitations in the ability of numerical model results to assess all potential failure modes (effect of faults, effect of rockmass variability, the effect of draw, and the stabilizing effect of leaving ore in the stope are some examples of factors beyond the explicit capabilities of the models run for this Project to date). Further work should be performed to reduce the uncertainties in the analysis and to examine other potential modes of failure. This could be done in more detailed engineering studies at a later stage of the Project and would include the following.

- An analysis to represent the effects of jointing on the rockmass and the effects of major faults either through the Ubiquitous Joint Rockmass Model in FLAC3D or by using 3DEC;
- An analysis to consider rockmass variability in which the entire spectrum of rockmass strengths is respected in the model in a statistically valid way; and
- If additional faults are found to intersect the region of the Renard 2 pipe, they should be explicitly considered in both the stress modelling and the hydrogeology modelling.

As underground experience is gained at this mine site, these analyses should be continually reviewed to determine if they are still valid over time and to ensure that new information is properly considered into the analysis.

16.3 Hydrogeology

Hydrogeological analyses were initiated by Itasca Denver Inc., in coordination with Itasca Consulting Canada Inc., in support of the 2011 Feasibility Study. The work included compiling the existing hydrogeologic data; constructing a numerical, three-dimensional (3D) groundwater flow model of the site; and estimating the potential groundwater inflows to the mine workings for the feasibility-case mine plan.

The groundwater flow model constructed for this investigation uses the numerical code MINEDW (Azrag et al., 1998) developed by Itasca, which solves 3D groundwater flow problems with an unconfined (or phreatic) surface using the finite-element method. MINEDW has several attributes that were specially developed to address conditions often encountered in mine dewatering. The modelling code has been used and verified for numerous mine dewatering projects throughout the world.

The geometry of the open pits and underground mine workings was provided by Stornoway. The boundaries of the groundwater flow model domain were selected far away from the mining area where the hydraulic stress induced by mining is insignificant. The water levels of the major lakes were obtained from Stornoway.

During 2010, WESA conducted packer testing in two existing boreholes that were selected by Stornoway and Itasca for the purpose of determining the hydraulic conductivity of the rock units. Data was also obtained from various sources, including the previous hydrogeologic field investigations that have been conducted at the Renard site (by GENIVAR, WESA and Roche) between 2005 and 2010. The hydraulic conductivity of the country rock generally decreases with depth, as expected, but shows high variability within 50 m of the ground surface.

In early 2012, the responsibility of the hydrogeological analysis was transferred to Golder who validated the work previously performed by Itasca. Three drilling campaigns were completed under their supervision since then. During the summer of 2012, packer testing was performed by Qualitas in existing holes to verify the hydraulic conductivity of fault 28, which is suspected to link Lake Lagopede with the surface and underground excavations. Monitoring wells were also installed at key locations to verify the variations in the water table as mining progresses. Then, during the fall of 2014, Qualitas completed pumping tests in three holes in the weak Zone 1, and the fresh and return air raise areas. Additional monitoring wells were installed in the soil, generally in the open pit areas. In December 2015, Golder assisted Stornoway during the investigation of a structure providing a significant amount of water to the face of the underground ramp. A total of eight holes were drilled and packer tests/geocamera surveys were completed in all these holes. Packer testing and geocamera surveying were also done on the fresh air raise pilot hole.

16.3.1 Prediction of Inflow

Itasca models are considered to be outdated and should only be used as reference. Golder is currently working on updating the hydrogeological model with the new information available in order to provide more updated water inflow volumes throughout the mine life.

The 2012 Itasca model considered the mining sequence and schedule, and estimated the inflows over time. The inflows reach an initial peak in 2018 with 2,300 m^3/day (421 USgpm) after 37 months of development, and then gradually drop down to a long-term inflow varying between 1,200 m^3/day (220 USgpm) and 1,600 m^3/day (294 USgpm) before reaching a life-of mine overall peak of 2,475 m^3/day (454 USgpm) late in the mine life when mining on R2 610L horizon begins.

It was found that the predicted inflow is highly sensitive to the hydraulic conductivity of the country rock. Increasing the hydraulic conductivity value by one order-of-magnitude would result in the predicted inflow rate three times greater than the value predicted from the base case.

The groundwater flow model was updated by Golder in early 2015 with the mining strategy and schedules as developed at that time. The purpose of this update was to determine the expected water infiltration flow during the ramp development, which was used for sizing dewatering pumps. In the 2012 model, it was shown that the maximal inflow in the ramp development should reach 1,400 m^3/day (256 USgpm) if it does not intersect fault 28. When the design was updated in the model, it was noticed that the new ramp could potentially intersect fault 28. In this case, the model showed that the peak inflow could reach 3,500 m^3/day (650 USgpm) and the nominal flow 2,050 m^3/day (376 USgpm). The ramp design has since been revised to avoid fault 28 and its associated water inflows.

As mentioned above, the model will shortly be updated again by Golder with the newly obtained data and new inflow rates will be proposed for various milestone periods in the open pit and underground operations. With the current mining strategy, underground excavations will break through the bottom of the pit in early 2018. At this time, precipitation falling on the Renard 2 pit will not be contained before flowing down to the underground operations. Work is underway to estimate the possible water volume from precipitation to be managed during 1:10 year and 1:100 year events. The mine design and dewatering strategies will be adapted to meet these parameters, in addition to managing the average water infiltration volumes.

16.3.2 Limitations

The confidence level of the groundwater flow model prediction depends on the available hydrogeologic data obtained at the site. Currently, there are limited data on the hydrogeologic conditions of both the overburden and the country rock at the site, the interaction between the surface water bodies and the groundwater system, and the hydrogeologic conditions of the faults. The model used the existing data in the predictive simulations, the results of which are subject to change depending on any new data collected in future investigations. These limitations should be addressed in future model updates.

16.4 Open Pit Design

The mining strategy for the Renard orebodies consists of recovering the upper portions of Renard 2, Renard 3 and Renard 65 through open pit mining methods, and the deeper portions of Renard 2, Renard 3 and Renard 4 using an underground blasthole shrinkage mining method. The size of the open pit was established through a trade-off study that optimized the profit from mining by open pit versus underground mining techniques.

The Renard 65 pit is of lower grade than Renard 2 and Renard 3 and is mined as supplemental feed to the underground operation once the R2/R3 OP is depleted. Waste rock mined from R65 will be used as underground backfill material.The Renard 65 OP will also be used to collect surface runoff water.

16.4.1 Open Pit Optimization

Open pit optimization was carried out using the Whittle software package implementing the Lerch-Grossman algorithm. The Whittle optimizations were performed using the Renard 2015 resource model provided by Stornoway. This 43-101 mining update considers mineralization classified in the Indicated Resource category, which limits open pit mining to the Renard 2, Renard 3 and Renard 65 kimberlite ore bodies.

A mining dilution factor was estimated for each pipe by including a 1 m envelope around the ore. The resulting dilution factor, which was employed in the optimization process, was used to calculate the diluted grade.

The optimization parameters used to generate the Whittle shells are summarized in Table 16.1. These parameters are alligned on the 2013 Optimization Study and adjusted with actual cost histories. For optimizations, an ore recovery of 96% was used for R2/R3 OP and 100% for R65 OP. However, for mineral reserve estimates, a uniform 98% was used for the open pits.

Table 16.1: Summary of Open Pit Optimization Parameters

Parameters		Renard 2	Renard 3	Renard 65
Diamond price	C$/carat	197	157	187
Marketing expense	%	3.0%	3.0%	3.0%
Royalty + IBA payment	%	3.0%	3.0%	3.0%
Metallurgical recovery	%	100%	100%	100%
Open pit mining dilution	%	2.7%	11.4%	3.5%
Open pit mining recovery	%	96.0%	96.0%	100%
Total processing & rehandle	C$/t processed	13.72	13.72	13.34
G&A cost	C$/t processed	15.50	15.50	15.50
Total ore based cost	C$/t processed	29.22	29.22	28.84
Cut-off grade	cpht	16.2	22.1	17.0
Reference mining cost	C$/t mined	Owner	Owner	Owner
Kimberlite ore	C$/t mined	4.13	4.13	4.13
Granite & gneiss waste rock	C$/t mined	4.13	4.13	4.13
Overburden	C$/t mined	3.45	3.45	3.45
Incremental bench cost	C$/t mined/10m	0.041	0.041	0.041
Whittle overall slope angles		Renard 2	Renard 3	Renard 65
Kimberlite ore	Degrees	45	45	49/53/54
Granite & gneiss waste rock	Degrees	45	45	49/53/54
Overburden	Degrees	20	20	20

Several optimization iterations and cases were performed for the R2/R3 OP imposing certain surface and pit depth constraints. The selected R2/R3 OP shell is not limited by surface constraints, but is limited to elevation 380 in the pit where the current underground mine is designed to recover the ore body. The R65 OP is constrained at surface due to the road and lake bounding the pit on the south side and does not consider that the ore body will be mined by underground methods. The optimum pit shell for R2/R3 OP and R65 OP are presented in Table 16.2.

Table 16.2: Summary of Optimal Shell used for Detailed Open Pit Design

Whittle Results		R2/R3	R65
Shell number		22	23
Revenue factor		0.74	0.9
Total tonnage	Mt	11.01	9.14
Waste tonnage	Mt	6.48	4.96
Ore tonnage	Mt	4.54	4.18
Diamonds	carats	2,465,000	1,265,000
Grade	cpht	54.4	30.3
Strip ratio	Mt : Mt	1.4 : 1	1.2 : 1

16.4.2 Bench Geometry

Bench heights of 10 m were selected to facilitate efficient drilling and blasting activities. The loading units will mine the benches in 10-m-high cuts. The pit slope profile was determined using geotechnical recommendations from Golder and Itasca.

The modelled overburden thickness varies from 5 m at the edges to > 27 m over the center of the orebodies. The slope configuration recommendations vary based on overburden thickness and are presented in Table 16.3.

Table 16.3: Overburden Slope Configuration Recommendations

OVB slope height (m)	Overburden Slope Configuration – R2/R3				
	Upper slope (max 10 m)	**Berm width (m) min.**	**Lower slope**	**Min. OVB slope crest offset**	**Min. pit rim offset (m)**
0 – 6	2H : 1V	---	---	10	8
6 – 10	2H : 1V	---	---	10	8
10 – 20	2H : 1V	5	2H : 1V	20	15
> 20	2H : 1V	5	2H : 1V	20	20
	Overburden Slope Configuration – R65				
Zone	**Upper slope (max 10 m)**	**Berm width (m) min.**	**Lower slope**	**Min OVB slope crest offset**	**Min pit rim offset (m)**
1	2H : 1V	---	---	4	3
2	2H : 1V	5	2H : 1V	4	3
3	2H : 1V	5	2H : 1V	4	5
4	2H : 1V	--	--	4	9

Itasca made recommendations for the hard rock slopes for the R2/R3 open pit. Itasca's pit slope recommendations are uniform for all pit walls. The recommended rock slope configuration is presented in Table 16.4. Golder was tasked to provide geotechnical parameters for the R65 pit. Three main zones were defined with azimuth angles where only catch bench width will vary to provide desired inter-ramp slope angles in the rock.

Table 16.4: Rock Slope Configuration Recommendations

Pit	Rock Slope Configuration				Azimuth
	Final bench height (m)	**Catch bench width (m)**	**Bench face angle (°)**	**Inter-ramp angle (crest to crest)**	
R2	20	9.0	80	57.9	
R3	20	9.0	80	57.9	
R65	20	11.6	90	60	235º to 045º
R65	20	12	90	59	045º to 170º
R65	20	14	90	55	170º to 235º

16.4.3 Ramp and Haul Road Design

The ramp was designed to accommodate double-lane traffic. The largest equipment considered for the Project is a 60-ton class rigid truck. The maximum ramp gradient is 10%. Single lane ramps are used to access the pit bottoms. Out-of-pit haul roads are designed for double-lane traffic. Double-lane ramps are 21.8 m wide and single-lane ramps are 13.7 m wide.

A road network connects the pits to the crusher and stockpile area, the waste dump, the overburden stockpile and the PK facility.

16.4.4 Open Pit Design

The Renard 2 and Renard 3 open pit excavations result in a single pit at surface with two pit bottoms centered on the respective orebodies. The excavation is therefore treated as a single pit and is referred to as the R2/R3 open pit. The R2/R3 final open pit is too small to allow for internal pit phases. The R65 OP is centered on one main kimberlite and will be mined in three phases to maximize value and sequence waste mining.

The dimensions of the pits are summarized in Table 16.5, and are presented in Figure 16.2 and Figure 16.3.

Table 16.5: Renard Pit Dimensions

Pit	Pit dimensions (m)	Pit depth (m)	Pit footprint surface area (ha)
R2/R3	455 x 320	130	12.34
R65	440 x 350	155	10.88

Figure 16.2: Final R2/R3 Open Pit Design

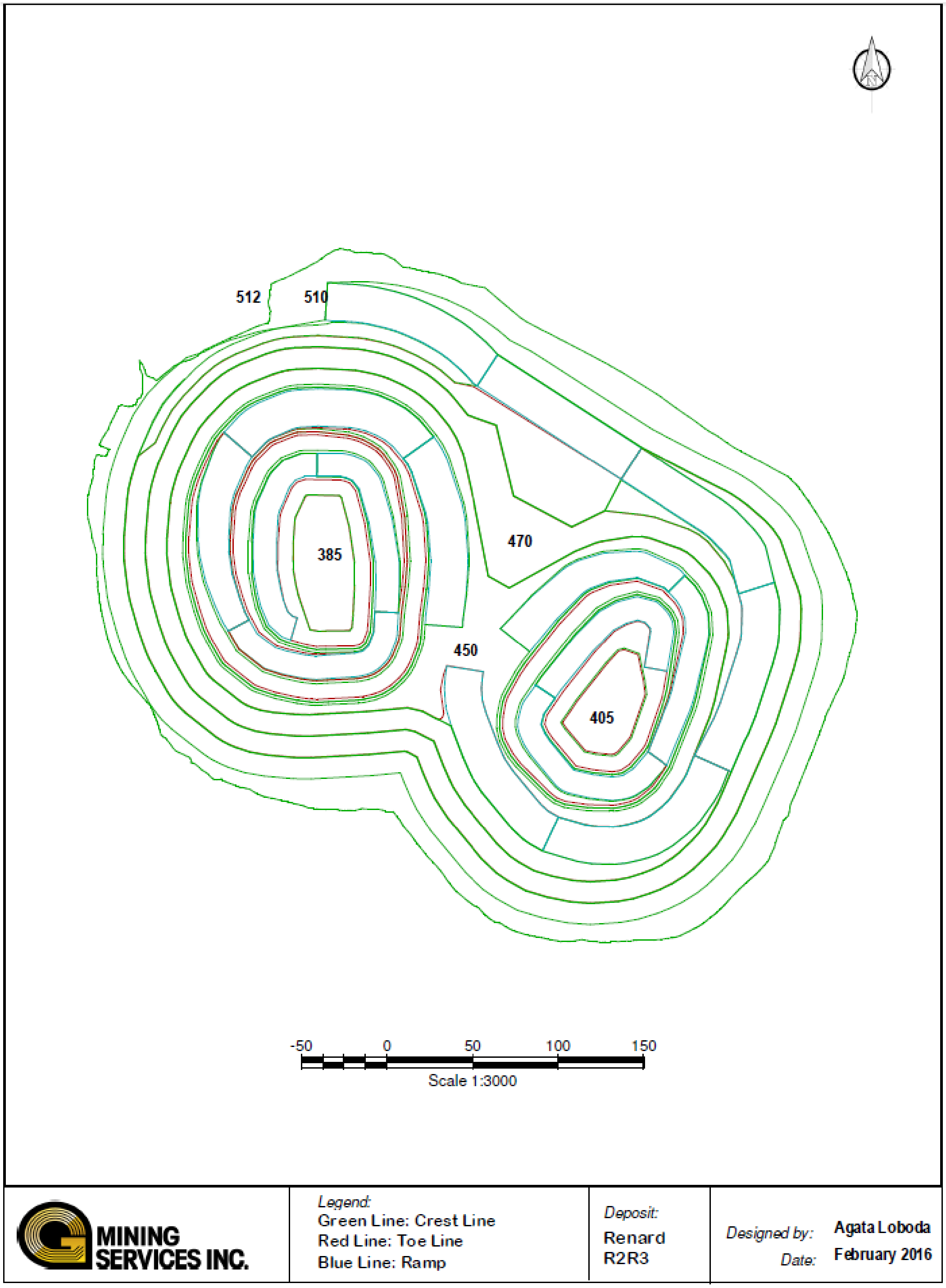
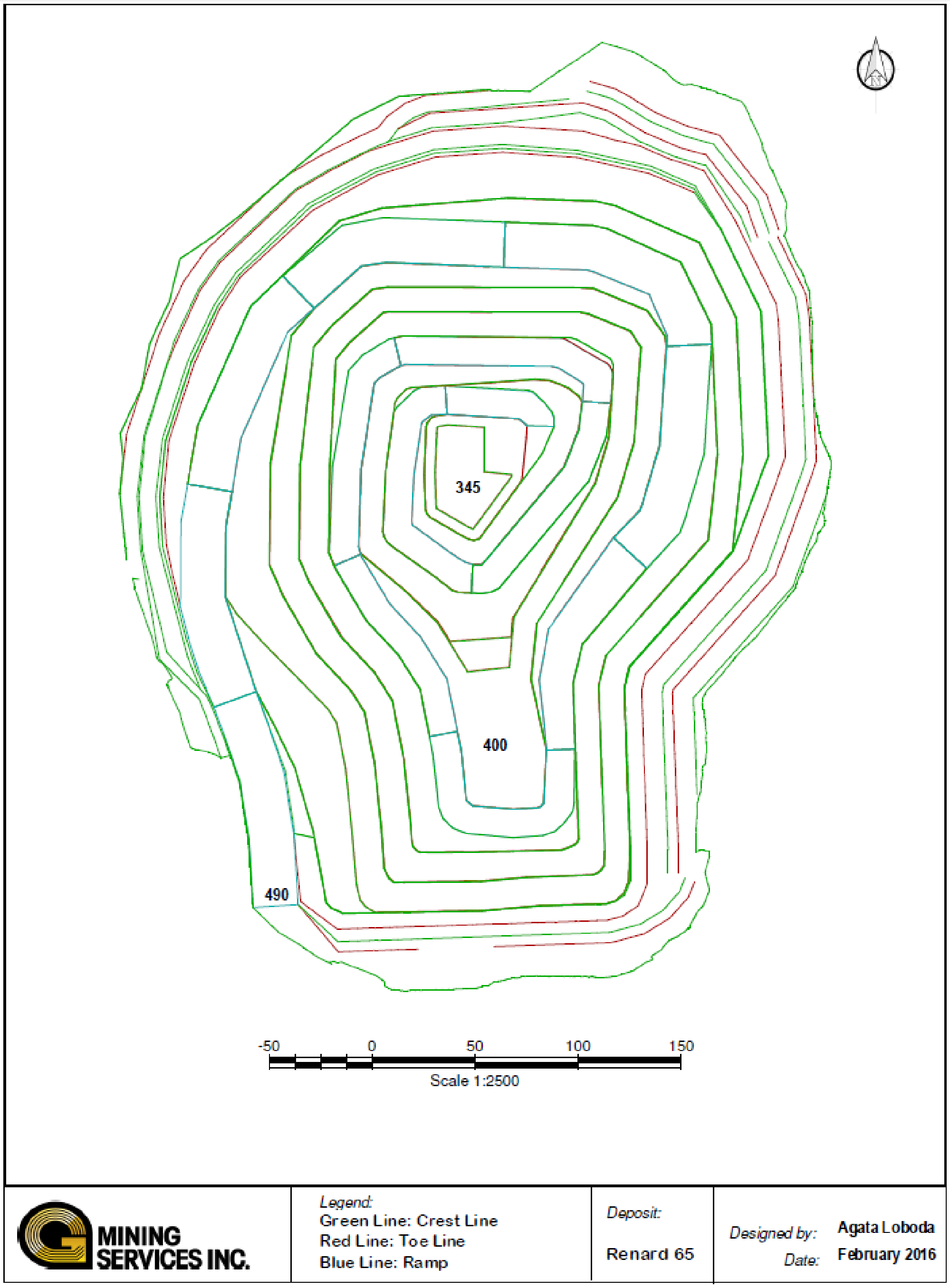


Figure 16.3: Renard 65 Pit Design



16.4.5 Drilling, Blasting and Surface Explosives Storage

Identical production drill patterns are proposed for both waste rock and kimberlite. The drill pattern proposed is presented in Table 16.6.

Table 16.6: Production Blast Pattern Parameters

Design ratios	Units	Blast Patterns (Kimberlite & Waste Rock)
Explosive type		Emulsion
Explosive density	g/cm^3	1.15
Hole diameter	mm	114
Burden	m	3.6
Spacing	m	3.6
Sub-drill	m	0.7
Stemming	m	2.5
Bench height	m	10.0
Blasthole length	m	10.7
Powder factor	kg/t	0.30

The pre-split design parameters are outlined in Table 16.7. The pre-split hole diameter is smaller at 102 mm (4 in) and are drilled with the same rig.

Table 16.7 Pre-split Design Parameters

Pattern yield	Units	Pre-split
Hole diameter	mm	102
Spacing	m	1.2
Stemming	m	3.4
Bench height	m	20
Blasthole length	m	20.6
Face area	m^2	24
Total explosive charge	kg	24.0
Toe charge	kg	5.0
Decoupled charge	kg	19
Decoupled charge length	m	15.7
Decoupled charge loading	kg/m	1.21

Drill rig requirements were established from the yield per metre drilled, as estimated from the proposed drill patterns, as well as from the instantaneous penetration rates and the actual drilling time described by an overall drilling factor. Table 16.8 provides the drilling productivity assumptions that have been validated by field results.

Table 16.8: Drill Productivity Assumptions

Drill productivity	Units	Production	Pre-split
Pure penetration rate	m/h	45.0	45.0
Overall drilling factor	%	50.0	50.0
Overall penetration rate	m/h	22.5	22.5
Drilling efficiency	t/h	685	-
Drilling efficiency	holes/h	2.0	1.1

Due to the limited tonnage to be mined, a single versatile drill rig model is proposed to drill pre-split and production holes. A diesel-powered, self-contained crawler-mounted top-hammer drilling rig with angle drilling capability has been selected.

Blast holes are loaded with a high-energy bulk emulsion explosive suitable for surface mines in wet or dry conditions. The bulk emulsion is booster sensitive with an average density of 1.15 g/cc that can be varied. A packaged pre-split explosive is used for blasting pre-split holes.

Blasting activities are planned on day shift only, and are outsourced to an explosives provider who is responsible for supplying and delivering explosives in the hole through a shot service contract.

The surface explosives storage site consists of an emulsion mixing plant, a 36,000 kg-capacity heated explosive magazine with stairway and loading ramp for storage of 1.1D class explosives, such as packaged product, detonating cord and boosters. An additional 12,500 kg capacity detonator magazine is required for storage of 1.1B class explosives, such as detonators.

16.4.6 Open Pit Mine Production Schedule

The mine production schedule was developed to initially supply the plant at a nominal rate of 2.16 Mt/a or 180,000 t/month. A plant ramp-up schedule is planned over a nine-month period starting in October 2016. The ramp-up includes three months of commissioning and is presented in Table 16.9. It is planned to increase the mill feed up to 210,000 t/month by July 2018 to reach 2.52 Mt/a.

Table 16.9: Plant Ramp-up Schedule

Month	Planned Tonnes (t)	Percent of Nameplate Capacity (%)
1 (October 2016)	35,154	Commissioning
2	82,004	Commissioning
3	108,000	Commissioning 60%
4	126,000	70%
5	135,000	75%
6	145,800	81%
7	162,000	90%
8	171,000	95%
9	180,000	100%

The open pit production rate in the R2/R3 pit is less than the mill capacity. An ore stockpile will be accumulated, which can be used to balance the open pit and plant schedule until such time as sufficient ore is available, produced by underground mining operations.

The Renard 65 pipe is of lower grade than Renard 2 and Renard 3 pipes and is mined as supplemental feed to the underground operation once the R2/R3 OP is depleted. Waste rock mined from R65 OP will also be used as underground backfill material and the Renard 65 OP will act as a water catchment basin to collect surface runoff water.

Mining of the R2/R3 OP commenced in March 2015 and will conclude in April 2018.

The production schedules were developed on a monthly basis until the end of 2019, quarterly until the end of 2023 and annually thereafter. The monthly scheduling provides a high level of detail on how the operation will transition from open pit to the underground. The annual OP production schedule is shown in Table 16.10.

Table 16.10: OP Production Schedule

		Total	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
R2/R3 OP																	
Total Tonnage	(t)	**11 179 370**	5 512 680	4 669 148	997 542	-	-	-	-	-	-	-	-	-	-	-	-
Waste Tonnage	(t)	**6 642 531**	3 694 308	2 317 363	630 860	-	-	-	-	-	-	-	-	-	-	-	-
Overburden	(t)	**203 819**	203 819	-	-	-	-	-	-	-	-	-	-	-	-	-	-
R2 Ore Tonnage	(t)	**1 489 134**	609 657	879 477	-	-	-	-	-	-	-	-	-	-	-	-	-
R2 In-situ Diamonds	(carats)	**1 380 993**	586 862	794 131	-	-	-	-	-	-	-	-	-	-	-	-	-
R2 Ore Grade	cpht	**1 567**	844	724	-	-	-	-	-	-	-	-	-	-	-	-	-
CRB2A Ore Tonnage	(t)	**474 703**	154 604	320 098	-	-	-	-	-	-	-	-	-	-	-	-	-
CRB2A In-situ Diamonds	(carats)	**149 246**	48 608	100 639	-	-	-	-	-	-	-	-	-	-	-	-	-
CRB2A Ore Grade	cpht	**534**	283	252	-	-	-	-	-	-	-	-	-	-	-	-	-
CRB Ore Tonnage	(t)	**1 575 465**	831 152	744 312	-	-	-	-	-	-	-	-	-	-	-	-	-
CRB In-situ Diamonds	(carats)	**318 672**	167 949	150 723	-	-	-	-	-	-	-	-	-	-	-	-	-
CRB Ore Grade	cpht	**384**	222	162	-	-	-	-	-	-	-	-	-	-	-	-	-
R3 Ore Tonnage	(t)	**793 719**	19 139	407 898	366 682	-	-	-	-	-	-	-	-	-	-	-	-
R3 In-situ Diamonds	(carats)	**732 579**	13 586	346 337	372 656	-	-	-	-	-	-	-	-	-	-	-	-
R3 Ore Grade	cpht	**1 230**	227	596	407	-	-	-	-	-	-	-	-	-	-	-	-
Total Ore Tonnage	(t)	**4 333 020**	1 614 553	2 351 785	366 682	-	-	-	-	-	-	-	-	-	-	-	-
Total In-situ Diamonds	(carats)	**2 581 491**	817 005	1 391 830	372 656	-	-	-	-	-	-	-	-	-	-	-	-
Total Ore Grade	cpht	**1 859**	657	795	407	-	-	-	-	-	-	-	-	-	-	-	-
R65 OP																	
Total Tonnage	(t)	**12 607 208**	633 412	-	1 149 971	1 218 381	1 436 878	1 446 684	1 441 684	1 400 000	1 362 825	1 215 379	962 151	339 844	-	-	-
Waste Tonnage (CCR +CR)	(t)	**6 073 570**	633 412	-	528 692	701 474	519 650	914 591	882 107	746 713	549 631	379 643	161 775	55 881	-	-	-
Waste Tonnage (CRB)	(t)	**1 114 383**	-	-	70 219	43 213	136 196	176 611	189 121	231 269	85 679	96 418	47 088	38 569	-	-	-
Overburden	(t)	**840 577**	-	-	274 323	60 588	495 111	10 555	-	-	-	-	-	-	-	-	-
R65 Kimberlite Tonnage	(t)	**4 578 679**	-	-	276 737	413 106	285 921	344 927	370 456	422 018	727 515	739 318	753 288	245 394	-	-	-
R65 In-situ Diamonds	(carats)	**1 376 242**	-	-	99 588	128 160	82 803	90 713	113 860	121 459	218 256	221 797	225 988	73 619	-	-	-
R65 Kimberlite Grade	cpht	**1 198**	-	-	356	255	123	107	123	115	30	30	30	30	-	-	-

16.4.7 Equipment Fleet

The primary loading tools will consist of one hydraulic excavator and one wheel loader. The hydraulic excavator is a Caterpillar 390F, fitted with a 5.0 m^3 bucket. The wheel loader is a Caterpillar 988K with a 6.4 m^3 bucket. Two Caterpillars 349F are also on site for construction purposes, secondary rock breakage, ditching and various support functions, and can act as a back-up loading unit as required.

Both rigid-body and articulated haul trucks are used for open pit production. The use of both types of trucks results in greater versatility and lower capital costs. A total of eight units are required: two Caterpillar 740 articulated trucks and six Caterpillar 775F rigid body off-highway trucks.

Major support equipment will consist of track dozers, motor graders and wheel dozers. Additional minor support equipment includes compactors, tool carriers, service trucks, fuel & lube trucks, water trucks and a loader.

16.5 Underground Design

16.5.1 Mining Methods

At the outset of the 2011 Feasibility Study mine design, a trade-off study was conducted to select the preferred underground mining method or methods to be used to extract the Renard 2, Renard 3 and Renard 4 pipes. In the study, potential mining methods were identified, evaluated and compared considering safety, dilution and recovery, production capacity, schedule, economics, risks and opportunities. As part of this study, a rock mechanics evaluation was conducted by Itasca, which determined that the Renard 2 and Renard 4 pipes were unlikely to support natural progressive caving, since the "caving" region is reached only for the lowest rock properties and largest hydraulic radius. As a result, block caving was eliminated as a potential mining method. Four underground mining methods were shortlisted and evaluated, including blasthole shrinkage (BHS), sublevel retreat (SLR), long hole panel mining with backfill (LHP), and blasthole shrinkage with pillars (BHSP). Based on this study, the blasthole shrinkage (BHS) mining method was selected to mine the underground portions of the Renard 2, Renard 3 and Renard 4 kimberlite pipes.

The BHS method consists of drilling and blasting the ore with long large-diameter blastholes, and extracting the ore at the base of the stopes from drawpoints. During the blasting phase, ore is left in the stope to support the walls until the complete stope has been blasted. Since ore expands when blasted due to the voids in the broken rock, it is necessary to draw out this swell on an ongoing basis, approximately 35% of the total in situ ore. Thus, with this method, there is a continual supply of ore production throughout the blasting period, and once the blasting is complete, the remaining broken ore can be drawn from the stope – generally at a very high rate with few restrictions.

The advantages of BHS are:

- High production capacity capable of producing 6,000 tpd;
- Minimizes overall development requirements with large sublevel spacing (60 m);
- Blasted ore left in the stope (shrinkage) provides support to stope walls until the blasted ore is drawn down;
- Modern large equipment (20t LHD's and ITH drills with 60+ m blasthole lengths) can be used, resulting in high equipment productivity;
- Initial production from the stopes can begin before completion of the R2/R3 OP;
- Over half of the blasted ore can be recovered with low dilution before higher dilution in the final drawdown phase begins.

The BHS method has been designed to progress in four stages from the top of the pipe (bottom of the open pit) downwards. As the ore is drawn from the stope, a large open void will be created if no backfill is used. The mine plan is to fill this void using open pit waste as backfill which will be placed into the stope by dumping it from trucks on surface onto the top of the broken ore column being drawn down in the pipe. For Renard 2 and Renard 3, wastes will be dumped over the side of the pit, while for Renard 4, wastes will be dumped into backfill raises leading to the top of the underground stope.

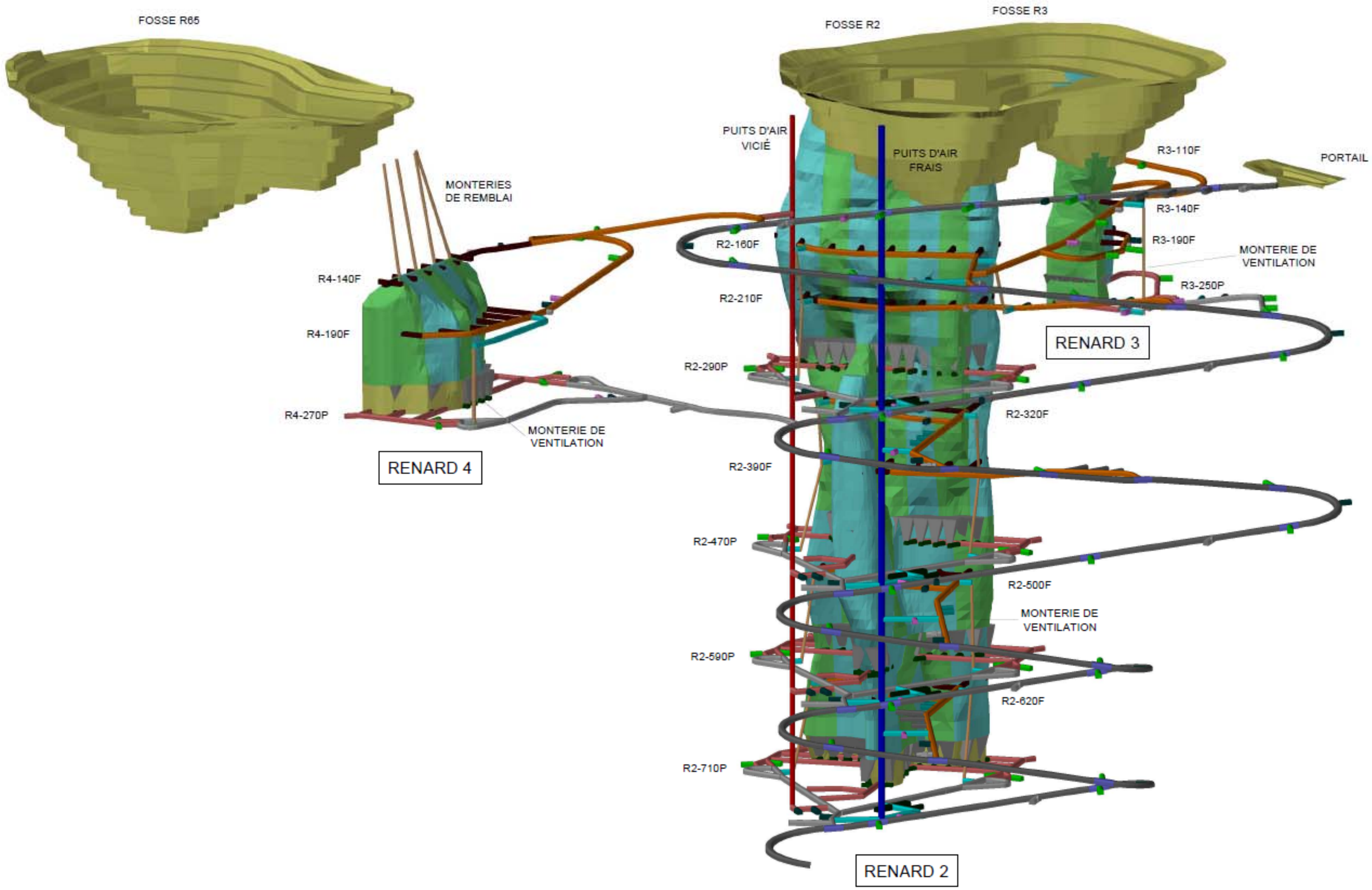
16.5.2 Mine Design

Access to the mine will be provided by a ramp to exploit the Probable Mineral Reserves that have been defined down to a depth of 710 m for the Renard 2 pipe, 250 m for the Renard 3 pipe and 270 m for the Renard 4 pipe. The underground mine has been planned so that a single common ramp will provide access to all three Renard pipes. The ramp has been sized to allow efficient ore extraction with an underground mobile equipment fleet, such as 60t haul trucks and 20t LHDs. It will be driven from surface to the 710L and will connect to all six drill levels and four production levels. Since only a ramp and a supported raise with a manway have access through to surface, all workers, equipment and materials will be transported in and out of the underground mine via the ramp facility.

As part of their scope of work, Itasca established two subsidence zone limits based on modelling with FLAC 3D. The first one considers that Renard 2 and Renard 3 will be filled with waste as mining progresses, and the second one considers no backfill (larger zone). The model considering no backfill was used to determine "no-go" zones for any long-term permanent structures, including a potential shaft, the main ramp and all other surface facilities. In addition, Itasca completed a global-scale numerical geotechnical model that identified areas from which long-term permanent underground infrastructure should be excluded. As part of this exercise, an ellipse was created within which a future shaft could potentially be located, and plans were produced showing horizontal strain and total displacement within that ellipse. Except for one corner of the ellipse closest to the Renard 2 pipe, all values were within acceptable limits. The global numerical model was also used to generate similar data at the various elevations and this data was used to guide the placement of the ramp, permanent levels and other underground infrastructure.

A 3D view of the planned development and stopes is shown in Figure 16.4.

Figure 16.4: 3D View of the Renard Mine



16.5.3 Stoping Design

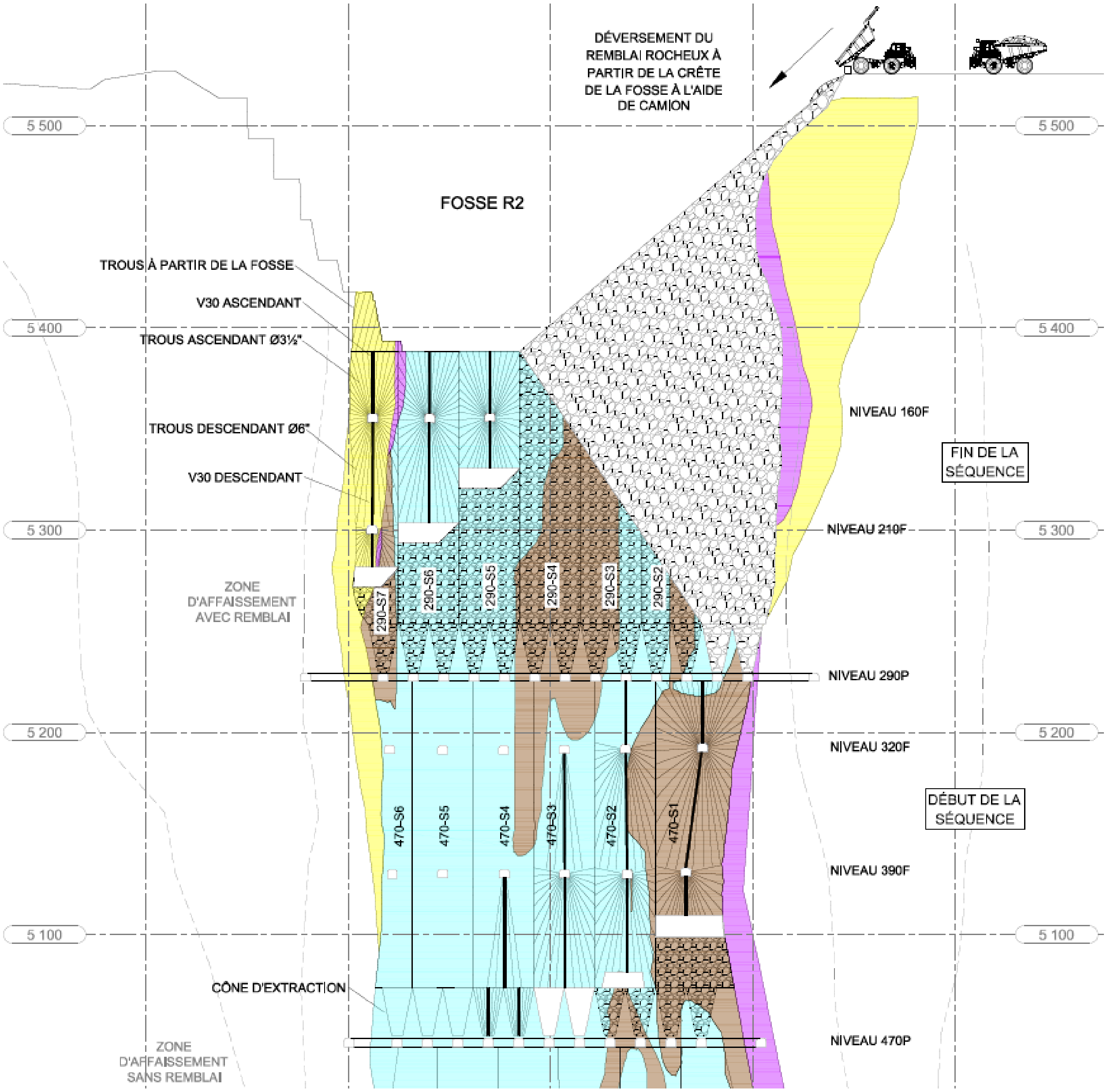
Since the largest proportion of the Project's Indicated Mineral Resources are contained within the Renard 2 kimberlite pipe, it has had the greatest influence on the selection of the stoping design in the underground mine. Indicated Mineral Resources extend from the surface of Renard 2 to a vertical depth of 710 meters

For production mining, the Renard 2 pipe has been divided vertically into four production stages (zones), each located at elevations where the ore body changes geometry and orientation, so that recovery and dilution factors could be optimized. The resulting production levels with this design are at level 290L, 470L, 590L and 710L (all level designations are depth below surface in meters). Their respective heights are 165 m (up to the pit bottom), 180 m and 120 m for the last two stages. Drawpoints and cones will be located at the base of each of the stages. A top-down mining sequence is planned for the four mining zones in Renard 2. First, the upper 290L zone will be mined, followed by the 470L and 590L zones, and finally the 710L zone. Each mining zone has been subdivided into vertical panels; seven panels on 290L, eight panels on 470L, nine panels on 590L and eight panels on 710L. Mining of the kimberlite will proceed horizontally from the North panels towards the South panels in a stair fashion. During the transition between two zones, production of the level below will start from the North panels as the South panels of the level above are in the progress of being emptied. On a panel scale, the shrinkage aspect of the BHS method comes into play, whereby the blasted lower section of the panel stope will be kept full of blasted ore during this period, thus limiting the potential for unravelling of the crown pillar. Once the crown/sill pillar level has been reached and blasted, full mucking activity of the panel can be initiated. As the blasting progresses towards the crown or sill pillar inside a particular panel, the blasting can start in the next adjacent stope to the south in a stair fashion. The blasting sequence for Renard 2 is shown in Figure 16.5.

The base of the Indicated Resource in the Renard 4 pipe is 235 m below surface (270 m from FAR collar) and will be mined below a 100-m-thick crown pillar for a vertical height of 135 m. The interpretation of the shape of the Renard 4 pipe is oblong, and very regular in its outline, so this height was deemed to be suitable for BHS mining. One production level and two drill levels will be used to extract all reserves from the Renard 4 orebody, which will also be subdivided into vertical panels as Renard 2. The bottom production level for Renard 4 pipe is at the 270L as shown in Figure 16.4.

The Renard 3 pipe is fairly large close to surface, but then necks down to a fairly small pipe below the open pit and further necks down at the base of the Indicated Mineral Resources at a depth of 250 m. Due to the very irregular shape of the orebody, the Renard 3 pipe was divided in two stopes. The stope on the West side of the pipe will first be mined bottom up by BHS as its grade and size are more important. This opens the opportunity to recover most of the carats in R3 without adding fill before the stope is depleted. Once the first stope filled with waste, the stope on the East side of the pipe (stope 2) will be mined using standard long hole stoping, as it is too narrow to accommodate drawcones. The bottom of the only production level for the Renard 3 pipe is at 250L, and the mining of two stopes will extend up to the bottom of the pit as shown in Figure 16.4. Due to the small size of the body, the spacing between the drill levels was decreased (40 m) compared to R2 in order to obtain more precision in the production drilling, which will result in better blasting result and consequently less dilution.

Figure 16.5: Typical Section Illustrating the BHS Method and Mining Zones



16.5.4 Drilling and Blasting

The kimberlite will be drilled and blasted by a combination of large-diameter downholes for the body of the stopes, and smaller diameter upholes for the undercut drawcones, sill pillars and drawpoint level pillar blasting.

The primary production drills will be in-the-hole (ITH) rigs drilling 165 mm diameter blastholes with a toe spacing of 4.0 m and a burden of 4.0 m for an average drill factor of approximately 22 t/m drilled. Typical hole lengths will be 60 to 75 m. The ITH drills will be compressed air-driven with each unit having its own booster compressor to raise the air pressure to 400 psi from the nominal mine compressed air pressure of 125 psi. The undercut drilling will be by a rubber tired electric hydraulic production drill unit drilling 75 mm diameter blastholes with a toe spacing of 3.0 m and a burden of 2.7 m, for an average drill factor of approximately 10 t/m drilled. Hole lengths will vary between 25 to 30 m.

Emulsion explosives will be used almost exclusively for both development and production blasting.

16.5.5 BHS Mining

With the BHS method, the kimberlite in a panel is blasted in successive vertical lifts over the height of the panels, in a benching fashion. One or two slot raises will be excavated over the height of the stope to provide initial free faces from which to initiate the blasting. These slot raises will be excavated during the drilling phase using a “Machine Roger” ITH drill head attached to an ITH drill rig. While the panel is being blasted, sufficient blasted ore will be left in the stope (shrinkage) to provide support to the walls and limit the extent of sloughing or caving from the back if that should be experienced. This is illustrated in Figure 16.5.

The blasting and production phase will start by blasting the undercut drawcones from the drawpoint level and advance in a retreat blasting and mucking sequence. After each blast, sufficient ore will be drawn (mucked out) to provide for a free face and ample void before proceeding with the next blast. The blasting strategy will be to blast out the drawcones below each panel before beginning the blasting of the downholes.

The downholes are typically 60 m long, drilled from the drill level above. Blasting will be done by vertical benching, initiated towards a central slot raise, in vertical lifts. Following each blast, the blasted ore will be mucked out from the drawpoints below, until the void or gap at the top of the blasted ore to the back is sufficient for choke blasting the next lift. During the blasting phase, 35% of the ore will be drawn from the stope through drawpoints to accommodate the swell of broken ore.

Once the crown pillar of the North panel of the 290 mining zone is blasted, the Renard 2 pipe will be open to surface and, at this point, production mucking in this panel can be ramped up to the full production rate. As blasting of the other panels progresses towards the South, the opening in the pit bottom will get bigger until the crown pillar of the last panel is blasted.

As the blasting inside the South panels of 290L is nearing completion, the blasting sequence can progress in the North panels of the 470L. The mass blasting of the 470L mining zone residual pillar (pillar below the 290L drawpoint level) will include not only the pillar, but also the underside of the drawpoint troughs on the 290L. Detailed modelling will confirm the thickness of the pillar to be kept in situ between the above mucking horizons and the blasted stope below. Mucking directly above the panel will have to be completed before the sill/crown pillar of the panel below can be removed. After this blast, the drawpoints on 290L will have been destroyed and mucking from the North panel on 470L can be ramped up to full production. This sequence will be repeated for each individual panel as mining progresses towards the South on this level. The blasting sequence on 590L and 710L will be the same as on 470L.

16.6 Backfilling

After the 290L North panel residual pillar is blasted through to the base of the open pit and production and full drawpoint production begins from the 290L drawpoint level, the column of blasted ore will be gradually drawn down, leaving an open hole. As the crown pillar of each panel is being blasted, ore drawdown will progress and void will be filled by the waste dumped from surface. Consequently, waste rock backfill will be used as an integral part of the mining plan. Waste rock for this purpose will be available from the stripping of the open pit. This use of the open pit waste will also enable the rehabilitation of the waste rock pile during mine operation rather than at the end of the mine life.

The method of placing the waste rock backfill into the mine is straightforward, it will be dumped over the edge of the open pit, cascade down to the stope opening at the base of the pit and then onto the blasted ore column. This concept is illustrated in the mining method drawing shown in Figure 16.5. The backfilling method for the Renard 4 pipe will be the same with the exception that four raises will be driven up from the base of the crown pillar, and the waste rock backfill placed into the stope through these raises.

As the ore column is drawn down during production, waste rock will be steadily dumped into the stope to maintain a constant level of backfill at the bottom of the pit. Waste rock for the backfill will be generated from the R2/R3 OP and R65 OP stripping. Overall, there is sufficient waste rock available for backfilling.

16.6.1 Drawpoint Mucking and Production

Ore production will be from blasted ore pulled from drawpoints situated at the base of each mining zone. With the BHS method, waste rock backfill will be placed on top of the ore column and consequently, it is critical that the drawpoint design provide good draw coverage and a draw management system be instituted that will result in an even drawdown of the blasted ore column. The drawpoint layout proposed for the R2 pipe at Renard is based upon a commonly used arrangement from block caving operations, which is the Offset Herringbone layout. The drawpoints are arranged on a 30 m x 15 m spacing with the drawpoint accesses having a 40° break angle from the access drift. With a drawcone spacing of 15 m centre to centre, and a height of roughly 30 m, the resulting drawpoint pattern is a 10-m pillar and a 5-m-wide drawpoint. Even distribution of drawpoints allows for maximum draw control and minimizes the likelihood of increased dilution and backfill ingress into the ore column. The drawpoint brows in the Renard 2 pipe will be sufficiently supported with cable bolts and shotcrete to ensure the durability of the drawpoint during the mining life cycle of the stope.

Ore drawdown management during production is critical to limiting dilution and maintaining high recovery. There are two aspects to managing drawdown: the first is to maintain an even drawdown over the whole blasted ore column in the stope, and the second is to promote mass flow of the ore. One of the big dangers resulting from poor draw management is the development of rat-holes above individual drawpoints, ahead of their neighbors, right up to the waste backfill placed on top of the blasted ore. A good drawdown plan in combination with strict draw procedures and regular monitoring is required for success. Identifiable markers placed in the blasted ore, either via placement in blastholes or on drill levels, will provide good data on the effectiveness of the drawdown system.

During stope production mucking, waste material that appears in a drawpoint will be sorted from the ore stream to reduce the amount of diluted material sent to the surface crusher. The sorting will be visually done by the geologist and scooptram operator as the surrounding country rock and waste inclusions in the pipe are significantly lighter in colour than the kimberlite. Sorted waste material will be mucked to surface for dumping to the waste pile.

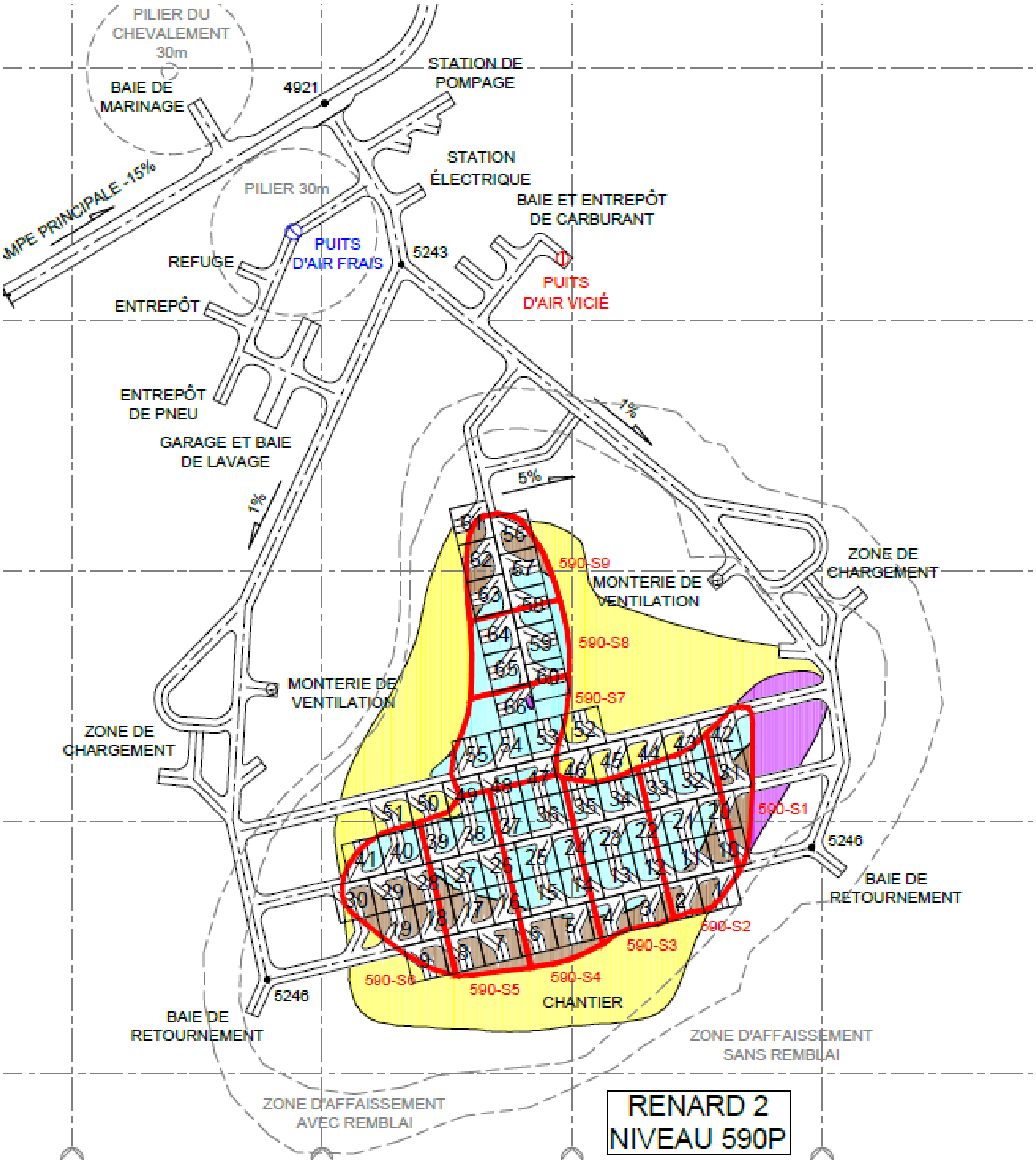
The objective on the drawpoint levels is to create and maintain a high production capacity through efficient mucking and ore transportation. A key element of the planned system is the use of high capacity mine haulage trucks to transport the ore to surface. To maximize the effectiveness of the trucking system, a series of truck loadouts are planned on each production horizon to facilitate the loading and queuing of trucks to assist in the prevention of production bottlenecks. The truck loadouts are placed to reduce the ore haul from the drawpoints by scooptrams to a minimum, and to use the haulage capacity of the mine trucks.

On the 290L, two truck loadouts are planned: one on the North side of the Renard 2 pipe and one on the South side. The average haul distance from drawpoint to loading station is approximately 150 m. On 470L, 590L and 710L, an additional truck loading station is planned for the west side of the Renard 2 pipe to minimize the haul distance from the West section of the pipe.

Drawpoint mucking will be done with 20 t capacity LHDs, and these will load from the drawpoints and haul directly to the truck loading station. To achieve 6,000 tpd, the LHDs will haul a total of 330 buckets per day or 165 per shift. Generally, this activity will be split between two producing haulage drifts each with between 6 and 20 drawpoints, depending on the advance of the stoping sequence. Based on the mucking and tramming simulation performed with Labrecques Technologies in 2014, up to three production LHDs will be required to achieve the planned production tonnage. Haulage of the ore from the production levels towards the surface crusher will be done using 60t trucks. The production haul truck fleet will increase as mining progresses towards the lower production zones, since the ramp distance will increase significantly between each level. The fleet should include 7 trucks on 290L and increase up to 12 trucks for the 710L.

The 590L drawpoint level design and location of the drawpoint drawdown cones are shown in Figure 16.6.

Figure 16.6: 590L Drawpoint Level Layout and Drawdown Coverage



16.6.2 Underground Mine Development

The mine will be accessed by a ramp driven at a size of 7.3 m wide by 5.6 m high, and at a grade of 15% from surface to 710L. When complete, the ramp will extend from surface down to the bottom of the 710 production level, with a total length of 4.9 km, connecting to all production and drill levels. To facilitate smooth traffic in the ramp during the ore haul, passing bays were incorporated into the ramp design to allow loaded ore trucks to travel the ramp with minimal interruptions.

All drill and production levels will be driven from the ramp, typically at a grade of +2%. Development waste and ore from the levels will be trucked to surface via the ramp. The sizes of the headings for ramps, main accesses on production/drill levels (5.6mW x 5.3mH), muck/drill/service drifts (5.3mW x 5.0mH), muckbays (6.0mW x 5.5mH) and drawpoints (5.0mW x 4.2mH) were selected based on the equipment to be used and to accommodate ventilation ducting during excavation. The haul drifts are aligned to minimize haulage distances on the levels, and the tramming distance between draw horizon and the truck loading zones is optimized to minimize the muck cycles.

The main production levels include a main refuge station, electrical substation, main pumping station, maintenance shop, fuel bay and reservoirs and storages. All of these are located in the fresh air raise/return air raise vicinity to allow better service distribution and be outside the main haul drifts to avoid production delays. Other service facilities (storages, sumps, refuge stations, electrical substations, etc.) are located on drill levels as required.

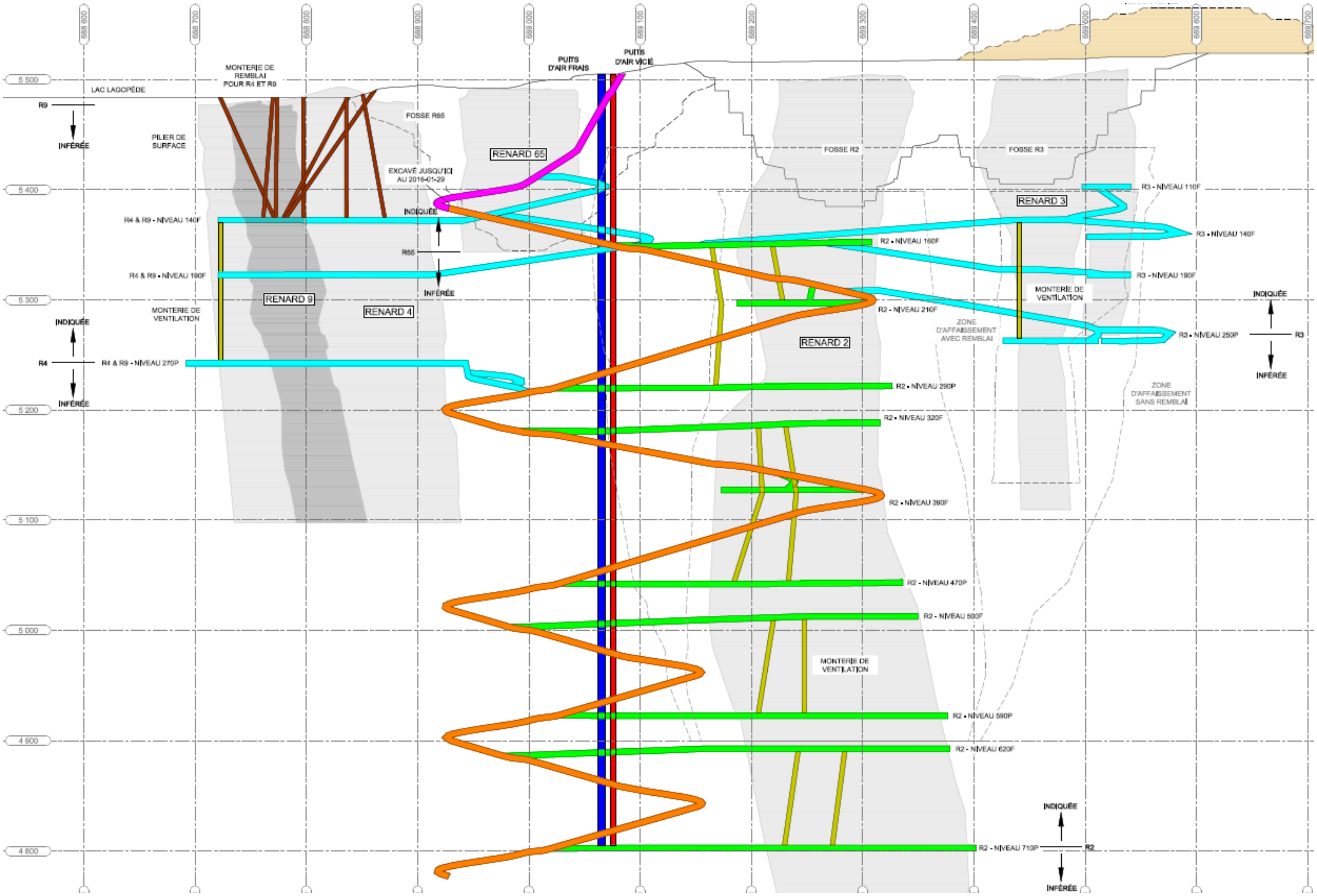
The Renard 3 and Renard 4 pipes will be accessed either from the main ramp or from existing levels initially developed to access R2, and the same design criteria has been applied.

The main exhaust raise will be a vertical and supported 5.0 m in diameter raise, extending from surface to 710L. It will be excavated in four sections: surface to 290L; 290L to 470L; 470L to 590L; and 590L to 710L. The main fresh air raise will be a vertical 6.5-m diameter supported raise, also extending from surface to 710L. It will be excavated in five sections: surface to 160L; 160L to 290L; 290L to 470L; 470L to 590L and 590L to 710L. The fresh air raise will also serve as the secondary egress from the mine in the event of emergency where the primary egress through the ramp is unavailable. Mine services such as electrical cables, fuel line, mine water, dewatering, compressed air and communications will be provided through the raise as well.

Many unsupported 3.0-m diameter ventilation raises will be excavated at various locations between the drill and production levels to allow the main air flow to circulate close to workplaces. These are generally located close to the orebodies on the abutments, and will only be required when mining the associated production horizon.

The main mine development infrastructures are shown in longitudinal projection in Figure 16.7.

Figure 16.7: Longitudinal Section showing the Main Mine Infrastructure



16.7 Ventilation and Heating System

Based on the planned diesel-operating equipment fleet, the mine ventilation requirement at full production is estimated to be 425 m^3/s (900,000 cfm). This quantity of fresh air will meet the requirements of Québec's *Occupational Health and Safety Act*. The mine will have one main fresh air intake, which will be a 6.5-m diameter supported raise with services and manway (secondary egress). The maximum ventilation requirement occurs in the years 2025–2026, when production will be transitioning to the deepest level of the mine, which will require a larger production truck fleet. Fresh air will be distributed on most levels directly from the fresh air raise. Drill levels which are not connected to the fresh air raise will be supplied via smaller vent raises connected to other levels with fresh air. The contaminated air will be exhausted either by the 5.0-m diameter supported return air raise (RAR) or by the main ramp up to the mine portal. Air distribution within the mine will be controlled by a combination of ventilation doors (SAS), regulators, plastic flaps and secondary fans installed in walls. This will allow large flow coverage of the levels with the use of auxiliary fans only locally (e.g., drill drifts).

The two main fans (1000 HP each) will be located on top of the fresh air raise in a push system. The raises will be connected to most levels to minimize the recirculation of contaminated air. A schematic of the primary ventilation network is shown in Figure 16.8.

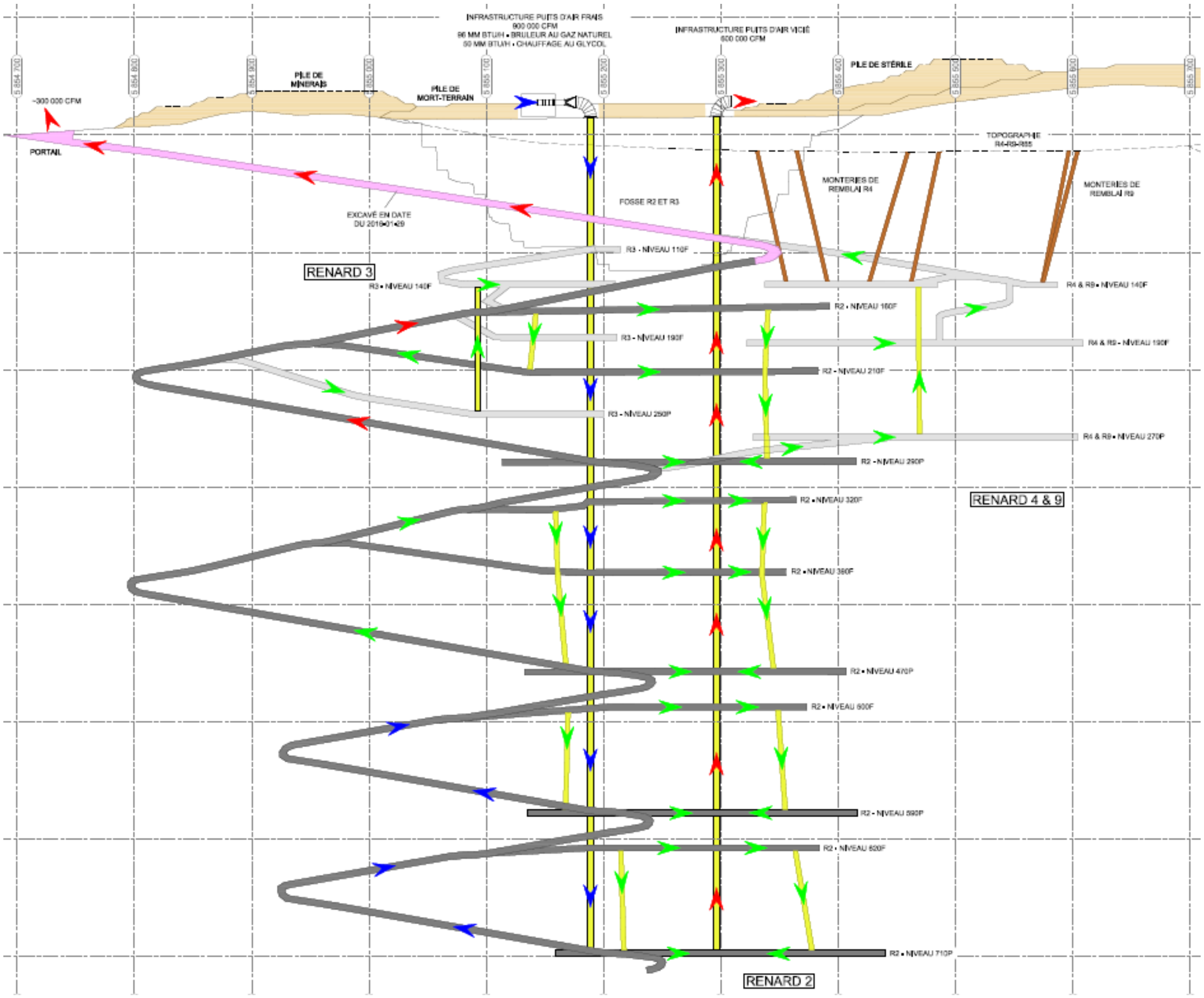
The R3 and R4 ore bodies will each be ventilated by 3.0-m diameter internal raises which will connect all related levels, allowing the main level air flow to circulate closer to the workplaces.

The main intake fans will be equipped with variable speed drive motors (VFD) to enable a high degree of control over the airflow rates produced by the fans. This degree of control will be used to maintain required airflow balances and ventilation pressure differentials throughout the mine. It should be noted that the fresh air requirement varies over the mine life, ranging from a minimum of 150 m^3/s in 2017 to a maximum of 425 m^3/s in 2025.

The main heating units will be installed ahead of the main fans on top of the fresh air raise, which includes a glycol heating system and natural gas burners. The glycol heating system will be installed in series ahead of the natural gas heaters near the main fresh air raise entrance. This system is estimated to have a heating capacity in the magnitude of 50.1 MBTU/hr, which should be sufficient to provide heat to the underground operations during most operating conditions (flow and temperature). The glycol heating system will use the waste heat from the power plant generators, which will be transported via a propylene glycol loop to the heater location just ahead of the propane heaters. The natural gas heating systems includes three units providing a total capacity of 96 MBTU/hr. This system should be required when the demand in fresh air will be higher and/or when the temperature will be in the lower values. The capacity of this system will also be enough to cover the demand if the glycol heating system is down or less efficient due to a decrease in the power demand at the site.

During the development phase, the ramp is ventilated by two 200 HP fans installed in series and 72" rigid ducting to the face from the mine portal. The first fan is installed on surface behind a 12 MBTU/hr propane heating unit and the second fan will be installed as a booster unit at the entrance of the 160 drill level entrance. Both fans will be managed by VFDs and the combination will provide up to 50 m^3/s (105,000 cfm) down to the 290 production level. The first leg of the fresh air raise from surface to the 160 level should become available in Q1–2017. At that time, the temporary ventilation setup at the portal will be removed, and the ramp development face will be ventilated via the 160 level until the second fresh air raise leg is completed, which will happen later in 2017.

Figure 16.8: Main Ventilation Network Schematic



16.8 Underground Infrastructure and Services

Electrical power will be provided throughout the mine for drills, jumbos, fans, pumps, lighting and other miscellaneous loads. The power will be distributed to the mine at 4.16 kV through three primary feeders, two located in the fresh air raise (3C 250 MCM) and the third in the ramp. All three feeders will be supplied from the power station electrical room. A substation located on surface will be fed from the power house electrical room containing BUS A and B and having 2000 amp capacity. From there, power will be distributed to the mine surface facilities, including the main intake fans.

Compressed air for production and development equipment such as ITH production drills, and small equipment such as hand-held drills, pumps, etc. will be provided using surface main compressor setup installed in Main Fans building. Development equipment, such as jumbos and bolters, will use onboard compressors for compressed air generation. Refuge stations will be fed using main surface compressors. Main feed lines will pass by the fresh air raise to feed levels. Compressed air flow will be monitored on surface and on every level to detect any leak. The total required compressor capacity is estimated at 6,000 cfm. However, provisions are made in the building to extend compressor capacity up to 8,500 cfm.

Process water is required underground for drilling, dust suppression, and wash-down of equipment and rock faces for geology and mining. The underground requirements have been estimated to be 1,300 m^3/day. Process water will be supplied from the Project process water distribution system on surface, and will be fed underground through 151 mm (6”) pipelines in the fresh air raise and 102 mm (4”) in the ramp. Provisions are made for water recirculation to reduce water consumption and water treatment. Potable water for underground use will be provided using a 51 mm (2”) line in the fresh air raise.

Water will enter the underground mine from several sources including ground water inflows into mine openings, precipitation into the R2 and R3 open pits, and process water. A hydrogeological study based on hydrogeological modelling was completed by Itasca (pre-2012) and Golder (2012–today) to estimate ground water inflows. In the 2012 Golder model, it was shown that the maximal inflow in the ramp development should reach 1,400 m^3/day (256 USgpm). The Itasca model considered the mining sequence and schedule, and estimated the inflows over time. The inflows reach an initial peak in 2018 with 2,300 m^3/day (421 USgpm) after 37 months of development, and then gradually drop down to a long-term inflow varying between 1,200 m^3/day (220 USgpm) and 1,600 m^3/day (294 USgpm) before reaching a life-of mine overall peak of 2,475 m^3/day (452 USgpm) late in the mine life when mining on the R2 610L horizon begins. The process water requirement is estimated at 1,300 m^3/day (240 USgpm). Precipitation is variable and could add 100 to 400 m^3/day. Based on these estimates, the pumping system has been designed with a nominal pumping capacity of 2,725 m^3/day (500 USgpm) operating at 60% utilization, so that peak inflows of up to 4,500 m^3/day could be accommodated on a short-term basis. Golder is currently working on updating the hydrogeological model with the new information available in order to provide more updated water inflow volumes throughout the mine life.

A four-stage dewatering system is planned. Main sumps and pumping stations will be located on 290L, 470L, 590L and 710L, with water pumped from the lower pump station being rehandled at the upper station. It is anticipated that approximately 30% of the mine water inflows will be intercepted and contained at 290L, 470L and 590L as mining progresses deeper. Consequently, the 710L lower pumping station has been designed for 1960 m^3/day (360 USgpm) while the 290L, 470L and 590L upper stations will be capable of pumping the full predicted flow of 2,725 m^3/day (500 USgpm). All pump stations will have two pump arrangements installed, one operational and the second on standby for maintenance. A gravity drainage system will be established whereby drainage holes will be drilled to connect the levels and channel mine water down to the sump on the nearest level.

The communication system for the underground operation will be installed in most tunnels and uses coaxial cable with modems, antennas and amplifiers. The system will be capable of transmitting data and voice, as well as high speed internet, telephone and equipment/personnel tracking. In addition, conventional telephone service will be provided to the refuge stations, maintenance facilities and fueling stations.

Emulsion explosives will be used almost exclusively for both development and production blasting. Most of the emulsion will be supplied as a bulk product with a minor amount as packaged explosive. The bulk emulsion will be delivered from surface storage to underground in 1,500 kg containers on specialty trucks provided for this purpose. From surface, the trucks will deliver loaded containers directly to the workplace or to one of the underground explosive magazines. Explosive storage areas include three excavations: the explosive truck parking, the explosive magazine and the detonator magazine. Three explosive storage areas are planned to provide explosives underground, all located near the Renard 2 production horizons. The main magazines will be located off the ramp close to surface, the second set at the entrance of the 390 drill level, and the third one at the same location on the 620 drill level. The explosive magazines have been designed to store 40 containers for a total capacity of 60,000 kg of emulsion and some packaged explosives.

A diesel fuel storage and distribution fuelling station will be provided on surface to service the site. All underground vehicles coming to surface via the ramp will be able to fuel up from this facility. In addition to this surface facility, a fuelling system to deliver fuel underground through a pipeline will be provided. On surface, the fuel delivery system will consist of a main tank pumping to a transfer/batching tank located in the vicinity of the fresh air raise. Underground fuel bays will be constructed on all main production levels: 290L, 470L, 590L and 710L. Each fuel bay will be equipped with a 20,000-litre receiving tank and dispensing equipment. The transfer of fuel underground will be done through a 25-mm diameter piping system on a batch basis, and will be fully instrumented to ensure safe operation. The pipeline on surface will be a buried line with at least a -1% grade to the fresh air raise intake structure, and will be installed down the fresh air raise with a valve located such that it can be delivered to any of the four fuelling stations. At each of the levels, the fuel line will be installed in the drift to the fuelling station location. The delivery system will be a “dry” system in which there is no fuel in the pipeline except when a batch of fuel is being transferred. The underground fuelling stations have been located close to the main exhaust raise as a fire safety precaution and will be equipped with fire doors and a fire suppression system. Mobile vehicles, such as service vehicles and trucks, will fuel directly at the fuelling stations, but for slow moving or more remote equipment, a fuel truck will be used to deliver fuel directly to the work heading as required.

The main maintenance facility for the site will be located on surface and will be used to service all surface and underground equipment. While major repairs and rebuilds will be performed in the surface shop, four satellite repair areas – one on 290L, a second on 470L, a third on 590L and a fourth on 710L – will be provided for minor repairs and preventive maintenance.

A total of 14 refuge stations will be located at various strategic locations throughout the mine according to Quebec’s regulations (15 min walk or 1000 m).

As the ramp is the primary egress, the secondary egress from the mine will be provided via a ladderway installed in the fresh air raise. The mine fresh air intake will be downcast so in the case of a fire, it will always be in fresh air and provide a safe escape route. In the rare case of a fire in or near the raise, personnel will report to the nearest refuge station.

A system to warn workers of fire or other emergency, in addition to the telephones and mobile radios, will be provided. The warning will be given by injecting ethyl mercaptan into the mine ventilation intake airway at the fresh air raise and into the main compressed air line.

16.9 Development Schedule

The total mine development requirement, including ramp, levels and raises, in the life-of-mine schedule is shown in Table 16.11. During the LOM, a total of 37.0 km of development will be excavated of which 4.0 km will be in initial or pre-production capital with the remainder in operating and sustaining capital. This development will produce 0.6 Mt of kimberlite ore and 2.0 Mt of waste.

The development is also categorized as CAPEX, sustaining and operating. CAPEX development considers all development done prior to the commercial production date. Sustaining includes all infrastructure development such as ramps, ventilation/backfill raises and main levels. Operating is development that is used to directly access the stopes from the levels and includes muck drifts, drawpoints and drill drifts.

Ramp development was initiated in December 2014 with an initial 8 m cut, and then recommenced in April 2015. As of December 31, 2015, 887 m of development had been completed. The main ramp will remain the highest priority task until 290L is reached, as it is part of the critical path to start the underground operation. The planned rates per development team are 5.2 m/d for the ramp; 5.5 m/d for a single face in standard size tunnels, and 6.5 m/d if there are multiple faces available in standard size tunnels. Except for the first 8 m, which were performed by a contractor, all lateral development will be done by Stornoway-employed crews with newly purchased equipment. There will be one development team until 160L is reached. At that time (August 2016), a second team will be added to develop drill levels in parallel with the ramp. Two development teams will be maintained until Q1-2020. At that time, development advancement will be ahead of the production and drilling schedules, and one team will be removed from the fleet. Development will progress towards the lower part of Renard 2 (710L), Renard 3 than Renard 4 with one development team until the beginning of 2027, when all development should be completed to recover all Mineral Reserves. With two teams, the advancement rates should reach 4,500 m/yr but drop down to 2,300 m/yr with one team. Stornoway will be responsible for the excavation of all production raises, but will retain outside contractors for the exhaust air, fresh air, secondary ventilation and backfill raises. Slot raises for stope blasting will be excavated by Stornoway crews using the Machine Roger ITH raise head.

Table 16.11: Underground Development Requirements and Schedule

Development type	Dimension	Capex	Sustaining	Operating	Total	Preproduction Period			Life of mine										
		Capex	Sustaining	Operating		2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
		(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)	(m)
Main ramp	**7.3m x 5.5m**	1 719	2 655	-	4 374	8	612	1 100	252	908	720	20	511	244	-	-	-	-	-
Passing bay	**9.7m x 5.5m**	300	502	-	802	-	100	200	60	161	121	20	80	60	-	-	-	-	-
Muckbays, sumps	**6.0m x 5.5m**	360	970	-	1 330	-	115	245	85	179	224	46	82	143	-	112	69	31	-
Haul drift, drill level	**5.6m x 5.3m**	1 019	8 448	-	9 467	-	10	1 009	793	1 532	1 335	224	611	916	194	1 229	1 432	182	-
Drill, muck, service drifts	**5.3m x 5.0m**	338	1 583	11 669	13 590	-	88	250	2 442	1 672	1 335	1 678	632	919	1 174	910	872	1 313	305
Drawpoint	**5.0m x 4.2m**	-	-	4 494	4 494	-	-	-	708	-	773	730	290	15	1 004	128	-	846	-
Fresh air raise	**6.5 m diameter**	165	547	-	712	-	-	165	129	125	150	23	-	120	-	-	-	-	-
Return air raise	**5.0 m diameter**	-	711	-	711	-	-	-	293	-	178	120	-	120	-	-	-	-	-
Secondary vent raise	**3.0 m diameter**	54	985	-	1 039	-	-	54	134	123	175	180	-	183	-	110	-	80	-
Backfill raise from surface	**3.0 m diameter**	-	460	-	460	-	-	-	-	-	-	-	-	-	-	-	-	460	-
Total ramp and lateral		**3 736**	**14 158**	**16 163**	**34 057**	**8**	**925**	**2 804**	**4 340**	**4 452**	**4 508**	**2 718**	**2 206**	**2 297**	**2 372**	**2 379**	**2 373**	**2 373**	**305**
Total raise		**219**	**2 703**	**-**	**2 922**	**-**	**-**	**219**	**556**	**248**	**503**	**323**	**-**	**423**	**-**	**110**	**-**	**540**	**-**
Grand total		**3 955**	**16 861**	**16 163**	**36 979**	**8**	**925**	**3 023**	**4 896**	**4 700**	**5 011**	**3 041**	**2 206**	**2 720**	**2 372**	**2 489**	**2 373**	**2 913**	**305**
Development and raise tonnages																			
Waste		355 048	1 293 382	324 935	1 973 365	3 200	89 545	262 303	207 899	296 043	293 271	100 449	136 553	184 511	47 427	146 929	143 118	55 490	6 627
Kimberlite		5 350	-	667 622	672 972	-	-	5 350	119 752	72 427	72 151	87 109	36 501	11 876	98 135	25 451	28 592	102 002	13 627
Total		**360 398**	**1 293 382**	**992 557**	**2 646 337**	**3 200**	**89 545**	**267 653**	**327 651**	**368 470**	**365 422**	**187 558**	**173 054**	**196 387**	**145 562**	**172 380**	**171 710**	**157 491**	**20 254**

16.10 Production Schedule

An annual summary of the underground production schedule by pipe and level is shown in Table 16.12. Initial production blasting in the Renard 2 pipe on 290L is scheduled to start in August 2017, at a modest production rate with the commencement of drawcone blasting. Production blasting in the first stope on 290L will be initiated in November 2017, with total production tonnage forecast to reach 60,000 t that month. The tonnage will then gradually increase of 15,000 t each month until May 2018 when it will reach 150,000 t. It is anticipated that the full production rate of 180,000 t (6,000 tpd) will be achieved in June 2018. Generally, each stope will provide between 1,000 tpd and 2,000 tpd in order to maintain a good ore/backfill interaction to optimize recovery and dilution factors.

The underground production schedule has been designed to coordinate with the open pit production schedule so that the mill feed continues at the full rate of 6,000 tpd during the transition period from the completion of the open pit through the build-up to full production from underground.

Underground production will continue from Q3-2017 until early 2029 for a mine life of 14 years. For the Renard 2 pipe, the 290L mining area will be completed in Q1-2021, the 470L area in Q2-2023, the 590L in 2025, and the final 710L area in 2028. The Renard 3 pipe has been scheduled in 2026 and 2027 during the transition of the production from the Renard 2-710 zone towards Renard 4. Due to its small body shape and confined design, the highest planned rate for Renard 3 is 3,000 tpd. The first production from Renard 4 starts in 2027 and continues to the end of the mine life in early 2029.

Table 16.12: Underground Production and Backfill Schedule

Mining Area		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	TOTAL
R2 pipe																
R2-290	Tonnes (t)	5,350	296,383	1,793,495	1,901,012	1,047,978	67,020	-	-	-	-	-	-	-	-	**5,111,238**
	Carats	2,919	183,319	1,134,231	1,234,445	640,798	40,231	-	-	-	-	-	-	-	-	**3,235,943**
	Grade (cpht)	54.6	61.9	63.2	64.9	61.1	60.0	-	-	-	-	-	-	-	-	**63.3**
R2-470	Tonnes (t)	-	-	66,505	253,889	1,024,735	1,643,246	1,549,377	206,497	-	-	-	-	-	-	**4,744,248**
	Carats	-	-	55,220	220,728	902,881	1,339,932	1,271,454	226,492	-	-	-	-	-	-	**4,016,707**
	Grade (cpht)	-	-	83.0	86.9	88.1	81.5	82.1	109.7	-	-	-	-	-	-	**84.7**
R2-590	Tonnes (t)	-	-	-	5,100	87,287	441,836	599,304	1,646,874	1,772,324	197,594	-	-	-	-	**4,750,319**
	Carats	-	-	-	5,270	80,628	360,229	570,087	1,432,150	1,583,044	239,045	-	-	-	-	**4,270,453**
	Grade (cpht)	-	-	-	103.3	92.4	81.5	95.1	87.0	89.3	121.0	-	-	-	-	**89.9**
R2-710	Tonnes (t)	-	-	-	-	-	7,898	11,319	306,622	362,225	1,933,814	1,790,787	520,000	140,589	-	**5,073,253**
	Carats	-	-	-	-	-	6,138	8,801	222,468	296,716	1,570,804	1,515,319	402,600	108,848	-	**4,131,695**
	Grade (cpht)	-	-	-	-	-	77.7	77.8	72.6	81.9	81.2	84.6	77.4	77.4	-	**81.4**
Subtotal R2	**Tonnes (t)**	**5,350**	**296,383**	**1,860,000**	**2,160,000**	**2,160,000**	**2,160,000**	**2,160,000**	**2,160,000**	**2,134,549**	**2,131,408**	**1,790,787**	**520,000**	**140,589**	**-**	**19,679,067**
	Carats	**2,919**	**183,319**	**1,189,451**	**1,460,443**	**1,624,307**	**1,746,530**	**1,850,342**	**1,881,110**	**1,879,760**	**1,809,850**	**1,515,319**	**402,600**	**108,848**	**-**	**15,654,798**
	Grade (cpht)	**54.6**	**61.9**	**63.9**	**67.6**	**75.2**	**80.9**	**85.7**	**87.1**	**88.1**	**84.9**	**84.6**	**77.4**	**77.4**	**-**	**79.6**
R3 Pipe																
R3-250	Tonnes (t)	-	-	-	-	-	-	-	-	25,451	-	191,276	1,005,934	-	-	**1,222,660**
	Carats	-	-	-	-	-	-	-	-	18,105	-	160,419	679,676	-	-	**858,200**
	Grade (cpht)	-	-	-	-	-	-	-	-	71.1	-	83.9	67.6	-	-	**70.2**
R4 Pipe																
R4-270	Tonnes (t)	-	-	-	-	-	-	-	-	-	28,592	177,937	634,066	2,019,411	598,337	**3,458,344**
	Carats	-	-	-	-	-	-	-	-	-	13,603	81,244	282,921	1,000,934	292,527	**1,671,229**
	Grade (cpht)	-	-	-	-	-	-	-	-	-	47.6	45.7	44.6	49.6	48.9	**48.3**
Total	**Tonnes (t)**	**5,350**	**296,383**	**1,860,000**	**2,160,000**	**2,160,000**	**2,160,000**	**2,160,000**	**2,160,000**	**2,160,000**	**2,160,000**	**2,160,000**	**2,160,000**	**2,160,000**	**598,337**	**24,360,071**
	Carats	**2,919**	**183,319**	**1,189,451**	**1,460,443**	**1,624,307**	**1,746,530**	**1,850,342**	**1,881,110**	**1,897,865**	**1,823,453**	**1,756,982**	**1,365,197**	**1,109,782**	**292,527**	**18,184,227**
	Grade (cpht)	**54.6**	**61.9**	**63.9**	**67.6**	**75.2**	**80.9**	**85.7**	**87.1**	**87.9**	**84.4**	**81.3**	**63.2**	**51.4**	**48.9**	**74.6**
Note: Development ore included in totals																
Backfill	**Tonnes (x 1,000)**	**-**	**-**	**1,452,511**	**1,452,350**	**1,435,641**	**1,464,396**	**1,425,600**	**1,457,649**	**1,452,704**	**1,436,299**	**1,425,600**	**1,425,600**	**1,425,600**	**394,902**	**16,248,852**

16.11 Equipment Fleet

The mine has been designed to take advantage of modern drilling and haulage equipment and efficient ore handling systems and mining methods in order to decrease operating costs. This includes the specification of 20 t capacity LHDs to load, haul and transfer stope production to the truck loading stations for direct loading onto 60t trucks. The same equipment will be used for waste development. High productivity down-the-hole (DTH) production drills for the BHS stopes and electric hydraulic top hammer drills for the stope undercut drilling are planned. In addition, a full suite of support equipment will be provided for development and service requirements.

The proposed underground mobile equipment fleet types for the mine are shown in Table 16.13. Most of the development equipment was purchased when underground mining activities started in Q2-2015 with the main ramp excavation, and the remainder will be added when the second development team starts in Q3-2016. However, the production equipment will not be required until mid-2017, when production activities begin.

Table 16.13: Underground Mobile Equipment Fleet

Typical Equipment Type	Supplier/Model	Total Number of Units
Scooptram 20t	CAT R3000H	5
Haulage Truck 60t	CAT AD60	14
Two-boom development Jumbo	Atlas Copco Boomer M2C	3
Scissorlift truck	Maclean Mine Mate SL-3	3
Scissor rockbolter	Maclean Series 928	4
Explosives loader	Maclean CS-3 with Orica loader	2
Mobile shotcrete sprayer mixer	Putzmeister PM 407	1
Boom truck	Maclean Mine Mate BT-3	2
ITH drill c/w rod handler, booster	Cubex Orion, rod handler PT50 and booster compressor S150	3
V30 Cubex reaming kit (up and down)	Machine Roger V30	2
Production longhole drill	Atlas Copco Simba 1354	2
Blockhole drill	Maclean Mine Mate BH-3	1
Water canon	Maclean Mine Mate WC-3	1
Fuel/lube truck	Maclean Mine Mate CS3	1
Compact service wheel loader	CAT 906H2	2
Road compactor	CAT CS56	1
Grader	CAT 140M	1
Backhoe	CAT 420F IT	2
Mobile compressor	Doosan P425/HP375WCU	2
Underground emergency vehicle	Toyota Landcruiser	1
Man carrier	Toyota Landcruiser	3
Mechanics service vehicle	Kubota M9960	1
Electricians service vehicle	Kubota M9960	1
Supervisor/technical vehicle	Toyota Landcruiser	9
Total		**67**

16.12 Material Balance Summary

The entire plant feed schedule is the culmination of ore production from the R2/R3 and R65 open pits, and the Renard 2, Renard 3 and Renard 4 pipes in the underground mine. Figures 16.9, 16.10, 16.11 and 16.12 illustrate the plant feed, diamond production schedules, open pit and underground mining schedules.

Figure 16.9: Plant Feed Schedule

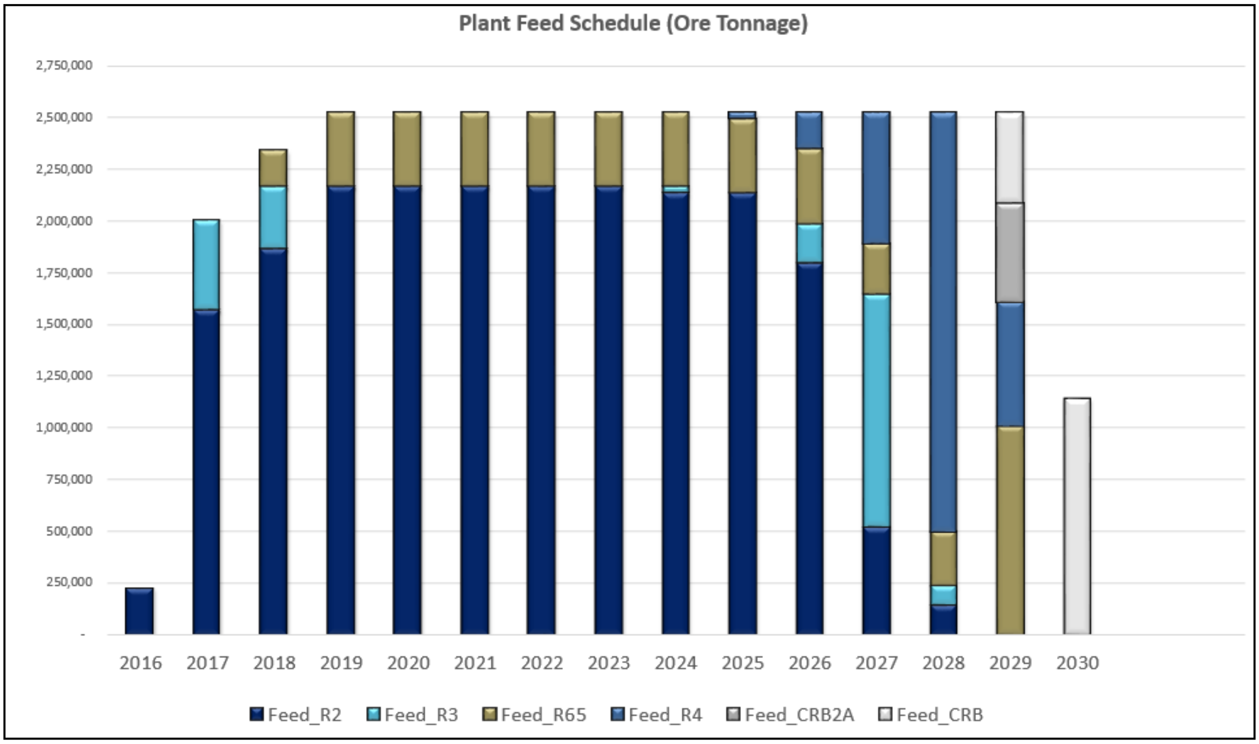


Figure 16.10: Mine Feed Diamond Production (Carats)

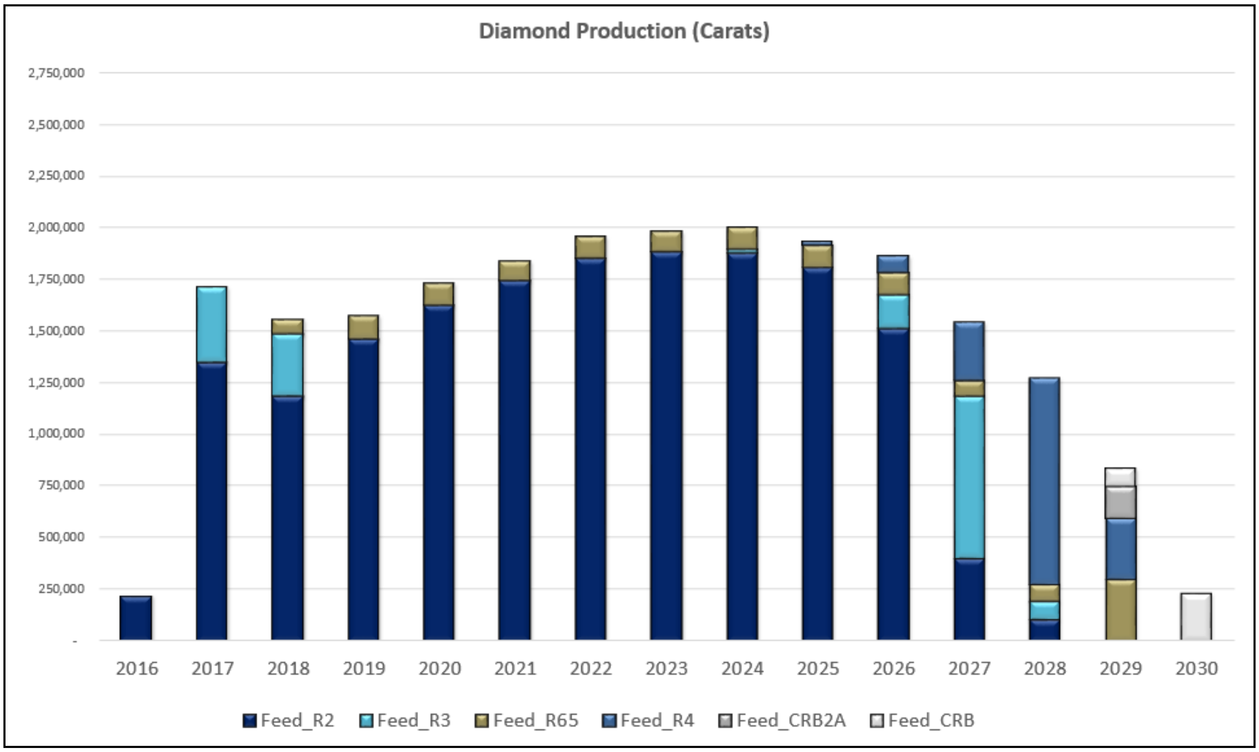


Figure 16.11: R2/R3 Open Pit Mining Schedule

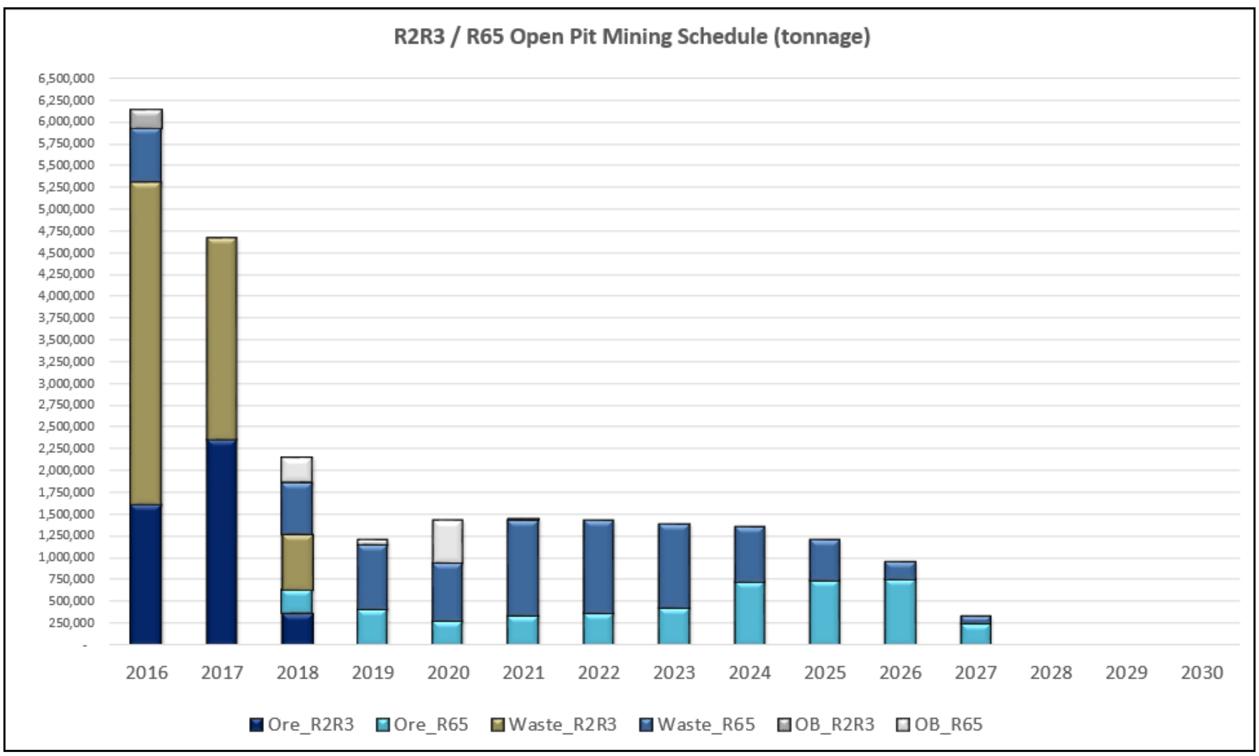
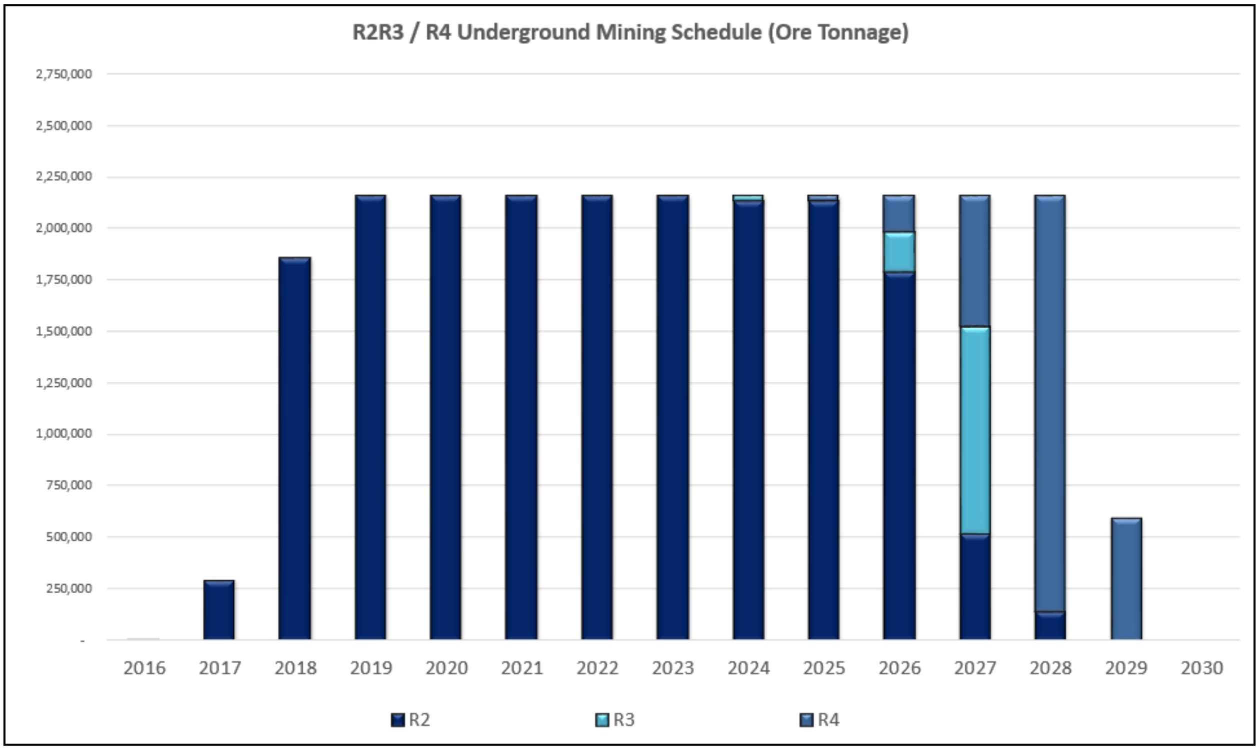


Figure 16.12: Underground Mining Schedule



17. RECOVERY METHODS

17.1 Process Flow Sheet

The process flow sheet in Figure 17.1 was developed based on the following key aspects:

- The material characteristics observed, and data obtained, from the treatment of underground samples from Renard 2, 3 and 4 in Stornoway's Lagopede bulk sample plant;
- Specific unit operation test work;
- Conventional diamond processing techniques as successfully employed in the diamond industry;
- Liberation effectiveness with focus on diamond breakage;
- Circuit simplicity; and
- Cost effectiveness.

17.2 Plant Design

To optimize footprint and capital costs, the diamond process plant design has a single process line for comminution and ore preparation. Capacity considerations dictate that two (2) DMS circuits for concentration and four (4) lines for fines dewatering are required. All process equipment including storage bins and materials handling equipment is housed within a heated building, heated transfer towers or heated conveyor galleries.

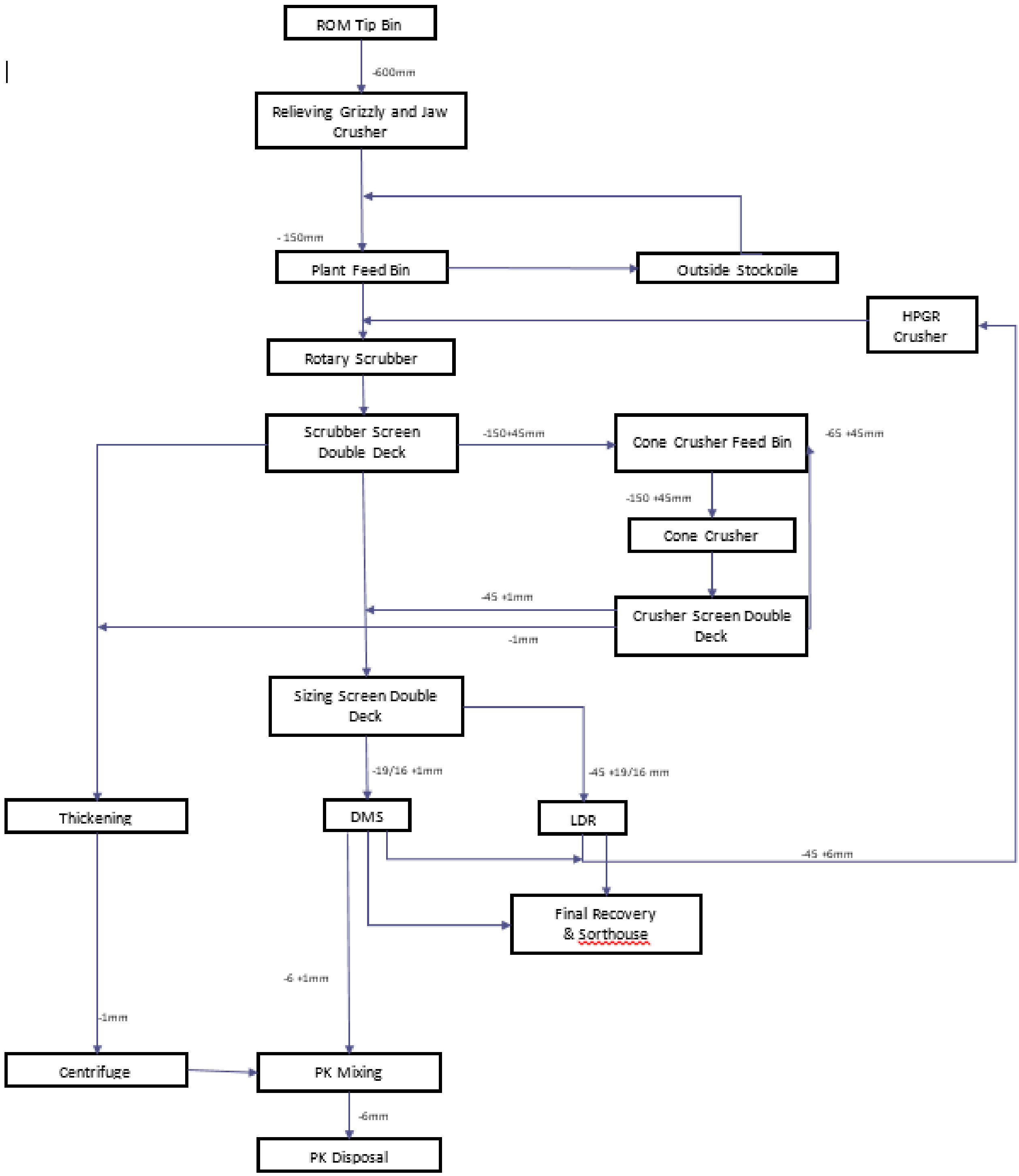
The process plant design capacity is 2,160,000 tpa (dry solids basis). The plant overall utilization is estimated to be 78%, equivalent to an operating time of 6,833 hours per year or 315 tph and 6,000 tpd. Stornoway expects to increase the plant throughput to 323 tph, already within initial plant equipment capacity, and increase the plant availability to 83.5% by optimizing plant maintenance sequences. Once the plant achieves sustainable nameplate throughput, further optimization work will be conducted focusing on liberation, diamond breakage and increasing the overall plant utilization. The ore processing system will liberate, concentrate and recover diamonds from 45 mm to 1mm.

Taking the above into consideration by July 2018, Stornoway expects the plant will operate at a throughput of 7,000tpd at an overall plant utilization of 83,5%.

The ore processing system will liberate, concentrate and recover diamonds from 45 mm to 1mm.

Once the plant achieves sustainable nameplate throughput, further optimization work will be conducted focusing on liberation, diamond breakage and increasing operating plant utilisation.

Figure 17.1: Process Flowsheet



Liberation design will use stage crushing where each crusher will operate with a large crushing gap to protect against potential diamond breakage. ROM material will be crushed, washed and sized at various stages to produce -45 mm ore, before all liberated diamonds are recovered within either a Dense Media Separation (DMS) or Large Diamond Recovery (LDR) concentration process. All rejected material larger than 6 mm from these processes is re-crushed within a High Pressure Grinding Roll (HPGR) crusher to maximize the liberation of trapped diamonds. The HPGR product is then returned to the scrubber circuit to wash and de-agglomerate the HPGR product. The HPGR promotes interparticle crushing used for tertiary crushing and is the principal diamond liberator generating a product size of 60% smaller than 6 mm. The majority of the unwanted fines (-1 mm) are separated from the ore in these circuits and then pumped to the thickening circuit for dewatering.

A surge bin with a capacity of 240 tonnes decouples the DMS from the ore preparation circuit. The process plant will produce a DMS and LDR concentrate which will be treated in a secure diamond recovery facility that uses diamond differentiation techniques based on magnetic, X-ray, laser Raman and ultra-violet technologies, with hand sorting as final de-falsing step to produce a nominally 98% diamond product.

Recovered diamonds will be processed through the cleaning facility and then prepared for valuation within the recovery facility. The diamond plant is expected to have a diamond recovery efficiency of not less than 97% by mass and 99% by value of liberated diamonds.

Liberation design uses three stages of crushing where each crusher operates with a large gap to protect against diamond breakage. Primary crushing is performed by a jaw crusher, which reduces the size of Run-of-Mine (ROM) ore to less than 220 mm. Secondary crushing is performed by a cone crusher to reduce ore to less than 45 mm.

Two 200 tph DMS circuits concentrate -19/16 +1 mm feed by separating liberated diamonds and heavy minerals from lighter kimberlite and waste constituent minerals, in a density controlled ferrosilicon medium. DMS rejects are sized into two streams; -19/16 +6 mm, which is combined with fresh feed reporting to the HPGR, and the -6 +1 mm, which is mixed with dewatered fines and discarded as dry stackable material to the processed kimberlite containment area.

Concentrate surge bins with a capacity of 60 tonnes decouples the recovery plant from the DMS circuits. The recovery plant is capable of processing at a maximum rate of 7.2 tph.

Concentrate fed to the recovery plant is sized into 3 size fractions, -3+1mm (fines), -8+3mm (middles) and -19/16 +8mm (coarse). The fines and middles size fraction are fed to a two stage magnetic separation stage. The non-magnetic material will then be sent via jet pump to the two stages of X-ray sorting and the magnetic material will flow to the rejects sump. X-ray concentrate is stored prior to the reconcentration circuit where the concentrate will be dried and fed to the Recon X-ray machines. The concentrate from the Recon X-ray will then be transfered to the Raven single particle sorting machines. The “super” concentrate (98% diamond by weight) will then be transferred to the sorthouse for further de-falsing. The rejects from the Recon X-ray, Raven and sorthouse (waste) will be collected and sent through to the auditing stream (scavenging) and processed through an UV Osprey machine. The coarse size fraction will transfer directly to a two stage X-ray sorting stage. The concentrate will transfer directly to the sorthouse and the rejects will be transferred and combined with the fines and middles rejects at the rejects sump. The recovery rejects will be jet pumped to a tails screen where the coarser fraction (-45+8mm) will be sent to the HPGR for further liberation and the -6+1mm will combine with the DMS tails for PK disposal.

Final diamond recovery is achieved by hand sorting diamonds from waste in a secure glove box, located within the sorthouse in the recovery plant. The diamonds will be chemically cleaned before preliminary valuation and export.

Security is a key element of the process plant design given the high value and volume of the final diamond product. A security management system to detect, deter and reduce the possibility of diamond theft in conjunction with an automated diamond recovery process is part of the design.

In the rejects disposal circuit, fines rejects (-1 mm) are collected in a header box prior to the thickening stage. The slimes fraction is thickened to recover water for reuse within the plant and the grits fraction is pump fed to a centrifuge stage for dewatering to produce dry stackable fines rejects. Fines rejects are mixed with DMS and recovery plant rejects are conveyed to PK mixer and further conveyed to a PK storage area where the PK will be fed to a truck via belt feeder for disposal to the processed kimberlite containment area.

The following facilities are included in the plant design:

- Main plant office, control room and electrical rooms;
- Recovery plant office, control room and storage room;
- Security offices, access control and search rooms;
- Metallurgical laboratory;
- Air compressor area;
- Mid size maintenance workshop;
- Diamond cleaning facility;
- Diamond stock room; and
- Diamond valuation room.

The level of automation within the process plant is appropriate in order to minimize operator intervention and optimize productivity and safety.

The process plant building has a main fire water ring supplying wall hydrants and fire water hose station distributed throughout the building. Conveyors are protected with automatic wet pipe sprinklers.

17.3 Materials Handling

Production ore extracted from the open pits and the underground mine will be delivered to the surface primary crushing facility by truck located adjacent to the R2/R3 OP and the underground mine portal. On surface, trucks from the open pit will direct tip into a plant feed bin at the primary crushing station or dump ROM ore on a working stockpile located adjacent to the primary crusher. Ore from underground will also be dumped on this stockpile if the plant is unable to receive ore. Stockpiled ore will be reclaimed by front-end loader and delivered to a 120 t capacity primary crusher feed bin fitted with a static grizzly. A hydraulic rock breaker will be provided to break oversize lumps or rocks unable to pass through the grizzly. Ore will be withdrawn by a variable speed apron feeder to feed a vibrating grizzly screen. Oversized material will discharge to a jaw crusher and crushed product will report to the same conveyor as the undersize from the grizzly screen.

The material from the jaw crusher and undersized material from the grizzly will be conveyed and introduced into a plant storage bin. Should the plant be unable to process the ROM material, the level in the plant feed bin will increase until it overflows on the stockpile feed conveyor. The excess/overspill material from the plant feed bin is then conveyed onto an open stockpile situated alongside the plant building. Should more stockpile volume be required, front end loaders will be employed to move the material to another location.

In the event that the plant is able to accept crushed ROM material, and neither mining or primary crusher circuit is unavailable, a front end loader will be used to discharge crushed ROM stockpile material onto the scrubber feed conveyor via a reload chute hopper. The reload chute is fitted with a static grizzly to prevent accidental oversize spillage or large, frozen agglomerated material from entering the plant.

Conventional belt conveyors will be used to transport +1 mm ore and waste within the plant. Conveyors linking the primary crusher, main process building and the transfer tower will be housed in tubular galleries, which are insulated and heated.

Plant rejects will be dewatered and mixed to produce a stackable waste, which will be trucked from a load-out bin to the processed kimberlite containment area.

17.4 Consumption

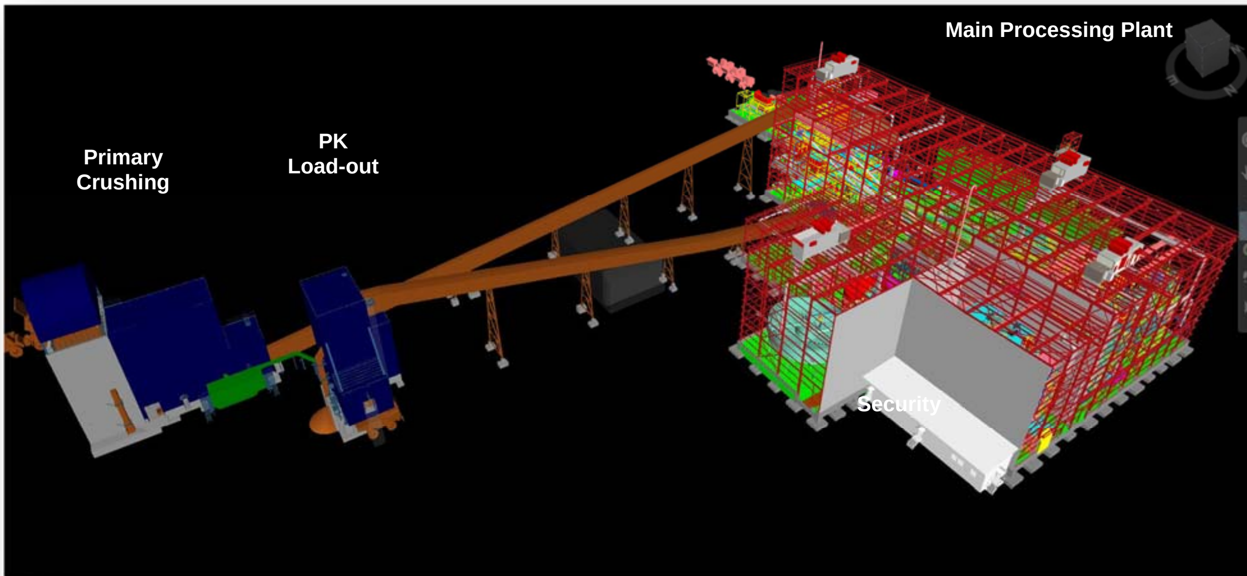
Kimberlite processing to recover diamonds is essentially a mechanical process, with limited use of chemicals or reagents. The estimated consumption of the chemicals/reagents required for a 2.16 Mt/a plant is shown in Table 17.1.

Table 17.1: Consumption of Chemicals/Reagents

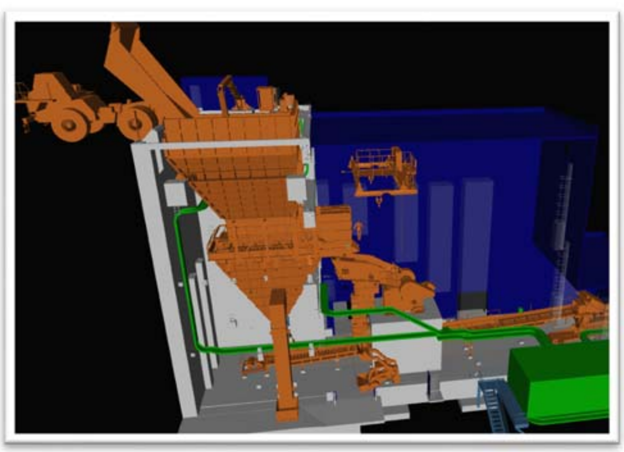
Chemical/Reagent	Consumption
Ferrosilicon	576 tonnes per year
Flocculant	10.8 tonnes per year
Caustic Soda	1.8 tonnes per year
Hydrochloric Acid	3,500 litres per year

17.5 Plant 3D model

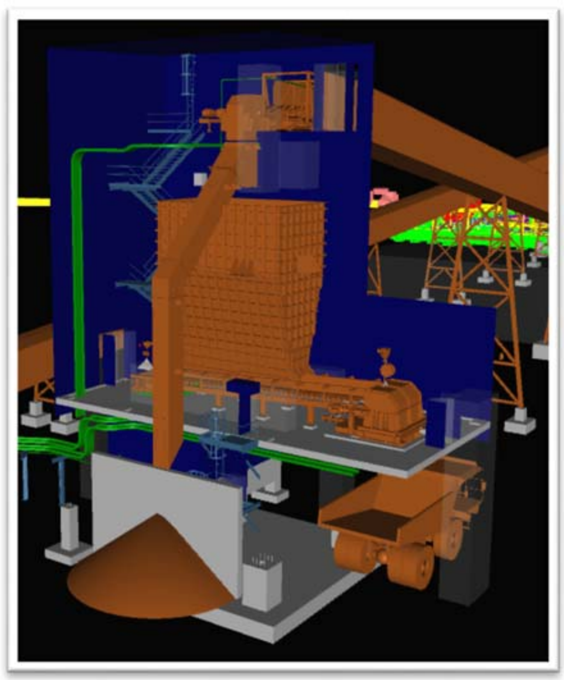
PROCESS PLANT:



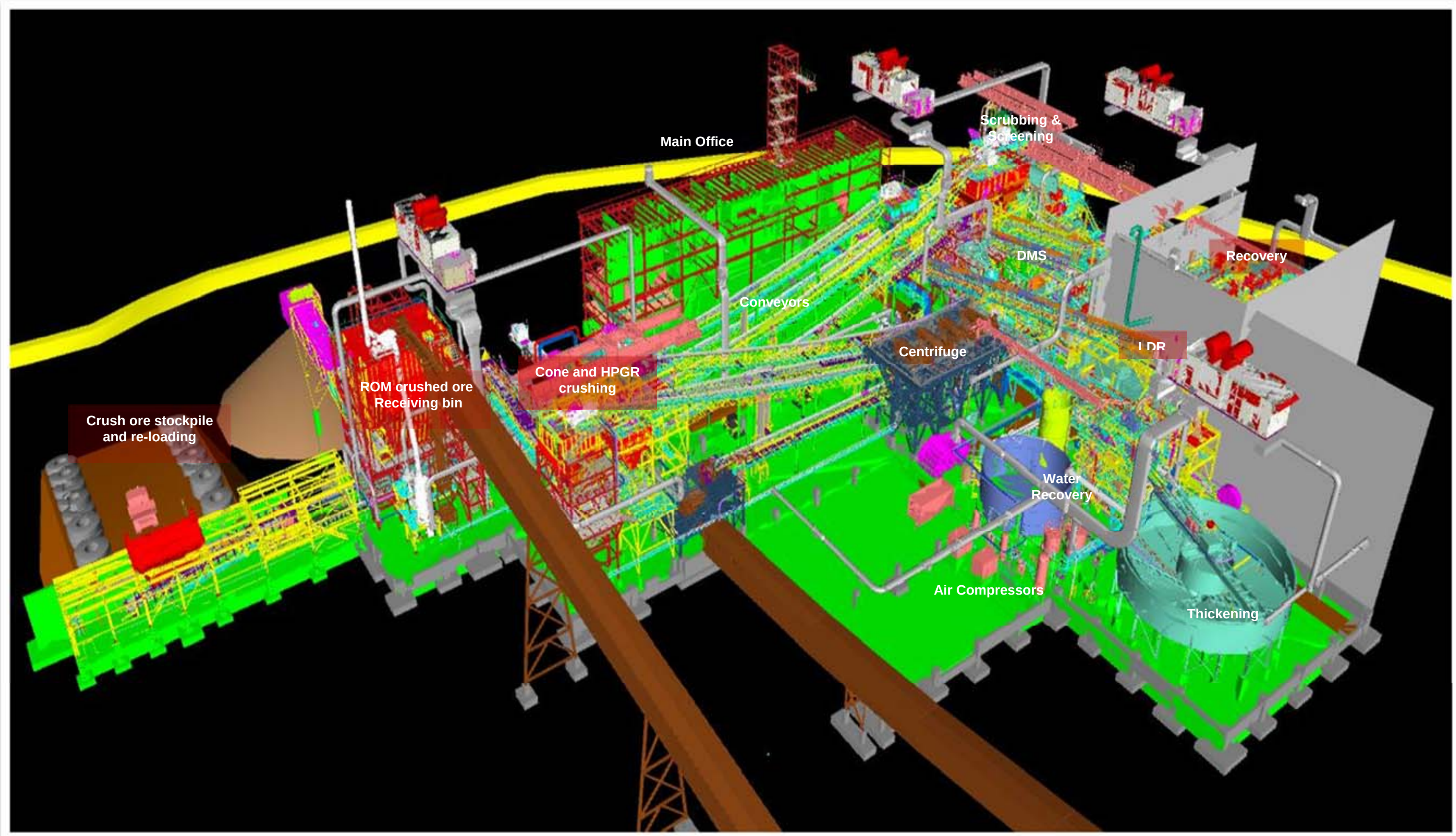
PRIMARY CRUSHER:



PK LOADOUT:



MAIN PLANT:



Main Office
Scrubbing & Screening
DMS
Recovery
Conveyors
Centrifuge
LDR
Cone and HPGR crushing
ROM crushed ore Receiving bin
Crush ore stockpile and re-loading
Water Recovery
Air Compressors
Thickening

18. PROJECT INFRASTRUCTURE

18.1 Geotechnical Investigations

This section provides a summary of the geotechnical investigations performed at the various project sites to determine the soil and bedrock conditions in order to provide input to infrastructure design and construction. A first set of geotechnical investigations was performed in the fall of 2010 and mainly focused on the process plant, airstrip, landfill and processed kimberlite containment (PKC) facility sites. Bedrock surface profiling in the plant site area was also performed using a drill rig. A second set of geotechnical investigation was completed during the summer of 2012 and included the permanent camp complex, process plant and associated infrastructure, domestic waste water treatment site, trench landfill site (TLFS), airstrip, collection ditches, explosive storages site and mine waste water treatment plant.

The soil and groundwater conditions discussed herein describe those encountered at the testing locations and at the time of testing; conditions may vary substantially from those encountered.

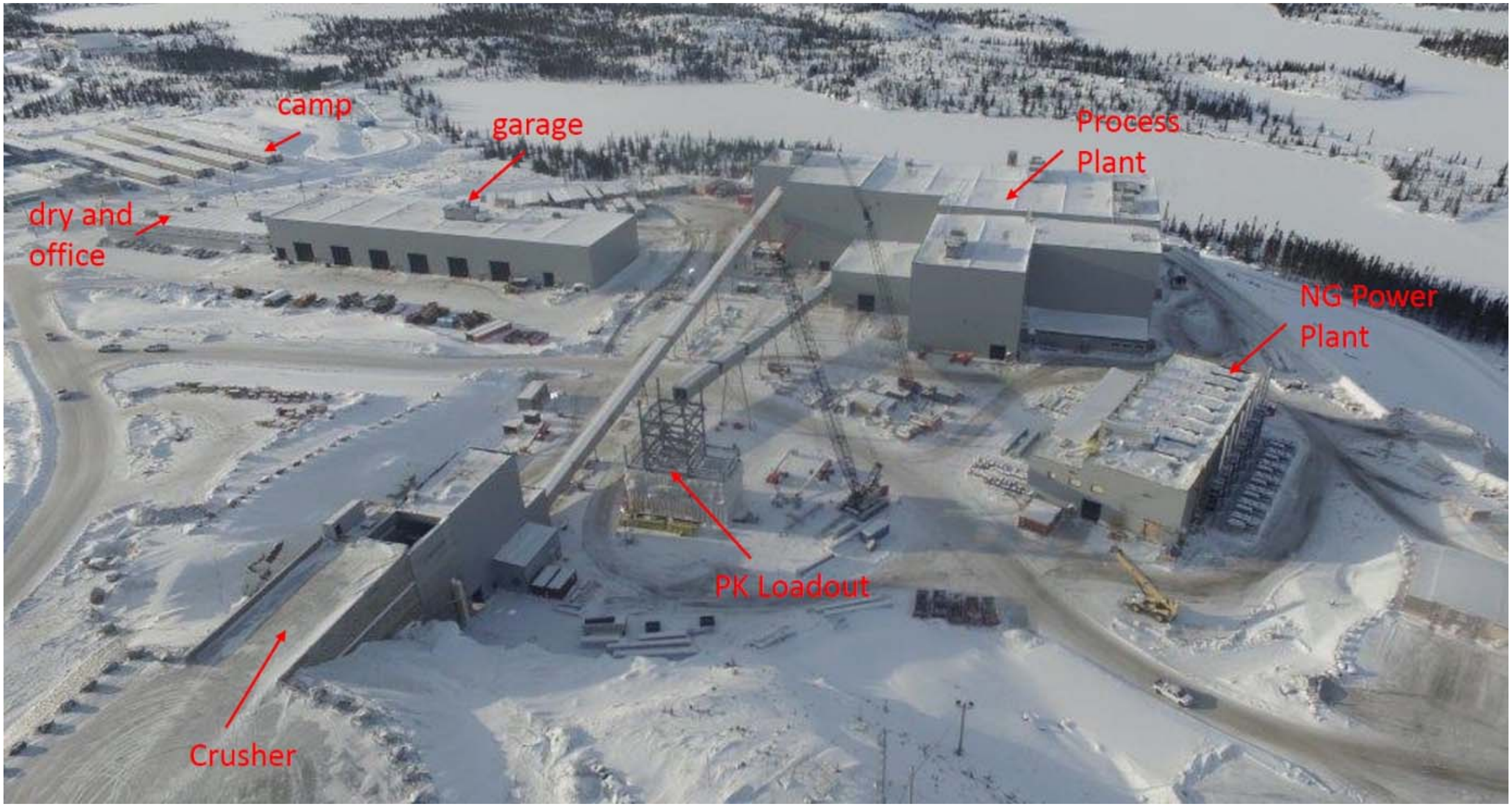
The general stratigraphy of the plant site, airstrip and landfill is topsoil over sand to sand and gravel over till over bedrock. Surficial peat deposits and outcropped bedrock are also present at the sites. The topsoil thickness varied from 0 m to about 1.2 m. The compactness condition of the sand to sand and gravel soils ranged from loose to very dense, whereas the condition of the till was very dense for that of the till. The depth to bedrock ranged from 1.0 m to 6.8 m at the plant site, 1.5 m to 8.5 m at the camp site, 16.8 m to 17.3 m at the airstrip, and 2.8 m to 3.8 m at the landfill at the drilling locations The bedrock depth could not be defined in the waste treatment plant site since no investigation has reached the soil/rock contact.

18.2 On-site Infrastructure

This section describes the on-site infrastructure required for the development of the Renard Project. The section discusses the necessary site preparation, water supply requirements, solid and liquid waste disposal, power requirements, fuel and LNG storage, service building requirements, communications requirements, airport facilities and explosives handling and storage. The locations of the plant site infrastructure and the accommodation complex are shown in Figures 18.1 and 18.2.

Figure 18.1: Plant Site Infrastructure

Figure 18.2: Plant Site Infrastructure



18.2.1 Civil Work

Tree clearing work was carried out over the total area and included the areas for the process plant and infrastructure camp, the wastewater treatment site, the trench landfill site and the airport. The work of stump removal and grubbing, top soil removal and rough grading was done only in areas where infrastructure work is planned.

Three types of roads were constructed, namely heavy-load roads (15 m wide), light-vehicle roads and access roads (7 m wide).

18.2.2 Water Distribution

A raw water intake is installed in Lake Lagopede and a pumping station was installed on the shore at hundred meters from the lake. It is located near the accommodation camp and a few hundred meters from the main process area; consequently, the lake is used as firewater storage. The same pumping station also supplies the potable water treatment plant. The treatment sequence comprises the following steps: sand & anthracite filtration, anionic ion exchange, membrane filtration, following by Nano filtration, pH correction and UV disinfection followed by chlorination.

18.2.3 Wastewater Treatment

The effluent standards for the Project are established according to the characteristics of the receiving environment, the loads of the sewage effluent and the use of the receiving environment. The selected treatment system is a membrane bioreactor (MBR). Sludge's are dehydrated into a disc press and waste bagged for disposal at the landfill.

The effluent from the sewage treatment plant is discharged after treatment into Lagopede Lake, downstream from the raw water intake.

18.2.4 Solid Waste Management

An outdoor sorting center was set up for the management of recyclable waste. Containers are used for sorting and transporting wastes. Sea containers are used for the storage of hazardous wastes. These containers are sent to southern Québec for recycling and/or disposal.

A trench landfill system with a volume of 40,000 m³ is in operation for domestic solid waste management. This volume takes into account the waste which will be produced during the construction, operation and closing phases. Granular materials are stockpiled in the vicinity of the landfill to be used for weekly covering of waste. Composting of kitchen waste is not considered for the moment due to the presence of bears in the sector.

18.2.5 Power Plant and Electrical Distribution

Power generation is provided by seven natural gas (NG) powered generators (2050 KW each) and three diesel-powered generators (1800 KW each). A combination of six NG-powered generators plus 2 diesel-powered generators will be operated at peak capacity while the remaining generators (1 NG + 1 diesel) will be on standby to replace generators undergoing planned maintenance or in the event of a breakdown. The generators operate at 4.16 kV, 3 phases and 60 Hz synchronized together on a common switchgear.

A heat recovery system is installed on the NG generators; heat recovered this way will be used to heat the fresh air for the underground mine.

The Power Plant building is shown in Figure18.3.

Figure 18.3: Power Plant

The main power generation plant feeds the medium voltage distribution network at 4.16 kV. This main switchgear is housed in the main electrical room located near the power plant. An aerial distribution network at 4.16 kV feeds the buildings on the site, and pole-mounted or pad-mounted step-down transformers, as required by the connected loads, reduce the voltage to 347/600 V, 3 phases, 60 Hz. Uninterruptible power supply (UPS) units are provided to the accommodation complex, mine office and airport for telecommunication equipment, fire alarm system, security system and closed circuit television (CCTV).

Emergency generators will be used to support critical loads during power outages at camp buildings.

18.2.6 Heating

Underground mine air will be heated at the fresh air raise (FAR) with heat recovered from the cooling systems of the NG-powered generators in the power plant. Natural gas burners will supplement mine air heating during winter.

Most of the buildings are heated by NG-fired furnaces or by electricity. The Mine Water Treatment Plant, however, is heated by propane heaters as it is too far from the natural gas distribution network.

18.2.7 Fuel and LNG Storage and Regasification Plant

A tank farm was constructed to store arctic grade diesel and unleaded gasoline in double-walled fuel tanks. Fuel stored is mostly used for mobile equipment and for diesel generators (3 mains and emergency generators).

An LNG storage and regasification plant was constructed to power the main generators (power plant). This facility consists of a truck offloading station, six storage tanks and a regasification station with its ancilliary.

In addition to fuels stored on-site, there is a jet-fuel and diesel storage facility at the Renard airport. This facility includes a double-walled tank equipped with integrated loading/unloading pumping systems.

The LNG storage and regasification plant is shown in Figure 18.4.

Figure 18.4: LNG Storage and Regasification Plant

18.2.8 Service Buildings

Accommodation Complex: the accommodation complex houses the reception area, which also serves as the main gate to the site, the service area for the cafeteria, dormitories, one leisure building, a utility building and a garage for emergency vehicles. It is made of wooden modular buildings, factory-fabricated and assembled on site.

Camp Reception Complex: this building is the main entrance to the site and consists of offices, a training room, a reception desk to register and direct people, and a closed room for lockers to store the belongings of workers in transit, and an X-ray scanner for luggage. It also houses the security control room for the site.

Dormitories: the dormitories consist of individual furnished rooms. Each room is equipped with a bed and night table, a desk and a chair, a closet, a bathroom with shower, a commode and a sink. Each room has an IP phone, an internet connection and a television. Two types of rooms are provided: those with private bathrooms and those with a shared bathroom between two rooms. There are 264 private rooms and 106 semi-private rooms.

Emergency Vehicle Garage & Potable Water Treatment Plant: this building is made of an assembly of containers and a light structural frame recovered by a PVC/polyester fabric. The potable water treatment plant is located inside the container area while the central area serves as a garage for the fire truck and ambulance. Heating is provided to prevent freezing.

Service Building: the service building provides the area for food service with a seating capacity of 224 people. The service is cafeteria-style, with a self-service area. It is equipped for food preparation, washing, all required storage and a reception area. The cafeteria is ventilated and air conditioned by a variable volume system. A make-up air system is provided to compensate for the kitchen exhaust hoods.

Leisure Building: the leisure building is fully equipped for physical training activities (weight lifting equipment, stationary bicycles, etc.) and leisure activities (billiards and cards tables, fireplace, television room, cinema and a veranda).

Heavy Vehicle Truck Shop, Light Vehicle Garage, Workshop and Warehouse: this building is used for maintenance of heavy and light vehicles (HME & LV), including a wash bay, welding shop, instrument and electrical shops and a mechanical maintenance shop. A two-storey service area, located above the electrical shop and tool crib, houses offices, a server room, a conference and training room, a canteen, lockers and ablution rooms.

A single-storey warehouse with racking is adjacent to the HV truck shop, LV garage and workshops.

Office Building: the office building is a two-storey building made of wooden structural modules, factory-fabricated and assembled on site. The ground floor houses the mine rescue area, the infirmary, a training room, the women's locker room, the men's locker room, a prevention office, the lamp room and the foremen area.The second floor contains the reception area, offices, the training room, the photocopy and paper room, a meeting room, a lunch area, the computer room, two storage rooms, the telecommunication room and a heating room.

The services buildings are shown in Figure 18.5.

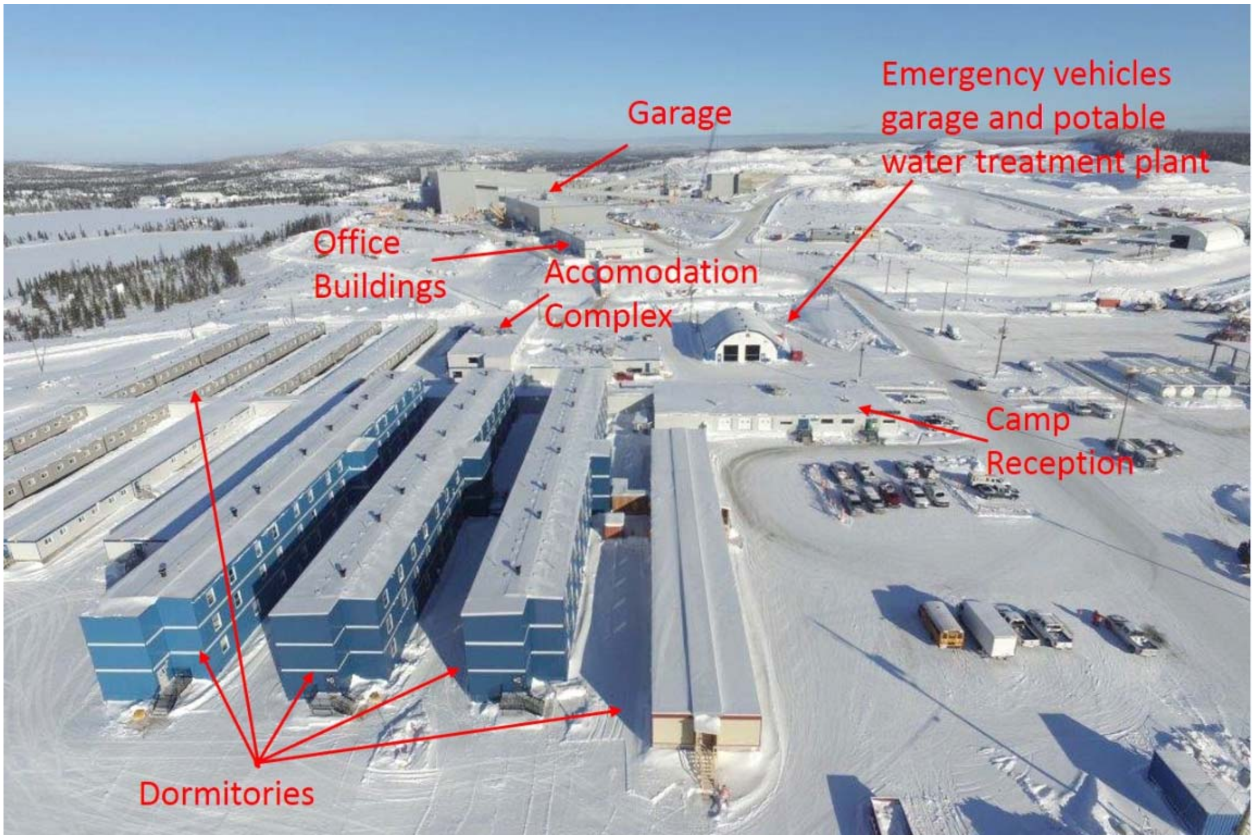
18.2.9 Technical Facilities

The following buildings were constructed as technical facilities:

Raw Water Intake & Pumps Station: the pump station houses two firewater pumps and a jockey pump, two raw water pumps to feed the potable water treatment plant, and two raw water pumps to feed the Process Plant. All pumps are fed through a raw water intake installed in Lake Lagopede.

Potable Water Treatment Plant: the potable water treatment plant is installed in a container located within the Emergency Vehicles Garage. It houses the water treatment equipment, pumps and a small office.

Figure 18.5: Services buildings



18.2.10 Telecommunications

A single satellite dish antenna, modems and related equipment have been installed at the mine site for recreational use (mainly TV signal to the rooms). Voice, video conferencing and data transmission are assured to the Renard mining site via a multi-megabit Microwave link going through seven repeater towers. The above Microwave link originates at the office building where MPLS connectivity to all other Stornoway locations is provided. The airport communication systems are serviced from the mine site by way of another multi-megabit Microwave link. This provides surveillance of the remote airport site while it is unmanned.

IP Telephone service with QoS (Quality of Service) is installed at all site locations where warranted for operation and/or emergency communication needs. The industrial areas of the site (mill, power plant, garages etc.) are linked via fiber optic cables and equipped with more robust telephone sets and cabling, where warranted. VLANs with Firewall rules are used in order to keep recreational, corporate, production, CCTV and other types of content isolated from each other.

18.2.11 Access Control

All vehicles accessing the site via the main road have to pass through the main gate, which is temporarily located on the main road near the Renard airport junction; this is where all access checks are conducted. Next spring, the main access gate will be relocated to the accommodation complex and a new gate will be installed at the LNG tank farm to control its access.

The CCTV provides coverage of the exterior of the mine complex, the accommodation complex, the airport and the Power Plant. The main monitoring station is installed in the video surveillance office at the camp reception complex.

18.2.12 Airstrip

The airstrip for the Renard Project is owned and operated by Stornoway with chartered aircrafts for its exclusive needs. The reference aircraft for the design of the runway is the DASH 8, Series 300 with a capacity of 50 passengers. The airstrip is certified by Transport Canada and is equipped to support night operations, including visual aids and lighting.

A gravel surface covers the runway, taxiway and apron areas.

The airport terminal is a two storey building, erected on site from an engineered wood structure. It includes an office for the airport operator, toilets and a general waiting area. A service shed, also built from an engineered wood structure, houses the generators and de-icing equipment. Jet fuel and diesel are stored in double-walled tanks.

A self-supporting pre-engineered steel clad structure building is the maintenance garage for the storage of materials and the wheeled equipment used for the summer runway maintenance and winter snow clearing activities. All equipment maintenance is performed at the mine site truck shop.

Water supply forthe terminal building and maintenance garage is provided by a water well. Bottled water is used for drinking water. Wastewater is treated by a small local septic installation.

The Airstrip is shown in Figure 18.6.

Figure 18.6: Airstrip

18.2.13 Explosives Storage and Handling Area

The Explosive Storage and Handling Area consists of storage containers for emulsion, one detonator magazine, one packaged explosive magazine and a garage / wash bay building.

Bulk emulsion is stored into 30,000 kg storage containers located at the Explosive Storage Area. These containers are electrically heated to prevent freezing. From these containers, emulsion is pumped into an emulsion truck for open-pit blasting or into 1,300 kg bins for underground operations.

The wash bay garage is used to park and wash the emulsion truck. It is a pre-fabricated fold-away type building assembled on site.

18.3 Off-Site Infrastructure

<u>*Overview*</u>
Land access to the Renard Project is provided by the extension of provincial highway Route 167 (built by the *Ministère des Transport du Québec*) and the Renard mining road (built by Stornoway).

This road infrastructure affords year-round access linking the project to the municipalities of Mistissini and Chibougamau. The Route 167 Extension was constructed to MTQ standards and is a two-lane gravel-topped road with two-lane bridges and a design speed limit of 70 km/h, while the mining road is a two-lane gravel topped Class III road with one-lane bridges and a design speed limit of 50 km/h.

The portion of the Route 167 Extension Certificate of Authorization (CA) that is applicable to the Renard mining road was transferred to Stornoway in 2012 in order to support the construction of this segment of the road.

Operating and maintenance costs for Route 167 Extension are covered by the Quebec Government, while operating and maintenance costs for the Renard mining road are entirely assumed by Stornoway.

Provincial Highway Route 167 and Mining Road Layout
Figures 18.7 and 18.8 below illustrate the selected road layout for Route 167, Route 167 Extension and the Renard mining road.

Figure 18.7: Road Location Map

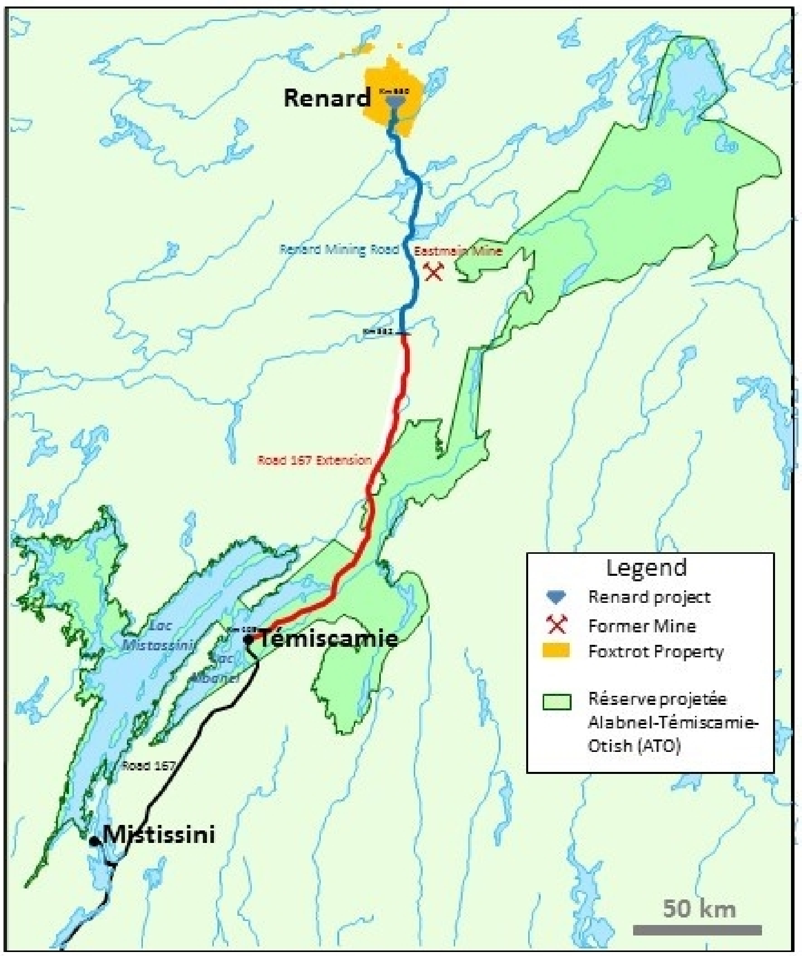


Figure 18.8: Road Access

18.4 Processed Kimberlite Containment Facility

This section provides a summary of the geochemical classification of the kimberlite, processed kimberlite (PK) and waste rock, and the design of the processed kimberlite containment (PKC) facility.

18.4.1 Facility Design

The PKC facility is the long-term storage facility for the processed kimberlite (PK) generated during operations. The facility has a total footprint area of about 72 hectares (ha) and is planned to store the 44.8 megatonnes (Mt) of PK (23 Mm3 of PK) generated during the planned 19-year mine life (Long Term Business Plan). The PKC facility at closure is shown in Figure 18.9 below.

The site, located in the Canadian subarctic, is permafrost-free and of low seismicity. The PKC facility is located on top of a watershed, which generally drains towards the processing plant located to the southwest, thus facilitating water management.

Geotechnical testing carried out to date classifies the PK as well-graded sand with some gravel to gravelly sand with some low plastic or non-plastic fines. Of concern is the potential plasticity of the PK; therefore the PKC facility was designed based on the potential undrained behaviour of the PK. Geochemical testing carried out to date showed that the PK and waste rock release low concentration of dissolved constituents; therefore, the PK and mine waste rock are classified as low risk materials. Furthermore, the waste rock is considered to be Category I and is suitable for use as a construction material without restriction. Geotechnical and geochemical testing of the PK will be performed throughout the development of the PKC facility.

The design objective of the PKC facility is that it receives all materials generated by the processing plant at all times such that neither mining nor processing operations are adversely affected. The design, including development and operating plans, has been developed in consideration of the expected variability of the PK and in consideration of the facility having a high consequence of failure as per the 2013 Dam Safety Guidelines published by the Canadian Dam Safety Guidelines. The PK will be dewatered at the processing plant and trucked to the PKC facility. The facility will be developed throughout operations as a stacked facility with an external slope configuration of 3H: 1V. Placement of the erosion protection layer on the exterior of the PKC facility will occur during operations, thereby progressively closing the facility during its development. No water will be allowed to pond within the PKC facility.

The development of the PKC facility is required to provide the required storage capacity for the PK produced during the operation of the mine. Therefore, it is of critical importance that the development of the PKC facility maximize the use of PK as the construction material. All non-PK materials placed in the facility reduce the available storage capacity for PK; their use must be minimized. A key objective of the early development of the PKC facility will be to develop PK management and construction methodologies such as the use of non-PK materials to be reduced to the practicable minimum.

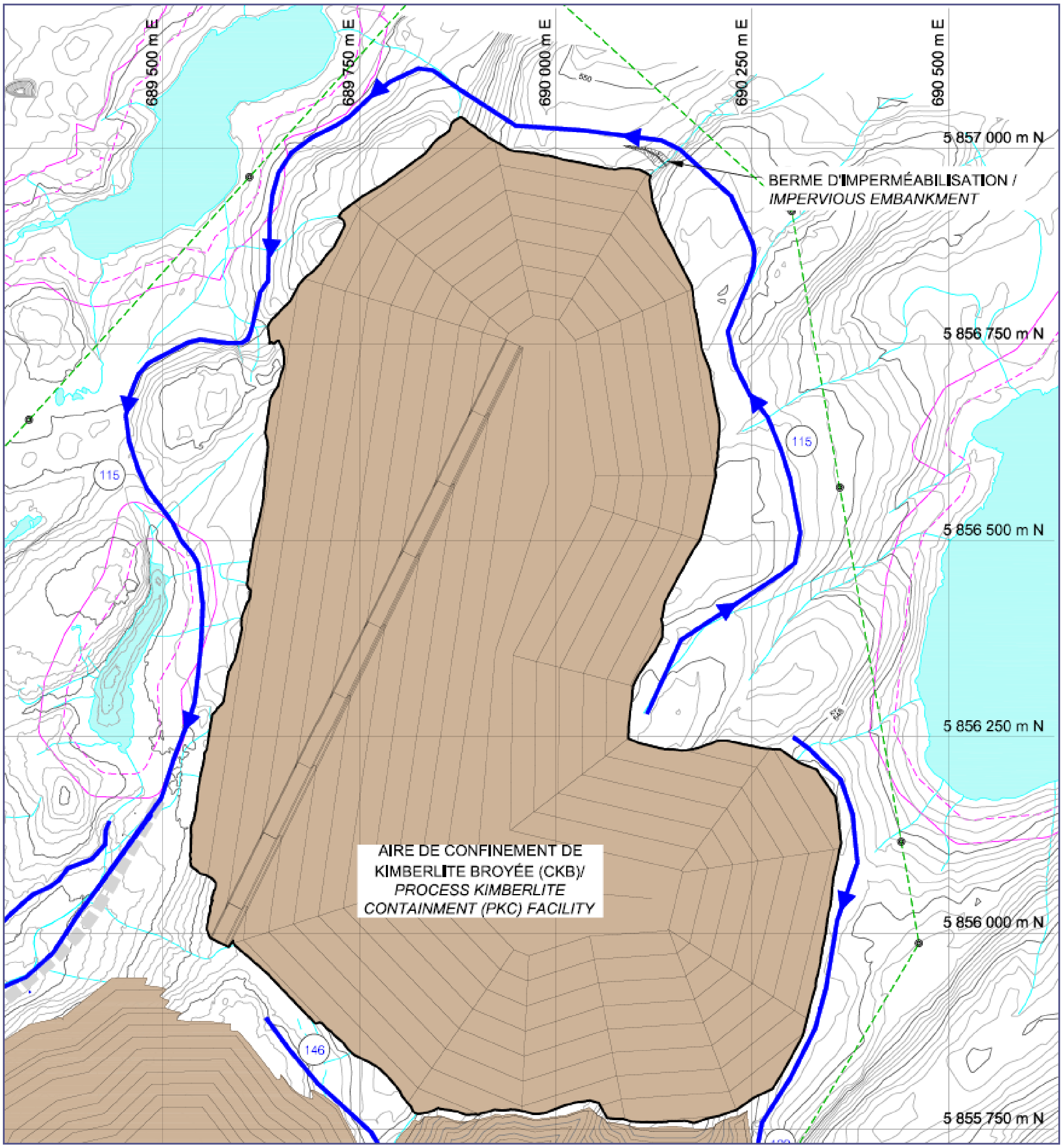
The PKC facility includes material placement zones (nominally compacted zone, engineered filled zone, and PK waste zone), a starter berm, a containment berm, an access ramp to allow haulage of PK, as well as a water management system, including internal drainage elements (rockfill blanket and foundation rock drains), a slope drainage channel for the access ramp, and a series of ditches and sumps around the perimeter of the facility. The engineered fill zone corresponds to the outer shell of the PKC facility and provides sufficient material strength to ensure stability of the facility. This zone will be constructed on a prepared foundation and placed as an engineered fill. The nominally compacted fill zone corresponds to the internal portion of the PKC facility. This zone is built with PK material and provides PK storage capacity into which the material will be deposited with nominal compaction control measures. During processing plant commissioning and early operations, the PK is expected to have higher water content than for the duration of operations. The PK waste zone contains PK with excessively high water contents requiring the construction of the containment berm to separate the PK waste from both the engineered and nominally compacted fill zones. A rockfill embankment, namely the starter berm, will be constructed prior to the deposition of PK to allow for the deposition of PK with high water content. The starter berm will be placed on the interior limit of the engineered fill zone.

The PK is expected to be variable over the life of mine; this will affect its geotechnical behaviour. The key parameter for the development of the PKC facility is the water content of the PK. The PK hauled and placed in the PKC facility is expected to be several percents above the geotechnical optimum water content. Mechanical reworking of the PK to dry it to enable the required compaction within the engineered fill zone is expected to be required. The facility development and PK management plans are developed with this in mind.

PK is an erodible material. Therefore, the crest of the facility will be sloped and crowned to promote drainage of surface water and precipitation towards water collection systems. The placement of the erosion protection layer on the external slopes will be concurrent with PKC facility development. This is consistent with the progressive closure concept aimed to provide environmental, financial and operational benefits for the mine. For facility closure, the PKC facility pile will be contoured and surfaced to mimic the surrounding landforms.

The conditions in the foundation and deposited PK will be monitored during the development of the PKC facility and into the closure period. The geotechnical instrumentation program includes the installation and the monitoring of piezometers (measurement of phreatic surface and porewater pressures within the PK), thermistors (temperature measurement within the PK), and survey monuments (displacement measurement of the PK).

Figure 18.9: Processed Kimberlite Containment Facility - At closure



18.4.2 Waste Rock, Ore & Overburden Piles

Overburden, waste rock and kimberlite ore will be stored in separate piles. The specifications for these piles are presented in Table 18.1 and Figure 18.10.

The overburden pile is sized to include the overburden material mined from the R2/R3 OP and the overburden mined from the Renard 65 borrow pit. The total overburden to be stored is 3.36 Mt, or 2.2 million loose cubic meters since some overburden will be used to surface slopes of the PKC facility. The overburden pile will be constructed in 3 metre lifts and will be 32 meters in height. The slopes of the overburden pile will be armoured with rip-rap material to increase slope stability.

The waste rock dump is sized to contain waste rock mined from the R2/R3 OP, R65 OP and underground development. The total waste rock pile at the end of March 2018 is 7.2 Mt.

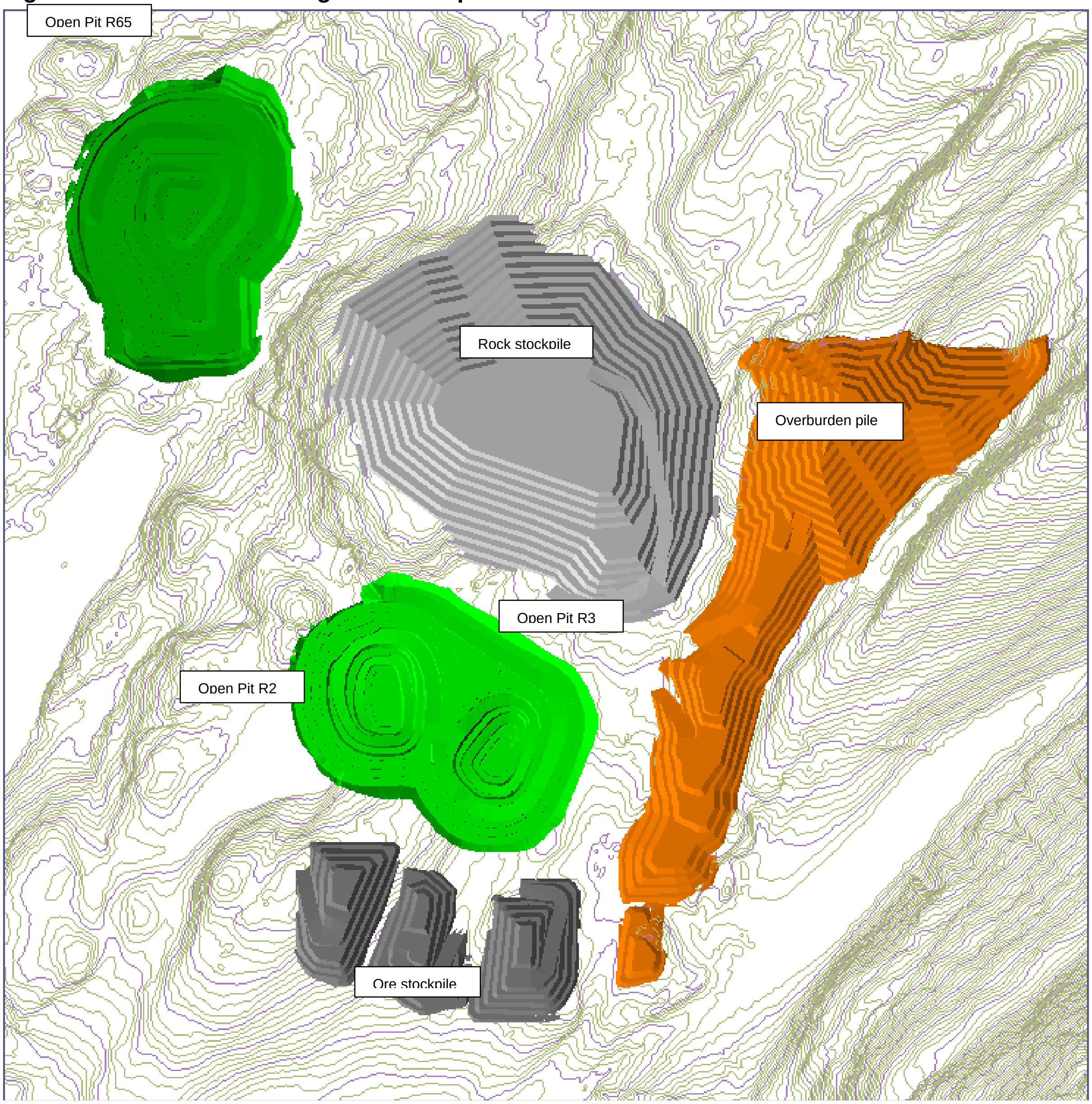
The waste will be re-handled and used as backfill into the R2/R3 OP, once the underground operation breaks through into the pit bottom, and into the Renard 4 mine through backfill raises.

The ore stockpile base is built with approximately 125,000 m3 of rock fill and is designed to accommodate approximately one Mt of ore.

Table 18.1: Pile Specifications

Pile Specifications	**Ore SP R2**	**Ore SP R3**	**Ore SP CRB2a**	**Ore SP R65**	**Ore SP CRB**
Lift height (m)	5	5	5	5	5
Lift offset or catch bench (m)	5	5	5	5	8.37
Pile face angle (deg)	37° (1.33H:1V)	37° (1.33H:1V)	37° (1.33H:1V)	37° (1.33H:1V)	37° (1.33H:1V)
Overall slope angle (deg)	Avg. 26°	Avg. 26°	Avg. 26°	Avg. 26°	Avg. 21°
Pile maximum height (m)	30	25	28	15	43
Elevation of pile surface (m)	543	533	536	528	565
Footprint surface area (ha)	2.428	2.236	2.686	2.428	7.465
Volume of pile (Mm^3)	0.230	0.162	0.226	0.039	1.252
Pile Specifications	**OVB Pile**	**Waste Dump**			
Lift height (m)	3	5			
Lift offset or catch bench (m)	8	8.37			
Pile face angle (deg)	27° (2H:1V)	37 (1.33H:1V)			
Overall slope angle (deg)	Avg. 12°	Avg. 26°			
Pile maximum height (m)	32	100			
Elevation of pile surface (m)	560	585			
Footprint surface area (ha)	23.1	25.9			
Volume of pile (Mm^3)	2.2	3.7			

Figure 18.10: General Arrangement of Open Pits and Piles



19. MARKET STUDIES AND CONTRACTS

19.1 Diamond Pricing and Market Studies

19.1.1 Diamond Market Fundamentals

Rough diamonds are a mined product characterised by a high degree of non-homogeneity in terms of size, colour, quality and shape. Rough diamonds are not exchange traded and have no terminal market. Rather, they are sold directly by mine producers to a wide range of clients within an industry pipeline that funnels their transformation into polished diamond jewellery through cutting and polishing, polished diamond wholesaling, diamond jewellery manufacturing, and end-product distribution to the retail markets of the world.

Most mine producers will undertake the sorting of their rough diamond production into parcels of like characteristics so as to maximise their attractiveness and value prior to sale. Sales mechanisms include contracted sales with a regular list of qualified rough market buyers, tenders and auctions, offtakes with retail end-users, and downstream beneficiation partnerships designed to capture value-add. Sales are typically transacted on a cash basis in US dollars.

Diamond pricing is set by a mine producer based on a regularly adjusted price book, or set on the basis of an achieved price achieved in a tender or auction sale. No benchmark price list for rough prices exist, although several rough market agencies, such as RoughPrices.com, publish a rough price index based on proprietary transactional data and public data compiled from the international Kimberley Process and producer country customs agencies.

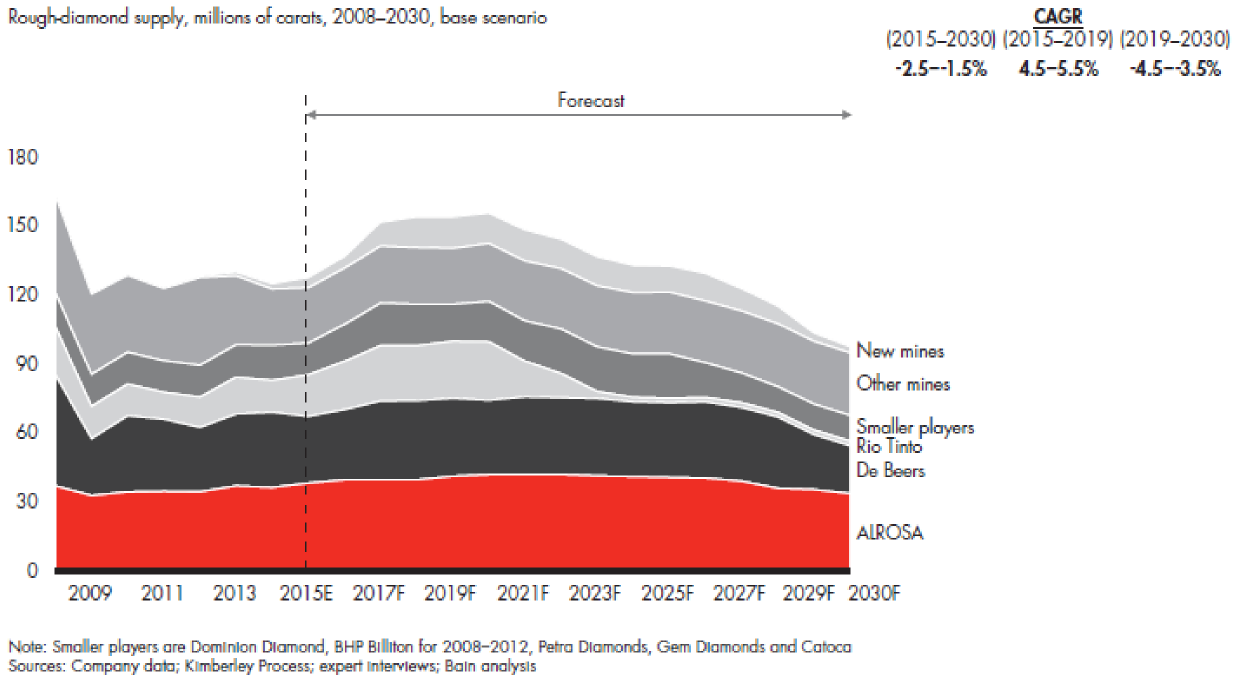
Stornoway intends to sell the diamond production from the Renard Project in 10 tender sales per year in Antwerp, Belgium. To this end, Stornoway's wholly owned subsidiary FCDC Sales and Marketing Inc. has entered into a sales and marketing agreement with the diamond industry broker and rough distributor Bonas-Couzyn, which will act as sales commissionaire and tender agent for arm's length market sales. Sales of diamonds under the Renard streaming agreement will be undertaken by Bonas-Couzyn, on an undivided basis, on behalf of FCDC and the streamers.

19.1.2 Diamond Supply and Demand Outlook

In most diamond supply and demand forecasts, future rough diamond supply is assessed on the basis of current and future production plans at the major producing mines. These supply projections are generally robust, as most of the world's major diamond mines are operating on a steady-state basis or transitioning to lower production rates as they age, and it typically takes between 8 and 12 years to find and develop a new diamond mine. In addition to the Renard Project, two new projects are expected to start production between 2016 and 2017: the Ghacho Kue Project in Canada, a joint venture between De Beers Canada and Mountain Province Diamonds Inc., and the Liqhobong Project in Lesotho of Firestone Diamonds.

Including these three new projects, Bain & Company forecast a potential maximum of 20 million carats of new diamond production between 2013 and 2019 on the basis that all currently known sources of rough supply are brought to the market. This yields a forecast Compound Annual Growth Rate ("CAGR") of 4.5% to 5.5% in carat terms (Bain, 2015). Thereafter, a CAGR of -4.5% to -3.5% is forecast based on the steady depletion of existing mines without new discoveries (Figure 19.1).

Figure 19.1: Bain & Co. (2015) Forecast for World Rough Diamond Supply (Carat Terms)



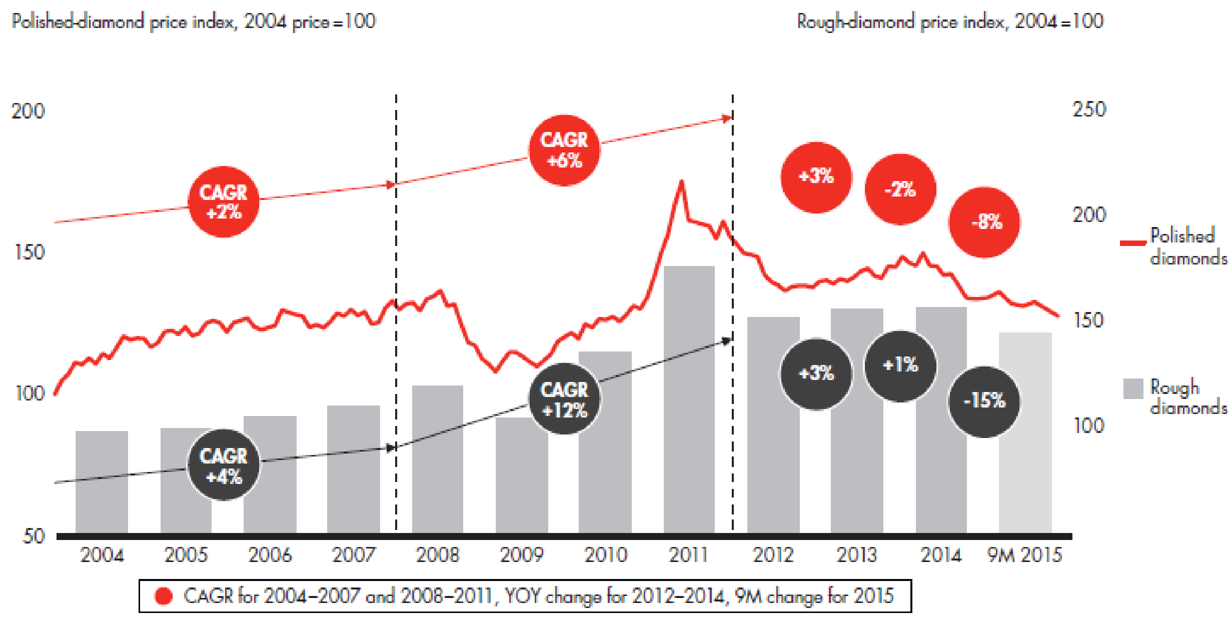
Diamond demand forecasting (expressed as either demand for rough diamonds in the cutting centers or demand for polished diamonds for diamond jewelry manufacture) is a more complex calculation based on, amongst other things:

- Long term Gross Domestic Product (GDP) growth forecasts in the main consumer markets;
- Diamond jewelry consumer growth trends in developing markets;
- Short term inventory draw-down or re-stocking trends within the diamond pipeline, often linked to levels of bank debt; and
- The impact of diamond recycling or synthetic diamond substitution on polished diamond demand.

The principal difference between published diamond demand forecasts is whether they are based purely on the first of these elements solely and assume steady state diamond demand linked to GDP growth or whether they also take into account the second and fourth of these demand factors, and allow for fundamental market changes outside of GDP growth and short term inventory movements.

2015 was a year of volatility in the world diamond markets which saw price declines for both rough and polished diamonds in the face of moderate sales growth for diamond jewellery in the US market, flat or negative growth in China, and low profitability for those in the transformative, middle part of the diamond pipeline (Figure 19.2). High stocking levels of polished diamonds along with high debt levels in the cutting centers created the impression of a short term over-supply of rough diamonds and low sell-through of polished diamond jewellery.

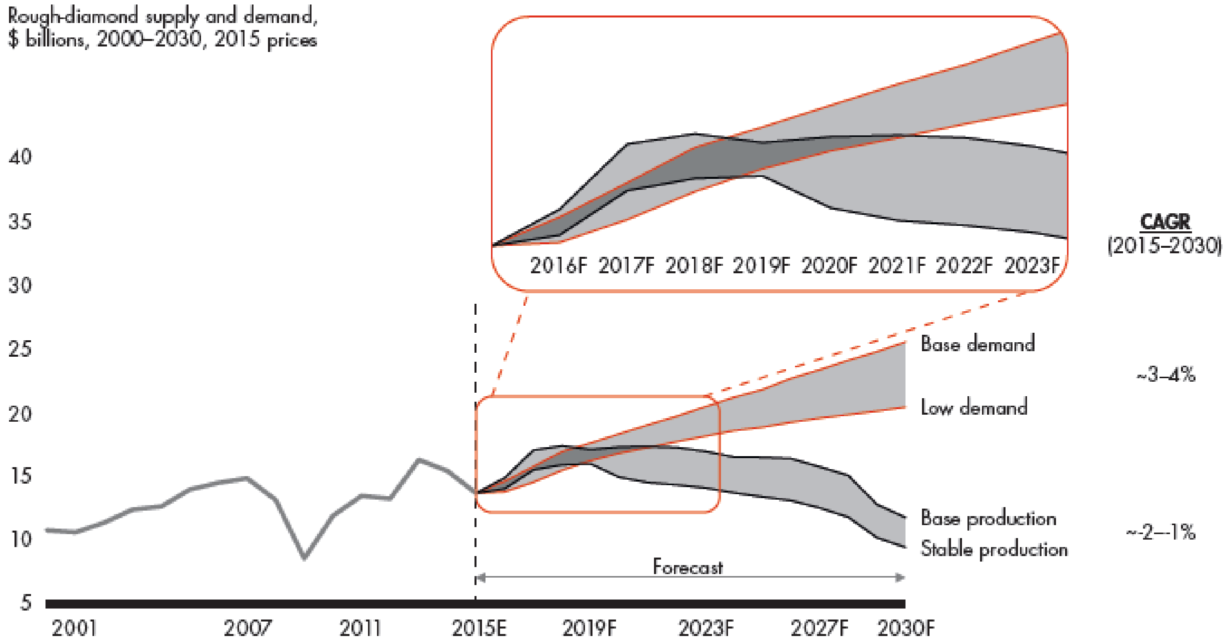
Figure 19.2: Bain & Co. (2015) Rough and Polished Diamond Prices 2004-2015



Note: The CAGR for polished-diamond prices is calculated as the growth rate for year-end or period-end prices
Sources: WWW Diamond Forecasts; Kimberley Process; company data; Bain analysis

On a longer term basis, Bain & Company forecast a global CAGR in rough diamond demand of 3% to 4% to 2030, with a period of supply balance between 2015 and 2019 as the new projects come into production (Figure 19.3). The differential between a long term positive demand outlook and negative supply outlook prompts the common adoption in the diamond mining industry of a real terms price escalation factor on rough diamond prices for the forward planning and valuation purposes, with real terms escalators of between 1% and 4% common. Stornoway has typically chosen to represent forward diamond prices based on a combination of “spot” diamond market pricing and a forward escalation factor of 2.5% CAGR in the base case, applied for a period of 10 years, with sensitivities of 0% and 5%.

Figure 19.3: Bain & Co. (2015) Forecast for World Rough Diamond Supply and Demand (Dollar Terms)



Note: Rough-diamond demand has been converted from polished-diamond demand using historical ratio of rough diamonds and polished diamonds values
Sources: Kimberley Process; Euromonitor; EIU; expert interviews; De Beers; IDEX, Tacy Ltd. and Chaim Even-Zohar (dynamics for 2000–2005); Bain analysis

19.2 Diamond Price Estimates

Between May 9th and May 13th, 2011, parcels of diamonds recovered during the 2007 bulk sampling program from the Renard 2, 3 and 4 kimberlite pipes (the “Valuation Samples”) were valued in Antwerp, Belgium under the supervision of WWW International Diamond Consultants Ltd. (Table 19.1). WWW also valued diamond parcels from the Lynx and Hibou kimberlite dykes. WWW is an internationally recognized independent diamond valuation and advisory service to diamond mining and exploration companies. In Canada WWW, through Diamonds International Canada (DICAN) Ltd., serves as the valuator for the Government of the Northwest Territories and the Government of Ontario.

In addition to performing its own valuation, WWW showed the Renard 2, 3 and 4 diamond samples to four other experienced rough diamond companies in order to obtain additional market based valuations (an “open market” valuation). In each case WWW’s own valuation was higher than the average of the five independent valuations and the average was used by WWW to construct a diamond price model, with “High” and “Minimum” sensitivities based on alternate interpretations of diamond quality and potential value. Diamond price models represent the true diamond price that might reasonably be expected for a kimberlite ore body based on standard commercial-scale recoveries of all diamond size classes. They differ from the achieved diamond valuation price principally through a correction which is applied for the absence of large diamonds which are typically under-represented in exploration scale samples. The choice of “Minimum” and “High” to describe the sensitivity limits is deliberate: in WWW’s view it is highly unlikely that an actual diamond price achieved for each kimberlite ore body upon production would fall below the “Minimum” sensitivity, but it is possible that the actual diamond price achieved may be higher than the “High” sensitivity, which is not a maximum price.

At the time of the May 2011 open market valuation, WWW recommended the adoption of a single diamond price model for the Renard 2 and Renard 3 valuation samples given the similarity of the diamonds in terms of diamond qualities and size distribution.

A separate diamond price model was adopted for the Renard 4 Valuation Sample given its apparently finer distribution of diamond sizes and marginally different diamond quality characteristics. However, independent studies on diamond breakage and plant performance during the processing of the Renard bulk samples have indicated that the size distribution of the Renard 4 sample was most likely been modified during its recovery. For this reason, an alternate diamond price model for the Renard 4 sample has been adopted for planning purposes since the May 2011 valuation exercise which assumes a diamond size distribution equal to the average Renard 2-Renard 3 size distribution.

The collection of a bulk sample of diamonds from the Renard 65 kimberlite in 2013 revealed a diamond population with a markedly different assortment of diamond qualities compared to any of the other kimberlite pipes. Accordingly, individual price models have been adopted for each kimberlite pipe at the Renard Project since this time on the basis that the small differences in diamond quality and size distribution that can be observed between the pipes should be treated as real. Updated diamond valuation exercises were conducted on this basis by WWW in March 2013 and March 2014 (Table 19.2). For the Renard 2, 3 and 4 valuation samples, the result of each WWW re-valuation was used to adjust the average valuations obtained in the May 2011 exercise from the five independent valuators, and a revised diamond price model with High and Minimum sensitivities generated. The Lynx and Hibou samples were not re-valued in 2013 and 2014 and the proposed mine plan for Renard presently does not include these kimberlite bodies. At the time of the most recent WWW re-valuation in March 2014, base case diamond price models of US$197/carat were calculated for Renard 2, US$157/carat for Renard 3, US$155/carat for Renard 4 and US$187/carat for Renard 65.

Table 19.1: May 2011 "Open Market" Diamond Valuation Results

Body	Valuation Sample (carats)	Open Market Valuation of Valuation Samples[1]					WWW Price Modeling		
		Number of Independent Valuations	Average of Independent Valuations (US$/carat)	Minimum of Independent Valuations (US$/carat)	Maximum of Independent Valuations (US$/carat)	WWW Valuation[2] (US$/carat)	Base Case Model (US$/carat)	"High" Model (US$/carat)	"Minimum" Model (US$/carat)
Renard 2	1,580	5	$173	$143	$195	$195	**$182**	$236	$163
Renard 3	2,753	5	$171	$137	$195	$190	**$182**	$205	$153
Renard 4	2,674	5	$100	$87	$107	$107	**$112 *($164)*[3]**	$185	$105
Lynx	537	1	n/a	n/a	n/a	$99	**$119**	$144	$99
Hibou	771	1	n/a	n/a	n/a	$71	**$118**	$136	$88

Notes

1) *As determined by WWW International Diamond Consultants Ltd. at a +1 DTC sieve size cut off.*
2) *Based on WWW alone. WWW's own valuation for the Renard 2, 3 and 4 samples was higher than the average of the five independent valuations.*
3) *Should the Renard 4 diamond population prove to have a size distribution equal to the average of Renard 2 and 3, WWW estimated that a base case diamond price model of US$164 per carat would apply based on May 2011 pricing.*

Table 19.2: March 2013 and March 2014 Diamond Valuation Results

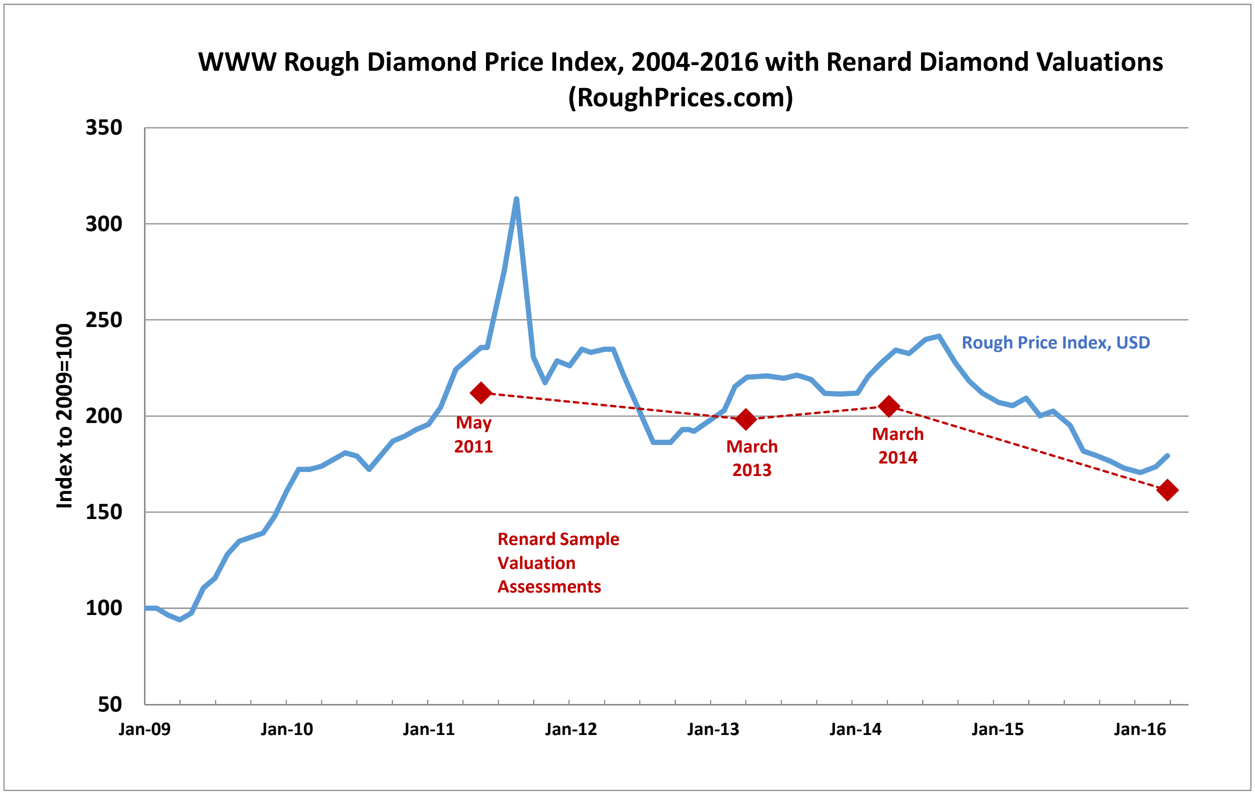
Body	Valuation Sample (carats)	March 2013 Re-Valuation of Valuation Samples[1]			March 2014 Re-Valuation of Valuation Samples[1]		
		WWW Valuation[2] (US$/carat)	Adjusted Average of Five Independent Valuations (US$/carat)	Diamond Price Model (US$/carat)	WWW Valuation[2] (US$/carat)	Adjusted Average of Five Independent Valuations (US$/carat)	Diamond Price Model (US$/carat)
Renard 2	1,580	$180	$161	**$190** *(Sens. $151 to $214)*	$187	$167	**$197** *(Sens. $178 to $222)*
Renard 3	2,753	$173	$157	**$151** *(Sens. $141 to $185)*	$179	$163	**$157** *(Sens. $146 to $192)*
Renard 4	2,674	$100	$93	**$104 *($150)*[3]** *(Sens. $98 to $168)*	$101	$96	**$106 *($155)*[3]** *(Sens. $100 to $174)*
Renard 65	997	$250	n/a	**$180** *(Sens. $169 to $203)*	$262	n/a	**$187** *(Sens. $160 to $190)*

Notes

1) *As determined by WWW International Diamond Consultants Ltd. at a +1 DTC sieve size cut off.*
2) *Based on WWW alone. In each case WWW's own valuation was higher than the average of the five independent valuations.*
3) *Should the Renard 4 diamond population prove to have a size distribution equal to the average of Renard 2 and 3, WWW have estimated that a base case diamond price model of US$150 per carat would apply based on March 2013 pricing and US$155 per carat based on March 2014 pricing.*

A tracking of world average rough diamond prices by RoughPrices.com (Figure 19.4), based on a market assortment maintained by WWW, indicates a -19% drop in average diamond pricing between March 2014 and March 2016. Stornoway has applied this market adjustment to the March 2014 WWW diamond price models to arrive at an estimate of "Spot" diamond pricing for each Renard kimberlite pipe for use in the Renard Project's economic analysis and the declaration of the project's Mineral Reserves.

Figure 19.4: WWW Rough Diamond Price Index, 2004-2016 with Renard Diamond Valuations (RoughPrices.com)



<u>Notes</u>

The valuation conducted on the Renard 2, 3 & 4 diamond samples in May 2011 resulted in value models based on the average of 5 separate diamantaires valuations. WWW's own valuation was on average 10% higher than the average of the five independent valuations. The sample valuation exercises in March 2013 and March 2014 were conducted by WWW for the purpose of measuring market movement in the intervening periods, and were used to adjust the previous price models upwards or downwards accordingly. The March 2016 assessment of "spot price" for this report has been derived by Stornoway by applying the world average rough price movement between March 2014 and March 2016 (as published by Roughprices.com which is based, in large part, on WWW market assortments) to the March 2014 valuation results.

Table 19.3: Estimated Diamond Price Adjustments, March 2014 to March 2016

Body	March 2014 Diamond Price Model[1] (US$/carat)	Estimated Market Price Adjustment March 2014 to March 2016	Adjusted Price Estimates March 2016: "Spot" Price Models[2] (US$/carat)
Renard 2	**$197** *(High $222, Min $178)*	*-19%*	**$160** *(High $181, Min $145)*
Renard 3	**$157** *(High $192, Min $146)*	*-19%*	**$128** *(High $156, Min $119)*
Renard 4	**$106 *($155)*[3]** *(High $174, Min $100)*	*-19%*	**$86 *($126)*[3]** *(High $141, Min $81)*
Renard 65	**$187** *(High $190, Min $160)*	*-19%*	**$152** *(High $155, Min $130)*

Notes

1) *As determined WWW International Diamond Consultants Ltd. at a +1 DTC sieve size cut off.*
2) *As determined by applying the world average rough price index of roughrices.com to the March 2014 price models, at a +1 DTC sieve size cut-off.*
3) *Should the Renard 4 diamond population prove to have a size distribution equal to the average of Renard 2 and 3, WWW have estimated that a base case diamond price model of US$155 per carat would apply based on March 2014 pricing, equivalent to US$126 per carat on a market price adjusted basis to March 2016.*

19.3 Contracts

19.3.1 Financing Transactions for the Renard Diamond Project

The construction of the Renard Diamond Project was fully financed following the closing by Stornoway and SDCI, on July 8, 2014, of a series of financing transactions totalling gross proceeds of $946 million (assuming a US$1.00:C$1.10 conversion) intended to provide a comprehensive funding package for the construction of the Renard Diamond Project. The financing consisted of a combination of senior and subordinated debt facilities, equity issuances, the forward sale of diamonds, and an equipment financing facility.

Purchase and Sale Agreement for the Forward Sale of Diamonds or "Streaming Agreement"

As part of the Financing Transactions, FCDC, a subsidiary of SDCI, entered into the Streaming Agreement for the forward sale of diamonds (the Forward Sale of Diamonds) with certain Streamers. Under the terms of the Forward Sale of Diamonds, FCDC agreed to sell to the Streamers a 20% undivided interest (the Subject Diamonds Interest) in each of the run of mine diamonds produced from the Project, until Streamers have been delivered the Subject Diamonds Interest in 30 million carats of diamonds, and following such time, from all run of mine diamonds derived from Renard 2, Renard 3, Renard 4, Renard 65 and Renard 9.

The Forward Sale of Diamonds provides for an up-front payment to FCDC, representing prepayment of a portion of the purchase price payable for the Subject Diamonds Interest, in an aggregate amount of US$250 million (the Deposit), disbursed in three installments. The last installement of the Deposit in the amount of US$90 million was paid on March 30, 2016.The Deposit must utilized for development, construction, and working capital requirements of the Renard Diamond Project.

The terms of the Forward Sale of Diamonds also provide that until the Deposit has been fully offset, the purchase price of the Subject Diamonds Interest will be equal to the gross proceeds from the sale of the Subject Diamonds Interest, payable by payment of a US$50 per carat cash price and the balance by offset against the Deposit. The US$50 per carat cash price is subject to an increase of 1% per annum beginning three years after the commencement of commercial production. Once the Deposit has been fully offset, the purchase price will be equal to US$50, indexed annually as mentioned above. The Streaming Agreement for the Forward Sale of Diamonds is available on Sedar under Stornoway's profile at www.Sedar.com.

Debt Financing Facilities

Senior Secured Loan

On July 8, 2014, SDCI also entered into a senior loan agreement with DIAQUEM, a wholly-owned subsidiary of Ressources Québec, which provides for an aggregate amount not to exceed $120 million plus an amount equal to the full outstanding principal amount of all loans under a loan agreement between DIAQUEM and SDCI dated October 1, 2013, that can be used for construction and development costs at the Renard Project, subject to certain conditions precedent (the Senior Secured Loan). The proceeds of the Senior Secured Loan can be used to finance, in part, the engineering, design, construction, plant, equipment, start-up, interest, financing fees and expenses, other Renard Project development costs and working capital requirements of the Renard Project. The Senior Secured Loan is available on Sedar under Stornoway's profile at www.Sedar.com.

Unsecured Cost Overrun Facility

On July 8, 2014, Stornoway executed with CDPQ an unsecured cost overrun facility (the COF) in an amount of $28 million. The proceeds of the COF can be used to finance, in part, the engineering, design, construction, plant, equipment, start-up, interest, financing fees and expenses, other Renard Diamond Project development costs and working capital requirements of the Renard Diamond Project, subject to certain conditions precedent. The COF is available on Sedar under Stornoway's profile at www.Sedar.com.

Equipment Facility

On July 25, 2014, SDCI (as lessee), Ashton and Stornoway (as guarantors) executed a master lease agreement with Caterpillar Financial Services Limited (as lessor) for the acquisition and financing of certain mine equipment items manufactured by Caterpillar and others, including the Renard Project's mobile mining fleet (the "Equipment Facility"). The Equipment Facility is available to SDCI in the maximum aggregate amount of US$75,000,000 of acquired equipment value (less the applicable upfront payments due by the lessee), being the sum of the following two tranches: (i) a tranche "A" facility in the maximum amount of US$50,000,000; and (ii) a tranche "B" facility, in the maximum amount of US$25,000,000. The Equipment Facility is available on Sedar under Stornoway's profile at www.Sedar.com.

19.3.2 Construction and Procurement Agreements for the Renard Diamond Project

As of December 31, 2015, all material construction agreements and equipment procurement agreements for the Renard Diamond Project had been executed, except with respect to the ventilation raise of the underground mine which are expected to be entered into during calendar 2016. Contract terms are consistent with industry norms.

19.3.3 Agreements for the Operation of the Renard Diamond Project

SDCI is a mining company organized to carry out the core activities of mining and processing to produce rough diamonds as the saleable product. Supporting functions within SDCI are similar to those in other mining companies and include maintenance, environmental, technical, health & safety, administration and corporate services. The operating scope of SDCI ends with the shipment of rough diamonds to its sale and marketing agent under a sale and marketing agreement. Beyond the core business of mining, processing and delivery of saleable product in exchange for revenue, the operation of a remote work site makes necessary a number of ancillary functions that – while non-core and non-mining – are equally essential. These non-mining support functions are well-served by organizations specializing in the delivery of such services. The client mining company benefits from the service providers' expertise, capability, and flexibility.

A number of the Renard Diamond Mine's contracts have been awarded to Cree businesses. This helps fulfilling the socio-economic goals of the Renard Diamond Mine to provide regional business opportunities and the goals of community and aboriginal stakeholders to participate in pursuing the opportunities to build skills and capacity. Some of the businesses existed before the mine's start-up, others are joint ventures between established "southern" firms and northern groups, while still others are/were new entities created expressly for participating in the Renard Diamond Mine.

Important contracts at the Renard mine include:

- explosives supply;
- management and personnel air transportation;
- supply of LNG;
- Transportation of LNG to the Renard mine;
- service agreement for catering, housekeeping and camp management;

Contract terms are consistent with industry norms.

19.3.4 Marketing and Sale Agreement of the Renard Diamond Project

The marketing and sale of rough diamonds from the Renard Diamond Project will be performed by Bonas-Couzyn (Antwerp) N.V., a third party marketing and sales agent, acting as commissionaire within the meaning of Belgian applicable laws, under a marketing and sales agreement entered into by FCDC on July 8, 2014 (the "Marketing and Sale Agreement").

The contract terms of the Marketing and Sale Agreement are consistent with industry norms.

20. ENVIRONMENTAL STUDIES, PERMITTING AND SOCIAL OR COMMUNITY IMPACT

The Renard Project was subject to environmental authorizations under various legislations, including the *Mining Act* (*Québec*) (R.S.Q., c. M-13.1) (Mining Act), the *Environment Quality Act* (R.S.Q. c. Q-2) (EQA), the *Canadian Environmental Assessment Act* (S.C., c. 37) (CEAA), and the *James Bay and Northern Québec Agreement 1975* (JBNQA).

On December 4, 2012, the Renard Project obtained the Global Certificate of Authorization (CA) from the *Ministère du Développement durable, de l'Environnement et de la Lutte contre les changements climatiques* (MDDELCC) for the development of the Project as outlined in the Environmental and Social Impact Assessment (ESIA) filed on December 28, 2011. In addition, the CEAA provided a positive environmental assessment decision in its report published on July 12, 2013.

The ESIA process was based on the most up-to-date and detailed information available on the physical, biological and human environment and on the project description. It included information on the physical, biological and human environment of the Project study area (127 km²). Collaboration between the ESIA and Engineering teams resulted in a project design that prevented and minimized environmental and social impacts and risks, and developed specific mitigation and compensation measures to address those impacts and risks. Table 20.1 lists the various baseline studies that were conducted and included in the ESIA (2002–2011).

From 2012 to 2015, in addition to the Provincial and Federal global authorizations, Stornoway obtained most of the specific permits, authorizations and leases required by the Federal, Provincial and Municipal levels for the Project. The permitting process is being done by splitting planned construction and operation activities into distinct phases so that work can begin and progress according to the Project timeline. Section 20.5.1 below provides information about the Québec and Canadian environmental assessment and review process and the legislative framework under which the Project is being carried out. Section 20.5.2 describes all of the sectoral permits that are required in addition to the global authorizations.

The following sections present:

1. A summary of environmental baseline studies;
2. A presentation of environmental and social issues;
3. Waste and tailings as well as water management plans;
4. The permitting process;
5. Social and community management plans;
6. A discussion on mine rehabilitation requirements and costs.

To date there are no identified environmental issues that could significantly impact Stornoway's ability to exploit the planned Renard diamond mine on the Foxtrot Property.

Table 20.1: Environmental and Social Baseline Studies Completed as part of the Renard Project ESIA (2011)

Year	Study	Reference
Biological Environment		
Ichthyofauna		
2003	Environmental Baseline Study	Roche. 2003. Environmental Baseline Study – Foxtrot Property. Report prepared for Ashton Mining of Canada Inc.
2005	Environmental Baseline Study	Roche. 2005a. Environmental Baseline Study (2004) – Foxtrot Property. Report prepared for Ashton Mining of Canada Inc. 44 pages + 12 appendices.
2009	Fish habitat assessment	Stantec. 2009. Fish Habitat Assessment - Renard Mine Property. Report prepared for Stornoway Diamonds Canada Inc.
2010	Environmental and social impact assessment	Roche – SNC-Lavalin. 2010. Étude d'impact sur l'environnement et le milieu social du prolongement de la route 167 Nord vers les monts Otish. Report prepared for the Ministère des Transports du Québec. 404 pages + 2 appendices.
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamond Corporation
Vegetation		
2003	Environmental Baseline Study	Roche. 2003. Environmental Baseline Study – Foxtrot Property. Report prepared for Ashton Mining of Canada Inc.
2005	Environmental Baseline Study	Roche. 2005a. Environmental Baseline Study (2004) – Foxtrot Property. Report prepared for Ashton Mining of Canada Inc. 44 pages + 12 appendices.
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamond Corporation
Benthos		
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamond Corporation
Herpetofauna (Amphibians and Reptiles)		
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamond Corporation
Avifauna		
2005	Environmental Baseline Study	Roche. 2005a. Environmental Baseline Study (2004) – Foxtrot Property. Report prepared for Ashton Mining of Canada Inc. 44 pages + 12 appendices.
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamond Corporation
Mammals		
2005	Environmental Baseline Study	Roche. 2005a. Environmental Baseline Study (2004) – Foxtrot Property Report prepared for Ashton Mining of Canada Inc. 44 pages + 12 appendixes.
2010	Consultations with Tallymen	Lussier, C. 2010a. Stornoway Diamond Corporation Renard Project. Consultations with land users of trapline M11. Preliminary report. Presented to ROCHE Ltd. 11 pages + maps.
2010	Environmental and social impact assessment	Roche – SNC-Lavalin. 2010. Étude d'impact sur l'environnement et le milieu social du prolongement de la route 167 Nord vers les monts Otish. Report prepared for the Ministère des Transports du Québec. 404 pages + 2 appendices.
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamonds Corporation Inc.

Physical Environment		
Surface deposits		
2002	Desktop study	Amyot, G. 2002. Programme de caractérisation environnementale préliminaire – Projet d'exploration Propriété Foxtrot. SOQUEM Inc. Ashton Mining Canada.
2003	Environmental Baseline Study	Roche. 2003. Environmental Baseline Study– Foxtrot Property. Report prepared for Ashton Mining of Canada Inc. 46 pages + 10 appendices.
2006	Hydrogeological and Geotechnical Study	GENIVAR. 2006. Étude hydrogéologique et géotechnique. GENIVAR report. 39 pages + appendices.
2010	Geological Study	Fitzgerald, C., C. Muntener et B. Kupsch. 2010. Geology of the Renard Kimberlites and the Lynx and Hibou Dyke - Foxtrot Property, Québec. Stornoway Diamond Corporation, February 10, 2010. 95 pages.
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamond Corporation
Hydrology		
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamond Corporation
2011	Climate and Hydrological Study	Golder 2011a. Renard Project Climate and Hydrological Analysis. Technical Memorandum No. 10-1427-0020/3050. Doc No. 19 Ver. 0. Submitted to Stornoway Diamonds Inc.
2011	Environmental and Social Impact Assessment	Roche. 2011. Renard Project. Environmental and Social Impact Assessment. Report presented to Stornoway Diamond Corporation
Hydrogeology		
2003	Environmental Baseline Study	Roche. 2003. Environmental Baseline Study – Foxtrot Property. Report prepared for Ashton Mining of Canada Inc.
2006	Hydrogeological and Geotechnical Study	GENIVAR. 2006. Étude hydrogéologique et géotechnique. Report by GENIVAR Québec prepared for GENIVAR Val-d'Or. 39 pages + appendices.
2011	Geotechnical Study	Golder 2011b. Geotechnical investigation at the processed kimberlite containment (PKC) facility-Renard Project – Québec – Technical Memorandum, February 4, 2011, 6 pages + appendix.
2011	Hydrogeological Study	Itasca. 2011. Prediction of Inflow Rates, Drawdown, and Particle Migration for the Business Case Mine Plan at Renard Diamond Mine, November 17, 2011, 17 pages + appendix.
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamond Corporation
2011	Environmental and Social Impact Assessment	Roche. 2011. Renard Project. Environmental and Social Impact Assessment. Report presented to Stornoway Diamond Corporation
Surface Water and Sediment Quality		
2003	Environmental Baseline Study	Roche. 2003. Environmental Baseline Study – Foxtrot Property. Report prepared for Ashton Mining of Canada Inc.
2005	Environmental Baseline Study	Roche. 2005a. Environmental Baseline Study (2004) – Foxtrot Property. Report prepared for Ashton Mining of Canada Inc. 44 pages + 12 appendices.
2005	Application for an attestation of exemption	Roche. 2005b. Mining Exploration Program 2006–2007 – Application for an attestation of exemption. Prepared for Ashton Mining of Canada Inc. 40 pages + appendices.
2009	Water Quality Study	Stantec. 2009. Programme d'échantillonnage de l'eau brute du lac Kaakus Kaanipaahaapisk – Projet Mine Renard. 8 pages + appendices.
2010	Water Quality Study	Stantec. 2010. Caractérisation de l'eau brute du lac Kaakus Kaanipaahaapisk – Projet Mine Renard. Rapport présenté à Stornorway Diamond Corporation. 9 pages + appendices.

2011	Static and Kinetic Testing Study	Golder. 2011c. Static and Kinetic Testing of Wate Rock, Kimberlite and Processed Kimberlite - Renard Project, Québec. Report No. 10-1427-0020. Doc. No. 069 Rev. 0., 41 pages + appendices
2011	Environmental and Social Impact Assessment	Roche. 2011. Renard Project. Environmental and Social Impact Assessment. Report presented to Stornoway Diamond Corporation
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamond Corporation
Soil Quality		
2003	Environmental Baseline Study	Roche. 2003. Environmental Baseline Study – Foxtrot Property. Report prepared for Ashton Mining of Canada Inc.
2010	Environmental Soil Study	Stantec. 2010a. Caractérisation environnementale des sols 2009 – Projet mine de diamant Renard. Report presented to Stornoway Diamond Corporation. 14 pages +appendices.
2011	Static Testing Study	Golder. 2011d. Static Testing of waste Rock, kimberlite and Processed kimberlite Renard Project, Québec, March 4, 2011, 16 pages and appendix.
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamond Corporation
Human Environment		
Socioeconomics		
2010	Consultations with Tallymen	Lussier, C. 2010a. Stornoway Diamond Corporation Renard Project. Consultations with land users of trapline M11. Preliminary report. Presented to Roche Ltd. 11 pages + maps.
2010	Consultations with Tallymen	Lussier, C. 2010b. Stornoway Diamond Corporation Preliminary consultations with land users of trapline located in the winter road study area. Final report. Presented to Roche Ltd, Consulting Group. 35 pages + appendices
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamond Corporation
Land and Natural Resource Use		
2003	Environmental Baseline Study	Roche. 2003. Environmental Baseline Study – Foxtrot Property. Report prepared for Ashton Mining of Canada Inc.
2010	Consultations with Tallymen	Lussier, C. 2010a. Stornoway Diamond Corporation Inc. Renard Project. Consultations with land users of trapline M11. Preliminary report. Presented to Roche Ltd. 11 pages + maps.
2010	Consultations with Tallymen	Lussier, C. 2010b. Stornoway Diamond Corporation Inc. Preliminary consultations with land users of trapline located in the winter road study area. Final report. Presented to Roche Ltd, Consulting Group. 35 pages + appendices
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamond Corporation
Archaeology		
2003	Environmental Baseline Study	Roche. 2003. Environmental Baseline Study – Foxtrot Property. Report prepared for Ashton Mining of Canada Inc.
2004	Review of Archaeological Knowledge	Pintal, J.-Y. 2004. Lac Emmanuel. Bilan des connaissances archéologiques. Unpublished report presented to Roche Ltd, Québec.
2010	Consultations with Tallymen	Lussier, C. 2010a. Stornoway Diamond Corporation Inc. Renard Project. Consultations with land users of trapline M11. Preliminary report. Presented to Roche Ltd. 11 pages + maps.
2010	Consultations with Tallymen	Lussier, C. 2010b. Stornoway Diamond Corporation Inc. Preliminary consultations with land users of trapline located in the winter road study area. Final report. Presented to Roche Ltd, Consulting Group. 35 pages + appendices
2011	Environmental Baseline Study	Roche. 2011. Renard Project. Environmental Baseline Study. Report presented to Stornoway Diamonds Corporation Inc.

20.1 Description of the Receiving Environment

20.1.1 Physical Environment

The Renard Project is located in a region with a subarctic climate. Winters are long and cold, while summers are short and relatively warm. Air quality is generally excellent due to the low intensity of human activities in the region.

The Foxtrot Property is located in a region with an elevation between 450 and 550 m above mean sea level (masl). The hills found on the Foxtrot Property have a southwest-northeast orientation and reach a maximum elevation of 670 m above masl. The Renard Project airstrip is located in a sector that is flat with an elevation of approximately 470 m.

Glacial deposits found in the study area have a very heterogeneous granulometric composition, often ranging from clay to large metre-sized boulders, and a high proportion of sand (Roche, 2011). On the Foxtrot Property, these deposits range in thickness from 0 to 30 m. Soils at the regional level are mainly composed of humo-ferric podzols and less often of brunisols. Soil drainage generally varies from good to moderate. There are also areas with poor drainage such as topographical depressions, where wetlands often form, as well as rocky outcrops. At several locations on the Foxtrot Property and elsewhere in the region, a thick ferruginous hardpan (called “ortstein”) developed in the soil. In bogs and swamps, organic soils from 30 cm to more than 1 m thick are found.

The Foxtrot Property is almost entirely located within the watershed of the upper Eastmain River. The Renard Project study area (127 km^2) is completely contained within this watershed, in the Misask River sub-watershed (1,510 km^2). This sub-watershed is further subdivided into over twenty secondary sub-watersheds that for the most part flow into the Lake Lagopede sub-watershed, the largest at 16.1 km^2. A small part of the Foxtrot Property located just north of the Renard Project is found within the upper Sakami River watershed, which runs northwest towards the La Grande-3 Reservoir.

The study area’s waterways have a pluvio-nival hydrological regime (Schetagne et al., 2005), which is characterized by a strong spring freshet in May-June, a rather severe low flow period in summer, an autumn rainfall freshet and a winter low flow period from November to May. The surface waters are generally acidic, well oxygenated, nutrient-poor, of low turbidity and low in mineral concentrations (Roche, 2011).

An aquifer is present in the rock and in the surface deposits. Groundwater flows mainly towards the different surface water bodies, which suggests that groundwater resurgence occurs in the various lakes whereas aquifer recharge begins along the topographical ridges (Roche, 2011).

20.1.2 Biological Environment

The Renard Project study area is located in the spruce-lichen bioclimactic domain of the James Bay region (CRRNTBJ, 2010).

According to baseline studies, plant communities found within the Renard Project study area (127 km^2) consisted of several types of forest stands, and was largely dominated by young black spruce-lichen stands (64%), old black spruce-lichen stands (12%) and old black spruce-moss stands (12%) (Roche, 2011). Old black spruce-sphagnum stands and old jack pine-lichen stands represented 4% and 3% of the plant community’s surface area, respectively, while the other forest stands each accounted for less than 1%.

Within the Renard Project study area, wetlands represented only 3% of the plant communities (excluding the hydrographical network). They consisted of small ombrotrophic bogs (open, forested or riparian), swamps and patterned fens that cover at the most a few hectares each.

A total of 36 species of freshwater fish from 11 families are found in the James Bay watershed. Brook Trout, lake Trout, walleye and northern pike are the most sought after game fish in this area (CRRNTBJ, 2010). The presence of 14 fish species has been confirmed within the Renard Project study area. Among these, 5 species are of interest for sport fishing: northern pike, lake whitefish, brook trout, lake trout and burbot. White sucker, pearl dace and brooktrout are the most abundant species in the area

Close to 25 species of fur-bearing animals are present within the James Bay region, and approximately ten have been trapped in the past by Cree trappers on Trapline M-11 in the sector of the Renard Project. The most commonly captured are American marten, American beaver, mink and muskrat (Roche, 2011). Trapping is a popular traditional activity among native populations. Caribou, moose and black bears are the main large mammals found within the James Bay region. Two ecotypes of the Woodland caribou are likely present in the Renard Project study area: forest-dwelling and barren-ground. The forest-dwelling ecotype is designated as threatened in Canada and vulnerable in Québec. The presence of the forest-dwelling ecotype was not confirmed during the 2011 mammal winter survey in an area covering 7,621 km^2.

The Renard Project study area is located on Trapline M-11, which covers the entire Foxtrot Property. Subsistence hunting is practiced by Crees in this area and targets mainly moose and waterfowl.

20.1.3 Human Environment

The Renard Project is located in a region that was inhabited some four thousand years ago. To date, the area has been occupied mainly by Amerindians, with infrequent visits over time by European and Canadian explorers. Furthermore, it is enclaved and was not, until recently with the extension of Route 167, accessible by road.

Land use around the mine is typical of traditional resource harvesting by the James Bay Crees; the immediate responsibility for it lies with the users of trapline M11 from the Cree community of Mistissini. This vast trapline, covers some 3,800 km^2, includes two main Cree camps and is intersected by a number of snowmobile trails, this being the preferred means of transportation for hunting, fishing and trapping. These activities are concentrated in certain favourite parts of the trapline, specifically around lakes Lagopede, Emmanuel and Bray. The tallymen spend most of the year on trapline M11 and approximately 25 other users visit trapline M11 on a regular basis. Activities such as waterfowl hunting, big game hunting, trapping of fur-bearing animals, collecting berries and wood, and fishing are practiced by family members using the M11 trapline. The trapline also houses a number of valued sites, specifically birthplaces and burial sites.

From an administrative point of view, the Renard Project is in an area that has seen many changes since Phase 1 of the James Bay hydropower development. At that time, in the mid 1970s, the James Bay and Northern Québec Agreement was signed. It was followed, in the early 2000s, by the Agreement Concerning a New Relationship between Le gouvernement du Québec and the Crees of Québec, commonly called the Paix des Braves. More recent is the Framework Agreement between the Crees of Eeyou Istchee and the gouvernement du Québec on Governance in the Eeyou Istchee James Bay Territory, which provides for negotiation of a merger between the Jamesian and Cree authorities in a new, shared regional governance entity that, will be known as the Eeyou Istchee James Bay Regional Government.

On a more local scale, the Project concerns the Cree community of Mistissini, which is located about 250 km south of the Renard Project, and the towns of Chibougamau and Chapais, located about 100 km farther south. In 2006, these municipalities had 3,500, 7,565 and 1,630 inhabitants respectively. They provide various services that can be used during the construction and operation of the Renard mine. Some of the labour required can also come from the region, generating a real opportunity for economic and social development.

Such development is even more desirable in the Cree community where socioeconomic conditions lag somewhat behind those of Quebec as a whole. The median personal income for the Mistissini Crees is about $22,000, compared to nearly $24,500 for Quebec. Unemployment exceeds 20%, while in Quebec it wavers around 7.5%. In recent years, the number of students who obtain a high school diploma has remained between 20% and 40% in Mistissini, while it is around 75% for Quebec.

The employment and education challenge among the Crees is all the more blatant and urgent because it is a young, growing society. Half of the population of Mistissini is under age 24 whereas in Quebec this age group accounts for about 30%.

There are no known archaeological sites within a 50 km radius of the Renard Project site. Nevertheless, based on available ethnological and geomorphological data, a total of 51 zones of archaeological potential were identified. These are sites where the probability of uncovering traces of human settlement is highest. None of these zones were located within the Renard Project site. However, ten or more of these zones of archaeological potential were located around Lake Lagopede.

20.2 Main Environmental Issues

The main environmental issues and public concerns that were identified and expressed during the environmental impact assessment process are presented in the section below. Environmental issues were addressed as part of the ESIA by developing environmental design criteria, construction best practices as well as mitigation and compensation measures that were intended to minimize the Project's environmental footprint. In addition, as required by the provincial and federal authorizations issued for the Renard Project, an environmental and social monitoring program was developed (ESMP). The ESMP is part of an environmental and social management framework based on the by ISO 14001 standard.

The overall objective of the ESMP is to measure, observe, and document any environmental changes (natural or project-related) relative to baseline conditions. It also aims to measure the accuracy of the impact assessment process and the effectiveness of mitigation measures. Since the start of the Renard Project construction phase (May 2013), scheduled environmental and social surveillance and monitoring activities were conducted as planned. During the construction phase, these activities include environmental surveillance as well as fish and large mammal population monitoring. Environmental monitoring procedures were prepared for:

- Air emissions and air quality monitoring and surveillance;
- Vibration and noise level monitoring and surveillance;
- Surface water and sediment quality monitoring and surveillance;
- Domestic influent and effluent monitoring and surveillance;
- Mine effluent monitoring and surveillance;
- Vegetation and wetland monitoring and surveillance;
- Benthos and fish monitoring and surveillance;
- Fish and fish habitat monitoring and surveillance;
- Fish habitat compensation monitoring and surveillance;
- Segments C and D of the route 167 extension (mine access road) monitoring and surveillance;
- Terrestrial wildlife and bird monitoring and surveillance;
- Meteorology, climate and hydrology monitoring and surveillance;
- Groundwater monitoring and surveillance.

As well, the fish habitat and compensation program activities scheduled as part of the DFO authorizations were launched. A wetland compensation program is currently being developed.

Table 20.2 lists some of the various monitoring reports that have been issued since the ESIA was completed in 2011.

Table 20.2: Environmental and Social Monitoring Studies conducted after the ESIA (2011–2015)

Year	Valued Environmental Component	Reference
2016	Groundwater	Norda Stelo inc. 2016. Projet diamantifère Renard - Suivi de la qualité des eaux souterraines. In preparation. Norda Stelo Inc. 2016. Projet diamantifère Renard - Détermination des teneurs de fond dans les eaux souterraines. February 2016.
2015–2016	Fish and fish habitat	Roche Ltd, Consulting Group. 2015. Rapport d'activité – Permis SEG N°2014-09-16-141-10-G-P; Projet diamantifère Renard. 12 pages + appendices. Norda Stelo inc. 2016. Projet diamantifère Renard - Programme de compensation de l'habitat du poisson - Aménagements pour l'omble de fontaine tels que construits. Progress report. 31 pages + appendices.
2016	Benthic invertebrates	Norda Stelo inc. 2016. Étude de suivi des effets sur l'environnement – État de reference des populations d'invertébrés du lac Lagopède – Année 2015. In preparation.
2016	Large mammals and land use	Roche, 2016. ENVS-3.3.11 - Procédure de suivi et surveillance de la faune terrestre et aviaire – Suivi de la grande faune. 38 pages + appendices.
2015	Environmental monitoring	Stornoway, 2015. Suivi environnemental, Projet Renard et campement minier, Rapport 2014.
2015	Suatainable Development Report	Stornoway, 2015, Sustainable Developpement Report for 2014.

The main environmental issues of the Renard Project are discussed below.

- *Surface water quality (in particular in Lake Lagopede) and groundwater contamination*

The streams located in the Renard Project area are low flow or intermittent and the lakes are characterized by oligotrohic waters. During the operation phase, the Project will generate a large quantity of mineral materials that will be stored in areas developed specifically for ore, waste rock, overburden, and processed kimberlite. Runoff and exfiltration water in contact with these materials have the potential to affect surface water quality. Following the extraction and the management of mine water and runoff, and of processed kimberlite and waste rock, an important environmental issue will thus be related to surface water and groundwater management.

To mitigate the impact on water quality a water management system was put in place on the site to limit residual impacts on water quality (see Section 20.4). All the water that comes in contact with the mine site is collected by means of a system of perimeter ditches, directed toward a sedimentation pond, then treated and tested before being discharged into Lagopede Lake. The treated mine effluent outfall has been positioned in Lake Lagopede in a way to enhance rapid effluent dispersion.

Modeling results show that the impacts on water quality will essentially be felt in the vicinity of the plumes of treated domestic wastewater and mine water effluents discharged into Lagopede Lake. The design criteria meet the criteria set out in Guideline 019 and the *Metal Mining Effluent Regulations*. Specific mine water and domestic effluent discharge objectives have been set by the MDDELCC for the Renard Project and will be monitored. The plume dispersion modelling results for the treated domestic effluent indicate that, at the edge of the mixing zone, the contaminant concentrations will be of the same order of magnitude as the concentrations currently measured in the receiving environment. Within the plumes, additional loads of nutrients and organic matter could increase local habitat productivity in the vicinity of the discharge point.

- *Power supply*

In October 2013, Stornoway released the results of a feasibility study undertaken on the use of liquefied natural gas (LNG) for power generation instead of the diesel option contained in the 2013 Optimization Study. Accordingly, energy will be supplied by 7 natural gas generators instead of the 12 diesel generators originally planned. Replacing diesel with natural gas reduces atmospheric emissions by 42% and improves air quality on the site. The volume of petroleum hydrocarbons used annually will be reduced by 74% to 85%, depending on the year of operation, reducing the risk of leaks and accidental spills that could have localized effects on soil, groundwater, surface water, and sediment quality. In addition, the use of natural gas will reduce the risk of fire and explosion at the mine site. The global CA modification request including an updated risk analysis was submitted to the COMEX, and the amended global authorization was obtained on October 20, 2015.

- *Plant communities and wetlands*

The main Project components that will affect vegetation during construction and operation are site preparation and development activities (clearing, excavation, earthworks, etc.). Through various iterations of the Project design, it was possible to minimize the Project footprint and to limit encroachment on wetlands and on old forest stands. A rehabilitation plan has been developed to stabilize disturbed land conditions and restore them by revegetating surfaces to allow local indigenous species to recolonize the area.

Under the *Act respecting compensation measures for the implementation of projects affecting wetlands or bodies of water* (chapter M-11.4), the loss of wetlands must be compensated. To determine what compensation strategy would satisfy MDDELCC requirements and guidelines, a committee bringing together the main stakeholders concerned with wetland conservation at the local and regional levels was setup by Stornoway. It was proposed to focus the compensation program on the establishment of a peatland knowledge acquisition program (biophysical and social) for the Renard Project area. If authorized by the MDDELCC Regional Division, this option would allow for future ecological valorization of these habitats and, at the regional level, a better ecological understanding of northern wetlands. Following a presentation made on October 7, 2014, this strategy received a principle agreement from the MDDELCC. In November 2015, a letter of intent was sent to the *Chaire de recherche sur la Dynamique des Écosystèmes Tourbeux et Changements Climatiques* of the Université du Québec to Montreal and a funding proposal to CRNSG was prepared and submitted in February 2016.

- *Fish habitat*

Several activities undertaken as part of the construction, operation and closure phases of the Project are likely to directly or indirectly affect surface water or sediment quality and alter fish habitat and affect some fish populations.

During the construction phase of the Renard Project, approximately 15,800 m² of lakes and 3,880 m² of streams were lost due to lake draining and stream diversion activities, for a total permanent loss of approximately 19,680 m² of fish habitat. To mitigate these impacts, 5,671 pearl dace, 30 burbots, and 12 brook trouts were captured and relocated in Lake Lagopede prior to draining the tiny lakes F3302 and F3303 in 2014. Only one stream was diverted and initial habitat conditions were restored. In order to compensate for fish habitat losses, a compensation program was developed and approved, which included five interventions distributed over two distinct geographic areas: the Renard mine area and the Mistissini area. This program was approved by the DFO. Program implementation began in 2015 by the development of brook trout spawning grounds in four streams located in the Renard mine area.

During the construction of the Mine Access Road, 17 bridges and 26 culverts were built and caused the loss of about 6,000 m^2 and the disturbance of 960 m^2 of fish spawning, nursery and feeding habitats. To minimize this impact, compensation work was done in 2014 at six (6) sites, totalling 1,012 m^2 of fish habitat. Compensation work included the removal of culverts and of road embankments limiting fish passage, the reconstitution of streams beds and riprap, shoreline stabilization and development of brook trout spawning grounds. A first monitoring study of compensation work was conducted in September 2015.

- *Wildlife habitat loss and increased hunting pressure on wildlife population in relation to increased land access*

The main potential impacts of the Project on wildlife during the construction and operation phases were related to site preparation and development activities (clearing, excavation, earthworks, etc.). These activities resulted in habitat loss. Most species in the vicinity of the mine site will be able to find similar habitat at the periphery of the mine and airstrip sites; however, such shifts in habitat may result in additional energy expenditure and competition for habitat and resources.

To limit the extent of these impacts, the Project design has been progressively optimized to limit the Project footprint on wildlife habitat, while avoiding encroachment on Trapline M-11 moose hunting areas. By design, the Project site roads were optimized to limit length and extent. Increased access to the area may lead to additional pressure on wildlife resources (e.g., moose or marten), but hunting and trapping by workers or contractors is prohibited.

In order to assess changes in the distribution of large mammal populations since the beginning of the construction phase and the commissioning of the mine access road, an aerial survey and consultations with land users were conducted in 2015. For moose, the 2015 survey results were similar to those of the surveys conducted previously to describe large mammal baseline conditions. According to land users, the number of moose in the traplines has been relatively stable since the beginning of the mine site construction phase and the commissioning of the mine access road. Nonetheless, moose present in the mine study area seem to have moved further south and west of the mine site.

During construction, the draining of two small, shallow lakes were the main sources of impact on waterfowl and aquatic birds. No significant impacts were expected on forest birds as a result of site preparation and development activities; however, the relative significance of an impact may vary depending on the species.

- *Woodland Caribou*

The Ministère de la Faune, des Forêts et des Parcs (MFFP), in collaboration with the Ministère des Transports du Québec (MTQ) and Stornoway, has conducted activities to monitor the Témiscamie caribou herd (i.e., the closest forest-dwelling caribou herd relative to the Renard Project) since 2002. In March 2015, the MFFP conducted activities related to the monitoring of the Témiscamie caribou herd (i.e., the closest herd relative to the Renard Project). These activities aimed to (1) survey woodland caribou groups in areas suitable for this species, and (2) capture caribou to fit them with GPS-Argos collars. Areas suitable for woodland caribou were identified using vegetation types mapped on a Landsat image (2013), as well as the location of individuals of the Témiscamie herd previously obtained using GPS-Argos collars. In surveyed areas, 164 caribou were observed. Two groups of caribou were observed 40 km south of the mine site as well as 10 km west of the mine access road. A total of 15 caribou were fitted with GPS-Argos collars; two of these collars were fitted on caribou located 5 and 40 km south of the mine site.

The presence of the forest-dwelling ecotype was not confirmed during the 2011 mammal winter survey in an area extending over 7,621 km^2. Given its protected status and since the Renard Project study area is an area that forest-dwelling caribou could potentially use, impact assessment and mitigation measures were recommended to promote this ecotype's long-term presence in the area. More recently, caribou GPS-Argos locations seem to indicate that forest-dwelling caribou are moving north and that individuals could now be found near the mine and airstrip areas (V. Brodeur, MFFP, comm. pers.). In the future, Stornoway will continue to participate in Woodland Caribou population monitoring.

20.3 Mine Waste Management

Geochemical Classification

The geochemical classification of kimberlite, processed kimberlite (PK), and waste rock was performed through static and kinetic testing programs on samples collected at the Project site. The geochemical characterization program followed Québec Provincial guidance of *Directive 019 sur l'industrie minière* (MDDELCC, 2005, revised in 2009 and 2012). The environmental geochemistry of the overburden was characterized by static testing using the same methods as waste rock and PK. The materials expected to be handled through the exploitation of the Renards 2, 3, 4, 9, and 65 ore bodies were tested. The samples subjected to kinetic testing were selected in consideration of the static testing results and focussed on rock types that have the largest representation in future waste rock, kimberlite, and PK.

The objectives of the geochemical classification program were to:

- Classify mine waste according to the intent of Québec's Directive 019; and
- Identify chemicals of potential interest in the framework of probable future mine water quality.

Results obtained to date show that all mine wastes are non-acid generating and that leachability is low. Kinetic testing shows that mine wastes release low concentrations of dissolved constituents; concentrations of chemicals in leachates are below Québec groundwater quality guidelines. As a result, mine waste rock and PK are classified as low risk.

As per the same types of testing, waste rock and country rock is considered to be Category I and is therefore suitable for use as construction material without restriction.

Static leaching test methods revealed some exceedances to water quality criteria but the concentration of these constituents met the same criteria under the more representative and longer-term kinetic test methods. The results of the kinetic tests met the water quality criteria.

Kinetic leaching tests are considered to be more representative than static leaching tests in terms of probably leaching behaviour and leachate concentrations generated from waste material that will be exposed to ambient conditions at the site due to:

- The nature of the contact between the infiltrating water and the waste rock that more closely simulate the infiltration of water through a rock pile or PKC facility as opposed to agitated mixing of rock and water in a short-term leach test;
- The lower solution-to-solid ratio used in the kinetic tests which more closely represents the climate conditions at site;
- The long-term (20-week) nature of the testing that allows for the evaluation of transient chemical processes such as mineral dissolution, sulphide oxidation and other weathering reactions; and
- Kinetic weathering test results form the basis of risk evaluation for the mine wastes, together with existing site water quality data, rather than the static leaching test results.

20.4 Water Management and Treatment

20.4.1 Water Management

The strategy for water management on site is to reduce the potential impact of the Renard Project on local and regional water resources while satisfying process water requirements. The two primary objectives of the water management strategy are to manage all contact water in an efficient and effective manner, and to reduce the amount of contact water that will require treatment prior to discharge to the environment.

The approaches that are used to achieve these objectives include:

- Implementing measures (i.e., clean water diversions) to avoid the contact of clean runoff water with areas affected by the mine or mining activities;
- Collecting, transporting and treating, as necessary, mine water and runoff water in contact with core project activities;
- Reducing the intake of fresh water from the environment by recycling and reusing water to the greatest extent possible;
- Monitoring quality of discharge; and
- Adjusting management practices if monitoring results indicate discharge quality does not meet discharge criteria.

For the purpose of the water management strategy, surface water has been grouped into two categories: contact water and non-contact water. Contact water is defined as any water that may have been affected by mining activities. Contact water includes:

- Surface runoff from the mining and milling areas;
- Groundwater inflows to mine workings;
- Surface runoff (and shallow drainage) from rock, overburden, and processed kimberlite disposal areas;
- Processed kimberlite transport water;
- Seepage water through the rock, overburden and processed kimberlite containment areas; and
- Dewatering volumes having high total suspended solids (TSS) concentrations.

All contact water will be intercepted, contained, re-used (to the extent practical), monitored, and if required, treated prior to discharge to the receiving environment. Non-contact water is limited to intercepted and diverted runoff originating from areas undisturbed by mining. As a general rule, 100-year return period design events apply to perimeter infrastructure operating during both the construction and mine operations phases. Ten-year return period events apply to the design of internal infrastructure or to infrastructure operating during the two year construction phase only. The water management infrastructure, comprising ditches and sumps, is constructed in phases based on the development of the site. The final configuration of the water management infrastructure is shown in Figure 20.1.

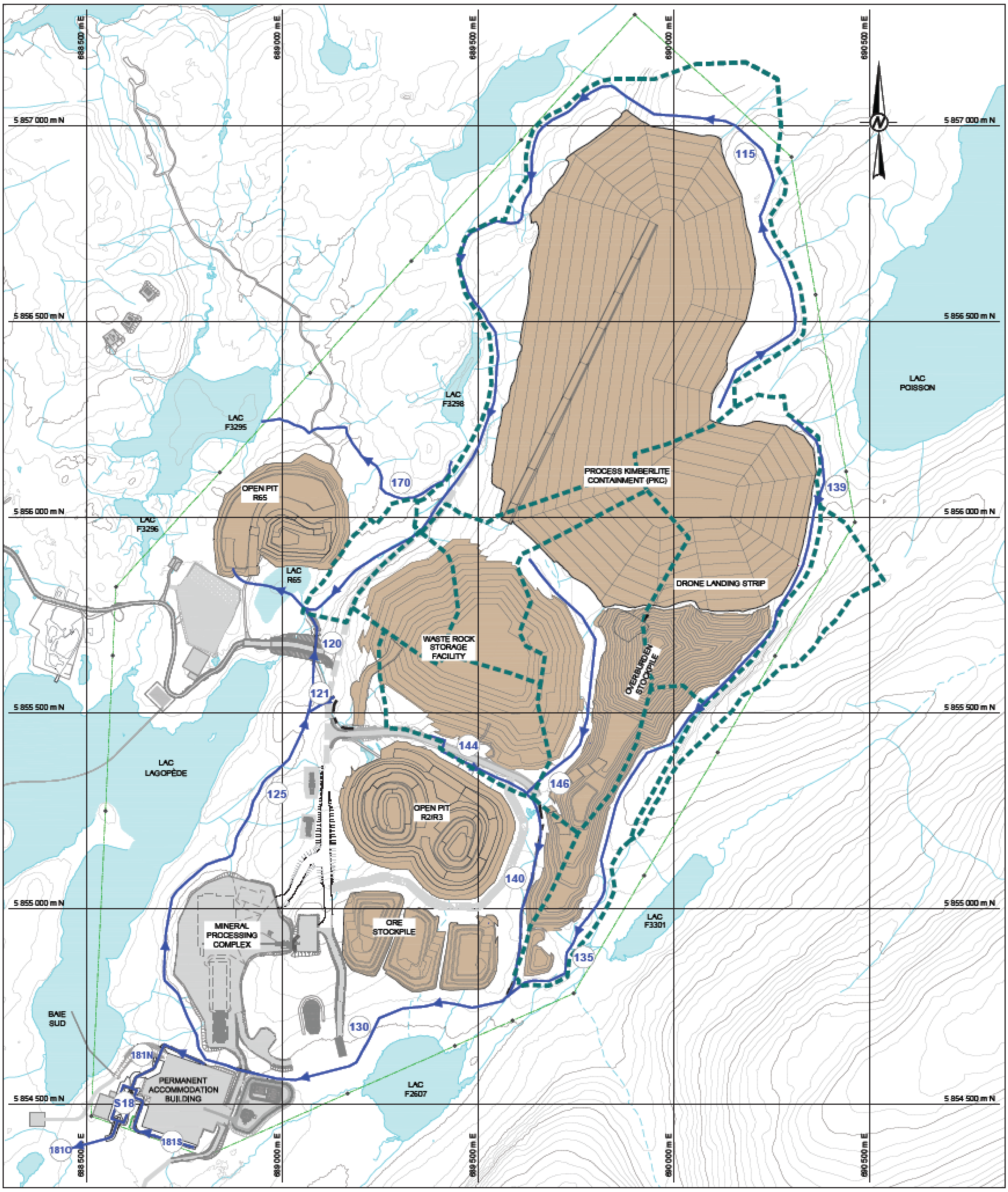
A ditch network will convey the surface water runoff to the Renard 65 settling pond by gravity. Once complete, the Renard 2, Renard 3, and Renard 4 underground developments will be inter-connected and drain to a low point of the ramp. Water from the ramp sump will be preferentially pumped to Renard 65 settling pond and treated to satisfy the demand for process water. Excess water pumped to the Renard 65 sump will be also treated if required before to discharge. It is anticipated that the groundwater inflows from the underground workings will fully satisfy process water demand.

The basis of the proposed water management plan is to collect, monitor, and treat (if required) all contact water from the site (with the exception of the camp) at the Renard 65 settling pond and treatment plant. After the required treatment, water will be discharged to Lake Lagopede.

20.4.2 Water Treatment

The water treatment evaluation focussed on water management during the active mining period; however, part of the system was put in place to manage water during the two-year construction period. The temporary water treatment system used during the construction phase will be replaced by a permanant one in April 2016.

Figure 20.1: Water Management Plan



LEGEND

- LAKE
- EXISTING STREAM
- CONTOUR LINE (m)
- LEASE LIMIT
- WATER MANAGEMENT INFRASTRUCTURE DESIGNED BY GOLDER
- CATCHMENT AREA DELINEATION

NOTES

1. COORDINATES SYSTEM: UTM NAD 83, ZONE 18
2. THE DESIGN OF SOME INFRASTRUCTURE IS NOT FINAL. THE FIGURE PROVIDES INFORMATION ON LOCATIONS AND STRUCTURES REQUIREMENTS. CONFIGURATION OF SUCH STRUCTURE MAY DIFFER FROM ACTUAL DESIGN BY THE TIME OF CONSTRUCTION.

REFERENCES

- PLAN # ROUTE #8_POUR_GOLDER.dwg PROVIDED BY CLIENT, NOVEMBER 2014.
- PLAN # PLAN GENERAL 26 AOUT 2014_Ben.dwg PROVIDED BY CLIENT.
- TOPOGRAPHIE PLAN # courbes_UTM-Nad83.dwg PROVIDED BY SNC, OCTOBER 28, 2010.

20.5 Permits and Authorizations

20.5.1 Environmental Assessment and Review Process

Provincial Environmental Assessment Process

The Renard Project was subject to the provincial and federal environmental and social assessment and review process under the *JBNQA,* the EQA and the CEAA. On December 28, 2011, the Corporation filed an Environmental and Social Impact Assessment ("ESIA") meeting the requirements of the provincial and federal ESIA guidelines and following that submission, public hearings on the project were held by the federal Canadian government and Québec government in June and August 2012, respectively. In December 2012, Stornoway received the global Certificate of Authorization for the Renard Diamond Project from the MDDELCC.(#3214-14-041). The Certificate of Authorization represents the principal regulatory approval that was required to commence mine construction.

Following the receipt of the global Certificate of Authorization, the Corporation continued the optimization of the Renard Diamond Project which resulted in amendments being made to the initial conditions stated in the Renard Diamond Project's global Certificate of Authorization, on June 9, September 19, October 7 and October 29, 2014, as well as on July 7 and October 20, 2015.

Mining Act

The Québec government is responsible for mining activities in the province. These activities are subject to the *Mining Act,* which defines ownership of the right to mineral substances (claims, mining exploration licenses, mining leases, mining concessions, etc.) and the rights and obligations of the claim holder or other mining rights granted by the State. With regard to the environment, it should be noted that the *Mining Act*, Chapter IV, Division III, specifies that the holder of mining rights has the responsibility to rehabilitate and restore the lands on which exploration and/or development activities took place. This work must be completed in accordance with a rehabilitation plan pre-approved by the MERN.

A Rehabilitation Plan for the Renard Project was prepared according to the requirements of the MERN outlined in the document entitled "*Guidelines for Preparing a Mining Site Rehabilitation Plan and General Mining Site Requirements*". The Plan was submitted on February 15, 2012 and approved by the MERN on December 21, 2012.

Under the Mining Act of Quebec, a financial guarantee must be submitted to the MERN to guarantee 100% of the rehabilitation costs of a mining project, in accordance with a payment schedule prescribed by applicable regulation. On August 29th 2014, the Corporation arranged for a surety bond of up to $15.2 million to provide a financial guarantee to the MERN with respect to the Closure Plan. The obligation to pay the first tranche of $7.6 million was met in August 2014; the second installment of $3.8 million was met in August 2015 and the third and last installement is due in August 2016.

Federal Environmental Assessment Process

The Canadian Environmental Assessment Act (CEAA 2012) came into force on July 6, 2012, replacing the Canadian Environmental Assessment Act (S.C. 1992, c.37) (the former Act). CEAA 2012 includes specific provisions concerning comprehensive studies initiated under the former Act, such as the Renard Project. In fact, such comprehensive studies are to be continued and completed under the former Act.

A federal environmental assessment under the former Act was necessary since, pursuant to the *Fisheries Act* and the Explosives Act, Fisheries and Oceans Canada and Natural Resources Canada needed to issue authorizations and approvals in order to allow the Renard Project to proceed. As such, the Project was subject to a comprehensive environmental assessment study, in accordance with subsection 10 of the Comprehensive Study List Regulations (SOR/94-638) under the former Act which reads as follows:

“The proposed construction, decommissioning or abandonment of a facilty for the extraction of 200,000 m³/year or more of groundwater or an expansion of such a facility that would result in production capacity of more than 35 per cent.”

The Canadian Environmental Assessment Agency conducted the comprehensive study in collaboration with the Federal Environmental Assessment Committee, composed of representatives from Fisheries and Oceans Canada, Natural Resources Canada and the Cree Regional Authority, which provided advices in their respective areas of expertise. The report was issued in May 2013. The report was based on the results of a detailed review of the ESIA that was submitted by Stornoway. The report concluded that “*Given the implementation of the proposed mitigation measures and follow-up program, the Canadian Environmental Assessment Agency concluded that the project is not likely to cause significant adverse effects*”. In July 2013, Stornoway received a positive environmental assessment decision for the Renard Diamond project from the CEAA.

20.5.2 Specific Permits, Approvals and Leases

In addition to the Provincial and Federal global authorizations, Stornoway has obtained, since 2012, many specific permits, authorizations and leases that are required at the Federal, Provincial and Municipal levels. Those are listed below and in Table 20.3:

Provincial Level

Certificate of Authorisations

Infrastructure or activities for which 21 certificates of authorization (CA) required under Sections 22 and/or 48 of the Environmental Quality Act have been granted:

- Site preparation and surface water management plan (2 CAs);
- Mining activities (mining pits, underground extraction, and ore and waste rock piles);
- Dewatering of two lakes, sediment excavation and diversion of a stream;
- Natural gas power station;
- Ore processing plant;
- Permanent mining wastewater treatment plant (platform and walls, operation, and outfall – 3 CAs);
- Extraction and crushing of R-65 waste rock for construction activities;
- Contaminated soil treatment site (construction);
- Exploitation activities of six borrow pits (6 CAs);
- Trench landfill;
- Mobile concrete plant at the waste pile;
- Explosives mixing factory;
- Fish habitat compensation program (brook trout developments in F3293, F3294, F2604 and F3301 lake outlets);
- Dust collectors of the ore processing plant.

Infrastructures or activities for which authorization requests have been submitted and that are under review by Provincial authorities:

- Mining activities (processed kimberlite confinement area);
- Wetland compensation program.

Note that the construction and operation of the airstrip are not subject to the obtention of a CA under Section 22 of the Environmental Quality Act (a certificate of exemption was delivered by the MDDELCC). Authorizations required under Sections 32 and/or 31.75 of the Environmental Quality Act have been obtained for the following infrastructures and activities:

- Hydrocarbon separator for the airfield garage;
- Contaminated soil treatment site by biopiles (operation);
- Temporary mining wastewater treatment plant;
- Permanent mining wastewater treatment system (operation);
- Water intake and drinking water treatment system for the construction camp (Lagopede camp);
- Water intake and drinking water treatment system for the construction camp (km-640 camp);
- Water intake and drinking water treatment system for the permanent camp;
- Domestic wastewater treatment system for the construction camp (Lagopede camp);
- Domestic wastewater treatment system for the construction camp (km-640 camp);
- Domestic wastewater treatment system for the permanent camp.

Authorization required under Section 31.16 which remains to be obtained

- Depollution attestation: The Depollution Attestation request is currently under preparation.

Permits

The following permits were required and have been obtained:

- Forest management permits (forest clearing) from MFFP;
- High-risk petroleum equipment operation permit from the Régie du Bâtiment du Québec;
- Explosives permit from the Sûreté du Québec.
- Permit of occupation for the infrastructures enabling water to be collected (drinking water intake) from the Centre d'Expertise Hydrique du Québec of the MDDELCC.
- Permit of occupation for the infrastructures enabling water to be evacuated (domestic wastewater outfalls) from the Centre d'Expertise Hydrique du Québec of the MDDELCC.

The following permit is required and remains to be obtained:

- Permit of occupation for the infrastructures enabling water to be evacuated (mine wastewater outfalls) from the Centre d'Expertise Hydrique du Québec of the MDDELCC.

Approvals

The following approvals were required and have been obtained:

Approval for the location of the ore processing plant and of the processed kimberlite confinement area; Rehabilitation and restoration plan of the Renard Diamond Project.

Federal Level

- ***Authorization to alter fish habitat:*** Under Section 35(2) b) of the Fisheries Act, "No person shall carry on any work or undertaking that results in the harmful alteration, disruption or destruction of fish habitat" without previous authorization from the Minister. The authorization request to alter fish habitat was submitted in 2011. A compensation program was included in the authorization request to modify fish habitat. The authorization was issued to Stornoway in 2014 (#2014-02).
- ***License for explosives factories and magazines*****:** Under Section 7(1)(a) of the *Explosives Act* (R.S.C., 1989 c. G17), licenses issued by the Minister of Natural Resources Canada are required for the operation of explosives factories and magazines in Canada. A licence was obtained in 2013 for the explosives factory (construction phase).

Municipal Level

- ***Certificate of compliance with municipal regulations:*** Under Section 8 of the Regulation respecting the application of the EQA, “A person who applies for a certificate of authorization shall also submit to the Minister a certificate attesting that the project does not contravene any municipal by-law. The certificate shall be issued by the clerk or the secretary-treasurer of a local municipality or, in the case of an unorganized territory, of a regional county municipality”. Requests for certificates for the Renard Project’s activities were submitted to the Municipality of James Bay at Matagami, except for those activities defined in Section 246 of the Act Respecting Land use Planning and Development (R.S.Q., c. A-19.1).
- ***Municipal Permits:*** Permits were required from the Municipality of Baie-James for certain activities that do not require a certificate of authorization under Section 22 of the EQA. This was the case for global construction work, groundwater catchment facilities (<75 m^3/day) and the airfield septic systems, among others.

Table 20.3: Necessary Authorzations, Permits, Approvals, Attestations and Leases, as of March 4, 2016.

I. MATERIAL PERMITS, CONSENTS AND APPROVALS ALREADY OBTAINED AND REQUIRED TO COMMENCE CONSTRUCTIONS

Governmental Authority	Description	Legal reference	Date obtained
MDDELCC	Global Certificate of authorization signed by the deputy minister and authorizing the entire Renard Project, following the successful completion of the environmental and social impact assessment and review process	Section 164 of the Environmental Quality Act. Permit obtained as a result of the successful completion of the environmental and social impact assessment and review process described in Chapter II, Division II, section 3 of the Environmental Quality Act	December 5, 2012
MDDELCC	Modification to the Global Certificate of authorization signed by the deputy minister and authorizing the entire Renard Diamond Project, <u>following the submission of the addendum for the LNG</u>	Section 164 of the Environmental Quality Act. Permit obtained as a result of the successful completion of the environmental and social impact assessment and review procedure described in Chapter II, Division II, section 3 of the Environmental Quality Act	Octobre 20, 2015
MDDELCC	Mining operation (Open pit, underground, ore and waste pile)	Section 22 of the Environmental Quality Act	March 02, 2015
MDDELCC	Construction and operation of an ore processing plant	Section 22 of the Environmental Quality Act	March 25, 2015
MDDELCC	Drainage of two waterbodies, sediment excavation and diversion of a creek	Section 22 of the Environmental Quality Act	September 25, 2014
MDDELCC	Operation of dust collectors	Section 22 of the Environmental Quality Act	February 17, 2016
MDDELCC	Site preparation and management of surface drainage	Section 22 of the Environmental Quality Act	August 16. 2013
MDDELCC	Construction and operation of an industrial wastewater treatment plant	Section 32 of the Environmental Quality Act	January 12, 2016
MDDELCC	Storage and treatment facility for contaminated soil	Section 22 of the Environmental Quality Act	April 24, 2015
MDDELCC	Construction and operation of a power plant	Section 22 of the Environmental Quality Act	May 8, 2015
MDDELCC	Extraction and crushing of waste rock from R-65 pit	Section 22 of the Environmental Quality Act	May 29, 2013

MDDELCC	Construction and operation for a trench landfill	Section 22 of the Environmental Quality Act	February 04, 2014
MDDELCC	Operation of a mobile batch plant	Section 22 of the Environmental Quality Act	November 17, 2014
MDDELCC	Explosive fabric	Section 22 of the Environmental Quality Act	January 6, 2015
MDDELCC	Implementation of compensation measures for fish habitat losses	Section 22 of the Environmental Quality Act	March 18, 2015
MDDELCC	Construction and operation of the aerodrome	Section 22 of the Environmental Quality Act	August 23, 2013
MDDELCC	Aerodrome garage Oil separator	Section 22 of the Environmental Quality Act	August 5, 2015
MDDELCC	Authorization regarding the supply and treatment of drinking water at the 640 km camp (80 bed_	Section 32 of the Environmental Quality Act	July 29, 2014
MDDELCC	Modification of the authorization regarding the supply and treatment of drinking water at the640 km camp (120 bed)	Section 32 of the Environmental Quality Act	January 15, 2015
MDDELCC	Modification of the authorization regarding domestic wastewater treatment at the Lagopede camp	Section 32 of the Environmental Quality Act	October 25, 2013
MDDELCC	Authorization regarding domestic wastewater treatment at the640 km camp (80 bed)	Section 32 of the Environmental Quality Act	July 29, 2014
MDDELCC	Modification of the authorization regarding domestic wastewater treatment at the640 km camp (120 bed)	Section 32 of the Environmental Quality Act	December 18, 2014
MDDELCC	Construction and operation of the permanent freshwater pumping station	Section 32 of the Environmental Quality Act	September 26, 2014
MDDELCC	Construction and operation of the permanent freshwater and drinking water treatment plant	Section 32 of the Environmental Quality Act	September 26, 2014
MDDELCC	Construction and operation of the permanent domestic wastewater treatment plant	Section 32 of the Environmental Quality Act	October 10, 2014
MDDELCC	Permit to occupy a part of the water property for non-profit purposes to install a pumping station in Lake Lagopede	Section 10 of the Regulation respecting the water property in the domain of the State	October 20, 2014
MDDELCC	Permit to occupy a part of the water property for non-profit purposes to install a pipe for the discharge of domestic wastewaters in Lake Lagopede	Section 10 of the Regulation respecting the water property in the domain of the State	October 20, 2014

MDDELCC	Notification the Minister regarding the establishment of a permanent camp for Renard	Section 2 of the Regulation respecting sanitary conditions in industrial or other camps	September 9, 2014
MDDELCC	Notification the Minister regarding the establishment of a permanent camp et km 640	Section 2 of the Regulation respecting sanitary conditions in industrial or other camps	July 29, 2014
MERN- MFFP	Location of the process plant and of the tailings and waste rock accumulation areas	Sections 240 and 241 of the Mining Act	October 31, 2012
MERN- MFFP	Rehabilitation and restoration plan of the Renard Diamond Project	Section 232.2 of the Mining Act	December 21, 2012
MERN- MFFP	Rehabilitation and restoration plan of the Lagopede camp	Section 232.2 of the Mining Act	July 8, 2015
MERN- MFFP	Forestry permit (to deforest geotechnical drilling sectors)	Section 73 of the Sustainable Forest Development Act	April 02, 2013
MERN- MFFP	Forestry permit (to deforest the mine site)	Section 73 of the Sustainable Forest Development Act	October 26, 2012
RBQ	Use of a high-risk petroleum equipment (Mine site)	Section 120 of the Safety Code (Régie du bâtiment du Québec)	April 30, 2015
RBQ	Use of a high-risk petroleum equipment (Aerodrome)	Section 120 of the Safety Code (Régie du bâtiment du Québec)	March, 23, 2015
RBQ	Use of a high-risk petroleum equipment (Lagopède camp)	Section 120 of the Safety Code (Régie du bâtiment du Québec)	January 22, 2014
SQ	General permit to have explosives (during construction)	Division II of the Regulation under the Act respecting explosives	February 18, 2014
SQ	Magazine permit and transport permit (during construction)	Division II of the Regulation under the Act respecting explosives	March 10, 2015
SQ	General permit to have explosives (during operation)	Division II of the Regulation under the Act respecting explosives	March 11, 2015
SQ	Magazine permit and transport permit (during Operation)	Division II of the Regulation under the Act respecting explosives	March 12, 2015
DFO/NRC	Positive Environmental Assessment Decision for the Renard Project from the Canadian Environmental Assessment Agency	Section 53 of the Canadian Environmental Assessment Act	9 avril 2014
DFO/NRC	Authorization to modify a fish Habitat	Section 35 of the Fisheries Act	9 avril 2014

DFO/NRC	Authorization to capture fish	Under the Fisheries Act	27 septembre 2014
DFO/NRC	Derogation to work in fish habitat for request 103, 124 and 125	Under the Fisheries Act	16 septembre 2014
DFO/NRC	Lease to operate an Explosive factory in construction	Under Natural Resources Explosive Act	31 March 2013
CNSC	Licence to possess radiating equipment	Nuclear Safety and control Act	January 26, 2016
Municipal	Municipal authorizations	Municipal regulation	20 mai 2014
	Municipal construction permit	Project of law #40	20 mai 2014

2. MATERIAL PERMITS, CONSENTS AND APPROVALS REQUESTED FOR THE COMMENCEMENT OF COMMERCIAL PRODUCTION REQUIRED AND THAT ARE EXPECTED TO BE OBTAINED IN THE ORDINARY COURSE OF CONSTRUCTION AS AND WHEN REQUIRED

Note: the following permits, consents and approvals are to be obtained pursuant to the plan and schedules contained in the Mine Plan

Governmental Authority	Description	Legal reference	Date of filing
MDDELCC	Operation of oil-water separators (garage and power plant)	Section 32 of the Environmental Quality Act	April 2016
MDDELCC	Processed Kimberlite Containment Facility	Section 22 of the Environmental Quality Act	February 4, 2016
MDDELCC B	Implementation of compensation measures for fish habitat losses	Section 22 of the Environmental Quality Act	
MDDELCC	Implementation of compensation measures for wetland losses	Section 22 of the Environmental Quality Act	8 janvier 2015
MDDELCC	Obtainment of a depollution attestation	Section 31.11 of the Environmental Quality Act	
MDDELCC	Permit to occupy a part of the water property for non-profit purposes to install a pipe for the discharge of industrial wastewaters in Lake Lagopede	Section 10 of the Regulation respecting the water property in the domain of the State	September 03, 2015

Surface leases are discussed in section 4 of this Technical Report.

20.6 Community Agreements

In March 2012, Stornoway completed negotiations with the Cree Nation of Mistissini, the Grand Council of the Crees (Eeyou Itschee) and the Cree Regional Authority on an Impacts and Benefits Agreement (IBA) designated the Mecheshoo Agreeement. The Mecheshoo Agreement is a binding agreement that will govern the long-term working relationship between Stornoway and the Cree parties during all phases of the Renard Diamond Project. It provides for training, employment and business opportunities for the Crees during project construction, operation and closure, and sets out the principles of social, cultural and environmental respect under which the project will be managed. The Mecheshoo Agreement includes a mechanism by which the Cree parties will benefit financially from the success of the Project on a long-term basis, consistent with the Mining Industry's best practices for engagement with First Nations communities.

On July 2012, Stornoway executed a Declaration of Partnership (the "Declaration") with the communities of Chibougamau and Chapais in the James Bay Region of Québec. The Declaration is a statement of cooperation between the partners for the responsible development of the Renard Diamond Project based on the principles of environmental protection, social responsibility and economic viability. The Declaration includes provisions to set up a Renard Liaison Committee that will address issues of mutual interest such as communication, employment, and the economic diversification of local communities. In particular, the committee will oversee initiatives to attract and retain new residents to the towns of Chibougamau and Chapais.

20.7 Mine Closure

Stornoway is committed to sound environmental practices in all of its activities through the implementation of its environmental policy. The policy includes a commitment to progressively rehabilitate disturbed areas, to develop a rehabilitation plan that can be continuously improved and to incorporate new technologies where practical.

The Rehabilitation Plan was prepared according to the requirements of the MERN outlined in the document titled "*Guidelines for Preparing a Mining Site Rehabilitation Plan and General Mining Site Requirements*."The Plan was submitted on February 15, 2012 and approved by the MERN on December 21, 2012.

Rehabilitation of the mine site will be carried out progressively throughout the life of mine (LOM). Acknowledging that Stornoway is only a temporary user of land that will be returned to users at the end of the mine life, the Project design has been optimized to reduce the footprint and to facilitate rehabilitation:

- The number of sub-watersheds of Lake Lagopede that will be affected by the development has been reduced;
- Materials and buildings have been designed to facilitate dismantling and/or recycling at the end of the mine life;
- The mine plan has been developed to ensure that all waste rock will be used (construction and backfill) to fill the underground mine or sent back to open-pits;
- The processed kimberlite containment (PKC) area has been designed to promote progressive revegetation and facilitate maintenance;
- Plans have been developed to progressively revegetate flat surfaces (slope benches) when they are no longer required for the PKC facility operation; and
- Environmental and geotechnical follow-ups have been designed to be carried out for a period of three to five years after closure and dismantling.

The following activities are part of the planned mine rehabilitation and were used to estimate the costs of Rehabilitation of the Renard Project:

- No building will be left in place. Whenever possible, buildings will be sold with the equipment they contain, completely or partially. During dismantling works, a reclamation/recycling program will be established. All non-contaminated waste will be sent to the project trench landfill. All equipment and machinery will be sent out of the mine site for sale or recycling;
- All hazardous waste will be managed according to applicable laws and regulations and will be sent out of the mine site;
- On-site roads and tracks will be scarified and revegetated;
- Revegetation will include hydroseeding followed by tree planting;
- Flat surfaces of the PKC facility will be revegetated. Revegetation will be carried out on 5-m-high benches on eight levels and on the upper surface. Bench revegetation will be carried out progressively during the operation phase. Flat surfaces will be covered with a 0.5-m overburden layer under 0.3 m to 0.5 m of topsoil (organic matter) and revegetated;
- During site construction, topsoil will be stockpiled for revegetation purposes. The overburden stockpile will be constructed in four 5-m lifts. Slopes will be covered with gravel. Flat surfaces will be revegetated. Eighty-five percent (85%) of flat surfaces will be located on the final plateau;
- All waste rock will be used (construction and backfill) to fill the underground mine or sent back to open-pits. The area previously occupied by the waste rock pile will be covered with a 0.5 m overburden layer under a 0.3 m to 0.5 m topsoil (organic matter) layer and revegetated;
- The area previously occupied by the temporary ore stockpile will be covered with a 0.5 m overburden layer under a 0.3 m to 0.5 m topsoil (organic matter) layer and revegetated;
- The sanitary wastewater treatment plant will be dismantled, and the sludge in the three lagoons will be emptied. Ponds will be either landscaped to develop a wetland or filled and revegetated;
- The outfall will be covered and left in place;
- The industrial wastewater treatment plant will be dismantled. The outfall will be covered and left in place;
- R2-R3 and Renard 65 open-pits will no longer be dewatered and will gradually become “lakes.” Surface hydraulic connectivity will be maintained between Renard 65 lake and Lake Lagopede The small stream derived to prevent surface runoff from entering the pit will be restored to its original route to re-establish the initial drainage;
- A hydraulic link will be constructed between the Renard 2 and Renard 3 pits and the surface water hydrological network to accelerate pit filling. Once filling is complete, a hydraulic link with the surface water hydrological system will be constructed;
- Security bunds will be constructed around the pits;
- A concrete slab will be installed to secure all underground access (shaft, ramp, vent raise and backfill raises);
- Whenever possible, surface water drainage will be re-established to conditions similar to the original hydrological system;
- The airstrip will be scarified and revegetated and its buildings dismantled;
- The explosives storage facilities will be dismantled;
- The sand pit will be covered with topsoil and revegetated;
- The trench landfill will be covered and revegetated;
- Revegetation will be monitored for three to five years and subsequent revegetation will be undertaken in spots where initial planting has not succeeded;
- Geotechnical monitoring of PKC facility, waste and overburden piles will be conducted for a three-to-five-year period; and
- The main access road, which belongs to Québec’s Ministry of Transport, and will be left in place.

Table 20.4 provides the October 2012 cost estimate for the rehabilitation work of the Renard Project. This estimate is taken from the Rehabilitation Plan prepared by Roche in 2011 (revised in 2012) and approved by Stornoway.

Table 20.4: October 2012 - Cost Estimate for Rehabilitation of the Renard Project

Item	Estimated cost
Rehabilitation of the PKC facility	$5,904,438
Revegetation of flat surfaces, other than the PKC facility	$2,472,437
Other rehabilitation activities	$6,091,680
TOTAL	**$14,468,555**

Under the Mining Act of Quebec, a financial guarantee must be submitted to the MERN to guarantee 100% of the rehabilitation costs of a mining project within a period of three years, in accordance with a payment schedule prescribed by applicable regulation. On August 29th 2014, the Corporation arranged for a surety bond of up to $15.2 million to provide a financial guarantee to the MERN with respect to the Closure Plan. The obligation to pay the first tranche of $7.6 million was met in August 2014; the second installment of $3.8 million was met in August 2015 and the third and last installement is due in August 2016.

20.8 Renard Mine Access Road

The Renard Mine Access Road Certificate of Authorization represented the principal regulatory approval required for Stornoway to begin road construction. The CA was received on December 17, 2012 from the MDDELCC. The transfer of this CA was consistent with the schedule contained in the Framework Agreement of November 15, 2012, between Stornoway and four provincial ministries: the MDDELCC, the MTQ, the Ministry of Natural Resources and the Ministry of Finance and Economy.

The list of permits and authorizations that were required for the construction of the Renard Mine Access Road is included in Table 20.5.

Table 20.5: List of authorizations, permits, leases and approvals related to the Renard Mine Road

Description	Department	Regulation/Act/Code
Provincial Government		
Authorizations		
Transfer of the CA from the MTQ to Stornoway to carry out a road project	MDDELCC	Section 22 of the *Environment Quality Act* (R.S.Q., c. Q-2)
Sand and gravel operation (borrow pits)	MDDELCC	Section 22 of the *Environment Quality Act* (R.S.Q., c. Q-2). For activities not subjected to the Regulation respecting standards of forest management for forests in the domain of the State.
Camps (potable and grey waters, waste disposal)	MDDELCC	Section 32 of the *Environment Quality Act* (R.S.Q., c. Q-2)
Sand and gravel operation (borrow pit) authorizations	COMEX	Conditions 5 and 6 of the CA. Chapter II of the *Environnement Quality Act (R.S.Q., c.Q-2).* Section 164.
Fish habitat authorizations	COMEX	Conditions 10 of the CA. Chapter II of the *Environnement Quality Act* (R.S.Q., c.Q-2). Section 164.
Permits		
Forest management permit for road activities All deforestation activities	MFFP	Chapter I, Divisions I and II of the *Forest Act* (R.S.Q., c. F-4.1) Regulation respecting standards of forest management for forests in the domain of the State (c. F-4.1, r. 7)
Leases		
Non-exclusive lease to mine surface minerals Borrow pits, quarries and sand pits	MERN-Mines	Section 140, Division VII of the *Mining Act* (R.S.Q., c. M-13.1)
Lease to occupy the domain of the State from km 143 to km 240	MERN-Territory	Section 239 of the *Mining Act* (R.S.Q., c. M-13.1). Section 47, Division II of the *Act respecting the lands in the domain of the State* (R.S.Q., c. T-8.1) Section 35 of the *Regulation respecting the sale, lease and granting of immovable rights on lands in the domain of the State* (c. T-8.1, r. 7)

Federal Government		
Authorizations		
Authorization to modify a fish habitat All activities affecting fish habitats	Fisheries and Oceans Canada	Section 35 of the *Fisheries Act* (R.S.C., 1985, c. F-14)
Approvals		
Approval of works in, over, under, through or across any navigable waterway	Transport Canada	Section 5 of the *Navigable Waters Protection Act* (R.S.C., 1985, c. N-22)
Municipal Government		
Conformity		
Certificate of compliance to municipal regulations	Municipality of Baie-James	Various municipal by-laws

21. CAPITAL AND OPERATING COSTS

21.1 Introduction

In the 2013 Optimization Study, SLI and its subcontractor AMEC were solely responsible for the capital cost estimate (CAPEX) of the infrastructure and process plant direct costs, as well as indirect and other costs related to these direct costs. SLI and its subcontractor AMEC Process was also responsible for providing the operating cost estimate (OPEX) as it relates to the diamond process plant and to provide the load list to Stornoway to be used as a basis for power consumption calculations during mine operation.

In July 2014, Stornoway entered into an Engineering, Procurement and Construction Management (EPCM) contract with SLI for a portion of the Renard Project scope of construction work, with subcontracted services from AMEC and DRA DRA relating to the project's diamond process plant and diamond recovery circuits. All open pit and underground equipment, development and infrastructure costs remained under the responsibility of Stornoway.

Effective March 9, 2015, Stornoway elected to assume additional construction management and procurement duties that had initially been under the EPCM and expanded its owner's team in these areas. Engineering work for the Renard Project under the agreement with SLI was completed on December 31, 2015 and Stornoway has assumed all construction management responsibilities for the Project at that time, Stornoway also elected to assume all cost control functions, including budgeting, change control and cost reporting. As a result, the cost reporting for SLI's managed scope was integrated into Stornoway's cost control system.

Table 21.1 shows the high-level responsibilities with respect to sub-areas in terms of engineering development.

Table 21.1: High-Level Responsibility Matrix

Sub Area – Description	Responsible
0000 – General, Multi section – ZONE 1	Stornoway
0000 – General, Multi section – ZONE 2	Stornoway
1100 – Mine mobile equipment, U/G	Stornoway
2100 – Mine mobile equipment, open pit	Stornoway
1200 – Off-shaft underground infrastructure	Stornoway
8100 – R2/R3 open pit pre-production	Stornoway
8100 – R2/R3 Underground ramp and shaft	Stornoway
1400 – Shaft and off shaft development (including hoists)	Stornoway
0310 – Renard 65 – Surface water drainage sump	Stornoway
2200 – Mine area surface facilities (explosives and detonator storage)	Stornoway
3000 – Mineral processing plant	Stornoway
4110 – Overburden, waste rock, pre-production ore stockpile	Stornoway
4310 – PKC facility	Stornoway & Golder
5000 – On-site utilities and infrastructure (Power Plant and LNG)	Stornoway
8000 – Off-site utilities and infrastructure (permanent access road and airport)	Stornoway

21.2 Capital Costs Estimate

In the 2013 Optimization Study, the capital cost estimate was qualified as a Class III type estimate by SLI per the Association for the Advancement of Cost Engineering (AACE), and the estimated accuracy was evaluated at -13% / +17%. The estimated labour costs were based on a 70-hour workweek (10 hrs/day @ 7 days/week), with a rotation of 2 weeks in / 1 week out. There was no allowance for double shifts in the estimate for above-ground work. Stornoway's self-performed activities for mine development were based on an 84-hour workweek (i.e., 12 hours per day, 7 days per week) with 2 weeks in / 2 weeks out rotation.

In the 2013 Optimization Study budgetary quotations, estimated costs and allowances were actualized to Q4-2012 Canadian dollars and included escalation beyond Q2-2011 up to and including Pre-Operations Verification scheduled for January 2016. Escalation was estimated per package based on this schedule as an additional cost item. Owner's costs were not escalated.

Direct Costs were based on engineering take-offs, and scope and quantity reviews were performed with engineering to confirm that the scope of work was entirely covered. Approximately 40% of the estimate is based on budget quotations while the remainder is based on in-house databases or factored historical data.

The portion of the mine development and production costs (underground costs and open pit costs) occurring prior to Commencement of Commercial Production was summarized and included in "Capital Cost – Mining". The operating costs of the process plant were also calculated as part of the OPEX and the portion occurring prior to Commencement of Commercial Production was summarized and included in "Capital Cost – Processing Plant". Owner's site operating costs (general and administrative, or G&A) were evaluated in detail in the OPEX, and the portion of the operating costs occurring prior to Commencement of Commercial Production was summarized and included in the "Capital Cost – Owner's costs".

Contingencies were evaluated for all packages on the basis of the engineering level of confidence.

In the LNG Feasibility Study for power generation at the Renard Project, a number of design changes relating to the replacement of 10 diesel generators for 7 natural gas generators at the power plant, and the addition of a LNG tank farm were included. This resulted in a net increase of $2.6 million to the project CAPEX.

In April 2014, an update on the CAPEX estimate was performed to establish a control budget for the pre-production capital costs, including latest engineering material take-off's (MTO's). This resulted in a net increase of $8.7 million to the project CAPEX, on top of which a late adjustment to add $565,000 for fuel was also made. For the purpose of the control budget, the $23.2 million in the Risk Account 9930, in the below table, was removed from the calculation total.

The net result of the CAPEX estimate from the 2013 Optimization Study, the November 2013 LNG Feasibility Study and the April 2014 control budgeting exercise was a CAPEX estimate of C$811 million for the commencement of project execution in July 2014. A summary of the evolution of the estimated total CAPEX is shown in Table 21.2.

Table 21.2: Summary of Total Estimated Capital Costs

Major Area / Sub-area (in thousands of Canadian Dollars)	**2013 Optimization Study Estimate [Feb. 2013]**	**LNG FS Option Estimate [Nov. 2013]**	**Estimate Update [Apr. 2014]**
DIRECT COSTS			
0000 –General, multi-Section	32,658	32,397	37,255
1000 – Mining	141,352	139,866	143,269
2000 – Mine area surface facilities	9,863	9,977	10, 974
3000 – Mineral processing plant	175,410	174,904	177,602
4000 – Stockpiles, PKC and waste rock management	0	0	0
5000 – Onsite utilities and infrastructures	114,753	118,517	121,702
8000 – Offsite utilities and infrastructures	0	0	0
Sub-Total Direct Costs	**474,035**	**475,661**	**490,801**
INDIRECT COSTS			
9300 – Owner's cost	94,656	93,535	98,446
9300 –Spares, fills, tools	7,071	7,364	7,688
9400 – EPCM services	47,862	49,980	46,860
9500 – Field indirects	54,693	55,563	58,786
9600 – Vendor reps	3,678	3,798	3,885
9700 – Freight (incl. duties)	5,476	5,476	5,585
9910 – Contingency	64,660	62,600	73,020
Sub-Total – Indirect costs	**278,096**	**278,315**	**294,271**
Total – excluding Escalation & Risk	**752,131**	**753,976**	**785,072**
9920 – Escalation	40,500	41,300	25,350
9930 – Risk	29,700	29,700	23,220

Total – including Escalation & Risk	**822,331**	**824,976**	**833,642**
Adjustment for fuel	0	0	565
Total – including fuel adjustment & excluding risk	**792,631**	**795,276**	**810,987**

In November 2015 Stornoway undertook a re-baselining exercise for the CAPEX on the basis of the Project's then percentage of construction completion. The re-baselining exercise resulted in a shortened overall construction schedule showing a 5-month improvement in the scheduled date of commencement of commercial production from May 29, 2017 to December 31, 2016.

As a result of the November 2015 re-baselining exercise, the estimate of CAPEX to complete the contruction of the Project to the date of Commencement of Commercial Production has been estimated at $775.4 million, including contingencies of $57.5 million and an escalation allowance of $25.3 million. A portion of the 2017 mine development costs and associated G&A costs that would have been incurred during the construction phase of the Project under the initial construction schedule was transferred to the 2017 sustaining capital. CAPEX is estimated at an accuracy of -13% and +17%. (Table 21.3)

Table 21.3: Estimate of Capital Costs

Site Preparation & General	$ 45.1
Mining	$ 57.1
Mineral processing plant	$ 137.7
Onsite utilities and infrastructures	$ 111.1
Network and Distribution	$ 15.9
Offsite utilities and infrastructures	$ 0.3
Pre-production and Ramp-up	$ 88.5
Project indirect costs	$ 153.9
Professional Services	$ 44.6
Construction indirect costs	$ 38.5
Contingency	$ 57.5
Escalation	$ 25.3
Total including escalation	**$ 775.4**

Note:
(1)Dollar amounts in $ million. Totals may not add due to rounding.

As of December 31, 2015, there was $24.6 million of contingency and $12.0 million of escalation allowance unallocated.

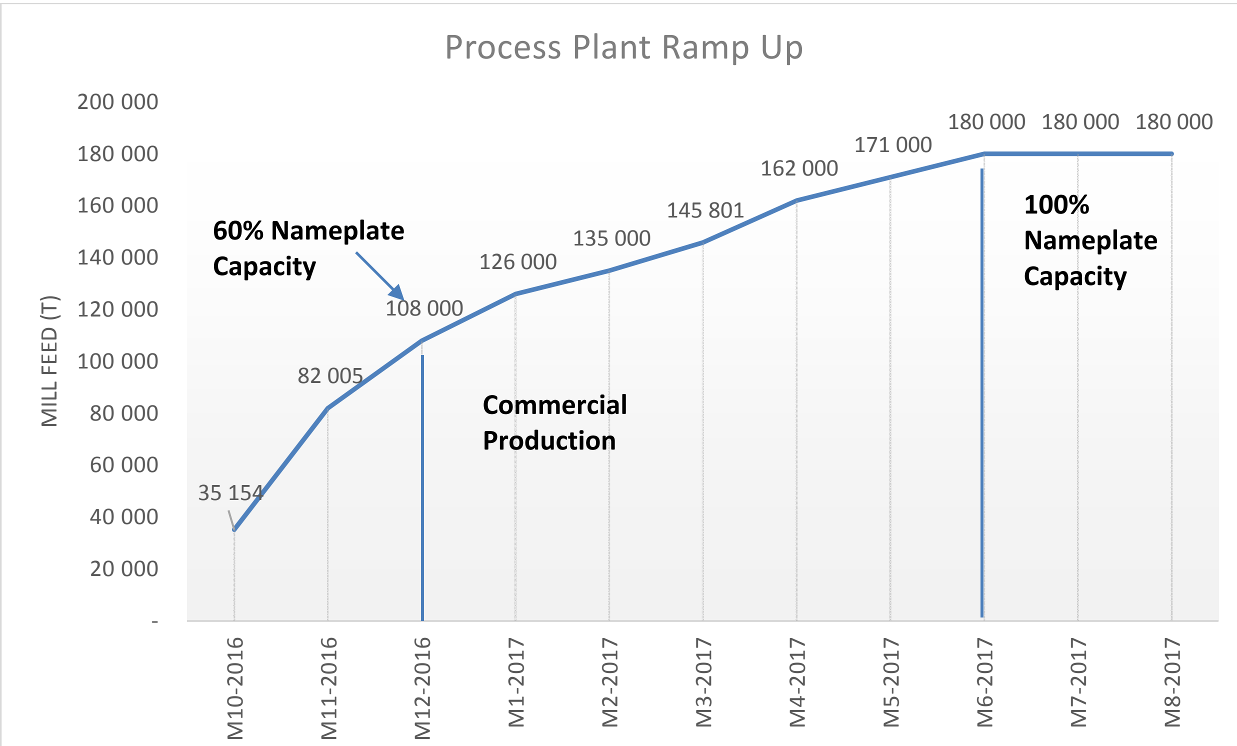
21.3 Operating Costs Estimate

OPEX is defined as those costs occurring after the Commencement of Commercial Production (defined as sustained ore processing at 60% or more of plant nameplate capacity for 30 consecutive days). Based on the re-baselined project schedule and estimated production ramp-up schedule, commercial production is scheduled to be achieved by December 2016, as illustrated in Figure 21.1.

OPEX was estimated by Stornoway and is subdivided into four areas as follows:

- Open Pit Operations;
- Underground Mine Operations;
- Process Plant and Auxiliaries;
- General, Administrative and Infrastructure.

Figure 21.1: Estimated Ramp-Up Production Curve



The open pit operations cost estimate is based on the updated open pit mine design and production schedule, and assumes standard open pit mine operating procedures. Kimberlite ore will be delivered initially to the ore stockpile or, once process plant operations start, to the surface primary crusher upstream of the process plant.

The underground mine operations cost estimate is based on the updated underground mine design and production schedule and estimated from first principles. Ore from the underground mine will be transported to the surface using truck hauling via ramp.

The process plant and auxiliaries cost estimate is based on the quantity of kimberlite ore feed to the process plant. For the purpose of the estimate, a process plant capacity of 78% plant utilization at a nameplate capacity of 2.16 Mt/a was assumed. The operating costs were estimated using standard manufacturer's cost data for power, fuel, equipment, consumables and hourly maintenance costs.

The general, administrative and infrastructure cost estimates were derived from budgetary quotes and project actuals established in the first year of mine operations. Labour costs are based on a salary scale and organizational chart developed by Stornoway.

A summary of the estimated OPEX, on a life of mine basis, for each of the four categories described above is shown in Tables 21.4 and 21.5.

Table 21.4: Summary of Estimated Total LOM OPEX ($ '000)

Description	Open Pit	Underground	Process plant	G&A and Infrastructure
Labour	64,751	263,780	188,863	236,559
Fuel	20,868	57,599	20,744	29,967
Explosives & supplies	8,038	21,368	-	-
Consum. & reagents	-	-	44,418	-
Equipment	26,099	143,175	-	-
Power	1,982	59,238	195,393	62,137
Maintenance	-	-	89,438	24,537
Heating	-	-	11,980	-
Accomodation	-	-	-	128,048
Transportation	-	-	-	54,961
Other	4,655	72,744	34,927	180,891
Totals	126,396	617,904	585,763	717,100

Total LOM OPEX is estimated at $1,878,431,631, excluding $168,728,602 of costs incurred prior to the attainment of commercial production and characterized as capitalized operating costs within the CAPEX estimate.

Table 21.5: Summary of Estimated OPEX LOM Unit Cost per Tonne Ore Processed ($)

Description	Open Pit	Underground	Process plant	G&A and Infrastructure
Labour	1.94	7.89	5.65	7.08
Fuel	0.62	1.72	0.62	0.90
Explosives & supplies	0.24	0.64	-	-
Consum. & reagents	-	-	1.33	-
Equipment	0.78	4.28	-	-
Power	0.06	1.77	5.85	1.86
Maintenance	-	-	2.68	0.73
Heating	-	-	0.36	-
Accomodation	-	-	-	3.83
Transportation	-	-	-	1.64
Other	0.14	2.18	1.04	5.41
Totals	3.78	18.49	17.53	21.45

Total life of mine OPEX unit cost is estimated at $56.20/tonne processed, excluding $5.05/tonne pre-production costs characterized as capitalized operating costs within the CAPEX estimate.

Sustaining Capital Costs

Sustaining Capital Costs are capital and replacement costs required to sustain operations against mine plan estimates, such as maintenance of the mobile mining fleet and power plant, or deferred capital costs occurring after the attainment of commercial production, such as the development of the underground mine. Sustaining Capital Costs are defined independently of the CAPEX estimate or OPEX. Annual estimates for the life of mine are shown in Table 21.6.

Table 21.6: Estimated Sustaining Capital and Capital Replacement Costs (in $M)

Description	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Open pit	1.8	0.5	0.7	3.2	1.5	1.5	1.3	3.0	2.7	0.0	0.0	0.0	0.0	0.0
Underground	43.7	45.2	27.8	18.4	11.8	31.8	5.8	18.5	13.6	5.0	0.5	0.1	0.1	0.0
Process plant	0.8	1.1	1.5	1.1	1.5	1.1	1.2	1.1	1.1	1.0	0.3	1.0	0.0	0.0
Adm. & Infra.	0.0	0.2	0.0	0.7	0.2	0.0	0.0	0.2	1.5	0.0	0.2	0.0	0.0	0.8
Power Plant & Airport	3.8	1.8	2.0	0.2	0.5	1.8	0.9	0.0	1.8	0.7	0.0	0.0	0.0	0.0
Totals	50.1	48.8	32.0	23.6	15.4	36.2	9.2	22.7	20.7	6.7	0.9	1.1	0.1	0.8

Basis and Accuracy Assessment of Estimate

The OPEX estimate was developed for the Project with the following key assumptions:

- Power unit cost was based on an estimate of fixed and variable costs. Power consumption was based on the average consumption of each motor included in the equipment list;
- Pricing is based on cost of supply, manpower performance and compensation actualized to the fourth quarter of 2015;
- No allowance was made for contingency, escalation and risk relating to OPEX;
- No allowance was made for final mine closure activities within the OPEX estimate, other than the progressive reclamation costs of the PKC facility. In the project Economic Analysis, the balance of mine closure costs are treated as a capital cost item.

The OPEX estimate was based on budgetary quotes and project actuals established in the first year of mine operations, and utlizes a Class III estimate methodology as defined by the type and quantity of engineering deliverables produced to support the estimate. The expected order of accuracy is in the range of -15% to +15%.

22. ECONOMIC ANALYSIS

The project economic analysis incorporates the estimated LOM diamond production profile based on the updated Mineral Reserves, diamond valuation and escalation estimates, CAPEX, OPEX, Sustaining Capital Costs, salvage value, working capital, closure and reclamation costs, taxation, costs under the Mecheshoo Agreement, royalties, the Renard diamond Streaming Agreement, and the project's financial parameter assumptions.

After-tax and pre-tax net present value (NPV) evaluations have been undertaken in real dollar terms (or constant dollars) utilizing an effective date ofJanuary 1, 2016. Valuations are presented on an un-levered basis.

The economic analysis employes the re-baselined estimates of cost and schedule to complete, being a CAPEX estimate of $775 million with the attainment of commercial production by December 31, 2016.

22.1 Financial Assumptions

The significant financial assumptions affecting the financial analysis of the Project include:

- Diamond selling prices;
- Energy, including diesel prices, LNG cost and all-in power costs;
- Process plant throughput, by mid 2018, at 2,500,000 tpa;
- CAD/USD exchange rates; and
- Diamond price escalation rates.

All quoted figures in the project economic analysis are quoted in Canadian $ terms unless stated otherwise.

22.2 Diamond Prices

"Spot" diamond valuation estimates used for revenue forecasting are based on a -19% estimated market adjustment to the March 2014 valuation of the Renard 2, 3, 4 and 65 valuation samples by WWW International Diamond Consultants, as described in section 19 of this report and summarized in Table 22.1.

Table 22.1: Diamond Valuation Results, Renard Kimberlite Pipes

Body	March 2014 Diamond Price Model[1] (US$/carat)	Estimated Market Price Adjustment March 2014 to March 2016[2]	Adjusted Price Estimates March 2016: "Spot" Price Models[2] (US$/carat)
Renard 2	**$197** *(High $222, Min $178)*	*-19%*	**$160** *(High $181, Min $145)*
Renard 3	**$157** *(High $192, Min $146)*	*-19%*	**$128** *(High $156, Min $119)*
Renard 4	**$106 *($155)*[3]** *(High $174, Min $100)*	*-19%*	**$86 *($126)*[3]** *(High $141, Min $81)*
Renard 65	**$187** *(High $190, Min $160)*	*-19%*	**$152** *(High $155, Min $130)*

Notes

1) *As determined WWW International Diamond Consultants Ltd. at a +1 DTC sieve size cut off.*
2) *As determined by applying the world average rough price index of roughrices.com to the March 2014 price models by Stornoway, at a +1 DTC sieve size cut-off.*

3) *Should the Renard 4 diamond population prove to have a size distribution equal to the average of Renard 2 and 3, WWW have estimated that a base case diamond price model of US$155 per carat would apply based on March 2014 pricing, equivalent to US$126 per carat on a market price adjusted basis to March 2016.*

Diamond prices are assumed to increase by 2.5% per annum in real terms (with sensitivities ranging from 0% to 5% in real terms), from January 1, 2016 until the end of 2026, as described in section 19 of this report.

22.3 Exchange Rate

The CAD/USD exchange rate is used to convert revenue from diamond sales into Canadian dollars and to calculate a delivered certain fuel price to site. The base case exchange rate assumption for financial modelling is 1.35 C$/US$ flat. All operating cost estimates are in Canadian dollars.

22.4 Energy

The delivered diesel price used in the model is $1/litre, and the delivered Liquified Natural Gas (LNG) price is $0.63/m^3 which includes transportation costs to site, federal excise tax and provincial fuel tax.

Power generation is via a combination of NG and diesel power plant located on site. The cost of generating power is therefore dependent on a mix between LNG and diesel and the price assumptions for both. The average mix of NG and diesel used to generate power over the life of mine is approximately 88% and 12% respectively.

The total energy cost (all-in power and fuels) represents 24.5% of total LOM OPEX.

22.5 Nominal Rates

Project cash flows have been calculated in real dollar terms (constant dollar terms). Where nominal cash flows have been calculated for comparable purposes, a general inflation factor of 2% per year has been applied to operating costs, revenues, deferred and sustaining capital expenditures, as well as closure costs and salvage values. The general inflation factor used is consistent with the monetary policy adopted by the Bank of Canada and the federal government at keeping total CPI inflation at 2% with a control range of 1% to 3% around the target (Bank of Canada Monetary Policy Report, July 2011).

22.6 Project Financing

On July 8, 2014 Stornoway completed a series of financing transactions, consisting of the issuance of common shares and warrants, convertible debentures, a diamond Streaming Agreement, a Senior Loan Agreement, cost over-run facilities and an Equipment Finance Facility. Proceeds from the financing transactions have been used and are being used for the construction of the Renard Project, and for working capital during the construction period, including interest and financing expenses. It is anticipated that proceeds from the financing transactions mentioned above will be sufficient to meet its capital requirements up to the commencement of commercial production at the Project.

22.7 Taxation, Duties, Royalty Summary

The Project is subject to three levels of taxation including provincial mining duties, provincial income tax and federal income tax. A 2% royalty on diamond production is payable to DIAQUEM as described below.

22.7.1 Québec Mining Duties

The Québec mining duties regime is based on the notion of mining profit. Mining duties aim to capture part of the profit derived from the value of ore. The mining duty calculation includes a processing allowance to provide for a financial return to the investor on the cost of property used to process the ore.

Under the mining duties regime an operator determines annual profit by subtracting from the gross value all deductions and allowances stipulated under the regime which include depreciation allowance, additional depreciation allowance, and additional allowance for a northern mine, the allowance for exploration, mineral deposit evaluation and mine development and the additional exploration allowance.

The Québec mining duties rate range between 16% and 28% based on the level of mining operation profit for 2015 and subsequent years.

The depreciation allowance an operator can claim in calculating its annual profit for a fiscal year regarding property class generally corresponds to the lesser of the following amounts:

- A percentage of the capital cost of the property of such class, for such fiscal year;
- The undepreciated capital cost of property of such class, at the end of the fiscal year.

Three property classes are relevant for the Renard Project:

- Mining roads, buildings and other equipment, depreciable at up to 30% per year (in certain circumstances up to 100% if acquired prior to March 31, 2011);
- Mineral deposit evaluation and pre-production mine development expenses, depreciable at up to 100% in any year; and
- Mineral deposit evaluation and post-production mine development expenses, depreciable at up to 30% in any year.

The mining duties paid to the province are estimated at $428.5 million less refundable credit for pre-production development expenses of $26.7 million for a net amount of $401.8 million.

22.7.2 Provincial Income Taxes

The Project is subject to Québec provincial income taxes as stipulated in the *Québec Tax Act*. The corporate income tax rate in Québec is 11.9% for 2015 and subsequent years. Taxes paid to the provincial government over the project life are estimated at $131.1 million.

22.7.3 Federal Income Taxes

The Project is also subject to federal corporate income taxes in Canada under the *Income Tax Act* (Canada) at a rate of 15% for 2015 and subsequent years. Taxes paid to the federal government over the project life are estimated at $165 million.

22.7.4 Royalty Agreement

A royalty agreement was executed on April 1, 2011 between Stornoway and DIAQUEM at the time of Stornoway's acquisition of DIAQUEM's 50% beneficial interest in the Renard Joint Venture Project.

The terms of the royalty agreement provide that SDCI shall grant and pay to DIAQUEM a royalty on all diamonds and other gemstones associated with kimberlite extracted from the property of the Renard Project (Diamond Royalty). The Diamond Royalty is calculated and payable in respect of diamonds sold by or on behalf of SDCI or any affiliate during each fiscal quarter of SDCI. The Diamond Royalty equals 2% of the aggregate of (i) the actual gross selling price for arm's-length sales and permitted non-arm's-length sales and (ii) the proceeds of insurance received by or for the benefit of SDCI or any affiliate during the quarter with respect to the loss or destruction of or damage to any diamonds, less any deductions, which shall be the lesser of the aggregate of diamond selling costs incurred by SDCI and 3% of the aggregate gross selling price in clause (i) above.

The Diamond Royalty is estimated at $109 million over the project life in real terms.

22.7.5 Streaming Agreement

The Streaming Agreement is described in section 19.3.1 of this Technical Report.

On April 30th, 2015, the Corporation announced that Blackstone Tactical Opportunities, an affiliate of Blackstone, acquired a meaningful equity position in Stornoway and a minority ownership interest in the Subject Diamonds Interest by way of a secondary market transaction with Orion. Following this transaction, the Subject Diamonds Interest is held in a proportion of 12% by Orion and/or one or more of its designated affiliates and/or respective limited partners or investors, 4% by CDPQ and 4% by Blackstone Tactical Opportunities.

22.7.6 Impacts and Benefits Agreement

SDCI signed an impacts and benefits agreement ("IBA"), the Mecheshoo Agreement, with the Grand Council of the Crees (Eeyou Istchee), the Cree Nation of Mistissini and the Cree Regional Authority in March 2012. The Mecheshoo Agreement provides for employment and business opportunities for the Crees, and fosters cultural, environmental and social protection. The project economic analysis incorporates the principal financial provisions of the Mecheshoo Agreement, which include joint contributions to training and business development established by the Cree Nation of Mistissini, contributions to a social and cultural fund, and additional financial provisions designed to ensure the Crees' participation in the Renard Project's long-term financial success.

22.8 Financial Analysis Results

22.8.1 Project Cash Flows

Projected revenues, operating costs and cash flow summaries are presented in Table 22.3 to 22.8, in real and nominal terms.

Life of mine gross revenue from diamond sales is estimated at $5,565 million in real dollar terms. Life of mine OPEX is estimated at $1,878 million in real terms to process 33.4 Mt of ore and produce 22.3 million diamond carats. The average life of mine operating cost is $56.20/t (US$41.63/t) of ore or $84.37/carat (US$62.50/ct) produced with the average annual profile presented.

An unlevered after-Tax NPV (7%) is estimated at $974 million, and $1,349 million on a pre-tax basis, in real dollar terms. Given the advanced nature of project construction, estimates of internal rate of return and payback period are not considered meaningful.

Table 22.2: Diamond Production and Revenues (Real $)

Year	Total	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
Production Summary (k t)																							
R2/R3 tonnage mined, ore	**25,387**	225	2,000	2,160	2,160	2,160	2,160	2,160	2,160	2,160	2,131	1,982	1,641	238	911	1,139	-	-	-	-	-	-	-
R4 tonnage mined, ore	**3,458**	-	-	-	-	-	-	-	-	-	29	178	634	2,019	598	-	-	-	-	-	-	-	-
R65 tonnage mined, ore	**4,579**	-	-	180	360	360	360	360	360	360	360	360	245	263	1,010	-	-	-	-	-	-	-	-
OP ore mined	**9,064**	1,615	2,352	643	413	286	345	370	422	728	739	753	245	-	-	-	-	-	-	-	-	-	-
UG ore mined	**24,360**	5	296	1,860	2,160	2,160	2,160	2,160	2,160	2,160	2,160	2,160	2,160	2,160	598	-	-	-	-	-	-	-	-
Total ore processed	**33,424**	225	2,000	2,340	2,520	2,520	2,520	2,520	2,520	2,520	2,520	2,520	2,520	2,520	2,520	1,139	-	-	-	-	-	-	-
Diamond Production (k carats)																							
Renard 2	**17,504**	218	1,349	1,189	1,460	1,624	1,747	1,850	1,881	1,880	1,810	1,515	403	109	238	230	-	-	-	-	-	-	-
Renard 3	**1,713**	-	365	300	-	-	-	-	-	18	-	160	783	87	-	-	-	-	-	-	-	-	-
Renard 4	**1,671**	-	-	-	-	-	-	-	-	-	14	81	283	1,001	293	-	-	-	-	-	-	-	-
Renard 65	**1,376**	-	-	66	116	108	95	110	103	108	108	108	74	79	302	-	-	-	-	-	-	-	-
Total Produced (k carats)	**22,264**	218	1,714	1,555	1,576	1,732	1,842	1,961	1,984	2,006	1,931	1,865	1,542	1,275	832	230	-	-	-	-	-	-	-
Diamond Sales (k carats)																							
Renard 2	**17,504**	-	1,357	926	1,436	1,552	1,748	1,775	1,916	1,855	1,833	1,615	778	208	194	233	78	-	-	-	-	-	-
Renard 3	**1,713**	-	-	665	-	-	-	-	-	12	6	106	573	322	29	-	-	-	-	-	-	-	-
Renard 4	**1,671**	-	-	-	-	-	-	-	-	-	9	58	215	759	532	99	-	-	-	-	-	-	-
Renard 65	**1,376**	-	-	21	123	112	98	102	107	107	108	108	85	77	227	102	-	-	-	-	-	-	-
Total Sales (k carats)	**22,264**	-	1,357	1,612	1,559	1,664	1,846	1,877	2,023	1,975	1,957	1,887	1,651	1,365	982	433	78	-	-	-	-	-	-
Prices (C$/carat)																							
Avg. Renard 2	**255**	-	223	229	234	240	246	252	259	265	272	278	284	291	296	296	296	-	-	-	-	-	-
Avg. Renard 3	**210**	-	-	181	-	-	-	-	-	212	215	223	228	232	236	-	-	-	-	-	-	-	-
Renard 4	**230**	-	-	-	-	-	-	-	-	-	215	220	225	231	233	233	-	-	-	-	-	-	-
Avg. Renard 65	**255**	-	-	219	222	228	234	240	245	252	258	264	271	278	281	281	-	-	-	-	-	-	-
Average	**250**	-	223	209	233	239	245	252	258	264	270	273	256	243	257	278	296	-	-	-	-	-	-
Revenues (k C$)																							
Renard 2	**4,468,605**	-	303,170	212,108	336,529	372,633	430,182	447,848	495,449	491,823	498,040	449,421	221,261	60,622	57,515	68,971	23,034	-	-	-	-	-	-
Renard 3	**360,296**	-	-	120,608	-	-	-	-	-	2,544	1,313	23,679	130,613	74,599	6,939	-	-	-	-	-	-	-	-
Renard 4	**385,139**	-	-	-	-	-	-	-	-	-	1,934	12,837	48,370	175,056	123,927	23,015	-	-	-	-	-	-	-
Renard 65	**351,098**	-	-	4,524	27,281	25,533	22,938	24,459	26,137	26,972	27,851	28,547	23,065	21,369	63,748	28,674	-	-	-	-	-	-	-
Total Gross Revenue (before Stream)	**5,565,138**	-	303,170	337,241	363,810	398,166	453,119	472,306	521,586	521,338	529,138	514,485	423,309	331,646	252,130	120,659	23,034	-	-	-	-	-	-
Net payments to Stream Purchasers	**(800,366)**	-	(42,320)	(45,689)	(51,713)	(57,044)	(65,312)	(68,464)	(76,024)	(76,368)	(77,910)	(75,696)	(60,638)	(46,261)	(35,855)	(17,641)	(3,433)	-	-	-	-	-	-
Net Marketing Fees	**(100,337)**	(149)	(6,234)	(6,041)	(6,476)	(7,038)	(7,937)	(8,251)	(9,057)	(9,053)	(9,181)	(8,941)	(7,449)	(5,950)	(4,910)	(2,773)	(679)	-	-	-	-	-	-
Diaquem Royalty	**(108,991)**	-	(5,919)	(6,607)	(7,127)	(7,800)	(8,877)	(9,253)	(10,218)	(10,214)	(10,366)	(10,079)	(8,293)	(6,497)	(4,933)	(2,357)	(449)	-	-	-	-	-	-
Net Revenue	**4,555,444**	(149)	248,697	278,904	298,493	326,283	370,994	386,339	426,287	425,703	431,681	419,768	346,929	272,939	206,433	97,888	18,474	-	-	-	-	-	-

***Note**: Totals include actuals from 2012 to 2015*.

Table 22.3: Diamond Production and Revenues (Nominal $)

Year	Total	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
Production Summary (k t)																							
R2/R3 tonnage mined, ore	**25,387**	225	2,000	2,160	2,160	2,160	2,160	2,160	2,160	2,160	2,131	1,982	1,641	238	911	1,139	-	-	-	-	-	-	-
R4 tonnage mined, ore	**3,458**	-	-	-	-	-	-	-	-	-	29	178	634	2,019	598	-	-	-	-	-	-	-	-
R65 tonnage mined, ore	**4,579**	-	-	180	360	360	360	360	360	360	360	360	245	263	1,010	-	-	-	-	-	-	-	-
OP ore mined	**9,064**	1,615	2,352	643	413	286	345	370	422	728	739	753	245	-	-	-	-	-	-	-	-	-	-
UG ore mined	**24,360**	5	296	1,860	2,160	2,160	2,160	2,160	2,160	2,160	2,160	2,160	2,160	2,160	598	-	-	-	-	-	-	-	-
Total ore processed	**33,424**	225	2,000	2,340	2,520	2,520	2,520	2,520	2,520	2,520	2,520	2,520	2,520	2,520	2,520	1,139	-	-	-	-	-	-	-
Diamond Production (k carats)																							
Renard 2	**17,504**	218	1,349	1,189	1,460	1,624	1,747	1,850	1,881	1,880	1,810	1,515	403	109	238	230	-	-	-	-	-	-	-
Renard 3	**1,713**	-	365	300	-	-	-	-	-	18	-	160	783	87	-	-	-	-	-	-	-	-	-
Renard 4	**1,671**	-	-	-	-	-	-	-	-	-	14	81	283	1,001	293	-	-	-	-	-	-	-	-
Renard 65	**1,376**	-	-	66	116	108	95	110	103	108	108	108	74	79	302	-	-	-	-	-	-	-	-
Total Produced (k carats)	**22,264**	218	1,714	1,555	1,576	1,732	1,842	1,961	1,984	2,006	1,931	1,865	1,542	1,275	832	230	-	-	-	-	-	-	-
Diamond Sales (k carats)																							
Renard 2	**17,504**	-	1,357	926	1,436	1,552	1,748	1,775	1,916	1,855	1,833	1,615	778	208	194	233	78	-	-	-	-	-	-
Renard 3	**1,713**	-	-	665	-	-	-	-	-	12	6	106	573	322	29	-	-	-	-	-	-	-	-
Renard 4	**1,671**	-	-	-	-	-	-	-	-	-	9	58	215	759	532	99	-	-	-	-	-	-	-
Renard 65	**1,376**	-	-	21	123	112	98	102	107	107	108	108	85	77	227	102	-	-	-	-	-	-	-
Total Sales (k carats)	**22,264**	-	1,357	1,612	1,559	1,664	1,846	1,877	2,023	1,975	1,957	1,887	1,651	1,365	982	433	78	-	-	-	-	-	-
Prices (C$/carat)																							
Avg. Renard 2	**321**	-	251	263	274	286	299	313	327	342	357	373	388	405	422	430	436	-	-	-	-	-	-
Avg. Renard 3	**273**	-	-	207	-	-	-	-	-	274	281	300	313	322	334	-	-	-	-	-	-	-	-
Renard 4	**323**	-	-	-	-	-	-	-	-	-	283	295	309	323	331	336	-	-	-	-	-	-	-
Avg. Renard 65	**336**	-	-	253	259	271	284	297	310	324	339	354	370	388	401	406	-	-	-	-	-	-	-
Average	**318**	-	251	239	272	285	298	312	326	340	356	366	351	339	365	403	436	-	-	-	-	-	-
Revenues (k C$)																							
Renard 2	**5,617,850**	-	340,827	243,220	392,937	443,820	522,605	554,984	626,087	634,035	654,782	602,495	301,754	84,333	81,968	100,112	33,891	-	-	-	-	-	-
Renard 3	**466,921**	-	-	137,555	-	-	-	-	-	3,289	1,715	31,858	179,081	103,610	9,814	-	-	-	-	-	-	-	-
Renard 4	**539,919**	-	-	-	-	-	-	-	-	-	2,551	17,259	66,298	244,743	175,870	33,199	-	-	-	-	-	-	-
Renard 65	**461,947**	-	-	5,218	31,842	30,402	27,856	30,311	33,024	34,770	36,619	38,285	31,525	29,821	90,913	41,362	-	-	-	-	-	-	-
Total Gross Revenue (before Stream)	**7,086,637**	-	340,827	385,993	424,779	474,222	550,461	585,295	659,111	672,094	695,667	689,896	578,658	462,506	358,564	174,673	33,891	-	-	-	-	-	-
Net payments to Stream Purchasers	**(1,104,666)**	-	(49,851)	(55,439)	(63,907)	(72,256)	(84,780)	(91,061)	(103,529)	(106,520)	(111,215)	(110,779)	(91,707)	(72,433)	(57,141)	(28,443)	(5,604)	-	-	-	-	-	-
Net Marketing Fees	**(124,778)**	(149)	(6,792)	(6,839)	(7,473)	(8,282)	(9,530)	(10,099)	(11,307)	(11,519)	(11,905)	(11,811)	(9,991)	(8,091)	(6,390)	(3,498)	(883)	-	-	-	-	-	-
Diaquem Royalty	**(138,810)**	-	(6,659)	(7,562)	(8,322)	(9,290)	(10,784)	(11,467)	(12,913)	(13,167)	(13,629)	(13,516)	(11,336)	(9,061)	(7,025)	(3,419)	(661)	-	-	-	-	-	-
Net Revenue	**5,718,384**	(149)	277,525	316,153	345,077	384,394	445,367	472,668	531,362	540,888	558,918	553,791	465,623	372,922	288,008	139,313	26,743	-	-	-	-	-	-

Note: *Totals include actuals from 2012 to 2015*.

Table 22.4: Operating Costs (real $)

Year	Total	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
Operating Costs (k C$)																							
Open Pit																							
Labour	**64,751**	7,934	7,244	4,886	3,921	3,961	3,934	3,675	3,661	3,450	3,422	3,362	2,520	1,750	2,385	1,220	-	-	-	-	-	-	-
Fuel	**20,868**	2,953	2,740	1,511	1,182	1,245	1,227	1,240	1,170	1,205	1,174	1,111	813	497	788	423	-	-	-	-	-	-	-
Explosives and accessories	**8,038**	1,843	1,556	681	346	374	429	431	418	464	403	313	102	-	-	-	-	-	-	-	-	-	-
Equipment	**26,099**	4,155	3,794	1,992	1,510	1,549	1,564	1,581	1,508	1,537	1,471	1,343	799	476	897	470	-	-	-	-	-	-	-
Power	**1,982**	176	258	174	93	94	94	94	95	99	100	100	94	94	94	-	-	-	-	-	-	-	-
Other	**4,655**	1,166	913	383	238	209	288	289	253	102	134	214	133	156	100	10	-	-	-	-	-	-	-
Transfer to capital	**(29,760)**	(18,227)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total Open Pit	**96,633**	-	16,506	9,627	7,290	7,432	7,537	7,311	7,106	6,858	6,703	6,443	4,461	2,973	4,263	2,124	-	-	-	-	-	-	-
Underground																							
Labour	**263,780**	-	9,729	16,504	18,621	21,321	21,433	20,046	25,504	23,310	23,998	25,632	20,671	19,459	8,457	4,179	-	-	-	-	-	-	-
Fuel	**57,599**	-	977	3,430	4,055	4,569	5,050	4,970	5,807	5,867	6,177	5,875	4,600	4,289	1,768	166	-	-	-	-	-	-	-
Supplies	**21,368**	-	1,014	1,788	2,065	1,997	1,776	1,764	2,009	1,739	1,604	1,932	1,616	1,614	451	-	-	-	-	-	-	-	-
Equipment	**143,175**	-	2,814	8,715	10,399	11,555	12,527	12,373	14,613	14,475	15,212	14,758	11,238	10,814	3,471	212	-	-	-	-	-	-	-
Power	**59,238**	-	1,477	2,809	3,071	4,632	4,423	3,893	5,991	6,336	6,904	7,598	4,035	4,035	4,035	-	-	-	-	-	-	-	-
Heating	**-**	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other	**72,744**	-	5,549	5,924	6,079	6,786	6,050	5,296	7,160	5,550	5,375	6,994	4,561	3,688	2,807	890	-	-	-	-	-	-	-
Transfer to capital	**(4,953)**	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total UG	**612,951**	-	21,559	39,170	44,290	50,860	51,259	48,342	61,084	57,277	59,269	62,787	46,720	43,898	20,989	5,448	-	-	-	-	-	-	-
Concentrator																							
Labour	**188,863**	9,638	12,934	13,396	13,078	13,058	13,117	13,104	13,114	13,210	13,195	13,149	13,221	13,168	13,192	6,252	-	-	-	-	-	-	-
Fuel	**20,744**	315	1,312	1,500	1,474	1,517	1,588	1,575	1,579	1,564	1,587	1,620	1,620	1,630	1,571	291	-	-	-	-	-	-	-
Consumables and reagents	**44,418**	377	3,045	3,280	3,280	3,280	3,280	3,280	3,280	3,280	3,280	3,280	3,280	3,280	3,280	1,640	-	-	-	-	-	-	-
Power	**195,393**	1,808	12,119	13,598	14,290	14,355	14,355	14,355	14,599	15,119	15,253	15,253	14,353	14,353	14,353	7,231	-	-	-	-	-	-	-
Maintenance	**89,438**	548	5,395	6,550	6,617	6,605	6,641	6,632	6,649	6,674	6,682	6,676	6,664	6,663	6,661	2,948	-	-	-	-	-	-	-
Heating	**11,980**	259	862	862	862	862	862	862	862	862	862	862	862	862	862	517	-	-	-	-	-	-	-
Other	**34,927**	1,496	1,404	1,833	2,624	2,624	2,624	2,624	2,624	2,624	2,624	2,624	2,624	2,624	2,624	1,312	-	-	-	-	-	-	-
Transfer to capital	**(17,323)**	(14,439)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total Concentrator	**568,440**	-	37,072	41,018	42,225	42,301	42,467	42,432	42,706	43,332	43,484	43,465	42,624	42,579	42,542	20,192	-	-	-	-	-	-	-
G&A																							
Labour	**236,559**	21,410	15,261	15,069	15,069	15,069	15,069	15,069	15,069	15,069	15,069	15,069	15,069	13,545	13,545	8,199	-	-	-	-	-	-	-
Fuel	**29,967**	1,016	1,971	1,971	1,971	1,971	1,971	1,971	1,971	1,971	1,971	1,971	1,971	1,971	1,971	548	-	-	-	-	-	-	-
Maintenance	**24,537**	653	1,640	1,678	1,703	1,683	1,685	1,685	1,690	1,700	1,699	1,692	1,682	1,678	1,677	947	-	-	-	-	-	-	-
Accommodation	**128,048**	9,551	7,510	7,510	7,510	7,510	7,510	7,510	7,510	7,510	7,510	7,510	7,510	7,510	7,510	3,738	-	-	-	-	-	-	-
Transportation	**54,961**	4,064	3,318	3,318	3,318	3,318	3,318	3,318	3,318	3,318	3,318	3,318	3,318	3,076	3,066	1,825	-	-	-	-	-	-	-
Power	**62,137**	4,479	4,349	4,323	4,347	4,365	4,352	4,520	4,423	4,587	4,615	4,615	4,349	4,349	4,349	114	-	-	-	-	-	-	-
Heating	**-**	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other	**180,891**	12,971	11,142	11,139	11,139	11,129	11,129	11,129	10,823	10,823	10,823	10,452	10,452	10,237	10,237	6,001	-	-	-	-	-	-	-
Route 167 maintenance	**-**	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Transfer to capital	**(116,693)**	(54,144)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total G&A	**600,407**	-	45,192	45,009	45,058	45,046	45,035	45,204	44,806	44,979	45,005	44,627	44,352	42,367	42,356	21,371	-	-	-	-	-	-	-
Cash Operating Costs	**1,878,432**	-	120,329	134,824	138,862	145,639	146,297	143,288	155,701	152,447	154,462	157,323	138,158	131,818	110,149	49,135	-	-	-	-	-	-	-
Operating Margin	**2,677,012**	(149)	128,368	144,080	159,631	180,644	224,697	243,051	270,586	273,256	277,219	262,445	208,772	141,121	96,284	48,753	18,474	-	-	-	-	-	-
Unit Operating Costs																							
Unit Cost ($/tonne processed)																							
Open pit	**2.89**	-	8.25	4.11	2.89	2.95	2.99	2.90	2.82	2.72	2.66	2.56	1.77	1.18	1.69	1.86	-	-	-	-	-	-	-
Underground	**18.34**	-	10.78	16.74	17.58	20.18	20.34	19.18	24.24	22.73	23.52	24.92	18.54	17.42	8.33	4.78	-	-	-	-	-	-	-
Concentrator	**17.01**	-	18.54	17.53	16.76	16.79	16.85	16.84	16.95	17.20	17.26	17.25	16.91	16.90	16.88	17.73	-	-	-	-	-	-	-
G&A	**17.96**	-	22.60	19.23	17.88	17.88	17.87	17.94	17.78	17.85	17.86	17.71	17.60	16.81	16.81	18.77	-	-	-	-	-	-	-
Total Unit Cost ($/t ore)	**56.20**	-	60.17	57.62	55.10	57.79	58.05	56.86	61.79	60.49	61.29	62.43	54.82	52.31	43.71	43.15	-	-	-	-	-	-	-
Unit Cost ($/carat)																							
Open pit	**4.34**	-	9.63	6.19	4.62	4.29	4.09	3.73	3.58	3.42	3.47	3.45	2.89	2.33	5.12	9.22	-	-	-	-	-	-	-
Underground	**27.53**	-	12.58	25.19	28.10	29.36	27.83	24.65	30.79	28.55	30.69	33.67	30.31	34.42	25.22	23.65	-	-	-	-	-	-	-
Concentrator	**25.53**	-	21.63	26.38	26.79	24.42	23.06	21.64	21.52	21.60	22.51	23.31	27.65	33.38	51.12	87.66	-	-	-	-	-	-	-
G&A	**26.97**	-	26.37	28.95	28.59	26.01	24.45	23.05	22.58	22.42	23.30	23.93	28.77	33.22	50.90	92.78	-	-	-	-	-	-	-
Total Unit Cost ($/carat produced)	**84.37**	-	70.20	86.71	88.10	84.08	79.43	73.08	78.47	76.00	79.97	84.36	89.62	103.35	132.36	213.30	-	-	-	-	-	-	-

***Note**: Totals include actuals from 2012 to 2015.*

Table 22.5: Operating Costs (nominal $)

Year	Total	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
Operating Costs (k C$)																							
Open Pit																							
Labour	74,340	8,295	7,726	5,315	4,350	4,483	4,542	4,328	4,397	4,227	4,276	4,284	3,276	2,321	3,225	1,683	-	-	-	-	-	-	-
Fuel	25,377	3,263	3,088	1,736	1,386	1,489	1,497	1,544	1,485	1,560	1,550	1,497	1,117	697	1,127	617	-	-	-	-	-	-	-
Explosives and accessories	9,447	2,037	1,753	782	406	446	523	536	531	601	532	422	140	-	-	-	-	-	-	-	-	-	-
Equipment	31,620	4,592	4,276	2,288	1,771	1,852	1,908	1,968	1,914	1,991	1,942	1,809	1,098	667	1,282	686	-	-	-	-	-	-	-
Power	2,400	196	291	200	110	112	114	117	121	128	132	134	129	132	134	-	-	-	-	-	-	-	-
Other	5,547	1,289	1,028	439	279	250	352	360	321	133	177	288	183	219	142	15	-	-	-	-	-	-	-
Transfer to capital	(31,745)	(19,671)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total Open Pit	116,985	-	18,163	10,759	8,302	8,633	8,937	8,852	8,769	8,639	8,609	8,436	5,943	4,034	5,910	3,001	-	-	-	-	-	-	-
Underground																							
Labour	318,212	-	10,375	17,953	20,661	24,131	24,742	23,603	30,630	28,556	29,986	32,669	26,872	25,803	11,438	5,766	-	-	-	-	-	-	-
Fuel	73,760	-	1,105	3,947	4,755	5,466	6,161	6,187	7,372	7,596	8,157	7,913	6,320	6,011	2,528	241	-	-	-	-	-	-	-
Supplies	27,024	-	1,146	2,058	2,420	2,390	2,165	2,197	2,550	2,251	2,119	2,602	2,220	2,262	644	-	-	-	-	-	-	-	-
Equipment	183,041	-	3,182	10,029	12,195	13,824	15,280	15,402	18,553	18,741	20,089	19,878	15,441	15,155	4,962	310	-	-	-	-	-	-	-
Power	76,413	-	1,672	3,230	3,602	5,545	5,393	4,847	7,603	8,204	9,117	10,234	5,543	5,654	5,767	-	-	-	-	-	-	-	-
Heating	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other	91,871	-	6,266	6,814	7,127	8,118	7,375	6,597	9,088	7,186	7,098	9,420	6,266	5,168	4,012	1,298	-	-	-	-	-	-	-
Transfer to capital	(5,064)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total UG	765,257	-	23,745	44,031	50,760	59,474	61,117	58,833	75,796	72,533	76,566	82,717	62,663	60,053	29,352	7,615	-	-	-	-	-	-	-
Concentrator																							
Labour	226,685	10,077	13,794	14,572	14,511	14,779	15,142	15,430	15,750	16,183	16,488	16,759	17,188	17,461	17,842	8,626	-	-	-	-	-	-	-
Fuel	26,481	350	1,480	1,724	1,729	1,815	1,937	1,960	2,004	2,025	2,096	2,182	2,226	2,284	2,245	423	-	-	-	-	-	-	-
Consumables and reagents	56,804	420	3,435	3,771	3,846	3,923	4,001	4,081	4,163	4,246	4,331	4,418	4,506	4,596	4,688	2,379	-	-	-	-	-	-	-
Power	250,254	2,013	13,667	15,636	16,757	17,170	17,513	17,863	18,530	19,574	20,144	20,546	19,720	20,114	20,516	10,491	-	-	-	-	-	-	-
Maintenance	114,337	610	6,090	7,531	7,760	7,901	8,102	8,253	8,439	8,641	8,824	8,993	9,156	9,338	9,521	4,278	-	-	-	-	-	-	-
Heating	15,289	288	971	990	1,010	1,030	1,051	1,072	1,093	1,115	1,137	1,160	1,183	1,207	1,231	750	-	-	-	-	-	-	-
Other	44,721	1,662	1,584	2,108	3,077	3,139	3,202	3,266	3,331	3,398	3,465	3,535	3,605	3,678	3,751	1,904	-	-	-	-	-	-	-
Transfer to capital	(18,420)	(15,419)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total Concentrator	716,152	-	41,021	46,332	48,689	49,755	50,948	51,925	53,311	55,181	56,486	57,593	57,585	58,678	59,796	28,850	-	-	-	-	-	-	-
G&A																							
Labour	279,951	22,386	16,275	16,392	16,720	17,054	17,395	17,743	18,098	18,460	18,829	19,206	19,590	17,962	18,321	11,311	-	-	-	-	-	-	-
Fuel	37,537	1,123	2,222	2,266	2,311	2,358	2,405	2,453	2,502	2,552	2,603	2,655	2,708	2,762	2,818	799	-	-	-	-	-	-	-
Maintenance	31,088	722	1,849	1,929	1,997	2,013	2,056	2,097	2,145	2,201	2,244	2,279	2,311	2,351	2,397	1,379	-	-	-	-	-	-	-
Accommodation	158,741	10,537	8,465	8,635	8,807	8,984	9,163	9,347	9,533	9,724	9,919	10,117	10,319	10,526	10,736	5,428	-	-	-	-	-	-	-
Transportation	68,291	4,481	3,740	3,815	3,891	3,969	4,048	4,129	4,212	4,296	4,382	4,469	4,559	4,309	4,380	2,651	-	-	-	-	-	-	-
Power	78,390	4,951	4,902	4,970	5,098	5,221	5,309	5,624	5,614	5,939	6,094	6,216	5,975	6,095	6,216	167	-	-	-	-	-	-	-
Heating	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other	224,717	14,314	12,559	12,807	13,063	13,312	13,578	13,850	13,738	14,013	14,293	14,080	14,361	14,347	14,634	8,779	-	-	-	-	-	-	-
Route 167 maintenance	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Transfer to capital	(125,296)	(58,514)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sub-Total G&A	753,419	-	50,011	50,813	51,887	52,909	53,954	55,243	55,843	57,185	58,363	59,022	59,823	58,351	59,502	30,512	-	-	-	-	-	-	-
Cash Operating Costs	2,351,813	-	132,940	151,935	159,638	170,771	174,957	174,853	193,719	193,538	200,024	207,768	186,015	181,117	154,560	69,978	-	-	-	-	-	-	-
Operating Margin	3,366,571	(149)	144,585	164,218	185,439	213,622	270,411	297,815	337,643	347,350	358,894	346,023	279,608	191,805	133,448	69,335	26,743	-	-	-	-	-	-
Unit Operating Costs																							
Unit Cost ($/tonne processed)																							
Open pit	3.50	-	9.08	4.60	3.29	3.43	3.55	3.51	3.48	3.43	3.42	3.35	2.36	1.60	2.35	2.63	-	-	-	-	-	-	-
Underground	22.90	-	11.87	18.82	20.14	23.60	24.25	23.35	30.08	28.78	30.38	32.82	24.87	23.83	11.65	6.69	-	-	-	-	-	-	-
Concentrator	21.43	-	20.51	19.80	19.32	19.74	20.22	20.61	21.16	21.90	22.42	22.85	22.85	23.28	23.73	25.33	-	-	-	-	-	-	-
G&A	22.54	-	25.01	21.71	20.59	21.00	21.41	21.92	22.16	22.69	23.16	23.42	23.74	23.16	23.61	26.79	-	-	-	-	-	-	-
Total Unit Cost ($/t ore)	70.36	-	66.48	64.93	63.35	67.77	69.43	69.39	76.87	76.80	79.37	82.45	73.82	71.87	61.33	61.45	-	-	-	-	-	-	-
Unit Cost ($/carat)																							
Open pit	5.25	-	10.60	6.92	5.27	4.98	4.85	4.51	4.42	4.31	4.46	4.52	3.86	3.16	7.10	13.03	-	-	-	-	-	-	-
Underground	34.37	-	13.85	28.32	32.21	34.34	33.18	30.00	38.20	36.16	39.64	44.35	40.65	47.08	35.27	33.06	-	-	-	-	-	-	-
Concentrator	32.17	-	23.93	29.80	30.89	28.73	27.66	26.48	26.87	27.51	29.25	30.88	37.35	46.00	71.85	125.24	-	-	-	-	-	-	-
G&A	33.84	-	29.18	32.68	32.92	30.55	29.29	28.17	28.14	28.51	30.22	31.65	38.80	45.75	71.50	132.46	-	-	-	-	-	-	-
Total Unit Cost ($/carat produced)	105.63	-	77.56	97.72	101.28	98.59	94.99	89.17	97.63	96.49	103.56	111.40	120.66	142.00	185.73	303.79	-	-	-	-	-	-	-

Note: *Totals include actuals from 2012 to 2015*.

Table 22.6: Cash Flow Summary (real $)

Year	Total	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
Summary (k C$)																							
Net Revenue	**4,555,444**	(149)	248,697	278,904	298,493	326,283	370,994	386,339	426,287	425,703	431,681	419,768	346,929	272,939	206,433	97,888	18,474	-	-	-	-	-	-
Cash Operating Costs	**(1,878,432)**	-	(120,329)	(134,824)	(138,862)	(145,639)	(146,297)	(143,288)	(155,701)	(152,447)	(154,462)	(157,323)	(138,158)	(131,818)	(110,149)	(49,135)	-	-	-	-	-	-	-
Operating Margin	**2,677,012**	(149)	128,368	144,080	159,631	180,644	224,697	243,051	270,586	273,256	277,219	262,445	208,772	141,121	96,284	48,753	18,474	-	-	-	-	-	-
Other costs	**(88,598)**	(200)	(1,447)	(2,581)	(2,530)	(2,475)	(2,418)	(10,853)	(11,799)	(12,468)	(12,530)	(12,259)	(10,380)	(6,857)	(4,892)	(3,533)	(1,582)	(768)	(467)	(461)	(350)	(65)	-
Working Capital	**12,392**	5,960	(50,029)	(5,782)	(1,062)	(1,757)	2,545	5,702	450	(422)	(346)	(548)	6,585	151	7,887	26,751	17,551	(907)	1	1	(18)	(256)	(65)
Fixed Capital	**(844,798)**	(282,495)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sustaining Capital	**(268,545)**	(250)	(50,099)	(48,812)	(31,987)	(23,639)	(15,445)	(36,187)	(9,172)	(22,695)	(20,658)	(6,664)	(891)	(1,083)	(129)	(832)	-	-	-	-	-	-	-
Closure Costs	**(1,035)**	(800)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	5,652	8,478	-	-	(7,482)	(4,582)	-
Pre-Tax Cash Flow	**1,486,429**	(277,933)	26,794	86,904	124,052	152,773	209,379	201,712	250,064	237,672	243,685	242,974	204,087	133,331	99,149	71,139	40,095	6,803	(466)	(460)	(7,850)	(4,903)	(65)
Total Taxes	**(697,989)**	1,300	9,780	(7,862)	(17,677)	(22,500)	(59,863)	(95,299)	(76,761)	(83,913)	(90,016)	(88,779)	(71,263)	(46,632)	(31,089)	(12,551)	(4,863)	-	-	-	-	-	-
After-Tax Cash Flow	**788,440**	(276,633)	36,574	79,042	106,375	130,272	149,515	106,413	173,303	153,759	153,668	154,195	132,824	86,699	68,060	58,588	35,232	6,803	(466)	(460)	(7,850)	(4,903)	(65)
Stream Disbursements	**328,860**	121,500	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Opening Cash Balance	**141,200**	141,200																					
Net After-Tax Cash Flow	**1,258,500**	(13,933)	36,574	79,042	106,375	130,272	149,515	106,413	173,303	153,759	153,668	154,195	132,824	86,699	68,060	58,588	35,232	6,803	(466)	(460)	(7,850)	(4,903)	(65)

***Note**: Totals include actuals from 2012 to 2015.*

Table 22.7: Cash Flow Summary (nominal $)

Year	Total	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037
Summary (k C$)																							
Net Revenue	**5,718,384**	(149)	277,525	316,153	345,077	384,394	445,367	472,668	531,362	540,888	558,918	553,791	465,623	372,922	288,008	139,313	26,743	-	-	-	-	-	-
Cash Operating Costs	**(2,351,813)**	-	(132,940)	(151,935)	(159,638)	(170,771)	(174,957)	(174,853)	(193,719)	(193,538)	(200,024)	(207,768)	(186,015)	(181,117)	(154,560)	(69,978)	-	-	-	-	-	-	-
Operating Margin	**3,366,571**	(149)	144,585	164,218	185,439	213,622	270,411	297,815	337,643	347,350	358,894	346,023	279,608	191,805	133,448	69,335	26,743	-	-	-	-	-	-
Other costs	**(116,552)**	(200)	(1,447)	(2,581)	(2,530)	(2,475)	(8,666)	(12,683)	(14,323)	(15,157)	(15,503)	(15,425)	(13,156)	(8,991)	(6,589)	(4,822)	(2,011)	(925)	(467)	(461)	(350)	(106)	-
Working Capital	**12,608**	5,849	(56,076)	(7,820)	(2,650)	(3,428)	7,760	(841)	(778)	(2,001)	(1,906)	(2,195)	7,424	(1,134)	9,555	37,378	25,047	(1,238)	1	1	(18)	(215)	(106)
Fixed Capital	**(844,798)**	(282,495)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sustaining Capital	**(323,595)**	(276)	(56,382)	(56,002)	(37,528)	(28,152)	(18,877)	(44,964)	(11,667)	(29,381)	(27,263)	(8,977)	(1,221)	(1,518)	(184)	(1,202)	-	-	-	-	-	-	-
Closure Costs	**(1,318)**	(886)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	8,482	12,978	-	-	(12,045)	(7,547)	-
Pre-Tax Cash Flow	**2,092,915**	(278,158)	30,681	97,814	142,731	179,567	250,627	239,326	310,874	300,810	314,222	319,427	272,654	180,161	136,231	100,689	58,261	10,815	(466)	(460)	(12,412)	(7,868)	(106)
Total Taxes	**(944,275)**	1,300	7,350	(14,233)	(25,901)	(33,986)	(104,500)	(79,465)	(103,495)	(112,680)	(121,226)	(120,758)	(98,324)	(65,512)	(44,790)	(20,047)	(8,009)	-	-	-	-	-	-
After-Tax Cash Flow	**1,148,640**	(276,858)	38,031	83,581	116,830	145,581	146,127	159,861	207,379	188,130	192,996	198,669	174,331	114,649	91,441	80,642	50,252	10,815	(466)	(460)	(12,412)	(7,868)	(106)
Stream Disbursements	**328,860**	121,500	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Opening Cash Balance	**141,200**	141,200																					
Net After-Tax Cash Flow	**1,618,700**	(14,158)	38,031	83,581	116,830	145,581	146,127	159,861	207,379	188,130	192,996	198,669	174,331	114,649	91,441	80,642	50,252	10,815	(466)	(460)	(12,412)	(7,868)	(106)

***Note**: Opening cash balance for 2016 of $141.2 million is net of payables of $45.3 million as at December 2015. Total include actuals from 2012 to 2015.*

22.8.2 Sensitivities

Valuation sensitivities based on revenue, OPEX and CAPEX are illustrated in Figures 22.1 and 22.2. Additional valuation sensitivities in Tables 22.9 to 22.13 are based on alternate diamond price scenarios, diamond price escalation factors, C$/US$ exchange, energy assumptions and recovered diamond grades. The sensitivities presented below are based on real dollar term cash flow projections.

Project NPV is most sensitive to revenue, specifically the diamond price assumption, followed by operating costs and finally by diamond ore grade and exchange rate.

Figure 22.1: After-Tax NPV 7% Sensitivities

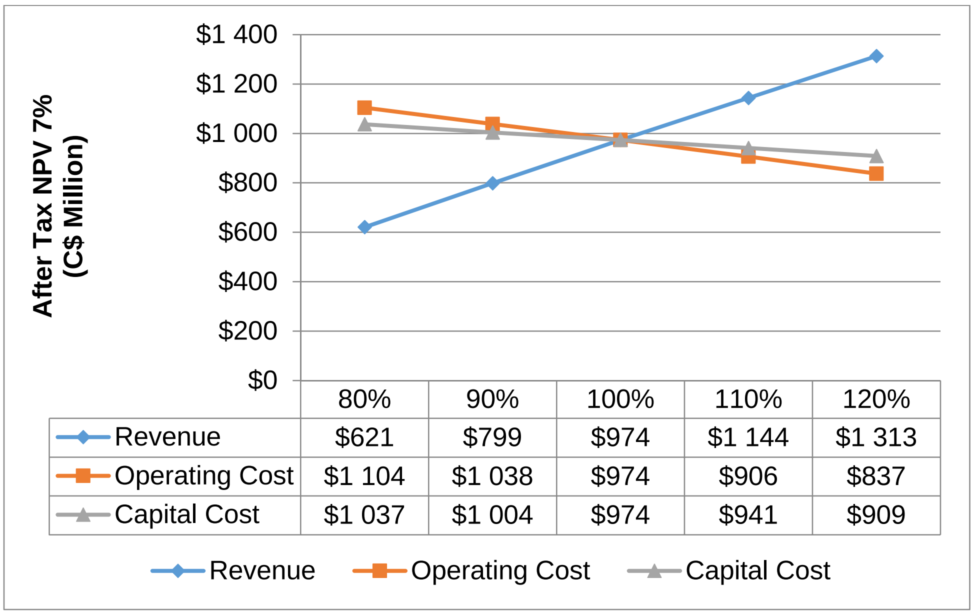


Figure 22.2: Pre-Tax NPV 7% Sensitivities

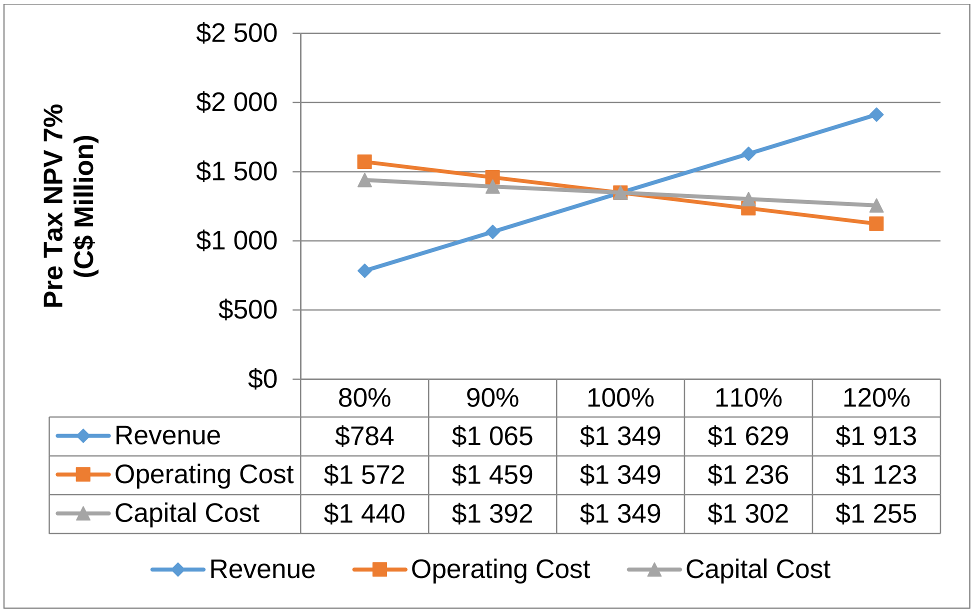


Table 22.8: Diamond Price Scenarios

Real C$ 000, After Tax

Diamond Price Scenarios[2]	0.0%	3.0%	5.0%	7.0%	8.0%	10.0%
WWW High	$1,902	$1,526	$1,329	$1,166	$1,095	$970
WWW Base	$1,604	$1,281	$1,113	$974	$913	$806
WWW Low	$1,324	$1,058	$919	$803	$753	$664
March 2014 No Escalation[3]	$1,672	$1,357	$1,191	$1,052	$991	$884

Real C$ 000, Pre Tax

Diamond Price Scenarios[2]	0.0%	3.0%	5.0%	7.0%	8.0%	10.0%
WWW High	$2,817	$2,226	$1,919	$1,666	$1,557	$1,364
WWW Base	$2,302	$1,811	$1,558	$1,349	$1,258	$1,099
WWW Low	$1,823	$1,436	$1,236	$1,070	$997	$871
March 2014 No Escalation[3]	$2,410	$1,928	$1,677	$1,468	$1,376	$1,216

Notes: *1) Grey shading presents base case assumption*

2) Represents “High” and “Minimum” sensitivities based on alternate interpretations of diamond quality and potential value when estimating a ‘Base” diamond price model. High, Base and Minimum models are from the March 2014 WWW diamond pricing exercise, adjusted by -19% by Stornoway applying the world average rough price movement between March 2014 and March 2016 (as published by Roughprices.com which is based, in large part, on WWW market assortments) to the March 2014 diamond price models (Table 19-3). The choice of “Minimum” and “High” to describe the sensitivity limits is deliberate: in the view of WWW International Diamond Consultants it is highly unlikely that an actual diamond price achieved for each kimberlite ore body upon production would fall below the “Minimum” sensitivity, but it is possible that the actual diamond price achieved may be higher than the “High” sensitivity, which is not a maximum price.

3) Represents the March 2014 WWW base case diamond price models, with no diamond price escalation.

Table 22.9: Diamond Price Escalation Rate Scenarios

Real C$ 000, After Tax

Diamond Price Escalation Rate	0.0%	3.0%	5.0%	7.0%	8.0%	10.0%
0.00%	$1,207	$974	$851	$749	$704	$625
1.25%	$1,398	$1,122	$978	$858	$805	$713
2.50%	$1,604	$1,281	$1,113	$974	$913	$806
3.75%	$1,825	$1,452	$1,258	$1,098	$1,028	$905
5.00%	$2,070	$1,641	$1,418	$1,234	$1,154	$1,014

Real C$ 000, Pre Tax

Diamond Price Escalation Rate	0.0%	3.0%	5.0%	7.0%	8.0%	10.0%
0.00%	$1,621	$1,290	$1,117	$974	$911	$802
1.25%	$1,944	$1,538	$1,327	$1,153	$1,077	$944
2.50%	$2,302	$1,811	$1,558	$1,349	$1,258	$1,099
3.75%	$2,691	$2,108	$1,807	$1,560	$1,453	$1,266
5.00%	$3,124	$2,437	$2,084	$1,794	$1,668	$1,450

Note 1) Grey shading presents base case assumption

Table 22.10: C$/US$ Exchange Rate Scenarios

Real C$ 000, After Tax

Exchange Rate $1.35 C$/US$		0.0%	3.0%	5.0%	7.0%	8.0%	10.0%
$1.49	90%	$1,889	$1,518	$1,324	$1,162	$1,092	$969
$1.42	95%	$1,738	$1,393	$1,213	$1,063	$998	$883
$1.35	100%	$1,604	$1,281	$1,113	$974	$913	$806
$1.28	105%	$1,480	$1,179	$1,022	$891	$835	$735
$1.22	110%	$1,366	$1,084	$937	$815	$762	$669

Real C$ 000, Pre Tax

Exchange Rate $1.35 C$/US$		0.0%	3.0%	5.0%	7.0%	8.0%	10.0%
$1.49	90%	$2,802	$2,216	$1,912	$1,661	$1,552	$1,361
$1.42	95%	$2,538	$2,002	$1,725	$1,496	$1,396	$1,223
$1.35	100%	$2,302	$1,811	$1,558	$1,349	$1,258	$1,099
$1.28	105%	$2,085	$1,636	$1,404	$1,213	$1,130	$986
$1.22	110%	$1,888	$1,477	$1,265	$1,090	$1,015	$882

Note: 1) Grey shading presents base case assumption

Table 22.11: Energy Scenarios

Real C$ 000, After Tax

Energy	0.0%	3.0%	5.0%	7.0%	8.0%	10.0%
80%	$1,647	$1,318	$1,146	$1,004	$942	$833
90%	$1,625	$1,300	$1,130	$989	$927	$819
100%	$1,604	$1,281	$1,113	$974	$913	$806
110%	$1,580	$1,261	$1,095	$957	$897	$792
120%	$1,557	$1,242	$1,077	$941	$882	$777

Real C$ 000, Pre Tax

Energy	0.0%	3.0%	5.0%	7.0%	8.0%	10.0%
80%	$2,383	$1,877	$1,616	$1,400	$1,306	$1,143
90%	$2,342	$1,844	$1,587	$1,374	$1,282	$1,121
100%	$2,302	$1,811	$1,558	$1,349	$1,258	$1,099
110%	$2,258	$1,776	$1,527	$1,321	$1,232	$1,076
120%	$2,216	$1,742	$1,496	$1,294	$1,206	$1,053

Note: 1) Grey shading presents base case assumption

Table 22.12: Diamond Grade Scenarios

Real C$ 000, After Tax

Ore Grade	0.0%	3.0%	5.0%	7.0%	8.0%	10.0%
80%	$1,119	$880	$756	$654	$609	$532
90%	$1,363	$1,082	$937	$816	$763	$671
100%	$1,604	$1,281	$1,113	$974	$913	$806
110%	$1,840	$1,476	$1,286	$1,127	$1,058	$937
120%	$2,079	$1,671	$1,458	$1,281	$1,204	$1,067

Real C$ 000, Pre Tax

Ore Grade	0.0%	3.0%	5.0%	7.0%	8.0%	10.0%
80%	$1,473	$1,145	$975	$836	$776	$671
90%	$1,886	$1,477	$1,265	$1,091	$1,016	$885
100%	$2,302	$1,811	$1,558	$1,349	$1,258	$1,099
110%	$2,713	$2,142	$1,846	$1,602	$1,496	$1,311
120%	$3,128	$2,477	$2,139	$1,859	$1,738	$1,526

Note 1) Grey shading presents base case assumption

23. ADJACENT PROPERTIES

23.1 Diamond Properties

There are no advanced stage diamond exploration properties in proximity to the Renard Project.

In 2004, Majescor Resources Inc. (Majescor) discovered two occurrences of diamond-bearing kimberlite float to the west of the Foxtrot Property claims. A sample of the float processed in 2004 indicated it was diamondiferous with a total of 32 diamonds greater than 0.075 mm recovered from 136 kg of kimberlite (Majescor, 2004). In 2005, Majescor drilled up-ice of the float and discovered the Remick dyke. Additional drilling delineated the dyke for a strike length of 900 m and the kimberlite dyke remains open to the northwest and down dip (Majescor, 2006).

During 2006, Forest Gate Resources, Majescor's joint venture partner at the time, identified a new kimberlite boulder dispersal train located near the Remick dyke. A 54.15 kg coarse kimberlite float sample (U0341-110) collected from this train returned a total of 83 diamonds (Forest Gate Resources, 2006; Majescor Resources Inc., 2007).

Other kimberlite discoveries in the region include:

- Lac Beaver kimberlite, located approximately 90 km south of the Renard kimberlites on claims held by Ditem Explorations Inc. (O'Hara, 2004);
- H1, H2, H3 and H4 kimberlites on the Tichegami property held by Ditem Explorations Inc. located approximately 75 km south of the Renard bodies (Robertson, 2003); and
- Hotish 1, 2 and 3 kimberlite dykes, within the Hotish Property held by Dios Exploration Inc., located about 100 km south of the Renard Kimberlites (Vaaldiam Resources Inc., 2006).

Stornoway is currently conducting diamond exploration on the 100% owned Adamantin Project, situated approximately 100km south-southwest of Renard and about 25km west of Route 167.

23.2 Other Commodities

A number of other commodities have been identified in the general region of the Renard kimberlites, primarily to the south. These include (Houle, 2006):

- Gold and copper mineralization in the Eastmain River greenstone belt (e.g., the Eastmain gold deposit, held by Eastmain Resources Inc. through its wholly-owned subsidiary, Eastmain Mines Inc.);
- Porphyry copper–molybdenum mineralization in the Opatica Subprovince (e.g., the Macleod Lake/Windy 1 deposit, held by Western Troy Capital Resources Inc.); and
- Uranium in the Otish Basin (e.g., the AM-15 zone, held by Strateco Resources Inc.).

Future work at the specific mineral occurrences listed above could have implications to the Renard Project in that they could lead to cost sharing for the Renard mining road maintenance.

24. OTHER RELEVANT DATA AND INFORMATION

24.1 Project Execution Plan

The initial project execution plan utilized an EPCM contract with SLI and its subcontractors AMEC and DRA, with Stornoway personnel managing mining operations, including development of the underground mine.

In early 2015, Stornoway elected under the terms of its EPCM contract to assume responsibility for procurement and construction management duties previously under the SLI's scope of work, with engineering remaining under the management of the SLI and its contractors.

Also upon the transfer of procurement and construction management duties from SLI to Stornoway, Stornoway started to assume primary responsibility for project cost control and schedule control.

SDCI is appropriately staffed to perform with diligence this revised allocation of work.

The DMS circuits #1 and #2 within the process plant have been identified as having the longest duration for mechanical, piping, electrical and automation work. Adding the upstream work required to have the building, foundations, concrete slab, secondary steel and equipment available for the DMS results in this sub-area defining the project's overall critical path.

The sequence of construction activities in the process plant area starts with the concrete work on the foundations and ends with the electrical and instrumentation installation. This is followed by pre-operational verification (POV) and commissioning activities. These activities are expected to take place simultaneously in several sub-areas inside the process plant area. Schedule activities preceding the process plant construction such as engineering and procurement activities and infrastructure-related activities were also taken into account in the critical path analysis. A high-level project schedule is shown in Table 24.1.

As of December 31, 2015, construction progress stood at 63.3% compared to (an initial) planned progress of 59.6%.

Table 24.1: Construction Activities

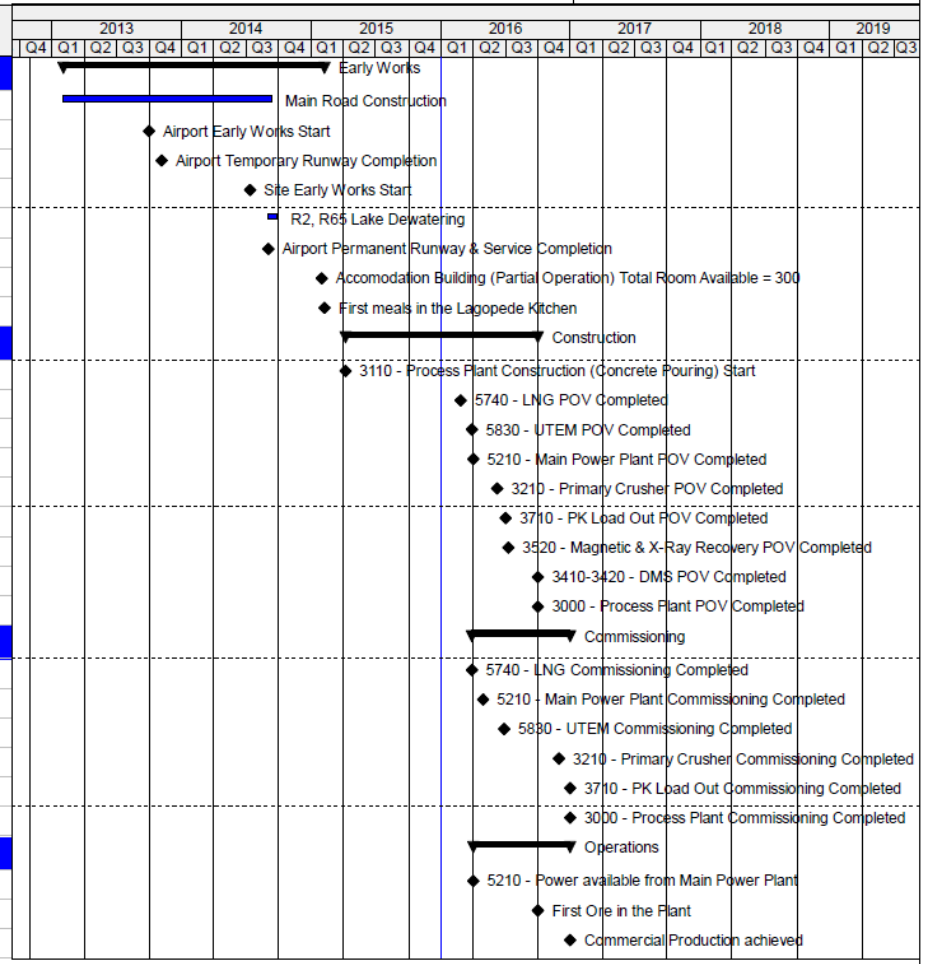
stornoway

Renard Stornoway Diamond - Project Execution Schedule

High Level Schedule

Data Date : 01-Jan-16

#	Activity Name	Start	Finish
1	**Early Works**		
2	Main Road Construction	01-Feb-13 A	12-Sep-14 A
3	Airport Early Works Start	01-Oct-13 A	
4	Airport Temporary Runway Completion		05-Nov-13 A
5	Site Early Works Start	14-Jul-14 A	
6	R2, R65 Lake Dewatering	01-Sep-14 A	25-Sep-14 A
7	Airport Permanent Runway & Service Completion		02-Sep-14 A
8	Accomodation Building (Partial Operation) Total Room Available = 3(	31-Jan-15 A	
9	First meals in the Lagopede Kitchen	08-Feb-15 A	
10	**Construction**		
11	3110 - Process Plant Construction (Concrete Pouring) Start	07-Apr-15 A	
12	5740 - LNG POV Completed		26-Feb-16
13	5830 - UTEM POV Completed		29-Mar-16
14	5210 - Main Power Plant POV Completed		31-Mar-16
15	3210 - Primary Crusher POV Completed		06-Jun-16
16	3710 - PK Load Out POV Completed		30-Jun-16
17	3520 - Magnetic & X-Ray Recovery POV Completed		09-Jul-16
18	3410-3420 - DMS POV Completed		30-Sep-16
19	3000 - Process Plant POV Completed		30-Sep-16
20	**Commissioning**		
21	5740 - LNG Commissioning Completed		29-Mar-16
22	5210 - Main Power Plant Commissioning Completed		30-Apr-16
23	5830 - UTEM Commissioning Completed		28-Jun-16
24	3210 - Primary Crusher Commissioning Completed		29-Nov-16
25	3710 - PK Load Out Commissioning Completed		31-Dec-16
26	3000 - Process Plant Commissioning Completed		31-Dec-16
27	**Operations**		
28	5210 - Power available from Main Power Plant	31-Mar-16	
29	First Ore in the Plant	01-Oct-16	
30	Commercial Production achieved		31-Dec-16*



24.2 Risk Assessment and Management

Risk identification is the process of examining the various project areas and each critical process, to identify and document any associated potential risks. All threats and opportunities, which are identified as having a potential financial impact on the Project, are evaluated and each risk is placed into the CAPEX or OPEX category. All risks are qualitatively assigned a level for consequence to CAPEX or OPEX should the risk materialize, and based on the probability of occurrence and manageability (Table 24.2).

Risks review meetings were held on a regular basis during the 2011 Feasibility Study and 2013 Optimization Study with key project members and with Stornoway, and a plan developed to identify, analyse, evaluate, communicate, mitigate, monitor and eventually treat the project risks.

Additional workshops were held in February 2016. Stornoway considered the Project's current risk profile to be consistent with those of other projects at a similar advanced stage of development.

Table 24.2: Levels of Probable Consequence, Probability of Occurrence and Manageability Matrix

Level	**Probable Consequence**	
Very High	0.65% and up	
High	0.35 to 0.65%	
Level	**Probability of Occurrence**	**Description**
High	50% to 70%	Might occur under most circumstances
Medium	30% to 50%	Might occur at some time
Level	**Manageability Matrix**	**Description**
High	60%	In most circumstances can be managed
Medium	40%	Can be managed
Low	20%	In most circumstances difficult to be managed

CAPEX and OPEX Threats

Eight (8) CAPEX threats have been identified with a "Very High" or "High" probable consequence. Five (5) OPEX threats have been identified with a "Very High" probable consequence. A summary of these risks is shown in Table 24.3.

Table 24.3: Summary of CAPEX and OPEX Threats

CAPEX Threats	Consequence	Probability	Manageability	Status
PKC permits on time	Very High	Low	Very High	Active
Loss a bridge on access road	High	Low	High	Active
Logistic Material control Loss or damage equipment	Very High	Low	High	Active
Planned electric load increase	High	Low	Very High	Active
Aggressive commissioning schedule	Very High	High	High	Active
Forest fire	High	Low	Low	Active
Re-work	Very High	Low	High	Active
Lack of LNG supply	High	Low	High	Active
Server or communication crash	Medium	High	High	Active
OPEX Threats	**Consequence**	**Probability**	**Manageability**	**Status**
Mudrush Potential	Very High	Medium	Low	Active
Ore Dilution	Very High	Medium	Medium	Active
Lack of Skilled Mining Labour	Very High	High	Medium	Active
ROM Production Gaps	Very High	Medium	Low	Active
Deep Underground Water Quality	Very High	Medium	Medium	Active

It will be very important to constantly monitor and review these risks during all phases of the Project until they are completely mitigated and retired, transferred and accepted.

Opportunities
Some of the most significant opportunities that were recognized in previous risk sessions were listed below:

- A LDR unit was designed in the 2013 Optimization Study as a project option. Prior to the commencement of project construction the LDR was added to the process plant execution strategy within the process plant CAPEX budget.
- Possible change from fiber optic cable to satellite link or microwave link or both links (ID-106): This is expected to reduce recurrent cost of telecommunication system between airstrip and the mining complex. This opportunity was implemented in 2015.
- The possibility to reduce cost by generating power with liquid natural gas (LNG) instead of diesel. This opportunity was assessed in the November 2013 LNG Feasibility Study and implemented in 2015 in the Project execution strategy.

25. INTERPRETATIONS AND CONCLUSIONS

This technical report was prepared by Stornoway Diamond Corporation for the purpose of National Instrument 43-101 to provide an updated mine plan and Mineral Reserves estimate for the Renard Diamond Project, as of December 31, 2015.

Stornoway, through its wholly-owned subsidiary SDCI, holds 100% of the Renard Project, subject to a 2% royalty in favour of DIAQUEM.

The Project is situated in the Otish Mountains region of the province of Québec, Canada, 820 km north of the city of Montreal and 360 km north-northeast of the mining town of Chibougamau. The Project is connected by land, by an all-weather gravel road, and by air, with the Renard Project airport, allowing year-round operations at the Renard mine.

The Renard Project comprises 650 claims (33,629.95ha), one mining lease BM 1021 (143.71 ha), a surface lease for the processed kimberlite area # 1303 10 000 (199.85 ha) and several other surface leases for infrastructure related to the Project (Table 4.2), all of which are expected to be sufficient for the operations of the Renard Project and are reported to be in good standing as of the date of this Technical Report.

The Project's environmental and social license is supported by the executions of the Mecheshoo Agreement and the grant of all global certificates of authorisations, approvals and permits by the applicable governmental authorities. Stornoway expects that sectorial permits, authorisations and approvals which have not yet been obtained will be obtained in due course of construction and development of the Project, in the ordinary course. Certificate of authorisations, approvals and permits obtained, or to be obtained in due course of construction and development of the Project, are expected to be sufficient to ensure that activities are conducted within the applicable regulatory framework. The Closure Plan of the Project has been approved by the applicable governmental authority and as of the date of this Technical Report, payment of financial guarantees for the Closure Plan are up to date and in compliance with applicable laws and regulations.

Knowledge of the deposit settings and lithologies, as well as the structural and alteration controls on mineralization and the mineralization style and setting, is sufficient to support Mineral Resource and Mineral Reserve estimation.

The core sampling, RC chip sampling, minibulk and bulk sampling, sample preparation and analysis, and the QA/QC and security protocols used for the Renard Project are adequate, and the data is valid and of sufficient quality to be used for Mineral Resource estimation. The results of Stornoway's QA/QC programs did not identify any significant analytical issues in any data used for the 2015 Mineral Resource Estimate. Considering the risks inherent in all kimberlite deposits, such as sampling for geological continuity, diamond grade and diamond revenue determination as discussed in Section 14, the Indicated portion of the Mineral Resources is considered suitable for the estimate of Probable Mineral Reserves.

All elements of the Project development plan, including the remaining required infrastructure, mine design, process plant design, waste disposal infrastructure and cost estimation, represent the current estimate for life of mine operations. The existing and planned infrastructure; staff availability; existing power, water, and communications facilities; the methods whereby goods will be transported to the mine; and any planned modifications or supporting studies for the Project are all well- established, or the requirements to establish such are well understood by Stornoway and can support the disclosure of Mineral Resources and Mineral Reserves.

A reasonable number of studies and investigations meeting best practices have been conducted to determine the most appropriate mining methods for Renard 2, 3, 4 and 65. A mine plan was developed to extract the Indicated Resources over a 14 year mine life in the Renard 2, Renard 3, Renard 4 and Renard 65 kimberlite pipes. The mine plan for Renard 2 and Renard 3 combines open pit and underground mining methods, with the pit depth determined through a trade-off study between open pit and underground mining costs. The Renard 65 mine plan employs open pit method only, whereas the Renard 4 mine plan employs underground method only. Since the Renard 4 kimberlite is located below Lake Lagopede, a 100-m-thick crown pillar will be maintained throughout the life of the mine.

The Probable Mineral Reserves for the Renard Project appropriately consider modifying factors, have been estimated using industry best practices, and conform to CIM definitions. The Project Probable Mineral Reserves can be summarised as follows:

Mine	Tonnes (k)	Grade (cpht)	Carats (k)	% Tonnes Total	% Carats Total	% Tonnes Method	% Carats Method
OPEN PIT							
R2	1,489	92.7	1,381	4.5%	6.2%	16.7%	34.9%
CRB2A	475	31.4	149	1.4%	0.7%	5.3%	3.8%
CRB	1,575	20.2	319	4.7%	1.4%	17.7%	8.1%
R2 Subtotal	**3,539**	**52.2**	**1,849**	**10.6%**	**8.4%**	**39.7%**	**46.7%**
R3 Subtotal	**794**	**92.3**	**733**	**2.4%**	**3.3%**	**8.9%**	**18.5%**
R65 Subtotal	**4,579**	**30.1**	**1,376**	**13.8%**	**6.2%**	**51.4%**	**34.8%**
TOTAL OP	**8,912**	**44.4**	**3,958**	**26.8%**	**17.9%**	**100%**	**100%**
UNDERGROUND							
R2-290	5,111	63.3	3,236	15.4%	14.6%	21.0%	17.8%
R2-470	4,744	84.7	4,017	14.3%	18.1%	19.5%	22.1%
R2-590	4,750	89.9	4,270	14.3%	19.3%	19.5%	23.5%
R2-710	5,073	81.4	4,132	15.2%	18.7%	20.8%	22.7%
R2 Subtotal	**19,679**	**79.6**	**15,655**	**59.1%**	**70.7%**	**80.8%**	**86.1%**
R3 Subtotal	**1,223**	**70.2**	**858**	**3.7%**	3.9%	**5.0%**	**4.7%**
R4 Subtotal	**3,458**	**48.3**	**1,671**	**10.4%**	7.5%	**14.2%**	**9.2%**
TOTAL UG	**24,360**	**74.6**	**18,184**	**73.2%**	**82.1%**	**100%**	**100%**
STOCKPILE	**153**	**73.5**	**113**				
TOTAL OP, UG & Stockpile	**33,424**	**66.5**	**22,255**	**100%**	**100%**		

Notes to accompany Probable Mineral Reserves Table

1) Probable Mineral Reserves have an effective date of December 31, 2015. The Probable Mineral Reserves were prepared under the supervision of Patrick Godin, Eng., Chief Operating Officer and Director of Stornoway Diamond Corporation. Mr. Godin is a Qualified Person within the meaning of NI 43-101.
2) Probable Mineral Reserves are reported on a 100% basis.
3) The reference point for the definition of Probable Mineral Reserves is at the point of delivery to the process plant.
4) Probable Mineral Reserves are reported at +1.0 mm diamond size (effective cut-off of 1.0 mm).
5) Probable Mineral Reserves that will be or are mined using open pit methods include Renard 2, Renard 3 and Renard 65. Mineral Reserves are estimated using the following assumptions: Renard 2 and Renard 3 open pit designs assuming external dilution of 4.3% and mining recovery of 98%; Renard 65 open pit design assuming external dilution of 3.5% and mining recovery of 98%.
6) Renard 2, Renard 3 and Renard 4 Probable Mineral Reserves are mined using underground mining methods. The Renard 2 Mineral Reserves estimate assumed an external dilution of 20% and mining recovery of 82%. The Renard 3 Mineral Reserves estimate assumed an external dilution of 14% and mining recovery of 85%. The Renard 4 Mineral Reserves estimate assumed an external dilution of 14% and mining recovery of 78%.
7) Tonnes are reported as thousand metric tonnes, diamond grades as carats per hundred tonnes, and contained diamond carats as thousands of contained carats.
8) Tables may not sum as totals have been rounded in accordance with reporting guidelines.

The process plant design capacity is 2,160,000 tpa (dry solids basis). The plant overall utilization is estimated to be 78%, equivalent to an operating time of 6,833 hours per year or 315 tph and 6,000 tpd. By mid 2018, Stornoway expects to increase the plant throughput to 323 tph, already within initial plant equipment capacity, and increase the plant availability to 83.5% by optimizing plant maintenance sequences. Taking the above into consideration, by July 2018, Stornoway expects the plant will operate at a throughput of 7,000tpd at an overall plant utilization of 83,5%.

Processed kimberlite will be disposed of through a dry stack facility currently under construction and the management of surface runoff water is facilitated through a system of peripheral drainage ditches designed to direct runoff water to an excavated catch basin for treatment, if required, before release to the environment. Appropriate metallurgical testing was completed to support the dry stacking technique. Water management will include the construction of a drainage network to collect surface water, and a catch basin that will be used to collect water for testing and treatment, if necessary, before release to the environment. Mine operations incorporate a program of progressive reclamation that minimizes costs and allows timely monitoring of performance. Waste rock generated by mining will be re-introduced to the underground as backfill.

The process plant and the mine operations have been optimized from the 2011 Feasibility Study by the inclusion of an LDR and NG fueled power plant.

FCDC entered into the Streaming Agreement for the forward sale of diamonds (the Forward Sale of Diamonds) with certain Streamers. Under the terms of the Forward Sale of Diamonds, FCDC agreed to sell to the Streamers for a fixed price a 20% undivided interest (the Subject Diamonds Interest) in each of the run of mine diamonds produced from the Project, until Streamers have been delivered the Subject Diamonds Interest in 30 million carats of diamonds, and following such time, from all run of mine diamonds derived from Renard 2, Renard 3, Renard 4, Renard 65 and Renard 9.

Stornoway intends to sell the Renard Project's diamond production in approximately 10 tender sales per year in Antwerp, Belgium. To this end, Stornoway's wholly owned subsidiary FCDC has entered into a sales and marketing agreement with the diamond industry broker and rough distributor Bonas-Couzyn, which will act as sales commissionaire and tender agent for arm's length market sales. Sales of diamonds under the Renard Streaming Agreement will be undertaken by Bonas-Couzyn, on an undivided basis, on behalf of FCDC and the Streamers.

As a result of a re-baselining exercise in November 2015, the estimate of CAPEX to complete the contruction of the Project to the date of Commencement of Commercial Production has been estimated at $775.4 million, including contingencies of $57.5 million and an escalation allowance of $25.3 million. A portion of the 2017 mine development costs and associated G&A costs that would have been incurred during the construction phase of the Project under the initial construction schedule was transferred to the 2017 sustaining capital. CAPEX is estimated at an accuracy of -13% and +17%.

The Project financial analysis incorporates the estimated LOM diamond production profile based on the updated Mineral Reserves, diamond valuation and escalation estimates, CAPEX, OPEX, Sustaining Capital Costs, salvage value, working capital, closure and reclamation costs, taxation, costs under the Mecheshoo Agreement, royalties, the Renard Streaming Agreement, and the project's financial parameter assumptions. After-tax and pre-tax net present value (NPV) evaluations have been undertaken in real dollar terms (or constant dollars) utilizing an effective date of January 1, 2016. Valuations are presented on an unlevered basis.

Life of mine gross revenue from diamond sales is estimated at $5,565 million in real dollar terms. Life of mine OPEX is estimated at $1,878 million in real terms to process 33.4 Mt of ore and produce 22.3 million diamond carats. The average life of mine operating cost is $56.20/t (US$41.63/t) of ore or $84.37/carat (US$62.50/ct) produced with the average annual profile presented.

An unlevered after-Tax NPV (7%) is estimated at $974 million, and $1,349 million on a pre-tax basis, in real dollar terms. Given the advanced nature of project construction, estimates of internal rate of return and payback period are not considered meaningful.

There is sufficient technical and scientific information and data to support the updated mine plan and Probable Mineral Reserves and initiate production in 2016.

25.1 Risks and Opportunities

This section identifies the significant internal risks, potential impacts and possible risk mitigation measures that could affect the economic outcome of the project. The list does not include the external risks that apply to all mining projects (e.g., changes in commodities prices, exchange rates, change in government regulations, etc.). Significant opportunities that could improve the economics, timing and permitting of the project are also identified. Further information and study supporting these opportunities could improve the project economics.

Risks

- ***Mineral reserve risks***

The risk to the open pit Mineral Reserves is minimal in that the open pits are small, relatively shallow and represent only 27% of the overall reserve tonnage and 18% of the reserve carats, including stockpiled material. Mining dilution remains a risk since ore/waste separation will rely on visual placement of dig limits by experienced geologists.

The main risks to the underground Mineral Reserves are higher dilution and lower recovery than planned. Ore drawdown management during production is critical to limiting dilution and maintaining high recovery. A key aspect of managing drawdown is maintaining an even drawdown over the entire blasted ore column in the stope to promote mass flow of the ore. One of the risks resulting from poor draw management is the development of rat-holes above individual drawpoints, ahead of their neighbours, right up to the waste backfill placed on top of the blasted ore. A good drawdown plan combined with strict draw procedures and regular monitoring is critical for success. Identifiable markers placed in the blasted ore, either via placement in blast holes or on drill levels, will provide good data on the effectiveness of the drawdown system.

- ***Crowns pillar risk***

The Renard 4 kimberlite pipe is located below a lake. Consequently, it is important to ensure the stability of the Renard 4 crown pillar since an uncontrolled failure could lead to water inflow into the mine during production. This would represent a significant risk to the safety of personnel working underground.

Empirical studies have been undertaken using the approach proposed by Carter (1992). This work suggested that a very thick crown pillar would be required for stability unless the spans could be reduced. Because of the importance of the stability of the crown pillars, a 3D global stress numerical modelling was undertaken, whereby Stornoway requested that the stability of a 100-m-thick crown pillar be evaluated.

The modelling that was conducted for the crown pillar in the Renard 4 pipe indicates the crown pillar is stable under base case assumptions. The models suggest that the peak of the arch should not be above the 100 m limit defined by the crown pillar. Additional work will be performed before mining R4 in order to better assess the crown pillar situation and the associated risks.

- ***Rock mechanics and Hydrogeology***

For any mining project, there are always unknown rock mechanics and hydrogeological conditions that cannot be predicted ahead of actual mining. These unknown conditions, such as faulting, zones of weak rock, or zones of unanticipated water inflow, may only be discovered during mining and may require significant changes to the mining plan. While additional laboratory testing and characterization work should always be ongoing at every stage of project development to reduce uncertainty, it is never possible to carry out enough drilling/characterization work ahead of time to identify all of these potential risks. This may in turn have significant effects on cost, as relates for example to slope angles, dewatering and/or underground development, excavation size, etc.

It must be understood that there are limitations in the ability of numerical model results to assess all potential failure modes (effect of faults, effect of rockmass variability, the effect of draw, and the stabilizing effect of leaving ore in the stope are some examples of factors beyond the explicit capabilities of the models run for this Project to date). Further work should be performed to reduce the uncertainties in the analysis and to examine other potential modes of failure. This could be done in more detailed engineering studies at a later stage of the Project, and would include the following:

Opportunities

The LDR has been designed and installed in the process plant with the expectation to recover diamonds from 30 to 45 mm. No revenues are assumed under the financial model in this regard.

26. RECOMMENDATIONS

The following section presents recommendations from the Qualified Persons.

26.1 Mine Design

It is recommended that further work be performed to reduce the uncertainties in the geomechanical and design analysis. This would include the following:

- A 3DEC model will be purpose built to test various sizes of panel backs. This model will incorporate jointing orientation and strength data collected from the site. The 3DEC model will also incorporate a bonded block model (BBM) approach to capture the strength of the intact rock between the discontinuities. The main goal of this exercise is to simulate the rock mass in the backs of the panels in the most realistic way possible, as the panel back stability will be very important to do a thorough evaluation for the method.
- Numerical models with FLAC3D will be constructed to evaluate the panel retreating sequence. The model will be used to evaluate a number of stability elements; stability of active workplaces (temporary sills), effect of sequence retreat (inclined mining front) and overall stability from beginning to end of mining. Careful consideration will be given to how the placement of backfill is implemented. Some sensitivity analyses on model inputs will also be done; i.e., rock mass strength, in situ stress orientation etc. Outputs from the model will help to validate support design used at Renard.
- Kimberlite weathering and mudrush assessment;
- Ongoing review of geostructural model;
- Continuous update of the hydrogeological model and modeling of structure fault causing water inflow
- Fragmentation studies and optimization;
- Panel sizing optimization;
- Draw point design optimization based on other similar operations;
- Additional REBOP flow modelling to refine recovery and dilution factors as design is updated;
- Update in the underground traffic simulation;
- R4 crown pillar stability analysis;

As underground experience is gained at this mine site, these analyses should be continually reviewed to determine if they remain valid over time and to ensure that new information is properly considered into the analysis.

26.2 Process Plant Design

No further recommendations for plant design. The plant will be further optimized based on experience gained during the operation phase. .

26.3 PKC Facility Design

The following recommendations for further work in support of the PKC facility design were provided by Golder:

- Testing of PK material (post-production) to include, but not necessarily limited to:
  - Grain size analyses;
  - Evaluation of solids content;
  - Atterberg limits;
  - Standard Proctor maximum dry density;
  - Specific gravity;
  - Direct shear;
  - Triaxial testing;
  - Hydraulic conductivity;
  - X-ray diffraction determination of clay mineralogy;
  - Determination of deposited dry density; and
  - Strength testing.
- Additional post-production testing of waste rock, quarried rock and PK to confirm the geochemical classification.

Costs estimated at $30,000/yr to test the PK material and to confirm the geochemical classification are included in the OPEX.

27. SIGNATURE PAGE

This report was prepared and signed by Robin Hopkins, P.Geol. (NT/NU), Patrick Godin, Eng. and Paul M. Bedell, M.E.Sc., P.Eng.

(Signed and sealed) Robin Hopkins, P.Geol. (NT/NU)

Robin Hopkins, P.Geol. (NT/NU)

(Signed and sealed) Patrick Godin, Eng.

Patrick Godin, Eng.

(Signed and sealed) Paul M. Bedell, M.E.Sc., P.Eng

Paul M. Bedell, M.E.Sc., P.Eng

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